

Lecture Notes: Mathematical Analysis of Machine Learning Algorithms

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Preface

This course material contains the lecture notes for MATH6450H/COMP6221F. The course is for students who are already familiar with basic machine learning algorithms such as support vector machines, decision trees, boosting, neural networks etc, and has learned the basic mathematical tools such as calculus, linear algebra, and probability. This is mainly a mathematical class, and the goal is to introduce the basic techniques in the theoretical analysis of machine learning algorithms, with motivations and applications.

Topics include the analysis of supervised learning algorithms, and online learning and bandits. The students are expected to learn the basic mathematical tools to analyze machine learning problems, and apply these tools to analyze various concrete algorithms.

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Chapter 1

Introduction

The goal of mathematical analysis of machine learning algorithms is to study the statistical and computational behaviors of machine learning algorithms, and understand their theoretical properties such as statistical rate of convergence (upper bound), statistical optimality (whether statistical upper bound matches lower bound), computational efficiency, for various learning models under different assumptions.

This course mainly focuses on the analysis of two common learning models: supervised learning and online learning.

In supervised learning, we train a machine learning model using training data, and then evaluate the model's prediction performance on unseen test data. In this case, we want to know the performance of this model on test data.

A mathematical theory for supervised learning answers the following basic questions, where we take linear model as example.

- Suppose that we learn a d -dimensional linear classifier with n training data by minimizing the training error. Assume that the training error is 10%. What is the classifier's test error on the test data? The test error in this setting is referred to as *generalization error*.
- Can we learn a linear classifier that has test error nearly as small as the optimal linear classifier?
- Can we find a computationally efficient procedure to find a linear classifier with small test error?

In online learning, we are interested in a sequential prediction problem, where we train a statistical model using historic data, and then test it on the data in the next step. It then observes the true outcome after prediction. This process is repeated in a sequential manner. In this case, we want to know the performance of this model in the sequential (online) prediction setting.

A mathematical theory for online learning needs to answer the following basic questions, where we again take linear model as example.

- In the online sequential prediction setting. Given a time step t , can we construct an online learning algorithm that predicts nearly as well as the optimal linear classifier up to time step t ?

This course develops the mathematical tools that can be used to answer the above questions.

1.1 Standard Model for Supervised Learning

In supervised learning, we have input random variable (feature vector) $X \in \mathbb{R}^d$ that represents the known information, and output variable (label) Y that represents the unknown information which we want to predict. The goal is to predict Y based on X .

As an example, we may want to predict whether an image (represented as input vector X) contains a cat or a dog (label Y).

In practice, the set of prediction rules are derived by parametrized functions $f(w, \cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^k$, where $w \in \Omega$ is the model parameter that can be learned on the training data. As an example, for k -class classification problem, where $Y \in \{1, \dots, k\}$, we predict Y using the following prediction rule given function $f(w, x) = [f_1(w, x), \dots, f_k(w, x)] \in \mathbb{R}^k$:

$$q(x) = \arg \max_{\ell \in \{1, \dots, k\}} f_\ell(w, x).$$

The prediction quality is measured by a loss function $L(f(x), y)$: the smaller the loss, the better the prediction accuracy.

The supervised learning approach is to estimate $\hat{w} \in \Omega$ based on observed (labeled) historical data $D_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$.

A supervised learning algorithm $\mathcal{A}$ takes a set of training data D_n as input, and outputs a function $f(\hat{w}, \cdot)$, where $\hat{w} = \mathcal{A}(D_n) \in \Omega$. The most common algorithm, which we will focus in this course, is *empirical risk minimization* (ERM):

$$\hat{w} = \arg \min_{w \in \Omega} \sum_{i=1}^n L(f(w, X_i), Y_i). \quad (1.1)$$

In the standard theoretical model for analyzing supervised learning, we assume that the training data $\{(X_i, Y_i) : i = 1, \dots, n\}$ are iid (independent and identically distributed) according to an unknown underlying distribution D . The loss of a classifier $\hat{f}(x) = f(\hat{w}, x)$ on the training data is the training error

$$\text{training-loss}(\hat{w}) = \frac{1}{n} \sum_{i=1}^n L(f(\hat{w}, X_i), Y_i).$$

Moreover, we assume that the test data (X, Y) (future unseen data) are also taken from the same distribution D , and we are interested in the generalization error of $\hat{f}$ on the test data, defined as:

$$\text{test-loss}(\hat{w}) = \mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y).$$

Since we only observe the training error of $\hat{f} = f(\hat{w}, \cdot)$, a major goal is to estimate the generalization error of $\hat{f}$ based on its training error, referred to as *generalization bound*, which is of the following form. Given $\epsilon \geq 0$, we want to determine $\delta_n(\epsilon)$ so that:

$$\Pr \left(\mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) \geq \frac{1}{n} \sum_{i=1}^n L(f(\hat{w}, X_i), Y_i) + \epsilon \right) \leq \delta_n(\epsilon),$$

where the probability is with respect to the randomness over the training data D_n . In general, $\delta_n(\epsilon) \rightarrow 0$ as $n \rightarrow \infty$.

In the literature, the above result is often stated in the following alternative form, where we want to determine a function $\epsilon_n(\delta)$ of δ , so that with probability at least $1 - \delta$ (over training data D_n):

$$\mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(\hat{w}, X_i), Y_i) + \epsilon_n(\delta). \quad (1.2)$$

We hope $\epsilon_n(\delta) \rightarrow 0$ as $n \rightarrow \infty$.

Another type of inequalities, often referred to as *oracle inequality*, is to show that with probability at least $1 - \delta$ (over training data D_n):

$$\mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) \leq \inf_{w \in \Omega} \mathbf{E}_{(X,Y) \sim D} L(f(w, X), Y) + \epsilon_n(\delta). \quad (1.3)$$

This shows that the test error achieved by the learning algorithm is nearly as small as that of the optimal test error achieved by $f(w, x)$ with $w \in \Omega$. We say the learning algorithm is consistent if $\epsilon_n(\delta) \rightarrow 0$ as $n \rightarrow \infty$. Moreover, the rate of convergence refers to the dependency of $\epsilon_n(\delta)$ on n .

Chapter 2 and Chapter 3 establish the basis mathematical tools for analyzing supervised learning. Chapter 4 and Chapter 6 further develops the techniques. Chapter 7 analyzes the kernel methods. regression, and boosting. Chapter 10 investigates the statistical lower and upper bounds for some learning problems.

1.2 Online Learning

In online learning, we consider observing (X_t, Y_t) one by one in a time sequence from $t = 1, 2, \dots$

In online learning, an online algorithm $\mathcal{A}$ learns a model parameter $\hat{w}_t$ at time t based on previously observed data $(X_1, Y_1), \dots, (X_t, Y_t)$: $\hat{w}_t =$

$\mathcal{A}(\{(X_1, Y_1), \dots, (X_t, Y_t)\})$. We then observe the next input vector X_{t+1} , and make prediction $f(\hat{w}_t, X_{t+1})$. After the prediction, we observe Y_{t+1} , and then compute the loss $L(f(\hat{w}_t, X_{t+1}), Y_{t+1})$. The goal of online learning is to minimize the aggregated loss

$$\sum_{t=0}^{T-1} L(f(\hat{w}_t, X_{t+1}), Y_{t+1}).$$

In the mathematical analysis of online algorithm, we are interested in the following inequality, referred to as *regret bound*, where the aggregated loss of an online algorithm is compared to the optimal aggregated loss:

$$\sum_{t=0}^{T-1} L(f(\hat{w}_t, X_{t+1}), Y_{t+1}) \leq \inf_{w \in \Omega} \sum_{t=0}^{T-1} L(f(w, X_{t+1}), Y_{t+1}) + \epsilon_T. \quad (1.4)$$

The regret ϵ_T , is the extra loss of the learning algorithm, compared to that of the optimal model at time T in retrospect.

As an example, we consider the stock price prediction problem, where the opening price of a certain stock at each trading day is $p_1, p_2, \dots$. At the beginning of each day t , we observe $p_1, \dots, p_t$, and want to predict p_{t+1} on day $t+1$, so that we use this prediction to trade the stock.

The input X_{t+1} is a d -dimensional real valued vector in $\mathbb{R}^d$ that represents the observed historical information of the stock on day t . The output $Y_{t+1} = \ln(p_{t+1}/p_t)$ will be observed on day $t+1$. We consider linear model with $f(w, x) = w^\top x$, with $\Omega = \mathbb{R}^d$. The quality is measured by the least squares error $L(f(w, X_{t+1}), Y_{t+1}) = (f(w, X_{t+1}) - Y_{t+1})^2$.

The learning algorithm can be empirical risk minimization, where

$$\hat{w}_t = \arg \min_{w \in \mathbb{R}^d} \frac{1}{t} \sum_{i=1}^t (w^\top X_i - Y_i)^2.$$

In regret analysis, we compare the prediction error

$$\sum_{t=0}^{T-1} (\hat{w}_t^\top X_{t+1} - Y_{t+1})^2$$

to the optimal prediction

$$\inf_{w \in \mathbb{R}^d} \sum_{t=0}^{T-1} (w^\top X_{t+1} - Y_{t+1})^2.$$

The online learning model will be studied in Chapter 11 and Chapter 12. . The related bandits problem will be investigated in Chapter 13 and Chapter 14. In the bandits problem, we investigate online problems with incomplete information. The goal is to take an action at each time step, but only observe the outcome of an action that is taken.

1.3 Computational Consideration

In the ERM method, the model parameter $\hat{w}$ is the solution of an optimization problem. If the optimization problem is convex, then the solution can be efficiently computed. If the optimization problem is non-convex, then its solution may not be obtained easily.

Theoretically, we separately consider two different types of complexity. One is statistical complexity, where we may ignore the complexity of computation, and try to derive bounds (1.3) and (1.4) even though the computational complexity of the underlying learning algorithm (such as ERM) may be high.

Another is computational complexity, where we are interested in computationally efficient algorithms with good generalization performance or regret bounds. For non-convex models, this kind of analysis can be rather complexity, with problem specific approaches.

One general approach is to use convex approximation (also referred to convex relaxation) to solve the non-convex problem approximately, and the theoretical question is that under what circumstances, the solution has statistical generalization performance comparable to that of the non-convex method. An example is the sparse learning problem, where the convex formulation with L_1 regularization is used as a proxy to the non-convex L_0 regularization. In this case, we are interested in establishing the condition under which one can obtain estimated solution from L_1 regularization close to the true sparse model.

The combined analysis of computational and statistical complexity is a major research direction in theoretical machine learning. The computational complexity will be considered throughout the course when practical algorithms are investigated.

1.4 Basic Concepts in Generalization Analysis

The goal of machine learning is to find a function $f(\hat{w}, x)$ that predicts well on unseen data (test data). However, we only observe the prediction accuracy of $f(\hat{w}, x)$ on the training data. In order to achieve high prediction accuracy, we need to balance the following two aspects of learning:

- The prediction function should fit the training data well; that is, achieving small training error. This requires a more expressive model, with a larger parameter space Ω .
- Performance of prediction function on the test data should match that on the training data. The difference is smaller for a less expressive model with a smaller parameter space Ω .

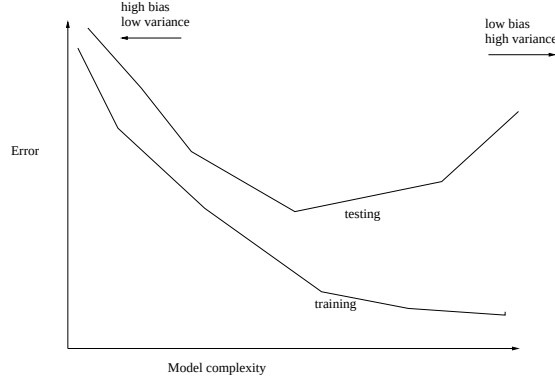


Figure 1.1: Training and test errors versus model complexity

The gap between the training and test errors depend on the model complexity, which characterizes how large the model parameter space Ω is. When Ω is too large, the training error becomes smaller, but the difference between training and test errors increases. Therefore in practice there is a trade-off in machine learning, and the best prediction performance is achieved with the right balance, often via a tuning parameter in the learning algorithm that characterizes model complexity. The phenomenon is described in Figure 1.1. Such a tuning process is often referred to as hyperparameter optimization.

When the class of prediction functions is too large (or complex), then the difference between training and test errors increases. This leads to so-called overfitting phenomenon. A simple example for overfitting can be described as follows. Let X be a one-dimensional feature uniformly distributed in $[-1, 1]$, with class label $Y = 1$ when $X \geq 0$ and $Y = -1$ when $X < 0$.

Given training data (X_i, Y_i) ($i = 1, \dots, n$), and assume X_i are all different. If we consider the prediction function class containing all possible functions, then the empirical risk minimization with the following solution can fit data perfectly:

$$\hat{f}(X) = \begin{cases} Y_i & \text{if } X = X_i \text{ for some } i \\ 1 & \text{otherwise} \end{cases}$$

The above model is very complex (model complexity characterized using learning theory). However, this prediction rule makes no meaningful prediction when X not in the training data. This is because although the training error of 0 is small, it is significantly different from the test error of 0.5.

In contrast, if we let the prediction model contains only one function $\{f(x) : f(x) \equiv 0\}$, then using the tail inequality of independent random variables of Chapter 2, we know that the difference between training and test error will small when n is large. However, since the training error of ≈ 0.5 is large, the test error is also large.

Let $I(x \in A)$ be the set indicator function that takes value 1 if $x \in A$, and 0 if $x \notin A$. Assume that we pick the function class $\{f(w, x) : f(w, x) = 2I(x \geq w) - 1\}$ parametrized by a parameter $w \in R$, and then finds a classifier $f(\hat{w}, x)$ that minimizes the training error. Using techniques of this course, it can be shown that both training and test error of this classifier converge to zero when $n \rightarrow \infty$. This model balances the training error and generalization performance. Learning theory can be used to estimate the generalization performance (prediction accuracy on unseen data) of learning algorithms, and quantify the degree of overfitting.

Chapter 2

Basic Probability Inequalities for Random Variables

We derive exponential tail probability inequalities for sums of independent variables. These inequalities are the basic tools that can be used to analyze machine learning algorithms.

Let $X_1, \dots, X_n$ be n iid random variables, with mean $\mu = \mathbf{E}X_i$. Let $\bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$. Given $\epsilon > 0$, we are interested in estimating the tail probability

$$\begin{aligned} \Pr(\bar{X}_n \geq \mu + \epsilon) \\ \Pr(\bar{X}_n \leq \mu - \epsilon). \end{aligned}$$

In machine learning, we can think $\bar{X}_n$ as training error observed on the training data. The unknown mean μ is the test error which we want to minimize. Therefore in machine learning, these tail inequalities can be interpreted as follows: with high probability, the test error is close to training error. Such results can be used to derive rigorous statements of generalization error bounds.

2.1 Gaussian Random Variable

Let X be a standard Gaussian random variable: $X \sim N(0, 1)$, with density function

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

For $\epsilon > 0$, we want to estimate the tail probability $\Pr(X \geq \epsilon)$ as follows.

$$\begin{aligned}\Pr(X \geq \epsilon) &= \int_{\epsilon}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x+\epsilon)^2/2} dx \leq \int_0^{\infty} \frac{1}{\sqrt{2\pi}} e^{-(x^2+\epsilon^2)/2} dx \\ &= 0.5e^{-\epsilon^2/2}.\end{aligned}$$

We also have the following bound:

$$\begin{aligned}\Pr(X \geq \epsilon) &= \int_{\epsilon}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &\geq \int_0^1 \frac{1}{\sqrt{2\pi}} e^{-(x+\epsilon)^2/2} dx \geq 0.34e^{-(2\epsilon+\epsilon^2)/2} \\ &\geq 0.5e^{-(\epsilon+1)^2/2}.\end{aligned}$$

Therefore we have

$$0.5e^{-(\epsilon+1)^2/2} \leq \Pr(X \geq \epsilon) \leq 0.5e^{-\epsilon^2/2}.$$

Now let $X_1, \dots, X_n$ be n iid Gaussians $X_i \sim N(\mu, \sigma^2)$, and let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$. Then

$$\Pr(\bar{X}_n \geq \mu + \epsilon) = \Pr(\sqrt{n}(\bar{X}_n - \mu)/\sigma \geq \sqrt{n}\epsilon/\sigma).$$

Since $\sqrt{n}(\bar{X}_n - \mu)/\sigma \sim N(0, 1)$, an estimate of the above tail bound can be obtained from the tail bound of the Gaussian variable:

$$0.5e^{-n(\epsilon+\sigma/\sqrt{n})^2/2\sigma^2} \leq \Pr(\bar{X}_n \geq \mu + \epsilon) \leq 0.5e^{-n\epsilon^2/2\sigma^2}.$$

This tail probability decays exponentially fast, and thus called exponential inequality. The upper bound is tight in the following sense. For any $\epsilon > 0$, we have

$$\lim_{n \rightarrow \infty} n^{-1} \ln \Pr(|\bar{X}_n - \mu| \geq \epsilon) = -\epsilon^2/2\sigma^2.$$

This is a large deviation result, meaning that the deviation ϵ from the mean is much larger than standard deviation $\sigma/\sqrt{n}$ of $\bar{X}$ when $n \rightarrow \infty$.

2.2 Markov Inequality

More generally, let $X_1, \dots, X_n$ be n iid random variables (not necessarily Gaussians) with mean μ . Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$, we are interested in estimating the tail bound $\Pr(\bar{X}_n \geq \mu + \epsilon)$ and $\Pr(\bar{X}_n \leq \mu - \epsilon)$ for some $\epsilon > 0$.

Generally, such estimates can be obtained through Markov inequality.

Theorem 2.1 (Markov Inequality) *Given any non-negative function $g(x) \geq 0$, we have for set S :*

$$\Pr(\bar{X}_n \in S) \leq \frac{\mathbf{E} g(\bar{X}_n)}{\inf_{x \in S} g(x)}.$$

Proof Since $g(x)$ is non-negative, we have

$$\mathbf{E} g(\bar{X}_n) \geq \mathbf{E}_{\bar{X}_n \in S} g(\bar{X}_n) \geq \mathbf{E}_{\bar{x}_n \in S} g_S = \Pr(\bar{X}_n \in S) g_S,$$

where $g_S = \inf_{x \in S} g(x)$. This leads to the desired bound. $\blacksquare$

In particular, we consider the choice of $g(z) = z^2$, which leads to the following equation

$$\mathbf{E} g(\bar{X}_n - \mu) = \mathbf{E}(\bar{X}_n - \mu)^2 = \frac{1}{n} \mathbf{Var}(X_1).$$

We obtain from Markov inequality with $S = \{|\bar{X}_n - \mu| \geq \epsilon\}$ the following Chebyshev inequality:

$$\Pr(|\bar{X}_n - \mu| \geq \epsilon) \leq \frac{\mathbf{Var}(X_1)}{n\epsilon^2}. \quad (2.1)$$

2.3 Exponential Tail Inequality

In order to get exponential tail bounds, we choose $g(z) = e^{\lambda n z}$ with some tuning parameter $\lambda > 0$. Given $\epsilon > 0$, then the Markov inequality of Theorem 2.1 with $S = \{\bar{X}_n - \mu \geq \epsilon\}$ becomes

$$\begin{aligned} \Pr(\bar{X}_n \geq \mu + \epsilon) &\leq \frac{\mathbf{E} e^{\lambda n \bar{X}_n}}{e^{\lambda n(\mu + \epsilon)}} = \frac{\mathbf{E} e^{\lambda \sum_{i=1}^n X_i}}{e^{\lambda n(\mu + \epsilon)}} \\ &= \frac{\mathbf{E} \prod_{i=1}^n e^{\lambda X_i}}{e^{\lambda n(\mu + \epsilon)}} = e^{-\lambda n(\mu + \epsilon)} \left[\mathbf{E} e^{\lambda X_1} \right]^n. \end{aligned}$$

Note that in order to use this estimate, we have to assume that $\mathbf{E} e^{\lambda X_1} < \infty$ for some $\lambda > 0$. By taking logarithm, we obtain the following theorem.

Theorem 2.2 *For any n and $\epsilon > 0$:*

$$n^{-1} \ln \Pr(\bar{X}_n \geq \mu + \epsilon) \leq \inf_{\lambda > 0} \left[-\lambda \epsilon + \ln \mathbf{E} e^{\lambda(X_1 - \mu)} \right].$$

Similarly

$$n^{-1} \ln \Pr(\bar{X}_n \leq \mu - \epsilon) \leq \inf_{\lambda < 0} \left[\lambda \epsilon + \ln \mathbf{E} e^{\lambda(X_1 - \mu)} \right].$$

The function $\Gamma_{X_1}(\lambda) = \ln \mathbf{E} e^{\lambda X_1}$ is called *logarithmic moment generating function* of a random variable X_1 .

Exponential tail inequality for sum of independent random variables can be obtained by applying Theorem 2.2 for specific situations. It simplifies the problem because the formula allows us to change the problem of estimating the exponential moment of the sum of independent random variables into the estimating of the exponential moment of a single random variable. Another way to write tail bound is

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp[-nI_{X_1}(\mu + \epsilon)],$$

where given a random variable X , the rate function $I_X(z)$ is defined as

$$-I_X(z) = \inf_{\lambda > 0} [-\lambda z + \ln \mathbf{E}_X e^{\lambda X}]. \quad (2.2)$$

Example 2.1 (Gaussian Random Variable) Assume that $X_i \sim N(\mu, \sigma^2)$, then the exponential moment is

$$\begin{aligned} \mathbf{E} e^{\lambda(X_1 - \mu)} &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{\lambda x} e^{-x^2/2\sigma^2} dx \\ &= \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{\lambda^2 \sigma^2/2} e^{-(x/\sigma - \lambda\sigma)^2/2} dx / \sigma = e^{\lambda^2 \sigma^2/2}. \end{aligned}$$

Therefore,

$$-I_{X_1}(\mu + \epsilon) = \inf_{\lambda > 0} [-\lambda\epsilon + \ln \mathbf{E} e^{\lambda(X_1 - \mu)}] = \inf_{\lambda > 0} \left[-\lambda\epsilon + \frac{\lambda^2 \sigma^2}{2} \right] = -\frac{\epsilon^2}{2\sigma^2},$$

where the optimal λ is achieved at $\lambda = \epsilon/\sigma^2$ below. This leads to the same probability bound as that in Section 2.1, which was derived by exact integration.

In the large deviation situation, the exponential Markov inequality is asymptotically tight in the following sense.

Theorem 2.3 For all $\epsilon' > \epsilon > 0$:

$$\underline{\lim}_{n \rightarrow \infty} n^{-1} \ln \Pr(\bar{X}_n \geq \mu + \epsilon) \geq -I_{X_1}(\mu + \epsilon').$$

Similarly,

$$\underline{\lim}_{n \rightarrow \infty} n^{-1} \ln \Pr(\bar{X}_n \leq \mu - \epsilon) \geq -I_{-X_1}(\mu + \epsilon').$$

Proof We only need to prove the first inequality. Consider $\Pr(X_i \leq x)$ as a function of x , and define a random variable X'_i with density at x as

$$d\Pr(X'_i \leq x) = e^{\lambda x - \Gamma_{X_1}(\lambda)} d\Pr(X_i \leq x).$$

This choice implies that

$$\frac{d}{d\lambda}\Gamma_{X_1}(\lambda) = \frac{\int x e^{\lambda x} d\Pr(X'_1 \leq x)}{\int e^{\lambda x} d\Pr(X'_1 \leq x)} = \mathbf{E}_{X'_1} X'_1.$$

Therefore if we take

$$\lambda = \arg \min_{\lambda} [-\lambda(\mu + (\epsilon' + \epsilon)/2) + \Gamma_{X_1}(\lambda)],$$

then

$$\mathbf{E}_{X'_1} X'_1 = \frac{d}{d\lambda}\Gamma_{X_1}(\lambda) = \mu + 0.5(\epsilon + \epsilon').$$

Let $\bar{X}'_n = n^{-1} \sum_{i=1}^n X'_i$. Then by the law of large numbers, we know that

$$\lim_{n \rightarrow \infty} \Pr(\bar{X}'_n - \mu \in [\epsilon, \epsilon']) = 1. \quad (2.3)$$

Since the joint density of $(X'_1, \dots, X'_n)$ satisfies

$$e^{-n\lambda \sum_i x_i + n\Gamma_{X_1}(\lambda)} \prod_i d\Pr(X'_i \leq x_i) = \prod_i d\Pr(X_i \leq x_i),$$

by using $I(\cdot)$ to denote the set indicator function, we obtain

$$\begin{aligned} \Pr(\bar{X}_n \geq \mu + \epsilon) &\geq \Pr(\bar{X}_n - \mu \in [\epsilon, \epsilon']) \\ &= \mathbf{E}_{X_1, \dots, X_n} I(\bar{X}_n - \mu \in [\epsilon, \epsilon']) \\ &= \mathbf{E}_{X'_1, \dots, X'_n} e^{-\lambda n \bar{X}'_n + n\Gamma(\lambda)} I(\bar{X}'_n - \mu \in [\epsilon, \epsilon']) \\ &\geq e^{-\lambda n \epsilon' + n\Gamma(\lambda)} \Pr(\bar{X}'_n - \mu \in [\epsilon, \epsilon']). \end{aligned}$$

Now we obtain the desired bound by letting $n \rightarrow \infty$ and applying (2.3). ■

For specific cases, one can obtain an estimate of $\Pr(\bar{X}'_n - \mu \in [\epsilon, \epsilon'])$ in (2.3) with finite n at $\epsilon' = \epsilon + \sqrt{\mathbf{Var}(X_1)/n}$. This means that the exponential tail inequality of Theorem 2.2 is generally loose by no more than $O(\sqrt{\mathbf{Var}(X_1)/n})$ in terms of deviation ϵ .

2.4 Sub-Gaussian Random Variable

Definition 2.1 A sub-Gaussian random variable X_1 has quadratic logarithmic moment generating function:

$$\ln \mathbf{E} e^{\lambda X_1} \leq \lambda \mu + \frac{\lambda^2}{2} b. \quad (2.4)$$

In this case, we have for any $z > \mu$:

$$-I_{X_1}(z) = \inf_{\lambda > 0} (-\lambda z + \lambda \mu + \frac{\lambda^2}{2} b).$$

Taking derivative with respect to λ , we obtain the following condition at the optimal λ_* :

$$-z + \mu = \lambda_* b,$$

which implies that $\lambda_* = (\mu - z)/b$ and

$$I_{X_1}(z) = \frac{(z - \mu)^2}{2b}.$$

Theorem 2.2 implies the following tail probability bound for sub-Gaussian random variables:

Theorem 2.4 *If X_1 is sub-Gaussian as in (2.4), then for all $\epsilon > 0$:*

$$\begin{aligned} \Pr(\bar{X}_n \geq \mu + \epsilon) &\leq e^{-n\epsilon^2/2b} \\ \Pr(\bar{X}_n \leq \mu - \epsilon) &\leq e^{-n\epsilon^2/2b}. \end{aligned}$$

Example 2.2 *Gaussian random variable $X_1 \sim N(\mu, \sigma^2)$ is sub-Gaussian with $b = \sigma^2$.*

Example 2.3 *Consider a bounded random variable: $X_1 \in [\alpha, \beta]$. Then X_1 is sub-Gaussian with $b = (\beta - \alpha)^2/4$.*

The tail probability inequality of Theorem 2.4 can also be expressed in a different form. Consider $\delta \in (0, 1)$ such that

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp(-n\epsilon^2/2b) = \delta.$$

We can solve for $\epsilon = \sqrt{(2b/n) \ln(1/\delta)}$. This means that we can alternatively express the bound of Theorem 2.4 as follows. With probability at least $1 - \delta$, we have

$$\bar{X}_n \leq \mu + \sqrt{\frac{2b \ln(1/\delta)}{n}}.$$

2.5 Chernoff Bounds

Chernoff bound, or Hoeffding's inequality [43], is an exponential tail inequality for bounded random variables. In this section, we consider a random variable $X \in [0, 1]$ and $\mathbf{E}X = \mu$.

Lemma 2.1 *We have the following inequality*

$$\mathbf{E}e^{\lambda X} \leq (1 - \mu)e^0 + \mu e^\lambda,$$

Moreover,

$$\ln[(1 - \mu)e^0 + \mu e^\lambda] \leq \lambda\mu + \lambda^2/8.$$

Proof Let $g_L(\lambda) = \mathbf{E}e^{\lambda X}$ and $g_R(\lambda) = (1 - \mu)e^0 + \mu e^\lambda$. We know that $g_L(0) = g_R(0)$. Moreover, $g'_L(\lambda) = \mathbf{E}Xe^{\lambda X} \leq \mathbf{E}Xe^{\lambda*1} = \mu e^\lambda = g'_R(\lambda)$. This proves the first inequality.

Now we let

$$g(\lambda) = \ln[(1 - \mu)e^0 + \mu e^\lambda],$$

then

$$g'(\lambda) = \frac{\mu e^\lambda}{(1 - \mu)e^0 + \mu e^\lambda},$$

and

$$\begin{aligned} g''(\lambda) &= \frac{\mu e^\lambda}{(1 - \mu)e^0 + \mu e^\lambda} - \frac{(\mu e^\lambda)^2}{[(1 - \mu)e^0 + \mu e^\lambda]^2} \\ &= |g'(\lambda)|(1 - |g'(\lambda)|) \leq 1/4. \end{aligned}$$

Using Taylor expansion, we obtain the inequality $g(\lambda) \leq g(0) + \lambda g'(0) + \lambda^2/8$, which implies the second inequality. $\blacksquare$

The lemma implies that the maximum logarithmic moment generating function of a random variable X_1 taking values in $[0, 1]$ is achieved with $\{0, 1\}$ valued random variable of the same mean. Moreover, the random variable X_1 is sub-Gaussian. We can then apply the sub-Gaussian tail-inequality in Theorem 2.4 to obtain the Chernoff bound.

Theorem 2.5 (Additive Forms of Chernoff Bounds) *Assume that $X_1 \in [0, 1]$. Then for all $\epsilon > 0$:*

$$\begin{aligned} \Pr(\bar{X}_n \geq \mu + \epsilon) &\leq e^{-2n\epsilon^2} \\ \Pr(\bar{X}_n \leq \mu - \epsilon) &\leq e^{-2n\epsilon^2}. \end{aligned}$$

In some applications, one often needs to employ a more refined form of Chernoff bound, which can be stated as follows.

Theorem 2.6 Assume that $X_1 \in [0, 1]$. Then for all $\epsilon > 0$, we have

$$\begin{aligned}\Pr(\bar{X}_n \geq \mu + \epsilon) &\leq e^{-n\text{KL}(\mu+\epsilon||\mu)} \\ \Pr(\bar{X}_n \leq \mu - \epsilon) &\leq e^{-n\text{KL}(\mu-\epsilon||\mu)},\end{aligned}$$

where $\text{KL}(z||\mu)$ is the Kullback-Leibler divergence (*KL divergence*) defined as

$$\text{KL}(z||\mu) = z \ln \frac{z}{\mu} + (1 - z) \ln \frac{1 - z}{1 - \mu}.$$

Proof Consider the case $z = \mu + \epsilon$.

$$-I_{X_1}(z) \leq \inf_{\lambda > 0} [-\lambda z + \ln((1 - \mu)e^0 + \mu e^\lambda)].$$

If the right hand side optimum is achieved at λ_* , by setting the derivative to zero, we obtain:

$$z = \frac{\mu e^{\lambda_*}}{(1 - \mu)e^0 + \mu e^{\lambda_*}},$$

which implies that

$$e^{\lambda_*} = \frac{z(1 - \mu)}{\mu(1 - z)}.$$

This implies that $-I(z) \leq -\text{KL}(z||\mu)$. The case of $z = \mu - \epsilon$ is similar. We thus obtained the desired bound from Theorem 2.2. ■

Theorem 2.6 can be simplified as in the following corollary. It is superior to Theorem 2.5 when $\mu \approx 0$, and this leads to better bounds in some applications.

Corollary 2.1 (Multiplicative Form of Chernoff Bounds) Assume that $X_1 \in [0, 1]$. Then for all $\epsilon > 0$:

$$\begin{aligned}P(\bar{X}_n \geq \mu + \epsilon) &\leq \exp[-n\epsilon^2/(2\mu + \epsilon)] \\ P(\bar{X}_n \leq \mu - \epsilon) &\leq \exp[-n\epsilon^2/(2\mu)].\end{aligned}$$

Proof The result follows from Theorem 2.6 and the estimate $\text{KL}(z||\mu) \geq (z - \mu)^2 / \max(z + \mu, 2\mu)$. ■

The multiplicative form of Chernoff bound can be expressed alternatively as follows. With probability at least $1 - \delta$:

$$\mu \leq \bar{X}_n + \sqrt{\frac{2\mu \ln(1/\delta)}{n}},$$

which implies that for any $\gamma \in (0, 1)$:

$$(1 - \gamma)\mu \leq \bar{X}_n + \frac{\ln(1/\delta)}{2\gamma n}.$$

Moreover, with probability at least $1 - \delta$:

$$\bar{X}_n \leq \mu + \sqrt{\frac{2\mu \ln(1/\delta)}{n}} + \frac{\ln(1/\delta)}{n},$$

which implies that for any $\gamma \in (0, 1)$:

$$\bar{X}_n \leq (1 + \gamma)\mu + \frac{(1 + 2\gamma) \ln(1/\delta)}{2\gamma n}.$$

2.6 Bennett's Inequality

In Bennett's inequality, we assume that the variable is upper bounded, and want to estimate its moment generating function using the variance information [16].

Lemma 2.2 *If $X - \mathbf{E}X \leq 1$, then $\forall \lambda \geq 0$:*

$$\ln \mathbf{E}e^{\lambda X} \leq \lambda \mathbf{E}X + (e^\lambda - \lambda - 1) \mathbf{Var}(X).$$

Proof Let $X' = X - \mathbf{E}X$. Using $\ln z \leq z - 1$, we have

$$\begin{aligned} \ln \mathbf{E}e^{\lambda X} &= \lambda \mathbf{E}X + \ln \mathbf{E}e^{\lambda X'} \leq \lambda \mathbf{E}X + \mathbf{E}e^{\lambda X'} - 1 \\ &= \lambda \mathbf{E}X + \lambda^2 \mathbf{E} \frac{e^{\lambda X'} - \lambda X' - 1}{(\lambda X')^2} (X')^2 \\ &\leq \lambda \mathbf{E}X + \lambda^2 \mathbf{E} \frac{e^\lambda - \lambda - 1}{\lambda^2} (X')^2, \end{aligned}$$

where the second inequality follows from the fact that the function $(e^z - z - 1)/z^2$ is non-decreasing and $\lambda X' \leq \lambda$. $\blacksquare$

The above lemma gives an estimate of the logarithmic moment generating function, which can be used to estimate the rate function.

Lemma 2.3 *If $X - \mathbf{E}X \leq 1$, then*

$$\begin{aligned} -I_{X_1}(\mathbf{E}X + \epsilon) &\leq \inf_{\lambda > 0} \left[-\lambda \epsilon + (e^\lambda - \lambda - 1) \mathbf{Var}(X) \right] \\ &= -\mathbf{Var}(X) \phi \left(\frac{\epsilon}{\mathbf{Var}(X)} \right) \leq -\frac{\epsilon^2}{2(\mathbf{Var}(X) + \epsilon/3)}, \end{aligned}$$

where $\phi(z) = (1 + z) \ln(1 + z) - z$.

Proof To compute $\inf_{\lambda>0}(\cdot)$, we can set the derivative of the objective function with respect to λ to zero at the minimum solution, and obtain the condition for the optimal λ :

$$-\epsilon + (e^\lambda - 1)\mathbf{Var}(X) = 0.$$

This gives the solution $\lambda = \ln(1 + \epsilon/\mathbf{Var}(X))$. Plugging this solution into the objective function, we obtain the equality.

Given $\lambda \in (0, 3)$, it is easy to verify the following inequality using the Taylor expansion of the exponential function

$$e^\lambda - \lambda - 1 \leq \frac{\lambda^2}{2} \sum_{m=0}^{\infty} (\lambda/3)^m = \frac{\lambda^2}{2(1 - \lambda/3)}.$$

Now by picking $\lambda = \epsilon/(\mathbf{Var}(X) + \epsilon/3)$, we obtain

$$-\lambda\epsilon + \frac{\lambda^2}{2(1 - \lambda/3)} = -\epsilon^2/[2\mathbf{Var}(X) + 2\epsilon/3].$$

This proves the desired bound. ■

The above bound implies the following result from Theorem 2.2.

Theorem 2.7 (Bennett's Inequality) *If $X_1 \leq \mu + b$, for some $b > 0$, then $\forall \epsilon > 0$:*

$$\Pr[\bar{X}_n \geq \mu + \epsilon] \leq \exp \left[\frac{-n\epsilon^2}{2\mathbf{Var}(X_1) + 2\epsilon b/3} \right].$$

Bennett's inequality can be expressed alternatively as follows. Given any $\delta \in (0, 1)$, with probability at least $1 - \delta$, we have

$$\bar{X}_n \leq \mu + \sqrt{\frac{2\mathbf{Var}(X_1) \ln(1/\delta)}{n}} + \frac{2b \ln(1/\delta)}{3n}.$$

Compared to the bound for Gaussian random variables, this form of Bennett's inequality has an extra term $2b \ln(1/\delta)/(3n)$, which is of higher order in $1/n$. Compared to the Chernoff bound, the Bennett's inequality is superior when $\mathbf{Var}(X_1)$ is small.

2.7 Bernstein Inequalities

In Bernstein inequalities, we obtain results similar to the Bennett's inequality, but using a moment condition[17]. There are several different forms of the inequalities, and we only consider one form, which relies on the following moment assumption.

Lemma 2.4 *If X satisfies the following moment condition with $b, V > 0$ for integers $m \geq 2$:*

$$\mathbf{E}[X - c]^m \leq m!(b/3)^{m-2}V/2,$$

where c is arbitrary. Then when $\lambda \in (0, 3/b)$:

$$\ln \mathbf{E}e^{\lambda X} \leq \lambda \mathbf{E}X + 0.5\lambda^2 V(1 - \lambda b/3)^{-1}.$$

Proof We have the following estimation of logarithmic moment generating function:

$$\begin{aligned} \ln \mathbf{E}e^{\lambda X} &\leq \lambda c + \mathbf{E}e^{\lambda(X-c)} - 1 \leq \lambda \mathbf{E}X + 0.5V\lambda^2 \sum_{m=2}^{\infty} (b/3)^{m-2} \lambda^{m-2} \\ &= \lambda \mathbf{E}X + 0.5\lambda^2 V(1 - \lambda b/3)^{-1}. \end{aligned}$$

This implies the desired bound. ■

In general we may take $c = \mathbf{E}[X]$ and $V = \mathbf{Var}[X]$.

We can use the similar proof as in Lemma 2.3 for the Bennett's inequality to obtain

$$-I_X(\mu + \epsilon) \leq -\frac{\epsilon^2}{2(V + \epsilon b/3)}.$$

This leads to the following tail probability from Theorem 2.2.

Theorem 2.8 (Bernstein's Inequalities) *Assume that X_1 satisfies the moment condition in Lemma 2.4. Then for all $\epsilon > 0$:*

$$\begin{aligned} \Pr[\bar{X}_n \geq \mu + \epsilon] &\leq \exp \left[\frac{-n\epsilon^2}{2V + 2\epsilon b/3} \right] \\ \Pr[\bar{X}_n \leq \mu - \epsilon] &\leq \exp \left[\frac{-n\epsilon^2}{2V + 2\epsilon b/3} \right]. \end{aligned}$$

Similar to Bennet's inequality, Bernstein's inequalities have alternatively expressions as follows. With probability at least $1 - \delta$,

$$\bar{X}_n \leq \mu + \sqrt{\frac{2V \ln(1/\delta)}{n}} + \frac{2b \ln(1/\delta)}{3n},$$

and with probability at least $1 - \delta$,

$$\mu \leq \bar{X}_n + \sqrt{\frac{2V \ln(1/\delta)}{n}} + \frac{2b \ln(1/\delta)}{3n}.$$

Example 2.4 *If the random variable X is bounded with $|X - \mu| \leq b$, then the moment condition of Lemma 2.4 holds with $c = \mu$ and $V = \mathbf{Var}(X)$.*

2.8 Non-identically Distributed Random Variables

If $X_1, \dots, X_n$ are independent but not identically distributed random variables, then a tail inequality similar to that of Theorem 2.2 holds. Let $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$, and $\mu = \mathbf{E}\bar{X}_n$, then we have the following bound.

Theorem 2.9 *We have*

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \inf_{\lambda > 0} \left[-\lambda n(\mu + \epsilon) + \sum_{i=1}^n \ln \mathbf{E} e^{\lambda X_i} \right].$$

For sub-Gaussian random variables, we have the following bound.

Corollary 2.2 *If X_i are sub-Gaussians with $\ln \mathbf{E} e^{\lambda X_i} \leq \lambda \mathbf{E} X_i + 0.5 \lambda^2 b_i$, then for all $\epsilon > 0$:*

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp \left[-\frac{n^2 \epsilon^2}{2 \sum_{i=1}^n b_i} \right].$$

The following is a useful example of the above bound for Rademacher average.

Example 2.5 *Let $\sigma_i = \{\pm 1\}$ be independent random Bernoulli variables, and let a_i be fixed numbers ($i = 1, \dots, n$). Then*

$$\Pr \left(n^{-1} \sum_{i=1}^n \sigma_i a_i \geq \epsilon \right) \leq \exp \left[-\frac{n \epsilon^2}{2 n^{-1} \sum_{i=1}^n a_i^2} \right].$$

One can also derive a Bennett style tail probability bound.

Corollary 2.3 *If $X_i - \mathbf{E} X_i \leq b$ for all i , then*

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp \left[-\frac{n^2 \epsilon^2}{2 \sum_{i=1}^n \mathbf{Var}(X_i) + 2nb\epsilon/3} \right].$$

2.9 Exercises

1. Assume that $X_1, X_2, \dots, X_n$ are iid random variables with pdf

$$p(x) = \frac{x^2}{\sqrt{2\pi}} \exp(-x^2/2).$$

Let $\mu = \mathbf{E}X_1$, and $\bar{X}_n = n^{-1} \sum_{i=1}^n X_i$.

- Estimate $\ln \mathbf{E} \exp(\lambda X_1)$
 - Estimate $\Pr(\bar{X}_n \geq \mu + \epsilon)$
 - Estimate $\Pr(\bar{X}_n \leq \mu - \epsilon)$
2. Fill in the missing proof of the multiplicative Chernoff bounds.
3. Assume that the density of D is $(1-p)U(-1, 1) + pU(-1/p, 1/p)$ for $p \in (0, 0.5)$, where $U(\cdot)$ denotes the uniform distribution. Let $X_1, \dots, X_n$ be iid samples from D . For $\epsilon > 0$, estimate the probability

$$\Pr \left(\frac{1}{n} \sum_{i=1}^n X_i \geq \epsilon \right)$$

using Bernstein's inequality.

Chapter 3

Uniform Convergence and Generalization Analysis

3.1 Probably Approximately Correct Learning

Probabilistic Approximately Correct (PAC) learning is a mathematical model for analyzing algorithms to learn Boolean functions [88], which is analogous to supervised learning.

In this model, we consider binary valued vector $X \in \{0, 1\}^d$. A Boolean function f maps a binary vector X to a binary output $Y \in \{0, 1\}$. Such a Boolean function is also referred to as a *concept*. A *concept class* C , is a set of such Boolean functions: $\{0, 1\}^d \rightarrow \{0, 1\}$. In the literature, a concept is also referred to as a *hypothesis*, and a concept class is also called a *hypothesis space*.

Example 3.1 (And Function Class) *Each member of And Function Class can be written as*

$$f(x) = \prod_{j \in J} x_j, \quad J \subset \{1, \dots, d\}.$$

Example 3.2 (Decision List) *A decision list is a function of the following form. Let $\{i_1, \dots, i_d\}$ be a permutation of $\{1, \dots, d\}$, and let $a_i, b_i \in \{0, 1\}$ for $i = 1, \dots, d + 1$. The function $f(x)$ can be computed as follows. if $x_{i_1} = a_1$ then $f(x) = b_1$; else if $x_{i_2} = a_2$ then $f(x) = b_2$, $\dots$, else if $x_{i_d} = a_d$ then $f(x) = b_d$; else $f(x) = b_{d+1}$.*

Assume now that there is an unknown true function $f_*(x) \in C$ which we want to learn. In the PAC learning model, the input X is taken from an unknown distribution D , but there is an oracle $\mathcal{O}$ that can sample from this distribution. Each call to $\mathcal{O}$ returns a sample $X \sim D$, together with the value $Y = f_*(X)$. The goal of a PAC learner is to learn this function

approximately up to an accuracy ϵ with respect to D , by randomly sampling its inputs with a polynomial (in ϵ^{-1} and d) computational complexity.

More formally, the (generalization) error of a learned function $f(x)$ is defined as

$$\text{err}_D(f) = \mathbf{E}_{X \sim D} I(f(x) \neq f_*(x)).$$

We may call $\mathcal{O}$ n times to form a training data $S_n = \{(X_i, Y_i) : i = 1, \dots, n\}$. The learner $\mathcal{A}$ takes S_n and returns a function $\hat{f} \in C$. Due to the randomness of S_n , the function $\hat{f}$ is also random. Therefore the quality of the learner should be stated in probabilistic terms.

Definition 3.1 (PAC Learning) *A concept class C is PAC learnable if there exists a learner $\mathcal{A}$ so that for all $f_* \in C$, distribution D , approximation error $\epsilon > 0$ and probability $\delta \in (0, 1)$, the following holds. With probability at least $1 - \delta$ over outputs of oracle $\mathcal{O}$ that samples over D , the learner produces a function $\hat{f}$*

$$\text{err}_D(\hat{f}) \leq \epsilon,$$

with the computational complexity polynomial in $(\epsilon^{-1}, \delta^{-1}, d)$.

The term *probabilistic approximately correct* is due to the fact that the statement is with probability at least $1 - \delta$, and the correctness is up to approximation error ϵ . The PAC learning model is similar to the supervised learning framework, except for the additional requirement that the computational complex should be polynomial. One may also extend the basic notation of PAC learning by assuming that the oracle $\mathcal{O}$ may take additional information, so that the learning algorithm can interact with the oracle sequentially.

In statistical complexity analysis of learning algorithms, the computational complexity requirement is de-emphasized unless needed. Statistical analysis will focus on the sample complexity, with an emphasis on the dependency of sample size n in terms of $(\epsilon^{-1}, \delta^{-1}, d)$ using the ERM method.

Define

$$\text{err}_{S_n}(f) = \frac{1}{n} \sum_{i=1}^n I(f(X_i) \neq Y_i),$$

If we use the ERM method, then it is equivalent to finding an $\hat{f} \in C$ so that

$$\text{err}_{S_n}(\hat{f}) = 0.$$

That is, we find a function achieving zero error. Such a function always exists because $\text{err}_{S_n}(f_*) = 0$. However, there may be more than one functions achieving zero-error, and we simply pick one such function. More generally, we may consider approximate ERM, which returns $\hat{f}$ so that

$$\text{err}_{S_n}(\hat{f}) \leq \epsilon' \tag{3.1}$$

for some accuracy ϵ' .

By using the union bound and Hoeffding's inequality, we can obtain the required sample complexity bound.

3.2 Analysis of PAC Learning

Consider m events $E_1, \dots, E_m$, the *union bound* refers to the following probability inequality:

$$\Pr(E_1 \cup \dots \cup E_m) \leq \sum_{j=1}^m \Pr(E_j).$$

In other words, with probability at least $1 - \sum_{j=1}^m \Pr(E_j)$, none of the events E_i ($i = 1, \dots, m$) occur. This is the key technique tool for generalization analysis.

Note that if we assume that the probability $\sum_j \Pr(E_j)$ is small, then the union bound is relatively tight when the events E_j are independent. In such case:

$$\Pr(E_1 \cup \dots \cup E_m) \geq \sum_{j=1}^m \Pr(E_j) - 0.5 \left(\sum_{j=1}^m \Pr(E_j) \right)^2.$$

If E_j are correlated, then the union bound may not be tight. For example when they are completely correlated: $E_1 = \dots = E_m$, then

$$\Pr(E_1 \cup \dots \cup E_m) = m^{-1} \sum_{j=1}^m \Pr(E_j).$$

We will use the union bound to derive the sample complexity of PAC learning with ERM. We assume that the concept class C contains N different functions.

Note that we cannot directly apply the Chernoff bound of Theorem 2.5 to the function $\hat{f}$ learned from the training data S_n , because it is a random function that depends on S_n . Instead, we can apply Theorem 2.5 to each fixed function $f(x) \in C$, and obtain the following inequality:

$$\Pr(\text{err}(f) \geq \hat{\text{err}}_{S_n}(f) + \epsilon) \leq \exp(-2n\epsilon^2).$$

Therefore

$$\begin{aligned} & \Pr \left(\sup_{f \in C} [\text{err}(f) - \hat{\text{err}}_{S_n}(f)] \geq \epsilon \right) \\ &= \Pr(\exists f \in C : \text{err}(f) \geq \hat{\text{err}}_{S_n}(f) + \epsilon) \leq N \exp(-2n\epsilon^2). \end{aligned}$$

Now by setting $N \exp(-2n\epsilon^2) = \delta$ and solving for ϵ to get

$$\epsilon = \sqrt{\frac{\ln(N/\delta)}{2n}},$$

we obtain the following equivalent statement. With probability at least $1 - \delta$, the following inequality holds for all $f \in C$:

$$\text{err}(f) < \hat{\text{err}}_{S_n}(f) + \sqrt{\frac{\ln(N/\delta)}{2n}}.$$

Such a result is called *uniform convergence*, because it holds for all $f \in C$, and thus it also holds for the output $\hat{f}$ of any learning algorithm.

By applying the uniform convergence result to the approximate ERM learner of (3.1), we obtain the following generalization bound. With probability at least $1 - \delta$, the following inequality holds for the ERM PAC learner (3.1) for all $\gamma > 0$:

$$\text{err}(\hat{f}) < (1 + \gamma) \sqrt{\frac{\ln(N/\delta)}{2n}}, \quad (3.2)$$

with

$$\epsilon' = \gamma \sqrt{\frac{\ln(N/\delta)}{2n}}.$$

Therefore if

$$n \geq \frac{(1 + \gamma)^2 \ln(N/\delta)}{2\epsilon^2},$$

then $\text{err}(\hat{f}) < \epsilon$ with probability at least $1 - \delta$. This implies a sample complexity bound of $O(1/\epsilon^2)$ for sample size n to achieve ϵ error. It is another form of statistical convergence rate of $O(1/\sqrt{n})$ in (3.2).

For the realizable case considered here, we can obtain a better result by applying the union bound, together with the multiplicative form of Chernoff bound in Corollary 2.1.

Theorem 3.1 *Consider a concept class C with N elements. With probability at least $1 - \delta$, the ERM PAC learner (3.1) with*

$$\epsilon' = \gamma^2 \frac{2 \ln(N/\delta)}{n}$$

for some $\gamma > 0$ satisfies

$$\text{err}(\hat{f}) \leq (1 + \gamma)^2 \frac{2 \ln(N/\delta)}{n}.$$

Proof Given any $f \in C$, we have from Corollary 2.1 that

$$\Pr(\text{err}(f) \geq \hat{\text{err}}_{S_n}(f) + \epsilon) \leq \exp(-n\epsilon^2/2\text{err}(f)).$$

Now by setting $\exp(-n\epsilon^2/2\text{err}(f)) = \delta/N$, and solve for ϵ :

$$\epsilon = \sqrt{\frac{2\text{err}(f) \ln(N/\delta)}{n}},$$

we obtain the following equivalent statement. With probability at least $1 - \delta/N$:

$$\text{err}(f) \leq \hat{\text{err}}_{S_n}(f) + \sqrt{\frac{2\text{err}(f) \ln(N/\delta)}{n}}.$$

The union bound thus implies the following statement. With probability at least $1 - \delta$, for all $f \in C$:

$$\text{err}(f) \leq \hat{\text{err}}_{S_n}(f) + \sqrt{\frac{2\text{err}(f) \ln(N/\delta)}{n}}.$$

Now by applying this inequality to the ERM PAC learner (3.1), we obtain

$$\text{err}(\hat{f}) \leq \gamma^2 \frac{2 \ln(N/\delta)}{n} + \sqrt{\frac{2\text{err}(\hat{f}) \ln(N/\delta)}{n}}.$$

The solution is

$$\text{err}(\hat{f}) \leq (0.5 + \sqrt{\gamma^2 + 0.25})^2 \frac{2 \ln(N/\delta)}{n},$$

which implies the desired bound. ■

Note that compared to (3.2), which predicts that the generalization error $\text{err}(\hat{f})$ decays at $O(1/\sqrt{n})$, Theorem 3.1 implies that the generalization error decays at $O(1/n)$, leading to a faster convergence rate as $n \rightarrow \infty$. This means the multiplicative form of Chernoff bound is preferred to Chernoff bound in the realizable case, where Y is generated by $f_*(X)$ for some $f_* \in C$.

Theorem 3.1 implies the following sample complexity bound. Given $\delta \in (0, 1)$. For all sample size

$$n \geq (1 + \gamma)^2 \frac{2 \ln(N/\delta)}{\epsilon},$$

we have $\text{err}(\hat{f}) < \epsilon$ with probability at least $1 - \delta$.

The generalization error bound has a logarithmic dependency $\ln N$ on the concept class size N . This logarithmic dependency is important for analyzing machine learning algorithms, and such a dependency requires the exponential tail probability bounds developed in Chapter 2. Since the generalization analysis does not depend on the underlying distribution D , the resulting bound is referred to as a *distribution free* generalization bound.

Example 3.3 *The concept class C of “And Functions” is PAC learnable. To see this, we show that ERM (3.1) can be solved in computationally efficient way with $\epsilon' = 0$. Given $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, we define $\hat{J} = \{j : \forall 1 \leq i \leq n, X_{i,j} \geq Y_i\}$ (where X_{ij} is the j -th component of the i -th training data X_i) and $\hat{f}(x) = \prod_{j \in \hat{J}} x_j$. This choice implies that $\hat{f}(X_i) = Y_i$ when $Y_i = 1$. It can be checked that if the true target is $f_*(x) + \prod_{j \in J} x_j$, then $\hat{J} \supset J$, and thus $\hat{f}(x) \leq f_*(x)$. This implies that $\hat{f}(X_i) = Y_i$ when $Y_i = 0$, and hence $\text{er}_{S_n}(\hat{f}) = 0$. Theorem 3.1 implies that C is PAC-learnable.*

3.3 Empirical Process

The analysis of realizable PAC learning can be generalized to deal with general non-binary-valued function classes which may contain an infinitely number of functions. It may also be generalized to handle the non-realizable case where $f_*(x) \notin C$ or when the observation Y contains noise. For such cases, the corresponding analysis requires the technical tool of *empirical processes*.

To simplify the notations, in the general setting, we may denote the observations $Z_i = (X_i, Y_i)$, prediction rule $f(X_i)$ and loss function $L(f(X_i), Y_i)$. Assume further that $f(x)$ is parametrized by $w \in \Omega$ as $f(w, x)$, and the hypothesis space is $\{f(w, \cdot) : w \in \Omega\}$. We use the simplified notation

$$g_w(z) = L(f(w, x), y) - L_*(x, y),$$

for $w \in \Omega$ and $z = (x, y)$, and a pre-chosen w independent function $L_*(x, y)$ of $z = (x, y)$. In general, we may simply take $L_*(x, y) = 0$. However, for some applications, we may choose a non-trivial $L_*(x, y)$ so that

$$L(f(w, x), y) - L_*(x, y)$$

has a smaller variance. Since $L_*(x, y)$ does not depend on w , it does not change the solution of the ERM method.

Example 3.4 *Consider linear model $f(w, x) = w^\top x$, and $L(f(w, x), y) = (w^\top x - y)^2$ is the least squares loss. Then with $L_*(x, y) = 0$, we have $g_w(z) = (w^\top x - y)^2$ for $z = (x, y)$.*

If we further assume that the problem is realizable by linear model, and w_ is the true weight: $\mathbf{E}[y|x] = w_*^\top x$. It follows that we may take $L_*(x, y) = (w_*^\top x - y)^2$, and*

$$g_w(z) = (w^\top x - y)^2 - (w_*^\top x - y)^2,$$

which has a small variance when $w \approx w_$: $\lim_{w \rightarrow w_*} g_w(z) = 0$.*

Assume $z \sim D$, then the function class $\{g_w(z) : w \in \Omega\}$ is indexed by $w \in \Omega$. Similar to the PAC learning analysis, we are interested in estimating $\hat{w}$ using the training data $S_n = \{Z_i = (X_i, Y_i) : i = 1, \dots, n\}$. Let

$$g_w(S_n) = \frac{1}{n} \sum_{i=1}^n g_w(Z_i),$$

then we call $\{g_w(S_n)\}_{w \in \Omega}$ an empirical process indexed by Ω . We are interested in the uniform convergence of this empirical process to

$$g_w(D) = \mathbf{E}_{Z \in D} g_w(Z).$$

Definition 3.2 (Uniform Convergence) *Given model space Ω , and distribution D . Let S_n be n iid examples sampled from D . The empirical process $g_w(S_n)$ converges to $g_w(D)$ uniformly in probability if for all $\epsilon > 0$:*

$$\lim_{n \rightarrow \infty} \Pr \left(\sup_{w \in \Omega} |g_w(S_n) - g_w(D)| > \epsilon \right) = 0,$$

where the probability is over S_n .

Note that the uniform convergence result is also referred to as the uniform law of large numbers. It says that the law of large number holds for all $\hat{w} \in \Omega$ that may depend on the training data S_n . It can thus be applied to the output of any learning algorithm.

Similar to the case of PAC learning, the uniform convergence result can be used to obtain an oracle inequality for approximate ERM, which can be expressed as follows

$$\hat{w} \approx \arg \min_{w \in \Omega} \frac{1}{n} \sum_{i=1}^n L(f(w, X_i), Y_i),$$

so that for some $\epsilon' > 0$, the following inequality holds:

$$\frac{1}{n} \sum_{i=1}^n L(f(\hat{w}, X_i), Y_i) \leq \inf_{w \in \Omega} \frac{1}{n} \sum_{i=1}^n L(f(w, X_i), Y_i) + \epsilon', \quad (3.3)$$

which is equivalent to

$$g_{\hat{w}}(S_n) \leq \inf_{w \in \Omega} g_w(S_n) + \epsilon'.$$

We can use uniform convergence to obtain oracle inequality as follows.

Theorem 3.2 *Assume that for any $\delta \in (0, 1)$, the following uniform convergence result holds. With probability at least $1 - \delta$,*

$$\forall w \in \Omega : g_w(D) \leq [g_w(S_n) + \epsilon_n(\delta, w)].$$

Moreover, $\forall w \in \Omega$, we have with probability at least $1 - \delta$,

$$[g_w(S_n) - g_w(D) - \epsilon'_n(\delta, w)] \leq 0.$$

Then we have the following statement. With probability at least $1 - \delta$, the approximate ERM method (3.3) satisfies the oracle inequality:

$$\begin{aligned} \mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) &\leq \inf_{w \in \Omega} [\mathbf{E}_{(X,Y) \sim D} L(f(w, X), Y) + \epsilon'_n(\delta/2, w)] \\ &\quad + \epsilon' + \epsilon_n(\delta/2, \hat{w}). \end{aligned}$$

Proof Consider an arbitrary $w \in \Omega$. We have with probability at least $1 - \delta/2$:

$$\begin{aligned} g_{\hat{w}}(D) &\leq g_{\hat{w}}(S_n) + \epsilon_n(\delta/2, \hat{w}) \\ &\leq g_w(S_n) + \epsilon' + \epsilon_n(\delta/2, \hat{w}), \end{aligned}$$

where the second inequality is due to (3.3). Moreover, with probability at least $1 - \delta/2$:

$$g_w(S_n) \leq g_w(D) + \epsilon'_n(\delta/2, w).$$

Taking the union bound of the two events, we obtain with probability at least $1 - \delta$:

$$g_{\hat{w}}(D) \leq g_w(D) + \epsilon' + \epsilon_n(\delta/2, \hat{w}) + \epsilon'_n(\delta/2, w).$$

Now by letting w approach the minimum of the right hand side, we obtain the desired bound. ■

The above result shows that the uniform convergence of empirical process can be used to derive oracle inequality for the ERM method. Next we will investigate the main technique to derive uniform convergence bounds.

3.4 Covering number

If Ω is finite, then we can use union bound to obtain uniform convergence of empirical processes. If Ω is infinite, then we can approximate the function class

$$G = \{g_w(z) : w \in \Omega\}$$

using a finite function class. We can then apply union bound to this finite approximation. Different types of approximations lead to different types of covering, and different covering numbers.

Definition 3.3 (Lower Bracketing Cover) Given a distribution D . A finite function class $G(\epsilon) = \{g_1(z), \dots, g_N(z)\}$ is an ϵ -lower bracketing cover of G if for all $w \in \Omega$, there exists $j = j(w)$ such that

$$\forall z : g_j(z) \leq g_w(z), \quad \mathbf{E}_{Z \sim D} g_j(Z) \geq \mathbf{E}_{Z \sim D} g_w(Z) - \epsilon.$$

The ϵ -lower bracketing covering number of G , denoted by $N_{LB}(\epsilon, G, D)$, is the smallest cardinality of such $G(\epsilon)$. The quantity $\ln N_{LB}(\epsilon, G, D)$ is referred to as the ϵ -lower bracketing entropy number.

We shall mention that the functions $g_j(z)$ may not necessarily belong to G . Next we show that the lower bracketing covering number can be used to obtain uniform convergence bounds for infinite function classes.

Theorem 3.3 Assume that $g_w(z) \in [0, 1]$ for all w and z . Then given $\delta \in (0, 1)$, with probability at least $1 - \delta$, the following inequality holds:

$$\forall w \in \Omega : \mathbf{E}_{Z \sim D} g_w(Z) \leq \left[\frac{1}{n} \sum_{i=1}^n g_w(Z_i) + \epsilon_n(\delta, G, D) \right],$$

where

$$\epsilon_n(\delta, G, D) = \inf_{\epsilon > 0} \left[\epsilon + \sqrt{\ln(N_{LB}(\epsilon, G, D)/\delta)/(2n)} \right].$$

Moreover, with probability at least $1 - \delta$, the following inequality holds:

$$\forall \gamma \in (0, 1), \forall w \in \Omega : (1 - \gamma)g_w(D) \leq g_w(S_n) + \epsilon_n^\gamma(\delta, G, D),$$

where

$$\epsilon_n^\gamma(\delta, G, D) = \inf_{\epsilon > 0} \left[(1 - \gamma)\epsilon + \frac{\ln(N_{LB}(\epsilon, G, D)/\delta)}{2\gamma n} \right].$$

Proof For any $\epsilon > 0$, let $G(\epsilon) = \{g_1(z), \dots, g_N(z)\}$ be an ϵ lower bracketing cover of G with $N = N_{LB}(\epsilon, G, D)$. We may assume that $g_j(z) \in [0, 1]$ for all j (otherwise, we may set $g_j(z)$ to $\min(1, \max(0, g_j(z)))$). In the following, we let $j = j(w)$ for simplified notation:

$$\frac{1}{n} \sum_{i=1}^n g_w(Z_i) - \mathbf{E}_{Z \sim D} g_w(Z) \geq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) - \mathbf{E}_{Z \sim D} g_j(Z) - \epsilon.$$

Let $\epsilon'' = \sqrt{\ln(N/\delta)/2n}$. It follows from the union bound on j that

$$\begin{aligned} & \Pr \left(\inf_{w \in \Omega} \left[\frac{1}{n} \sum_{i=1}^n g_w(Z_i) - \mathbf{E}_{Z \sim D} g_w(Z) + \epsilon + \epsilon'' \right] \leq 0 \right) \\ & \leq \Pr \left(\inf_j \left[\frac{1}{n} \sum_{i=1}^n g_j(Z_i) - \mathbf{E}_{Z \sim D} g_j(Z) + \epsilon'' \right] \leq 0 \right) \\ & \leq \sum_{j=1}^N \Pr \left(\frac{1}{n} \sum_{i=1}^n g_j(Z_i) - \mathbf{E}_{Z \sim D} g_j(Z) + \epsilon'' \leq 0 \right) \\ & \leq N \exp(-2n(\epsilon'')^2) = \delta. \end{aligned}$$

where the last inequality is due to the Chernoff bound. This leads to the first desired bound.

Moreover, using the multiplicative Chernoff bound and union bound, we obtain the following statement. With probability at least $1 - \delta$, the following inequality holds for all j :

$$\begin{aligned} \mathbf{E}_{Z \sim D} g_j(Z) &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \sqrt{\frac{2\mathbf{E}_{Z \sim D} g_j(Z) \ln(N/\delta)}{n}} \\ &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \gamma \mathbf{E}_{Z \sim D} g_j(Z) + \frac{1}{2\gamma} \frac{\ln(N/\delta)}{n}. \end{aligned}$$

This implies that for all j :

$$(1 - \gamma) \mathbf{E}_{Z \sim D} g_j(Z) \leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \frac{1}{2\gamma} \frac{\ln(N/\delta)}{n}.$$

Therefore for all $w \in \Omega$, let $j = j(w)$, we obtain

$$\begin{aligned} (1 - \gamma) \mathbf{E}_{Z \sim D} g_w(Z) &\leq (1 - \gamma) \mathbf{E}_{Z \sim D} g_j(Z) + (1 - \gamma) \epsilon \\ &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \frac{1}{2\gamma} \frac{\ln(N/\delta)}{n} + (1 - \gamma) \epsilon \\ &\leq \frac{1}{n} \sum_{i=1}^n g_w(Z_i) + \frac{1}{2\gamma} \frac{\ln(N/\delta)}{n} + (1 - \gamma) \epsilon. \end{aligned}$$

The first and the third inequalities are due to the definition of lower bracketing cover. This leads to the second desired bound. $\blacksquare$

The uniform convergence bounds imply generalization bounds as follows.

Corollary 3.1 *If $L(f(w, x), y) \in [0, 1]$. Let $G = \{L(f(w, x), y) : w \in \Omega\}$. With probability at least $1 - \delta$, the approximate ERM method (3.3) satisfies the (additive) oracle inequality:*

$$\begin{aligned} \mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) &\leq \inf_{w \in \Omega} \mathbf{E}_{(X,Y) \sim D} L(f(w, X), Y) + \epsilon' + \sqrt{\frac{\ln(2/\delta)}{2n}} \\ &\quad + \inf_{\epsilon > 0} \left[\epsilon + \sqrt{\ln(2N_{LB}(\epsilon, G, D)/\delta)/(2n)} \right]. \end{aligned}$$

Moreover, with probability at least $1 - \delta$, we have the following (multiplicative) oracle inequality for all $\gamma \in (0, 1)$:

$$\begin{aligned} (1 - \gamma) \mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) &\leq (1 + \gamma) \inf_{w \in \Omega} \mathbf{E}_{(X,Y) \sim D} L(f(w, X), Y) \\ &\quad + \epsilon' + \frac{(2\gamma + 1) \ln(2/\delta)}{2\gamma n} + \inf_{\epsilon > 0} \left[(1 - \gamma) \epsilon + \frac{\ln(2N_{LB}(\epsilon, G, D)/\delta)}{2\gamma n} \right]. \end{aligned}$$

Proof Consider Theorem 3.2 with $g_w(Z) = L(f(w, x), y)$. From Chernoff bound, we can set

$$\epsilon'_n(\delta/2, w) = \sqrt{\frac{\ln(2/\delta)}{2n}},$$

or

$$\begin{aligned} \epsilon'_n(\delta/2, w) &= \sqrt{\frac{2g_w(D) \ln(2/\delta)}{n}} + \frac{\ln(2/\delta)}{n} \\ &\leq \gamma g_w(D) + \frac{\ln(2/\delta)}{2\gamma n} + \frac{\ln(2/\delta)}{n}. \end{aligned}$$

Now combine with the two estimates in Theorem 3.3, we obtain the two bounds. ■

3.5 A Simple Example

Consider one dimensional classification problem, with x uniformly distributed in $[0, 1]$, and $y \in \{\pm 1\}$ generated according to

$$\Pr(y = 1|x) = \begin{cases} p & \text{if } x \geq w_* \\ (1 - p) & \text{otherwise} \end{cases}$$

for some unknown $w_* \in [-1, 1]$ and $p \in (0.5, 1]$. Since we don't know the true threshold w_* , we can consider a family of classifiers

$$f(w, x) = 2I(x \geq w) - 1, \quad w \in \Omega = [0, 1]$$

with w being the model parameter to be learned from training data. Here $I(\cdot)$ is the set indicator function. In this example, we consider the classification error loss function

$$L(f(x), y) = I(f(x) \neq y).$$

In this case, the optimal Bayes classifier is $f_*(x) = 1$ when $x \geq w_*$ and $f_*(x) = -1$ when $x < w_*$. The Bayes error is

$$L(f(w_*, x), y) = 1 - p.$$

Since for this example, it is easy to find a model parameter $\hat{w}$ to minimize the empirical risk (the solution may not be unique), we may set $\epsilon' = 0$ in (3.3).

Lower bracketing cover

Given any $\epsilon > 0$, we let $w_j = -1 + j\epsilon$ for $j = 1, \dots, \lceil 1/\epsilon \rceil$. Let

$$g_j(z) = \begin{cases} 0 & \text{if } x \in [w_j - \epsilon, w_j] \\ g_{w_j}(z) & \text{otherwise,} \end{cases}$$

where $z = (x, y)$. Note that $g_j \notin G$.

It follows that for any $w \in [0, 1]$, if we let w_j be the smallest j such that $w_j \geq w$, then we have $g_j(z) = 0 \leq g_w(z)$ when $x \in [w, w_j]$, and $g_j(z) = g_w(z)$ otherwise, where $z = (x, y)$. Moreover,

$$\mathbf{E}_{Z \sim D}[g_j(Z) - g_w(Z)] = \mathbf{E}_{X \in [w, w_j]}[0 - g_w(Z)] \geq -\epsilon.$$

This means that $\{g_j(z)\}$ is an ϵ lower bracketing cover of G , and thus

$$N_{LB}(\epsilon, G, D) \leq 1 + \epsilon^{-1}.$$

Oracle inequalities

We have (by picking $\epsilon = 2/n$):

$$\inf_{\epsilon > 0} \left[\epsilon + \sqrt{\ln(2N_{LB}(\epsilon, G, D)/\delta)/(2n)} \right] \leq \frac{2}{n} + \sqrt{\frac{\ln((n+2)/\delta)}{2n}}.$$

This implies the following additive oracle inequality from Corollary 3.1. With probability at least $1 - \delta$,

$$\mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) \leq (1-p) + \sqrt{\frac{\ln(2/\delta)}{2n}} + \frac{2}{n} + \sqrt{\frac{\ln(n/\delta)}{2n}}.$$

In addition, we have (by picking $\epsilon = 2/n$):

$$\inf_{\epsilon > 0} \left[(1-\gamma)\epsilon + \frac{\ln(2N_{LB}(\epsilon, G, D)/\delta)}{2\gamma n} \right] \leq \frac{2(1-\gamma)}{n} + \frac{\ln((n+2)/\delta)}{2\gamma n}.$$

This implies the following multiplicative oracle inequality from Corollary 3.1. With probability at least $1 - \delta$, for all $\gamma \in (0, 1)$:

$$\mathbf{E}_{(X,Y) \sim D} L(f(\hat{w}, X), Y) \leq \frac{1+\gamma}{1-\gamma}(1-p) + \frac{(2\gamma+1)\ln(2/\delta)}{2\gamma(1-\gamma)n} + \frac{2}{n} + \frac{\ln((n+2)/\delta)}{2\gamma(1-\gamma)n}.$$

The multiplicative bound is superior to the additive bound when the Bayes error is small, such as when $p = 1$. In this case, we have a convergence rate of $\ln n/n$ instead of $\sqrt{\ln n/n}$ as in the additive form.

3.6 Uniform Bernstein's Inequalities

In this section, we show that better bounds can be obtained with Bernstein's inequalities under the following condition.

Definition 3.4 *Given a function class G . We say it satisfies the (α, β) -variance condition if for all $g(z) \in G$:*

$$\mathbf{Var}_{Z \sim D}(g(Z)) \leq \alpha + \beta \mathbf{E}_{Z \sim D} g(Z).$$

The following are examples that satisfies the variance condition of Definition 3.4.

Example 3.5 *Let $G = \{g(\cdot) : \forall z, g(z) \in [0, 1]\}$. Then G satisfies the (α, β) -variance condition with $\alpha = 0$ and $\beta = 1$.*

Example 3.6 *Consider the least squares method $L(f(x), y) = (f(x) - y)^2$, with bounded response: $L(f(x), y) \leq M^2$ for some $M > 0$. Let F be a convex function class, and*

$$f_* = \arg \min_{f \in F} \mathbf{E}_{(x,y) \sim D} L(f(x), y).$$

Let $z = (x, y)$, and

$$G = \{g(\cdot) : g(z) = L(f(x), y) - L(f_*(x), y), f(x) \in F\}.$$

Then G satisfies the variance condition of Definition 3.4 with $\alpha = 0$.

Under the variance-condition of Definition 3.4, we can obtain the following uniform convergence result using Bernstein's inequalities.

Theorem 3.4 *Assume that for all $\epsilon > 0$, $G = \{g_w(z) : w \in \Omega\}$ has an ϵ lower-bracketing cover $G(\epsilon)$ with $N_{LB}(\epsilon, G, D)$ members so that $G \cup G(\epsilon)$ satisfies the $(\alpha(\epsilon), \beta(\epsilon))$ -variance condition.*

Moreover, assume for all $g \in G \cup G(\epsilon)$, The random variable $g(z)$ satisfies the conditions of Lemma 2.4 with $c = \mathbf{E}(g(Z))$, $V = \mathbf{Var}(g(Z))$ and $b = b(\epsilon)$.

Then $\forall \delta \in (0, 1)$, with probability at least $1 - \delta$, the following inequality holds:

$$\forall \gamma \in (0, 1), w \in \Omega : (1 - \gamma)g_w(D) \leq [g_w(S_n) + \epsilon_n^\gamma(\delta, G, D)],$$

where

$$\begin{aligned} \epsilon_n^\gamma(\delta, G, D) = \inf_{\epsilon > 0} & \left[(1 - \gamma)\epsilon + \sqrt{\frac{2\alpha(\epsilon) \ln(N_{LB}(\epsilon, G, D)/\delta)}{n}} \right. \\ & \left. + \frac{\ln(N_{LB}(\epsilon, G, D)/\delta)}{\frac{6\gamma}{3\beta(\epsilon) + 4\gamma b(\epsilon)} \cdot n} \right]. \end{aligned}$$

Proof For any $\epsilon > 0$, let $G(\epsilon) = \{g_1(z), \dots, g_N(z)\}$ be an ϵ lower bracketing cover of G with $N = N_{LB}(\epsilon, G, D)$.

Using Bernstein's inequalities and the union bound, we obtain the following statement. With probability at least $1 - \delta$, the following inequality holds for all j :

$$\begin{aligned} \mathbf{E}_{Z \sim D} g_j(Z) &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \sqrt{\frac{2 \mathbf{Var}_{Z \sim D} g_j(Z) \ln(N/\delta)}{n}} + \frac{2b \ln(N/\delta)}{3n} \\ &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \sqrt{\frac{2\alpha \ln(N/\delta)}{n}} + \sqrt{\frac{2\beta \mathbf{E}_{Z \sim D} g_j(Z) \ln(N/\delta)}{n}} + \frac{2b \ln(N/\delta)}{3n} \\ &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \sqrt{\frac{2\alpha \ln(N/\delta)}{n}} + \gamma \mathbf{E}_{Z \sim D} g_j(Z) + \left(\frac{\beta}{2\gamma} + \frac{2b}{3} \right) \frac{\ln(N/\delta)}{n}. \end{aligned}$$

The second inequality uses the fact that $g_j(z)$ satisfies the (α, β) variance condition. Therefore for all $w \in \Omega$, let $j = j(w)$, we obtain

$$\begin{aligned} &(1 - \gamma) \mathbf{E}_{Z \sim D} g_w(Z) \\ &\leq (1 - \gamma) \mathbf{E}_{Z \sim D} g_j(Z) + (1 - \gamma)\epsilon \\ &\leq \frac{1}{n} \sum_{i=1}^n g_j(Z_i) + \sqrt{\frac{2\alpha \ln(N/\delta)}{n}} + \frac{3\beta + 4\gamma b}{6\gamma} \cdot \frac{\ln(N/\delta)}{n} + (1 - \gamma)\epsilon \\ &\leq \frac{1}{n} \sum_{i=1}^n g_w(Z_i) + \sqrt{\frac{2\alpha \ln(N/\delta)}{n}} + \frac{3\beta + 4\gamma b}{6\gamma} \cdot \frac{\ln(N/\delta)}{n} + (1 - \gamma)\epsilon. \end{aligned}$$

This leads to the desired bound. ■

Similar to Corollary 3.1, we can obtain from Theorem 3.4 and Theorem 3.2 the following oracle inequality.

Corollary 3.2 *Assume that $g_w(x) = L(f(w, x), y) - L_*(x, y)$ for appropriately defined $L_*(x, y)$, and assume that the conditions of Theorem 3.4 hold. Moreover,*

With probability at least $1 - \delta$, the approximate ERM method (3.3) satisfies the following inequality for all $\gamma \in (0, 1)$:

$$\begin{aligned} &(1 - \gamma) \mathbf{E}_{(X, Y) \sim D} [L(f(\hat{w}, X), Y) - L_*(x, y)] \\ &\leq (1 + \gamma) \inf_{w \in \Omega} \mathbf{E}_{(X, Y) \sim D} [L(f(w, X), Y) - L_*(x, y)] + \epsilon' \\ &\quad + \inf_{\epsilon > 0} \left[(1 - \gamma)\epsilon + \sqrt{\frac{8\alpha(\epsilon) \ln(2N_{LB}(\epsilon, G, D)/\delta)}{n}} + \frac{\ln(2N_{LB}(\epsilon, G, D)/\delta)}{\frac{3\gamma}{3\beta(\epsilon) + 4\gamma b(\epsilon)} \cdot n} \right]. \end{aligned}$$

Proof Consider Theorem 3.2 with $g_w(Z) = L(f(w, x), y) - L_*(x, y)$. From Bernstein's inequality, and assumptions of Theorem 3.4, we can set

$$\begin{aligned} \epsilon'_n(\delta/2, w) &= \sqrt{\frac{2\text{Var}(g_w(Z)) \ln(2/\delta)}{n}} + \frac{2b(\epsilon) \ln(2/\delta)}{3n} \\ &\leq \gamma g_w(D) + \sqrt{\frac{2\alpha(\epsilon) \ln(2/\delta)}{n}} + \frac{\beta(\epsilon) \ln(2/\delta)}{2\gamma n} + \frac{2b(\epsilon) \ln(2/\delta)}{3n} \\ &\leq \gamma g_w(D) + \sqrt{\frac{2\alpha(\epsilon) \ln(2N_{LB}(\epsilon, G, D)/\delta)}{n}} + \frac{\ln(2N_{LB}(\epsilon, G, D)/\delta)}{\frac{6\gamma}{3\beta(\epsilon)+4\gamma b(\epsilon)} \cdot n}. \end{aligned}$$

Now combine with the estimate in Theorem 3.4, we obtain the desired bound. $\blacksquare$

We use the following example to illustrate Theorem 3.2.

Example 3.7 Consider the example of Section 3.5. We replace the definition of $g_w(z)$ by

$$L(f(w, x), y) - L(f(w_*, x), y),$$

and the functions $g'_j(z) = g_j(z) - L(f(w_*, x), y)$ with $g_j(z)$ defined in Section 3.5 form an ϵ lower-bracketing cover of $\{g_w(z) : w \in [0, 1]\}$. It can be shown that for this cover, the conditions of Theorem 3.4 holds with $\alpha(\epsilon) = \alpha_p \epsilon$, $\beta(\epsilon) = \beta_p$ where α_p and β_p are independent of ϵ but depends on p , and $b(\epsilon) = 2$. By taking $\gamma = 0.5$, we obtain the generalization bound with $\epsilon = 2/n$ that with probability at least $1 - \delta$:

$$\mathbf{E}_D I(f(\hat{w}, X) \neq Y) \leq (1-p) + \left[1 + \sqrt{\alpha_p \ln((n+2)/\delta)} + \frac{\ln((n+2)/\delta)}{\frac{6}{3\beta_p+8}} \right] \frac{8}{n}.$$

This shows the ERM method has generalization error converging to the Bayes error at a rate of $O(\ln n/n)$. The result is better than those in Section 3.5.

In general, for parametric function classes such as functions with d parameters, such as d dimensional least squares expression, we expect the entropy to behave as

$$\ln N_{LB}(\epsilon, G, D) = O(d \ln(1/\epsilon)).$$

Similar to the previous example, if the variance condition of Definition 3.4 holds with $\alpha(\epsilon) = O(\epsilon)$, then the generalization bound in Corollary 3.2 implies a convergence rate $O(d \ln n/n)$, which can be further improved to $O(d/n)$ using a technique called “peeling”.

For nonparametric function classes such as kernel methods, we generally expect the entropy to behave like

$$\ln N_{LB}(\epsilon, G, D) = O(1/\epsilon^q)$$

for some $q > 0$. Another way to view nonparametric models is that the intrinsic “dimension” of nonparametric models depends on the scale ϵ increases as $d \approx 1/\epsilon^q$. In this case, Corollary 3.2 can be improved by using a technique called “chaining”. These techniques will be described later.

3.7 Other Covering Numbers

We may be interested in two-sided uniform convergence:

$$\sup_w |g_w(S_n) - \mathbf{E}g_w(D)|.$$

For such uniform convergence, we may define two-sided bracketing cover as follows.

Definition 3.5 Let $G = \{g_w(\cdot) : w \in \Omega\}$ be the function class of the empirical process. We say $G(\epsilon) = \{g_1(z), \dots, g_N(z)\}$ is a ϵ -bracketing cover of G if for all $w \in \Omega$, there exists $j = j(w)$ such that $\forall z$:

$$g_j^L(z) \leq g_w(z) \leq g_j^U(z) \quad \mathbf{E}_D g_j^L(z) \geq \mathbf{E}_D g_j^U(z) - \epsilon.$$

The ϵ -bracketing covering number is the smallest cardinality $N_{BK}(\epsilon, G, D)$ of such $G(\epsilon)$.

If $G_f(\epsilon_1)$ is an ϵ_1 bracketing cover for $\{f_u(\cdot)\}$, and $G_g(\epsilon_2)$ is an ϵ_2 bracketing cover for $\{g_w(\cdot)\}$. Then $G = \{\alpha f(z) + \beta g(z) : f \in G_f(\epsilon); g \in G_g(\epsilon)\}$ is a $|\alpha|\epsilon_1 + |\beta|\epsilon_2$ bracketing cover for $\{\alpha f_u(z) + \beta g_w(z)\}$. The size of G is $|G| \leq |G_f| \cdot |G_g|$.

We may also define covering numbers for general metric space as follows.

Definition 3.6 Let (V, d) be a metric space with metric $d(\cdot, \cdot)$. A finite set $G(\epsilon) \subset V$ is an ϵ cover (or ϵ net) of $G \subset V$ if for all $g \in G$, there exists $g' \in G(\epsilon)$ so that $d(g', g) \leq \epsilon$. The ϵ -covering number of G with metric D is the smallest cardinality $N(\epsilon, G, d)$ of such $G(\epsilon)$. The number $\ln N(\epsilon, G, d)$ is called the ϵ -entropy.

In particular, given a distribution D , we define p -norm in function space as

$$d(f, g) = [\mathbf{E}_{Z \sim D} |f(Z) - g(Z)|^p]^{1/p},$$

then it induces a covering number, denoted by $N(\epsilon, G, L_p(D))$, and referred to as the $L_p(D)$ -covering number.

For $p = \infty$, the $L_\infty(D)$ covering can also be defined with the metric

$$d(f, g) = \inf \{\omega : \Pr_D[|f(z) - g(z)| \leq \omega] = 1\}.$$

The $L_\infty(D)$ cover is stronger than bracketing cover in the following sense. If $\{g_j\}$ is an ϵ $L_\infty(D)$ cover of G , then $g_j^L = g_j - \epsilon$ forms a 2ϵ lower bracketing cover; g_j^U forms a 2ϵ upper bracketing cover; and $[g_j^L, g_j^U]$ forms a 2ϵ bracketing cover. The reverse is not necessarily true. For example, the example in Section 3.5 has a finite bracketing cover, but does not have a finite $L_\infty(D)$ -cover.

Because of the relationship, the analysis of bracketing cover can be used with L_∞ cover with the following estimate:

$$N_{BK}(\epsilon, G, D) \leq N(G, \epsilon/2, L_\infty(D)).$$

3.8 Exercises

1. Assume that E_j ($j = 1, \dots, m$) are independent events. Prove

$$\Pr(E_1 \cup \dots \cup E_m) \geq \sum_{j=1}^m \Pr(E_j) - 0.5 \left(\sum_{j=1}^m \Pr(E_j) \right)^2.$$

2. Describe a computationally efficient learning algorithm for Decision List, and show that it is PAC learnable.
3. In Example 3.6, finish the proof and derive β .
4. In Example 3.7, compute β_p .
5. In Section 3.4, if we assume that all functions $g \in G \cup G(\epsilon)$ are sub-Gaussians. That is, there is a constant V so that

$$\ln \mathbf{E}_Z \exp(\lambda g(Z)) \leq \lambda \mathbf{E}g(Z) + \frac{\lambda^2}{2} V.$$

Derive an oracle inequality for approximate ERM in the form of Corollary 3.1.

6. If we replace the (α, β) variance condition of Definition 3.4 by the following condition

$$\mathbf{Var}_{Z \sim D}(g(Z)) \leq \beta \sqrt{\mathbf{E}_{Z \sim D} g(Z)}$$

with a constant β . Find a new definition of $\epsilon_n(G, \delta)$ so that the uniform convergence statement in Theorem 3.4 holds.

Chapter 4

Generalization Analysis with Validation Data

This chapter investigates the method for partitioning data into training and validation subsets, which is a technique heavily used in empirical evaluations of machine learning algorithms. We consider uniform convergence and oracle inequalities when the data are partitioned into two parts of roughly equal size of training versus validation data, as well as the relationship to oracle inequalities for the true test data. Results for general cross-validation methods with unequal training/validation partition are more subtle, which we will not consider here.

Using notations in Chapter 3, we let $Z = (X, Y)$. Consider the setting that we observe $S_{n+m} = \{Z_1, \dots, Z_{n+m}\}$ data for training a model. We randomly partition S_{n+m} to two disjoint parts: training data S_n of size n , and validation data S'_m of size m . Here we focus on the situation that $n \approx m$.

Consider a function class $G = \{g_w(Z) : w \in \Omega\}$, we are interested in bounding the difference of

$$g_w(S_n) = \frac{1}{n} \sum_{Z \in S_n} g_w(Z),$$

and

$$g_w(S'_m) = \frac{1}{m} \sum_{Z \in S'_m} g_w(Z)$$

for each partition (S_n, S'_m) . The statement will be probabilistic, with the randomness coming from the random partition of training/validation.

With a partition S_n , we may regard the value $g_w(S_n)$ as the empirical mean of n samples, drawn without replacement from the finite population $\{g_w(Z_1), \dots, g_w(Z_{n+m})\}$. Therefore we need to develop tools to bound the deviation of the empirical mean of n random variables to its population

mean, assuming the random variables are sampled without replacement from the underlying finite population.

4.1 Probability Inequalities with Random Permutation

Consider $n + m$ real numbers $\tilde{S} = \{x_1, \dots, x_{n+m}\}$, and define

$$\mu = \frac{1}{n+m} \sum_{i=1}^{n+m} x_i.$$

If we randomly partition the data into two parts of size n and m respectively, then it can be achieved as follows. Let $X_1, \dots, X_{n+m}$ be a random permutation (random sampling without replacement) of $\tilde{S}$, we let the first n random variables $\{X_1, \dots, X_n\}$ as the first (training) partition, and the last m random variables $\{X_{n+1}, \dots, X_{n+m}\}$ as the second (validation) partition. In this setting, we are interested in training loss, represented as

$$\bar{X}_n = n^{-1} \sum_{i=1}^n X_i,$$

and validation loss, represented as

$$\bar{X}'_m = \frac{1}{m} \sum_{i=n+1}^{n+m} X_i.$$

It was shown in [43] that the following moment inequality still holds for random permutations (sampling without replacement):

$$\mathbf{E} e^{\lambda n \bar{X}_n} \leq \left[\mathbf{E} e^{\lambda \bar{X}_1} \right]^n,$$

which implies the following theorem as a counter part of Theorem 2.2.

Theorem 4.1 *For any n and $\epsilon > 0$:*

$$n^{-1} \ln \Pr(\bar{X}_n \geq \mu + \epsilon) \leq \inf_{\lambda > 0} \left[-\lambda \epsilon + \ln \mathbf{E} e^{\lambda(X_1 - \mu)} \right].$$

Similarly

$$n^{-1} \ln \Pr(\bar{X}_n \leq \mu - \epsilon) \leq \inf_{\lambda < 0} \left[\lambda \epsilon + \ln \mathbf{E} e^{\lambda(X_1 - \mu)} \right].$$

This implies that Chernoff bounds and Bernstein inequalities hold for sampling without replacement as well. For the case that $m \ll n$ are highly unbalanced (which corresponds to the case that validation set is much smaller than training set), then it is possible to obtain sharper bounds [75, 12]. However, for our purpose, with $m \approx n$, such bounds only improve constants. Therefore to be consistent with Chapter 2, we use the direct consequence of Theorem 4.1, although they may be loose in constants. These bounds are for random samples without replacement (SWOR).

Theorem 4.2 (Additive Chernoff Bound (SWOR)) *Let $a = \min_i x_i$ and $b = \max_i x_i$, then*

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp\left(\frac{-2n\epsilon^2}{(b-a)^2}\right),$$

and

$$\Pr(\bar{X}_n \leq \mu - \epsilon) \leq \exp\left(\frac{-2n\epsilon^2}{(b-a)^2}\right).$$

Theorem 4.3 (Multiplicative Chernoff Bound (SWOR)) *Let $a = \min_i x_i$ and $b = \max_i x_i$, then*

$$\Pr(\bar{X}_n \leq \mu - \epsilon) \leq \exp\left(\frac{-n\epsilon^2}{2(b-a)(\mu-a)}\right),$$

and

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp\left(\frac{-n\epsilon^2}{(b-a)(2(\mu-a) + \epsilon)}\right).$$

Theorem 4.4 (Bernstein's Inequality (SWOR)) *Let $a = \min_i x_i$ and $b = \max_i x_i$, then*

$$\Pr(\bar{X}_n \geq \mu + \epsilon) \leq \exp\left(\frac{-n\epsilon^2}{2V + 2(b-a)\epsilon/3}\right),$$

and

$$\Pr(\bar{X}_n \leq \mu - \epsilon) \leq \exp\left(\frac{-n\epsilon^2}{2V + 2(b-a)\epsilon/3}\right),$$

where $V = \frac{1}{n+m} \sum_{i=1}^{n+m} (x_i - \mu)^2$.

These inequalities directly imply deviation bounds between the training loss and validation loss. In the following, we state relevant results for the Chernoff bounds.

Corollary 4.1 *Let $a = \min_i x_i$ and $b = \max_i x_i$. Then with probability at least $1 - \delta$, we have the following inequality.*

$$\bar{X}'_m \leq \bar{X}_n + (b - a)\sqrt{(m^{-1} + n^{-1})\ln(1/\delta)}.$$

Moreover, with probability at least $1 - \delta$, we have the following inequality.

$$\bar{X}_n \leq \bar{X}'_m + (b - a)\sqrt{(m^{-1} + n^{-1})\ln(1/\delta)}.$$

Proof Due to the symmetry. We can assume that $n \leq m$. We know that with probability at least $1 - \delta$:

$$\begin{aligned} \bar{X}_n &\leq \mu + (b - a)\sqrt{\frac{\ln(1/\delta)}{2n}} \\ &= \frac{n}{m+n}\bar{X}_n + \frac{m}{m+n}\bar{X}'_m + (b - a)\sqrt{\frac{\ln(1/\delta)}{2n}}. \end{aligned}$$

This implies the first bound. The second bound is similar. ■

Similarly, we have the following result, which is suitable for the situation $m \approx n$ considered here. Note that the bound is not symmetric in m and n .

Corollary 4.2 *Let $a = \min_i x_i$ and $b = \max_i x_i$. Then with probability at least $1 - \delta$, we have the following inequality for all $\gamma \in (0, 1)$:*

$$\bar{X}'_m \leq a + \frac{m + \gamma n}{(1 - \gamma)m}(\bar{X}_n - a) + \frac{(b - a)(m + n)\ln(1/\delta)}{2\gamma(1 - \gamma)mn}.$$

Proof We know that with probability at least $1 - \delta$:

$$\mu \leq \bar{X}_n + \sqrt{\frac{2(b - a)(\mu - a)\ln(1/\delta)}{n}} \leq \bar{X}_n + \gamma(\mu - a) + \frac{(b - a)\ln(1/\delta)}{2\gamma n}.$$

This implies that

$$\frac{(1 - \gamma)n}{m + n}\bar{X}_n + \frac{(1 - \gamma)m}{m + n}\bar{X}'_m \leq \bar{X}_n - \gamma a + \frac{(b - a)\ln(1/\delta)}{2\gamma n},$$

which implies the desired bound. ■

While it is possible to state a similar result for the Bernstein's inequality, this requires additional treatment of empirical variance. We will thus skip it for simplicity.

4.2 Empirical Covering Number

In order to obtain the generalization analysis on validation data, we need the following definition of empirical covering number.

Definition 4.1 (Empirical and Uniform Covering Number) *Given an empirical distribution $S_n = \{Z_1, \dots, Z_n\}$, we define the metric $d = L_p(S_n)$ as*

$$d(g, g') = \left[\frac{1}{n} \sum_{i=1}^n |g(Z_i) - g'(Z_i)|^p \right]^{1/p}.$$

The corresponding metric covering number $N(\epsilon, G, L_p(S_n))$ is referred to as the empirical L_p covering number. Given n , the largest L_p covering number over empirical distribution S_n is referred to as the uniform L_p covering number

$$N_p(\epsilon, G, n) = \sup_{S_n} N(\epsilon, G, L_p(S_n)).$$

We will later show that the empirical L_∞ covering number can be used to obtain uniform convergence and oracle inequalities. First, we show that it is easy to obtain an estimate of the empirical L_∞ covering number for linear classifiers.

Example 4.1 *Consider $\{0, 1\}$ valued linear classifiers in d dimension of the form $f_w(x) = I(w^\top x \geq 0)$, where $w, x \in R^d$. Let $Y \in \{0, 1\}$, then classification error is $g_w(z) = I(f_w(x) \neq y)$, where $z = (x, y)$. Note that it is difficult to obtain bracketing cover for such problems with arbitrary D . However it is easy to obtain L_∞ empirical covering number.*

Lemma 4.1 *Consider the class of linear classifiers G in Example 4.1. We have $N_\infty(G, \epsilon, n) \leq (2n)^d$ with $\epsilon = 0$.*

Proof Consider $\{X_1, \dots, X_n\}$. Consider any w , and let $\tilde{w}(J)$ be defined as the unique solution to the following optimization problem for all $J \subset J_n = \{1, \dots, n\}$:

$$\begin{aligned} \tilde{w}(w, J) = \arg \min_u & \|u\|_2^2 \\ \text{subject to} & \begin{cases} u^\top X_i \geq 0 & \text{if } w^\top X_i \geq 0 \\ u^\top X_i \leq -1 & \text{if } w^\top X_i < 0 \end{cases} \text{ for } i \in J. \end{aligned}$$

Moreover, let $\tilde{J}$ be the smallest cardinality of subsets J of J_n such that $\tilde{w}(w, J) = \tilde{w}(w, J_n)$. We simplify the corresponding $\tilde{w}(w, \tilde{J})$ as $\tilde{w}$. We know that the following facts hold true.

- For all $i \in \tilde{J}$: $\tilde{w}^\top X_i = 0$ if $w^\top X \geq 0$ and $\tilde{w}^\top X_i = -1$ if $w^\top X < 0$. This is a direct consequence of convex optimization and KKT condition, where inactive constraints can be removed.
- $|\tilde{J}| \leq d$. Otherwise, $\{X_i : i \in \tilde{J}\}$ is collinear, and thus we can find an index $i \in \tilde{J}$ and coefficients $\{\alpha_j\}$, so that

$$u^\top X_i = \sum_{j \in \tilde{J}, j \neq i} \alpha_j u^\top X_j.$$

It follows $\tilde{w}(w, \tilde{J} - \{i\}) = \tilde{w}(w, \tilde{J})$, which is a contradiction to the assumption that $\tilde{J}$ is the smallest.

Now $\tilde{w}$ is uniquely determined by $\tilde{J}$ and the value $I(\tilde{w}^\top X_j \geq 0)$ for each $j \in \tilde{J}$. Therefore there will be no more than $n^d \cdot 2^d$ choices. Since the function values of $I(w^\top X_i \geq 0)$ is identical to the values of one of such $\tilde{w}$, it follows that the function values of $\{I(w^\top X_i \geq 0) : 1 \leq i \leq n\}$ can be no more than $(2n)^d$. This proves the result. ■

A more general bound of uniform L_∞ covering numbers can be obtained using the concept of VC-dimension, which we will discuss later.

4.3 Uniform Convergence

We can obtain the following uniform convergence bounds, which are analogous to the results of Theorem 3.3.

Theorem 4.5 *Given a dataset S_{n+m} . Consider $G = \{g_w(z) : w \in \Omega\}$ so that $g_w(z) \in [0, 1]$. With probability at least $1 - \delta$ over random partitions of S_{n+m} into disjoint subsets $S_n \cup S'_m$, we have*

$$\forall w \in \Omega : g_w(S'_m) \leq g_w(S_n) + \epsilon_{n,m}(\delta, G, S_{n+m}),$$

where

$$\epsilon_{n,m}(\delta, G, S_{n+m}) = \inf_{\epsilon > 0} \left[2\epsilon + \sqrt{\frac{(m+n) \ln(N(\epsilon, G, L_\infty(S_{n+m}))/\delta)}{mn}} \right].$$

Moreover, with probability at least $1 - \delta$, the following inequality holds $\forall \gamma \in (0, 1)$:

$$\forall w \in \Omega : g_w(S'_m) \leq \gamma_{n,m} g_w(S_n) + \epsilon_{n,m}^\gamma(\delta, G, S_{n+m}),$$

where $\gamma_{n,m} = (1 + \gamma(n/m))/(1 - \gamma)$ and

$$\epsilon_{n,m}^\gamma(\delta, G, S_{n+m}) = \inf_{\epsilon > 0} \left[(1 + \gamma_{n,m})\epsilon + \frac{(m+n) \ln(N(\epsilon, G, L_\infty(S_{n+m}))/\delta)}{mn \cdot 2\gamma(1 - \gamma)} \right].$$

Proof Let $G(\epsilon) = \{g_1(z), \dots, g_N(z)\}$ be an ϵ empirical L_∞ cover of G on S_{n+m} with $N = N(\epsilon, G, L_\infty(S_{n+m}))$. We may assume that $g_j(z) \in [0, 1]$ for all j (otherwise, we may set $g_j(z)$ to $\min(1, \max(0, g_j(z)))$).

In the following, we let $j = j(w)$ for simplified notation:

$$g_w(S'_m) - g_w(S_n) \leq \frac{1}{m} \sum_{Z \in S'_m} g_j(Z) - \frac{1}{n} \sum_{Z \in S_n} g_j(Z) + 2\epsilon.$$

Let $\epsilon'' = \sqrt{(m^{-1} + n^{-1}) \ln(N/\delta)}$. It follows from the union bound on j that with probability at least $1 - \delta$:

$$\forall j : \quad \frac{1}{m} \sum_{Z \in S'_m} g_j(Z) \leq \frac{1}{n} \sum_{Z \in S_n} g_j(Z) + \epsilon''.$$

It follows that

$$\sup_{w \in \Omega} [g_w(S'_m) - g_w(S_n)] \leq \frac{1}{m} \sum_{Z \in S'_m} g_j(Z) - \frac{1}{n} \sum_{Z \in S_n} g_j(Z) + 2\epsilon \leq \epsilon'' + 2\epsilon.$$

This leads to the first desired bound.

Moreover, using the multiplicative Chernoff bound and union bound, we obtain the following statement. With probability at least $1 - \delta$, the following inequality holds for all j :

$$\frac{1}{m} \sum_{Z \in S'_m} g_j(Z) \leq \gamma_{m,n} \frac{1}{n} \sum_{Z \in S_n} g_j(Z) + \frac{(m^{-1} + n^{-1}) \ln(N/\delta)}{2\gamma(1 - \gamma)n}.$$

Therefore for all $w \in \Omega$, we obtain

$$\begin{aligned} g_w(S'_m) &\leq \frac{1}{m} \sum_{Z \in S'_m} g_j(Z) + \epsilon \\ &\leq \gamma_{m,n} \frac{1}{n} \sum_{Z \in S_n} g_j(Z) + \frac{(m^{-1} + n^{-1}) \ln(N/\delta)}{2\gamma(1 - \gamma)n} + \epsilon \\ &\leq \gamma_{m,n} g_w(S_n) + \frac{(m^{-1} + n^{-1}) \ln(N/\delta)}{2\gamma(1 - \gamma)n} + (1 + \gamma_{m,n})\epsilon. \end{aligned}$$

The first and the third inequalities are due to the definition of L_∞ cover. This leads to the second desired bound. $\blacksquare$

4.4 Oracle Inequalities

We can establish oracle inequalities using the uniform convergence results, both with respect to the validation data S'_m , and with respect to the true test distribution D .

The statement for the validation data is with respect to the random partition probability, and the statement for the true test distribution is with respect to the random sampling of training data S_n .

Given a data S_n , we define

$$L(w; S_n) = \frac{1}{n} \sum_{(X,Y) \in S_n} L(f(w, x), y).$$

The following result shows that we can obtain a solution from approximate ERM on the training data that has a validation loss nearly as small as the optimal validation loss. It is suitable for the situation that $m \approx n$.

Theorem 4.6 *Assume $L(f(w, x), y) \in [0, 1]$. Let $G = \{L(f(w, x), y) : w \in \Omega\}$. Consider S_{n+m} randomly partitioned to $S_n \cup S'_m$. Let $\hat{w}$ be approximate ERM solution so that*

$$L(\hat{w}, S_n) \leq \inf_{w \in \Omega} L(w, S_n) + \epsilon'.$$

With probability at least $1 - \delta$ over random partitions, we have the following (additive) oracle inequality on validation set:

$$L(\hat{w}, S'_m) \leq \inf_{w \in \Omega} L(w, S'_m) + \epsilon' + 2\epsilon_{n,m}(\delta/2, G, S_{n+m}),$$

where

$$\epsilon_{n,m}(\delta, G, S_{n+m}) = \inf_{\epsilon > 0} \left[2\epsilon + \sqrt{\frac{(n+m) \ln(N(\epsilon, G, L_\infty(S_{n+m}))/\delta)}{mn}} \right].$$

Moreover, with probability at least $1 - \delta$, for all $\gamma \in (0, 1)$, we have the following (multiplicative) oracle inequality on validation set:

$$\begin{aligned} L(\hat{w}, S'_m) &\leq \gamma_{n,m} \gamma_{m,n} \inf_{w \in \Omega} L(w, S'_m) + \gamma_{n,m} \epsilon' + \epsilon_{n,m}^\gamma(\delta/2, G, S_{n+m}) \\ &\quad + \gamma_{n,m} \epsilon_{m,n}^\gamma(\delta/2, G, S_{n+m}), \end{aligned}$$

where $\gamma_{n,m} = (1 + \gamma(n/m))/(1 - \gamma)$ and

$$\epsilon_{n,m}^\gamma(\delta, G, S_{n+m}) = \inf_{\epsilon > 0} \left[(1 + \gamma_{n,m})\epsilon + \frac{(m+n) \ln(N(\epsilon, G, L_\infty(S_{n+m}))/\delta)}{mn \cdot 2\gamma(1 - \gamma)} \right].$$

Proof Let $g_w(Z) = L(f(w, x), y)$. Let $\hat{w}'$ be an arbitrary $w \in \Omega$ that depends on S'_m .

From the additive bound of Theorem 4.5, we obtain with probability at least $1 - \delta/2$:

$$L(\hat{w}, S'_m) \leq L(\hat{w}, S_n) + \epsilon_{n,m}(\delta/2, G, S_{n+m}).$$

Similarly, with probability at least $1 - \delta/2$:

$$L(\hat{w}', S_n) \leq L(\hat{w}', S'_m) + \epsilon_{n,m}(\delta/2, G, S_{n+m}).$$

By taking the union bound, it follows that the above two inequalities hold with probability at least $1 - \delta$. In such case,

$$\begin{aligned} L(\hat{w}, S'_m) &\leq L(\hat{w}, S_n) + \epsilon_{n,m}(\delta/2, G, S_{n+m}) \\ &\leq L(\hat{w}', S_n) + \epsilon' + \epsilon_{n,m}(G, \delta/2, S_{n+m}) \\ &\leq L(\hat{w}', S'_m) + \epsilon' + 2\epsilon_{n,m}(\delta/2, G, S_{n+m}). \end{aligned}$$

This proves the first desired inequality of the theorem.

From the multiplicative bound of Theorem 4.5, we obtain with probability at least $1 - \delta/2$:

$$L(\hat{w}, S'_m) \leq \gamma_{n,m} L(\hat{w}, S_n) + \epsilon_{n,m}^\gamma(\delta/2, G, S_{n+m}).$$

Similarly, with probability at least $1 - \delta/2$:

$$L(\hat{w}', S_n) \leq \gamma_{m,n} L(\hat{w}', S'_m) + \epsilon_{m,n}^\gamma(\delta/2, G, S_{n+m}).$$

By taking the union bound, it follows that the above two inequalities hold with probability at least $1 - \delta$. In such case,

$$\begin{aligned} L(\hat{w}, S'_m) &\leq \gamma_{n,m} L(\hat{w}, S_n) + \epsilon_{n,m}^\gamma(\delta/2, G, S_{n+m}) \\ &\leq \gamma_{n,m} [L(\hat{w}', S_n) + \epsilon'] + \epsilon_{n,m}^\gamma(\delta/2, G, S_{n+m}) \\ &\leq \gamma_{n,m} \gamma_{m,n} L(\hat{w}', S'_m) + \gamma_{n,m} \epsilon' + \epsilon_{n,m}^\gamma(\delta/2, G, S_{n+m}) \\ &\quad + \gamma_{n,m} \epsilon_{m,n}^\gamma(\delta/2, G, S_{n+m}). \end{aligned}$$

This proves the second desired inequality of the theorem. ■

Using uniform L_∞ entropy, we can also obtain oracle inequality with respect to the true distribution D if S_n are n iid examples from D .

Theorem 4.7 *Assume $L(f(w, x), y) \in [0, 1]$. Let $G = \{L(f(w, x), y) : w \in \Omega\}$. Consider the approximate ERM method (3.3). With probability at least $1 - \delta$:*

$$\begin{aligned} \mathbf{E}_D L(f(\hat{w}, X), Y) &\leq \inf_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y) + \epsilon' + \sqrt{2 \ln(4/\delta)/n} \\ &\quad + \inf_{\epsilon > 0} \left[2\epsilon + \sqrt{2 \ln(2N_\infty(\epsilon, G, 2n)/\delta)/n} \right]. \end{aligned}$$

Moreover, with probability at least $1 - \delta$, the following inequality holds for all $\gamma \in (0, 1)$:

$$\begin{aligned} (1 - \gamma)^2 \mathbf{E}_D L(f(\hat{w}, X), Y) &\leq (1 + \gamma)^2 \inf_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y) + (1 + \gamma) \epsilon' \\ &\quad + \frac{3 \ln(4/\delta)}{\gamma n} + \inf_{\epsilon > 0} \left[2\epsilon + \frac{\ln(2N_\infty(\epsilon, G, 2n)/\delta)}{\gamma n} \right]. \end{aligned}$$

Proof Consider an arbitrary $w \in \Omega$. We consider additional $m = n$ iid samples drawn from $D \{(X_{n+1}, Y_{n+1}), \dots, (X_{n+m}, Y_{n+m})\}$.

From Theorem 4.5, we obtain with probability at least $1 - \delta/2$, the following additive bound holds:

$$L(\hat{w}, S'_m) \leq L(\hat{w}, S_n) + \inf_{\epsilon > 0} \left[2\epsilon + \sqrt{(2/n) \ln(2N_\infty(\epsilon, G, 2n)/\delta)} \right].$$

Moreover, since $\hat{w}$ does not depend on S'_m , we know from Chernoff bound that with probability at least $1 - \delta/4$:

$$\mathbf{E}_D L(f(\hat{w}, X), Y) \leq L(\hat{w}, S'_m) + \sqrt{\ln(4/\delta)/(2n)},$$

and with probability at least $1 - \delta/4$:

$$L(w, S_n) \leq \mathbf{E}_D L(f(w, X), Y) + \sqrt{\ln(4/\delta)/(2n)}.$$

Taking the union bound of the above three events, we know the above three inequalities hold with probability $1 - \delta$. Combine these inequalities, with the definition of approximate ERM, we obtain the first desired bound.

From Theorem 4.5, we obtain with probability at least $1 - \delta/2$, the following multiplicative bound holds:

$$(1 - \gamma)L(\hat{w}, S'_m) \leq (1 + \gamma)L(\hat{w}, S_n) + 2 \inf_{\epsilon > 0} \left[\epsilon + \frac{\ln(2N_\infty(\epsilon, G, 2n)/\delta)}{2\gamma n} \right].$$

Moreover, since $\hat{w}$ does not depend on S'_m , we know from the multiplicative Chernoff bound that with probability at least $1 - \delta/4$:

$$(1 - \gamma)^2 \mathbf{E}_D L(f(\hat{w}, X), Y) \leq (1 - \gamma)L(\hat{w}, S'_m) + \frac{(1 - \gamma) \ln(4/\delta)}{2\gamma n},$$

and with probability at least $1 - \delta/4$:

$$(1 + \gamma)L(w, S_n) \leq (1 + \gamma)^2 \mathbf{E}_D L(f(w, X), Y) + \frac{(1 + \gamma)(1 + 2\gamma) \ln(4/\delta)}{2\gamma n}.$$

Taking the union bound, we know the above three inequalities hold with probability $1 - \delta$. Combine these inequalities, we obtain the second desired bound. $\blacksquare$

Example 4.2 Consider the linear classifier example in Example 4.1. We have $\ln N_\infty(\epsilon, G, n) \leq d \ln(2n)$ according to Lemma 4.1. It follows that for the approximate ERM method, we have the following oracle inequalities. With probability at least $1 - \delta$:

$$\mathbf{E}_D I(f_{\hat{w}}(X) \neq Y) \leq \inf_{w \in \Omega} \mathbf{E}_D I(f_w(X) \neq Y) + \sqrt{\frac{8(\ln(2/\delta) + d \ln(2n))}{n}}.$$

Moreover, by optimizing the multiplicative bound over γ , we can obtain the following inequality. With probability at least $1 - \delta$:

$$\begin{aligned} \mathbf{E}_D I(f_{\hat{w}}(X) \neq Y) &\leq \text{err}(f_*) \\ &+ C \left[\sqrt{\text{err}(f_*) \frac{\ln(\delta^{-1}) + d \ln(n)}{n}} + \frac{\ln(\delta^{-1}) + d \ln(n)}{n} \right], \end{aligned}$$

where C is an absolute constant and

$$\text{err}(f_*) = \inf_{w \in \Omega} \mathbf{E}_D I(f_w(X) \neq Y).$$

4.5 Vapnik-Chervonenkis Dimension

Let $G = \{g_w(z) : w \in \Omega\}$ be a $\{0, 1\}$ valued binary function class indexed by w . Given an arbitrary n samples $S_n = \{Z_1, \dots, Z_n\}$. We are interested in the number of functions (uniform L_∞ cover of the function class at $\epsilon = 0$) that $G(S_n) = \{[g_w(Z_1), \dots, g_w(Z_n)] : w \in \Omega\}$ can achieve.

We have the following definition from [90].

Definition 4.2 (VC-dimension) We say that G shatters S_n if the number of elements $|G(S_n)|$ is 2^n . That means we can always find w so that $g_w(z)$ matches any possible choices of $\{0, 1\}$ values at the n points. The maximum n such that G shatters at least one instance of S_n , denoted by $\text{VC}(G)$, is called the VC-dimension of G .

Note that the maximum number of functions in $G(S_n)$ is 2^n . If $n > d$, then for any n samples S_n , $G(S_n)$ contains fewer than 2^n elements. Surprisingly, if a binary-valued function class G has VC dimension d , then when $n > d$, the size of set $G(S_n)$ can grow only polynomially in n . This gives an $O(\ln n)$ upper bound on the uniform entropy of the function class G with a finite VC-dimension [90, 71].

Lemma 4.2 (Sauer's Lemma) If $\text{VC}(G) = d$, then for any $n \geq d$, we have for all empirical samples $S_n = \{Z_1, \dots, Z_n\}$:

$$|G(S_n)| \leq \sum_{\ell=0}^d C_n^\ell \leq (en/d)^d.$$

Proof First, we prove the statement under the assumption that $|G(S_n)|$ is upper bounded by the number of subsets of S_n that are shattered by G . Under this assumption, since any subset shattered by G cannot be larger than d by the definition of VC-dimension, and the number of subsets of size

ℓ is C_n^ℓ , we know that the number of subsets shattered by G cannot be more than $\sum_{\ell=1}^d C_n^\ell$, which leads to the following bound for $n \geq d$:

$$\sum_{\ell=1}^d C_n^\ell \leq dn^d/d! \leq (en/d)^d.$$

In the following, we only need to prove the statement that $|G(S_n)|$ is upper bounded by the number of subsets of S_n that are shattered by G . This can be proved by induction on n . When $n = 1$, one can check that the claim holds trivially.

Now assume that the claim holds for all empirical sample of size no more than $n - 1$. Consider n samples $\{Z_1, \dots, Z_n\}$. We define $g_w(S_k) = [g_w(Z_1), \dots, g_w(Z_k)]$, and let

$$G_{n-1}(S_n) = \{[g_w(S_{n-1}), 1] : [g_w(S_{n-1}), 0], [g_w(S_{n-1}), 1] \in G_n(S_n)\}.$$

Using the induction hypothesis, we know that $|G_{n-1}(S_n)|$ is bounded by the number of shattered S in S_{n-1} ; for each shattered S , $S \cup \{Z_n\}$ is shattered by $G(S_n)$ because both $[g_w(S_{n-1}), 1]$ and $[g_w(S_{n-1}), 0]$ belong to $G(S_n)$. Therefore $|G_{n-1}(S_n)|$ is no more than the number of shattered subsets of S_n that contains Z_n .

Moreover, since for $g_w \in G(S_n) - G_{n-1}(S_n)$, $g_w(Z_n)$ is uniquely determined by its values at S_{n-1} (if not, then both $[g_w(S_{n-1}), 0]$ and $[g_w(S_{n-1}), 1]$ can be achieved in $G(S_n) - G_{n-1}(S_n)$, which is impossible because by definition, we should have put $[g_w(S_{n-1}), 1]$ in $G_{n-1}(S_n)$), it follows that $|G(S_n) - G_{n-1}(S_n)|$ is no more than $|G(S_{n-1})|$. By induction hypothesis, $|G(S_{n-1})|$ is no more than the number of shattered subsets of S_n that does not contain Z_n . By combining the above two facts, $|G(S_n)|$ is no more than the number of shattered subsets of S_n . ■

Sauer's Lemma implies the following oracle inequalities for problems with finite VC dimensions. It is a direct consequence of Theorem 4.7.

Theorem 4.8 *Assume $L(\cdot, \cdot) \in \{0, 1\}$ is a binary valued loss function. Let $G = \{L(f(w, x), y) : w \in \Omega\}$, with a finite VC-dimension $\text{vc}(G) = d$. Given $n \geq d/2$, and consider the approximate ERM method (3.3). With probability at least $1 - \delta$:*

$$\begin{aligned} \mathbf{E}_D L(f(\hat{w}, X), Y) &\leq \inf_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y) + \epsilon' + \sqrt{2 \ln(4/\delta)/n} \\ &\quad + \sqrt{2(d \ln(2en/d) + \ln(2/\delta))/n}. \end{aligned}$$

Moreover, with probability at least $1 - \delta$, the following inequality holds for

all $\gamma \in (0, 1)$:

$$(1 - \gamma)^2 \mathbf{E}_D L(f(\hat{w}, X), Y) \leq (1 + \gamma)^2 \inf_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y) + (1 + \gamma) \epsilon' + \frac{3 \ln(4/\delta)}{\gamma n} + \frac{d \ln(2en/d) + \ln(2/\delta)}{\gamma n}.$$

Example 4.3 Consider d -dimensional $\{0, 1\}$ valued linear classifiers of the form $F = \{f_w(x) = I(w^\top x \geq 0), w \in \mathbb{R}^d\}$, we have $\text{VC}(F) = d$. This implies that $G = \{I(f_w(X) \neq Y), w \in \mathbb{R}^d\}$ has a VC dimension d .

Since it is easy to find $n = d$ points shattered by F , we only need to show that any $n = d + 1$ points cannot be shattered by linear functions.

Let the $d + 1$ points be $x_1, \dots, x_{d+1}$. Then we know that they are linearly dependent. Therefore there exists $d + 1$ real valued coefficients $a_1, \dots, a_{d+1}$ that are not all zeros, such that $a_1 x_1 + \dots + a_{d+1} x_{d+1} = 0$ and we can assume that there exists at least one a_i such that $a_i > 0$.

In order to show that $x_1, \dots, x_{d+1}$ cannot be shattered by n points, we only need to show that there is no $w \in \mathbb{R}^d$ such that

$$I(w^\top x_i \geq 0) = 0 \quad (a_i > 0); \quad I(w^\top x_i \geq 0) = 1 \quad (a_i \leq 0),$$

which implies that a particular set of function value on these points cannot be achieved. We prove this by contradiction. Assume the above function values can be achieved, then $a_i w^\top x_i \leq 0$ for all i . Since there is at least one $a_i > 0$, we know that for this i , $a_i w^\top x_i < 0$. Therefore

$$\sum_{i=1}^{d+1} a_i w^\top x_i < 0.$$

However, this is a contradiction to the fact that $a_1 x_1 + \dots + a_{d+1} x_{d+1} = 0$.

Note that results of Theorem 4.8 holds uniformly for all distributions D . Therefore concept classes with finite VC dimensions are statistically learnable by ERM (assuming the solution of ERM is computationally efficient). On the other hand, if the VC dimension of a concept class is infinity, then for any sample size n , there exists a distribution D with n samples, so that the concept class can achieve all possible binary values of 2^n on D . Therefore on such a distribution, the learning of this concept class cannot be better than random guessing for sample size smaller than n . The following is an example of infinite VC dimension.

Example 4.4 The binary-valued function class $G = \{I(\cos(wz) \geq 0) : w \in \mathbb{R}\}$ has infinite VC-dimension.

Given any d , we consider $\{z_j = 10^{-j} \pi : j = 1, \dots, d\}$. Let $w = \sum_{j=1}^d (1 - b_j) 10^j$, with $b_j \in \{0, 1\}$. It is easy to verify that $I(\cos(w^\top z_j) \geq 0) = b_j$. It follows that the set can be shattered by G .

4.6 Discrete Model Selection

One common usage of training/validation partition is model selection. In our setting, a model is simply a learning algorithm $\mathcal{A}$, that maps training data S_n to a prediction function $f(\cdot) = \mathcal{A}(S_n)$. For the ERM algorithm, the model is determined by the parametrized family $\{f_w(x) : w \in \Omega\}$, which can be characterized by the parameter family Ω .

Consider a countable sequence of models $\{\mathcal{A}_j : j = 1, 2, \dots\}$, and let $\{q_j \geq 0\}$ be a sequence of numbers so that $\sum_j q_j \leq 1$. Consider the following model selection algorithm, which finds a model $\hat{j}$ to approximately minimize the following criterion:

$$Q(\mathcal{A}_{\hat{j}}, S_n, S'_m) \leq \inf_j Q(\mathcal{A}_j, S_n, S'_m) + \tilde{\epsilon}, \quad (4.1)$$

where

$$Q(\mathcal{A}_j, S_n, S'_m) = \frac{1}{|S'_m|} \sum_{(X,Y) \in S'_m} L(\hat{f}_j(X), Y) + r_m(q_j), \quad \hat{f}_j = \mathcal{A}_j(S_n).$$

We have the following generic model selection result using additive Chernoff bound. Similar results can be obtained for multiplicative Chernoff bounds, and Bernstein's inequalities.

Theorem 4.9 (Discrete Model Selection) *Assume $L(\cdot, \cdot) \in [0, 1]$, and the validation data S'_m contains m iid samples from D . Given training data S_n , consider (4.1) with*

$$r_m(q_j) = \sqrt{\frac{\ln(1/q_j)}{2m}}.$$

Then with probability at least $1 - \delta$ over the random selection of S'_m :

$$\mathbf{E}_D L(\hat{f}_{\hat{j}}(X), Y) \leq \inf_j Q(\mathcal{A}_j, S_n, S'_m) + \tilde{\epsilon} + \sqrt{\frac{\ln(1/\delta)}{2m}}.$$

This implies the following oracle inequality. With probability at least $1 - \delta$ over the random sampling of S'_m :

$$\mathbf{E}_D L(\hat{f}_{\hat{j}}(X), Y) \leq \inf_j [\mathbf{E}_D L(\hat{f}_j(X), Y) + r_m(q_j)] + \tilde{\epsilon} + \sqrt{\frac{2 \ln(2/\delta)}{m}}.$$

Proof For each j , we have from the additive Chernoff bound that with probability at least $1 - q_j \delta$:

$$\begin{aligned} \mathbf{E}_D L(\hat{f}_j(X), Y) &\leq \frac{1}{m} \sum_{(X,Y) \in S'_m} L(\hat{f}_j(X), Y) + \sqrt{\ln(1/(q_j \delta)) / (2m)} \\ &\leq \frac{1}{m} \sum_{(X,Y) \in S'_m} L(\hat{f}_j(X), Y) + \sqrt{\frac{\ln(1/q_j)}{2m}} + \sqrt{\frac{\ln(1/\delta)}{2m}}. \end{aligned}$$

Taking the union bound over j , we know that the above claim holds for all j with probability at least $1 - \delta$. This result, combined with the definition of $\hat{j}$ in (4.1), leads to the first desired bound.

Now by applying Chernoff bound for an arbitrary j that does not depend on S'_m , we obtain with probability at least $1 - \delta/2$:

$$Q(\mathcal{A}_j, S_n, S'_m) \leq \mathbf{E}_D L(\hat{f}_j(X), Y) + r_m(q_j) + \sqrt{\frac{\ln(2/\delta)}{2m}}.$$

By combining this inequality with the first bound of the theorem, we obtain the second desired inequality. $\blacksquare$

Theorem 4.9 can be combined with Theorem 4.5 to obtain a more precise statement for ERM methods, showing that a near optimal generalization bound can be obtained using the method of (4.1). This is an oracle inequality for model selection with ERM learners.

Corollary 4.3 *Consider a family of approximate ERM algorithms $\{\mathcal{A}_j\}$, each characterized by its parameter space Ω_j , and returns a function $\hat{f}_j(\cdot) = f_{\hat{w}_j}(\cdot)$ for some $\hat{w}_j \in \Omega_j$ such that*

$$L(\hat{w}, S_n) \leq \inf_{w \in \Omega_j} L(w, S_n) + \epsilon',$$

where we use the notation of Theorem 4.6.

Assume further that we use (4.1) to select the best model $\hat{j}$ on the validation data S'_m , with the choice of $r_m(\cdot)$ in Theorem 4.9, and we take $m = n$. Then the following result holds with probability at least $1 - \delta$ over random selections of $S_{n+m} = S_n \cup S'_m$:

$$\begin{aligned} \mathbf{E}_D L(\hat{f}_{\hat{j}}(X), Y) &\leq \inf_{j; w \in \Omega_j, \epsilon > 0} [\mathbf{E}_D L(f_w(X), Y) + r_m(q_j) + 2\epsilon \\ &\quad + \sqrt{\frac{2 \ln(N_\infty(\epsilon, G_j, 2n))}{n}}] + \epsilon' + \tilde{\epsilon} + 2\sqrt{\frac{2 \ln(3/\delta)}{n}}. \end{aligned}$$

Proof Consider any j and $w \in \Omega_j$: We have from Theorem 4.9 that with probability $1 - \delta/3$,

$$\mathbf{E}_D L(\hat{f}_{\hat{j}}(X), Y) \leq [L(\hat{w}_j, S'_m) + r_m(q_j)] + \tilde{\epsilon} + \sqrt{\frac{\ln(3/\delta)}{2m}}.$$

Moreover, from Theorem 4.5, we have with probability at least $1 - \delta/3$:

$$\begin{aligned} L(\hat{w}_j, S'_m) &\leq L(\hat{w}_j, S_n) + \epsilon_{n,m}(\delta/3, G_j, S_{n+m}) \\ &\leq L(w, S_n) + \epsilon' + \epsilon_{n,m}(\delta/3, G_j, S_{n+m}), \end{aligned}$$

where $\epsilon_{n,m}(\cdot)$ is defined in Theorem 4.5. Moreover, from the Chernoff bound, we have with probability at least $1 - \delta/3$:

$$L(w, S_n) \leq \mathbf{E}_D L(f_w(X), Y) + \sqrt{\frac{\ln(3/\delta)}{2n}}.$$

Taking the union bound, all of the above inequalities hold with probability at least $1 - \delta$, which leads to the desired bound. $\blacksquare$

Example 4.5 Consider binary classification problems, with binary classifiers $\{f_w^j(x) : w \in \Omega_j\}$ of VC-dimension d_j . Take $q_j = 1/(j+1)^2$. Then we have from Corollary 4.3 that

$$\begin{aligned} \mathbf{E}_D I(\hat{f}_{\hat{j}}(X) \neq Y) &\leq \inf_{j, w \in \Omega_j} \left[\mathbf{E}_D I(f_w^j(X) \neq Y) + \sqrt{\frac{2 \ln(j+1)}{2n}} \right. \\ &\quad \left. + \sqrt{\frac{2d_j \ln(2en/d_j)}{n}} \right] + \epsilon' + \tilde{\epsilon} + 2\sqrt{\frac{2 \ln(3/\delta)}{n}}. \end{aligned}$$

This result shows that the model selection algorithm of (4.1) can automatically balance the model accuracy $\mathbf{E}_D I(f_w^j(X) \neq Y)$ and model dimension d_j , and chooses the optimal model, up to a penalty of $O(\sqrt{\ln(j+1)/n})$.

4.7 Exercises

1. Consider $z \in \mathbb{R}^d$, let $w = [w_1, \dots, w_d, w_{d+1}, \dots, w_{2d}]$. Find the VC dimension of the function class

$$f_w(z) = I(z \in C(w)),$$

where $C(w) = \{z = [z_1, \dots, z_d] : z_j \in [w_j, w_{d+j}]\}$.

2. In Theorem 4.8. Assume that $\epsilon' = 0$. Try to optimize γ in the multiplicative bound to obtain a sharp oracle inequality. State results with explicit constants in a form similar to Example 4.2.
3. In Theorem 4.9, if we consider model selection with

$$r_m(q_j) = \frac{\ln(1/q_j)}{2\gamma n}$$

for a fixed $\gamma > 0$, use multiplicative Chernoff bound to obtain oracle inequalities similar to those of Theorem 4.9 and Corollary 4.3.

4. Consider $G = \{g_w(z) : w \in \Omega\}$ with $g_w(z) \in [0, 1]$. Given a dataset S_{n+m} , we consider random partitions of S_{n+m} into disjoint subsets $S_n \cup S_m$. Using notations in the lecture notes, we define

$$R(G, S_n, S'_m) = \sup_{w \in \Omega} [g_w(S_{n+m}) - g_w(S_n)].$$

- If $|G| = N$ is finite, derive an upper bound of

$$\mathbf{E}_{S_n \cup S'_m} R(G, S_n, S'_m)$$

by using the proof technique of Theorem 5.4, where the expectation is with respect to the random partition.

- For general G , with covering number $N(\epsilon, G, L_\infty(S_{n+m}))$, estimate

$$\mathbf{E}_{S_n \cup S'_m} R(G, S_n, S'_m)$$

by using the chaining technique of Theorem 5.5, where the expectation is with respect to the random partition.

Chapter 5

Covering Number Estimates

This chapter derive covering number estimates of certain function classes, including some parametric and nonparametric function classes.

5.1 Empirical L_p covering numbers of VC-class

We have obtained empirical L_∞ covering number bounds for VC classes, and the dependency depends logarithmically on the sample size n . It is also possible to obtain the empirical L_p covering number for VC classes for $p < \infty$ which is independent of n . Such estimate can be used with Chaining to estimate Rademacher complexity, as we will show later.

Recall that given empirical distribution S_n , the empirical L_p cover is the number of functions needed to cover g_w based on the empirical L_p metric:

$$d_p(g, g') = \left[\frac{1}{n} \sum_{i=1}^n |g(Z_i) - g'(Z_i)|^p \right]^{1/p}.$$

We have the following estimate.

Theorem 5.1 *If $G = \{g_w(Z) : w \in \Omega\}$ is a VC class, then*

$$\ln N(\epsilon, G, L_1(S_n)) \leq d \ln(4e \ln(4/\epsilon)/\epsilon).$$

Proof Given $S_n = \{Z_1, \dots, Z_n\}$. Let $Q = \{g_1, \dots, g_m\}$ be a maximal subset of G such that

$$\frac{1}{n} \sum_{i=1}^n |g_j(Z_i) - g_k(Z_i)| > \epsilon$$

for all $j \neq k$. By maximality, Q is an L_1 ϵ -cover of G . We know that for the empirical distribution $D(S_n)$ that puts a probability of $1/n$ on each S_i . For $j \neq k$, we have

$$\Pr_{Z \sim S_n} [g_j(Z) = g_k(Z)] < 1 - \epsilon.$$

Now consider random sample with replacement from S_n for T times to obtain samples $\{Z_{i_1}, \dots, Z_{i_T}\}$. We have

$$\Pr(\{\forall \ell : g_j(Z_{i_\ell}) = g_k(Z_{i_\ell})\}) < (1 - \epsilon)^T \leq e^{-T\epsilon}.$$

That is, with probability larger than $1 - e^{-T\epsilon}$,

$$\exists \ell : g_j(Z_{i_\ell}) \neq g_k(Z_{i_\ell}).$$

Taking the union bound for all $j \neq k$, we have with probability larger than $1 - C_m^2 \cdot e^{-T\epsilon}$, for all $j \neq k$:

$$\exists \ell : g_j(Z_{i_\ell}) \neq g_k(Z_{i_\ell}).$$

If we take $T \geq \ln(m^2 - m)/\epsilon$, then $e^{-T\epsilon}C_m^2 \leq 1$. Then there exists T samples $\{Z_{i_\ell} : \ell = 1, \dots, T\}$ such that $g_j \neq g_k$ for all k when restricted to these samples. Since $\text{vc}(G) = d$, we obtain from the Sauer's lemma:

$$m \leq \max[2, eT/d]^d \leq \max[2, 2e \ln(m)/(d\epsilon)]^d.$$

By solving this inequality we obtain the bound $m \leq d \ln(4e \ln(4/\epsilon)/\epsilon)$. This implies the desired result. $\blacksquare$

It is possible to prove a slightly stronger result of the form

Theorem 5.2 *If $\text{vc}(G) = d$, then*

$$\ln N(\epsilon, G, L_1(S_n)) \leq \ln 2 + d \ln(4e/\epsilon).$$

Proof We choose $i_1, \dots, i_T$ uniformly from $1, \dots, n$ as before, and also choose g and g' uniformly at random from Q .

Lemma 5.1 *For any k : if $T \geq d$, then*

$$\Pr(g(Z_{i_k}) \neq g'(Z_{i_k}) | g(Z_{i_j}) = g'(Z_{i_j}) \text{ for all } j \neq k) \geq (1 - (eT/d)^d/m)\epsilon.$$

Moreover,

$$\frac{1}{T} \sum_{k=1}^T \Pr(g(Z_{i_k}) \neq g'(Z_{i_k}) | g(Z_{i_j}) = g'(Z_{i_j}) \text{ for all } j \neq k) \leq 2d/T.$$

Combine the two inequalities, we obtain for all $T \geq d$:

$$(1 - (eT/d)^d/m)\epsilon \leq 2d/T.$$

This implies that with $T = \lfloor 4d/\epsilon \rfloor$:

$$m \leq (eT/d)^d / (1 - 2d/(\epsilon T)) \leq 2(4e/\epsilon)^d.$$

This leads to the desired bound. ■

Note that if we replace S_n by any distribution D over Z , then we still have

$$\ln N(\epsilon, G, L_p(D)) \leq \ln 2 + d \ln(4e/\epsilon^p),$$

because any D can be approximated by empirical distribution drawn from D with sufficiently large n . We thus have the following result.

Corollary 5.1 *If $\text{vc}(G) = d$, then for all distributions D over Z , we have*

$$\ln N(\epsilon, G, L_p(D)) \leq \ln 2 + d \ln(4e/\epsilon^p)$$

for $p \in [1, \infty)$.

Proof We just note that an $\epsilon^p L_1(D)$ cover of G is an $\epsilon L_p(D)$ cover of G . This is because

$$\mathbf{E}_D |g(Z) - g'(Z)| \leq \epsilon^p$$

implies that

$$[\mathbf{E}_D |g(Z) - g'(Z)|^p]^{1/p} \leq \epsilon,$$

where we may assume that $g(Z) \in [0, 1]$ and $g(Z') \in [0, 1]$. ■

5.2 VC-subgraph Class

One may extend the concept of VC dimension to real valued functions by introducing the definition of VC subgraph class.

Definition 5.1 *A real valued function class of $z \in \mathcal{Z}$*

$$G = \{g_w(Z) : w \in \Omega\}$$

is a VC-subgraph class, if the binary function class

$$G_{\text{subgraph}} = \{I(t < g_w(z)) : w \in \Omega\}$$

defined on $(z, t) \in \mathcal{Z} \times \mathbb{R}$ is a VC class. The VC dimension (some times called Pseudo-dimension) of G is $\text{vc}(G) = \text{vc}(G_{\text{sub-graph}})$.

Example 5.1 *The d dimensional linear functions of the form $f_w(x) = w^\top x$ is VC subgraph class of VC dimension $d+1$. This is because $w^\top x - t$ is linear function in $d+1$ dimension, and we have shown that it has VC dimension $d+1$.*

Moreover, if $F = \{f(w, x) : w \in \Omega\}$ is a VC subgraph class and h is monotone function, then $h \circ F = \{h(f(w, x)) : w \in \Omega\}$ is VC subgraph: $\text{vc}(h \circ F) \leq \text{vc}(F)$.

Theorem 5.3 *Assume that G is a VC subgraph class, with VC dimension d , and all $g \in G$ are bounded: $g(Z) \in [0, 1]$. Then for any distribution D over Z , we have*

$$\ln N(\epsilon, G, L_p(D)) \leq \ln 2 + d \ln(4e/\epsilon^p).$$

Proof Let U be a random variable distributed uniformly over $[0, 1]$. Then for all $a \in (0, 1)$: $\mathbf{E}_U I(U \leq a) = a$. Thus for all $g, g' \in G$:

$$\begin{aligned} & \mathbf{E}_D |g(Z) - g'(Z)|^p \\ &= \mathbf{E}_D |\mathbf{E}_U [I(U \leq g(Z)) - I(U \leq g'(Z))]|^p \\ &\leq \mathbf{E}_D \mathbf{E}_U |I(U \leq g(Z)) - I(U \leq g'(Z))|^p. \end{aligned}$$

Therefore

$$\ln N(\epsilon, G, L_p(D)) \leq \ln N(\epsilon, G_{\text{subgraph}}, L_p(D \times U(0, 1))).$$

This leads to the desired bound. ■

5.3 Convex Hull Class

Convex hull of a function class is frequently encountered in applications, and is related to L_1 regularization. We can define the convex hull of a function class as follows.

Definition 5.2 *The convex hull of a function class $\mathcal{F} = \{f(\theta, x) : \theta \in \Theta\}$ is defined as*

$$\text{CONV}(\Psi) = \left\{ \sum_{j=1}^m w_j f(\theta_j, x) : m > 0, \|w\|_1 = 1, w_j \geq 0, \theta_j \in \Theta \right\}.$$

We may also include the closure of the finite sum functions above in the convex hull.

If $\mathcal{F}$ is finite, then we have the following covering number estimates.

Theorem 5.4 *Consider finite class $\mathcal{F} = \{f_1, \dots, f_d\}$, and assume that for a distribution D , $\sup_{f \in \mathcal{F}} \|f\|_{L_p(D)} \leq A$. Then for $\epsilon \leq A$:*

$$\ln N(\epsilon, \text{CONV}(\mathcal{F}), L_p(D)) \leq d \ln(3A/\epsilon).$$

Proof For simplicity, we assume that $A = 1$. Let $B_1 \in R^d$ be the L_1 -ball:

$$B_1 = \{q \in R^d : \|q\|_1 \leq 1\}.$$

Consider a maximum subset of $Q \subset B_1$ such that $Q = \{q^1, \dots, q^N\}$ such that $\|q^j - q^k\|_1 > \epsilon$ for $j \neq k$. Then Q is an ϵ cover of B_1 .

It follows that $q^j + (\epsilon/2)B_1$ are non-overlapping for $j = 1, \dots, N$. Since each $q^j + (\epsilon/2)B_1 \subset (1 + \epsilon/2)B_1$. Thus

$$N \cdot (\epsilon/2)^d \text{vol}(B_1) = N \text{vol}((\epsilon/2)B_1) \leq \text{vol}((1 + \epsilon/2)B_1) = (1 + \epsilon/2)^d \text{vol}(B_1).$$

Therefore $N(\epsilon/2)^d \leq (1 + \epsilon/2)^d$. Thus $N \leq (3/\epsilon)^d$.

Note that for any $q \in B_1$, let $q^k \in Q$ so that $\|q - q^k\|_1 \leq \epsilon$, then $|\sum_{j=1}^d q_j^k f_j - \sum_{j=1}^d q_j f_j| \leq \epsilon$. Therefore $\{\sum_{j=1}^d q_j^k f_j : k = 1, \dots, N\}$ is an ϵ cover of $\text{CONV}(F)$. The result follows. ■

The above result is linear in the dimension d . However, for high dimensional problems, one would like to obtain a bound that is logarithmic in the dimensionality d . The following result gives such a bound for L_2 cover, but with a polynomial dependency on $1/\epsilon$ instead of logarithmic dependency as in the previous result.

Theorem 5.5 Consider any class $\mathcal{F} = \{f_1, \dots, f_d\}$, and assume that for a distribution D , $\sup_{f \in \mathcal{F}} \|f\|_{L_2(D)} \leq A$. Then

$$\ln N(\epsilon, \text{CONV}(\mathcal{F}), L_2(D)) \leq \lceil A^2/\epsilon^2 \rceil \ln[e + ed\epsilon^2/A^2].$$

Proof For simplicity, we assume that $A = 1$. Given any $f = \sum_j \alpha_j f_j \in \text{CONV}(\mathcal{F})$, with $\sum_j \alpha_j = 1$ and $\alpha_j \geq 0$, we can regard p_α as a probability measure on $\{1, \dots, d\}$ with $p_\alpha(j) = \alpha_j$ ($j = 1, \dots, d$). Now, let $j_1, \dots, j_k$ be k iid samples from p_α , then $f = \sum_j \alpha_j f_j = E_{j_s} f_{j_s}$.

$$\begin{aligned} & \mathbf{E}_{j_1, \dots, j_k} \left\| k^{-1} \sum_{s=1}^k f_{j_s} - f \right\|_{L_2(D)}^2 = \mathbf{Var}_{j_1, \dots, j_k} \left(\left\| k^{-1} \sum_{s=1}^k f_{j_s} \right\|_{L_2(D)} \right) \\ &= k^{-2} \sum_{s=1}^k \mathbf{Var}_{j_s} (\|f_{j_s}\|_{L_2(D)}) \\ &\leq k^{-2} \sum_{s=1}^k \mathbf{E}_{j_s} (\|f_{j_s}\|_{L_2(D)}^2) \leq 1/k. \end{aligned}$$

It means that there exists $j_1, \dots, j_s$ such that

$$\left\| k^{-1} \sum_{s=1}^k f_{j_s} - f \right\|_{L_2(D)} \leq 1/\sqrt{k}.$$

Now, consider

$$Q_k = \left\{ k^{-1} \sum_{j=1}^d n_j f_j : \sum_{j=1}^d n_j = k; n_j \geq 0 \right\}.$$

Since $k^{-1} \sum_{s=1}^k f_{j_s} \in Q_k$, we know that Q_k is an $1/\sqrt{k}$ cover of $\text{CONV}(\mathcal{F})$. Take $k = \lceil 1/\epsilon^2 \rceil$, we know that Q_k is an ϵ cover. Moreover, since

$$|Q_k| \leq C_{d+k-1}^k \leq e^k (1 + d/k)^k,$$

we obtain the bound in the theorem. ■

We can also estimate the L_2 covering number for the convex hull of a parametric function class, such as VC-subgraph class. Specifically, in a parametric function class, the covering number is given by

$$\ln N(\epsilon, \mathcal{F}, L_2(D)) \leq V \ln(c/\epsilon), \quad (5.1)$$

where c is a constant, and V is the dimensionality (such as VC-dimension) of the $\mathcal{F}$.

We have the following covering number estimates for the convex hull of (5.1).

Theorem 5.6 *Consider a function class $\mathcal{F}$ with covering number given by (5.1) for some $c \geq 1$ and $V > 0$. Let $A = \sup_{f \in \mathcal{F}} \|f\|_{L_2(D)}$, then we have*

$$\ln N(\epsilon, \text{CONV}(\mathcal{F}), L_2(D)) \leq 10(2c/\epsilon)^{2V/(V+2)} \ln \max[12A/\epsilon, 3 + 6(2c/\epsilon)^V].$$

Proof We let $\mathcal{F}_\epsilon$ be an $\epsilon/2$ cover of $\mathcal{F}$ in $L_2(D)$. Then any $\epsilon/2$ cover of $\text{CONV}(\mathcal{F}_\epsilon)$ gives an ϵ cover of $\text{CONV}(\mathcal{F})$.

Moreover, we consider $\mathcal{F}_{\epsilon'}$ as an $\epsilon'/2$ cover for some $\epsilon' \geq \epsilon$, and decompose each $f_j \in \mathcal{F}_\epsilon$ as

$$f = f' + \Delta f,$$

where $f' \in \mathcal{F}_{\epsilon'}$ and $\|\Delta f\|_{L_2(Q)} \leq \epsilon'/2$.

Let $\Delta\mathcal{F}_\epsilon = \{\Delta f : f \in \mathcal{F}_\epsilon\}$, then using this decomposition, we know that

$$\text{CONV}(\mathcal{F}_\epsilon) \subset \text{CONV}(\Delta\mathcal{F}_\epsilon) + \text{CONV}(\mathcal{F}_{\epsilon'}).$$

It follows that

$$\begin{aligned} \ln N(\epsilon, \text{CONV}(\mathcal{F}), L_2(D)) &\leq \ln N(\epsilon/2, \text{CONV}(\mathcal{F}_\epsilon), L_2(D)) \\ &\leq \ln N(\epsilon/4, \text{CONV}(\mathcal{F}_{\epsilon'}), L_2(D)) + \ln N(\epsilon/4, \text{CONV}(\Delta\mathcal{F}_\epsilon), L_2(D)). \end{aligned}$$

Since

$$|\mathcal{F}_{\epsilon'}| \leq (2c/\epsilon')^V,$$

we have from Theorem 5.4

$$\ln N(\epsilon/4, \text{CONV}(\mathcal{F}_{\epsilon'}), L_2(D)) \leq (2c/\epsilon')^V \ln(12A/\epsilon).$$

Moreover, since

$$|\Delta\mathcal{F}_{\epsilon}| \leq |\mathcal{F}_{\epsilon}| \leq (2c/\epsilon)^V,$$

we have from Theorem 5.5 that

$$\ln N(\epsilon/4, \text{CONV}(\Delta\mathcal{F}_{\epsilon}), L_2(D)) \leq (3\epsilon'/\epsilon)^2 \ln[e + e(2c/\epsilon)^V (2\epsilon/\epsilon')].$$

Now let $\epsilon'/2c = (\epsilon/2c)^{2/(V+2)}$, we have

$$\begin{aligned} & \ln N(\epsilon/4, \text{CONV}(\mathcal{F}_{\epsilon'}), L_2(D)) + \ln N(\epsilon/4, \text{CONV}(\Delta\mathcal{F}_{\epsilon}), L_2(D)) \\ & \leq (2c/\epsilon')^V \ln(12A/\epsilon) + (3\epsilon'/\epsilon)^2 \ln[e + e(2c/\epsilon)^V (2\epsilon/\epsilon')] \\ & \leq (2c/\epsilon)^{2V/(V+2)} \ln(12A/\epsilon) + 9(2c/\epsilon)^{2V/(V+2)} \ln[e + 2e(2c/\epsilon)^V]. \end{aligned}$$

This proves the theorem. ■

Note that for a finite dimensional class $d = |\mathcal{F}| < \infty$, it's VC dimension is no more than $\log_2 d$ because any $\lfloor \log_2 |\mathcal{F}| \rfloor + 1$ points cannot be shattered. It means that we can take $V = \log_2 d$, and obtain a result

$$\ln N(\epsilon, \text{CONV}(\mathcal{F}), L_2(D)) = O\left(\epsilon^{-2\log_2 d / \log_2(4d)} \log_2 d \ln(1/\epsilon)\right),$$

which is slightly stronger (in its dependency of ϵ) than that of Theorem 5.5, which has an entropy growth rate of $O(\epsilon^{-2})$.

Using a similar proof technique, but with a more careful analysis, it is possible to get rid of $\ln(1/\epsilon)$ in Theorem 5.6, and obtain the following result. The details can be found in [89].

Theorem 5.7 *Let $A = \sup_{f \in F} \|f\|_{L_2(D)}$. If $\ln N(F, \epsilon, L_2(D)) \leq V \ln(cA/\epsilon)$ for some $c \geq 1$ and $V > 0$, then when $\epsilon \leq 1$, we have*

$$\ln N(\text{CONV}(F), \epsilon, L_2(D)) \leq K(c, V)(A/\epsilon)^{2V/(V+2)}$$

for some $K(c, V)$ that depends on c and V .

The convex hull of a parametric function class has entropy growth rate with a polynomial $(1/\epsilon)^r$ dependency on $1/\epsilon$. Function class with such a growth behavior is referred to as *non-parameteric function classes*. Moreover, since $r < 2$, the entropy integral

$$\int_0^\infty \sqrt{N(\epsilon, \text{CONV}(\mathcal{F}), L_2(D))} d\epsilon < \infty.$$

Such a function class is referred to as *Donsker class*, for which central limit theorem holds for empirical processes [89].

Example 5.2 Consider neural networks with $x \in \mathbb{R}^d$. Let $h(z) = 1/(1 + \exp(-z))$ be the sigmoid transfer function. Let

$$\mathcal{F} = \{h(\theta^\top x) : \theta \in \mathbb{R}^d\} \cup \{-h(\theta^\top x) : \theta \in \mathbb{R}^d\}$$

be the function class of one-layer neurons, then $\mathcal{F}$ is a VC sub-graph class and thus has parametric covering number

$$N(\epsilon, \mathcal{F}, L_2(D)) \leq (c/\epsilon)^{d+1}$$

uniformly for all distributions D . The L_1 regularized two-layer neural network of the form

$$G = \left\{ \sum_{j=1}^m w_j h(\theta_j^\top x) \quad \|w\|_1 \leq A \right\}$$

has entropy number of

$$\ln N(\epsilon, G, L_2(D)) = O(\epsilon^{-2(d+1)/(d+3)}).$$

5.4 Exercises

1. Assume that $\mathcal{F} = \{f_w(x) : w \in \Omega \subset \mathbb{R}^d\}$ is a Lipschitz in w :

$$\forall x : \quad |f_w(x) - f_{w'}(x)| \leq \gamma \|w - w'\|_2,$$

and $d_2(\Omega) = \sup_{w, w' \in \Omega} \|w - w'\|_2 < \infty$. Derive a bound on

$$N(\epsilon, \mathcal{F}, L_\infty)$$

in terms of d , $d_2(\Omega)$, and γ .

Chapter 6

Rademacher Complexity and Concentration Inequalities

In Chapter 3 and Chapter 4, we obtained uniform convergence results using covering numbers and exponential probability inequalities. This Chapter considers a different, although highly related method. In this approach, we first bound the expectation of the supremum of an underlying empirical process using the so-called *Rademacher complexity*, and then use *concentration inequalities* to obtain high probability bounds. This approach simplifies various derivations in generalization analysis. The introduction of Rademacher complexity in machine learning is due to [49, 50, 15]. The treatment presented here mainly follows [15] and [61].

6.1 Rademacher Complexity

Using the notations from Section 3.3, we are given a function class $G = \{g_w(z) : w \in \Omega\}$, and are interested in the uniform convergence of training error

$$g_w(S_n) = \frac{1}{n} \sum_{i=1}^n g_w(Z_i)$$

on a training data $S_n = \{Z_1, \dots, Z_n\}$, to the test error

$$g_w(D) = \mathbf{E}_{Z \sim D} g_w(Z)$$

on the test data D . In particular, in the general analysis of learning algorithms, we want to estimate

$$\epsilon(G, S_n) = \sup_{w \in \Omega} [g_w(D) - g_w(S_n)].$$

Let its expectation be

$$\epsilon_n(G, D) = \mathbf{E}_{S_n} \epsilon(G, S_n).$$

The smaller this quantity is, the closer the gap between training error and test error is, which implies that we have less overfitting. In Chapter 3, we obtained large probability convergence result of $\epsilon(G, S_n)$, and then derived oracle inequalities in large probabilities. In the following, we show that if average convergence can be obtained then we can derive oracle inequalities in the average sense.

Theorem 6.1 *Consider $g_w(Z) = L(f(w, X), Y)$ with $Z = (X, Y)$, where $Z \sim D$. Let S_n be n iid samples from D . Then the approximate ERM method of (3.3) satisfies*

$$\mathbf{E}_{S_n} g_{\hat{w}}(D) \leq \inf_{w \in \Omega} g_w(D) + \epsilon' + \epsilon_n(G, D).$$

Proof Given any $w \in \Omega$, we have for each training data S_n

$$\begin{aligned} g_{\hat{w}}(D) &\leq g_{\hat{w}}(S_n) + \sup_{w \in \Omega} [g_w(D) - g_w(S_n)] \\ &\leq g_w(S_n) + \epsilon' + \sup_{w \in \Omega} [g_w(D) - g_w(S_n)]. \end{aligned}$$

Taking expectation with respect to S_n , and note that w does not depend on S_n , we obtain

$$\mathbf{E}_{S_n} g_{\hat{w}}(D) \leq g_w(D) + \epsilon' + \mathbf{E}_{S_n} \sup_{w \in \Omega} [g_w(D) - g_w(S_n)].$$

This implies the desired bound. ■

We are now ready to define Rademacher complexity. While the standard definition is two-sided where the supremum is over the absolute value of the sum, we consider one-sided bound which is more convenient for our purpose.

Definition 6.1 *Given $S_n = \{Z_1, \dots, Z_n\}$, the (one-sided) empirical Rademacher complexity of G is defined as*

$$R(G, S_n) = \mathbf{E}_{\sigma} \sup_{w \in \Omega} \frac{1}{n} \sum_{i=1}^n \sigma_i g_w(Z_i),$$

where $\sigma_1, \dots, \sigma_n$ are independent uniform $\{\pm 1\}$ -valued Bernoulli random variables. Moreover, the expected Rademacher complexity is

$$R_n(G, D) = \mathbf{E}_{S_n} R(G, S_n),$$

where S_n are iid samples from D .

The following result shows that the quantity $\epsilon_n(G, D)$ can be upper bounded by Rademacher complexity. It follows that an average oracle inequality can be obtained using Rademacher complexity. The proof uses a technique called *symmetrization*.

Corollary 6.1 *We have*

$$\epsilon_n(G, D) \leq 2R_n(G, D).$$

Consequently, the approximate ERM method of (3.3) satisfies

$$\mathbf{E}_{S_n} g_{\hat{w}}(D) \leq \inf_{w \in \Omega} g_w(D) + \epsilon' + 2R_n(G, D).$$

Proof Let $S'_n = \{Z'_1, \dots, Z'_n\}$ be n iid samples from D that are independent of S_n . We have

$$\begin{aligned} \epsilon_n(G, D) &= \mathbf{E}_{S_n \sim D} \sup_{w \in \Omega} [g_w(D) - g_w(S_n)] \\ &= \mathbf{E}_{S_n \sim D} \sup_{w \in \Omega} [\mathbf{E}_{S'_n} g_w(S'_n) - g_w(S_n)] \\ &\leq \mathbf{E}_{S_n, S'_n \sim D} \sup_{w \in \Omega} [g_w(S'_n) - g_w(S_n)] \\ &= \mathbf{E}_{S_n, S'_n \sim D} \mathbf{E}_{\sigma} \sup_{w \in \Omega} \frac{1}{n} \sum_{i=1}^n [\sigma_i g_w(Z'_i) - \sigma_i g_w(Z_i)] \\ &\leq \mathbf{E}_{S_n, S'_n \sim D} [R(G, S_n) + R(G, S'_n)] = 2R_n(G, D). \end{aligned}$$

This proves the desired bound. ■

One reason to introduce Rademacher complexity is that it can be estimated on the training data. Moreover, it is often easier to estimate this quantity theoretical for many problems.

6.2 Concentration Inequality

We showed that using Rademacher complexity, we may obtain an oracle inequality in expectation. By using concentration inequality, we can also obtain high probability uniform convergence and oracle inequality statements.

The simplest concentration inequality is a generalization of Chernoff bound, due to McDiarmid[58].

Theorem 6.2 (McDiarmid's Inequality) *Consider n independent random variables $X_1, \dots, X_n$, and a real-valued function $f(X_1, \dots, X_n)$ that satisfies the following inequality*

$$\sup_{x_1, \dots, x_n, x'_i} |f(x_1, \dots, x_n) - f(x_1, \dots, x_{i-1}, x'_i, x_{i+1}, \dots, x_n)| \leq c_i$$

for all $1 \leq i \leq n$. Then for all $\epsilon > 0$:

$$\Pr[f(X_1, \dots, X_n) \geq \mathbf{E}f(X_1, \dots, X_n) + \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Similarly:

$$\Pr[f(X_1, \dots, X_n) \leq \mathbf{E}f(X_1, \dots, X_n) - \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right).$$

Proof Let $X_k^l = \{X_k, \dots, X_l\}$, and consider X_1^n with $1 \leq k \leq n$, and $\tilde{X}_1^n = \{X_1, \dots, X_{k-1}, \tilde{X}_k, X_{k+1}, X_n\}$. We have

$$|\mathbf{E}_{X_{k+1}^n} f(X_1^n) - \mathbf{E}_{X_{k+1}^n} f(\tilde{X}_1^n)| \leq c_k.$$

We treat $\mathbf{E}_{X_{k+1}^n} f(X_1^n)$ as a random variable depending on X_k , with X_1^{k-1} fixed. It follows from the derivation of Chernoff bound that we have the following logarithmic moment generating function estimate:

$$\ln \mathbf{E}_{X_k} \exp[\lambda \mathbf{E}_{X_{k+1}^n} f(X_1^n)] \leq \lambda \mathbf{E}_{X_k} f(X_1^n) + \lambda^2 c_k^2 / 8.$$

Now we may exponentiate the above inequality, and take expectation with respect to X_1^{k-1} to obtain

$$\mathbf{E}_{X_1^k} \exp[\lambda \mathbf{E}_{X_{k+1}^n} f(X_1^n)] \leq \mathbf{E}_{X_1^{k-1}} \exp[\lambda \mathbf{E}_{X_k^n} f(X_1^n) + \lambda^2 c_k^2 / 8].$$

By taking logarithm, we obtain

$$\ln \mathbf{E}_{X_1^k} \exp[\lambda \mathbf{E}_{X_{k+1}^n} f(X_1^n)] \leq \ln \mathbf{E}_{X_1^{k-1}} \exp[\lambda \mathbf{E}_{X_k^n} f(X_1^n)] + \lambda^2 c_k^2 / 8.$$

By summing from $k = 1$ to $k = n$, and canceling redundant terms, we obtain

$$\ln \mathbf{E}_{X_1^n} \exp[\lambda f(X_1^n)] \leq \lambda \mathbf{E}_{X_1^n} f(X_1^n) + \lambda^2 \sum_{k=1}^n c_k^2 / 8.$$

Let

$$\delta = \Pr[f(X_1^n) \geq \mathbf{E}_{X_1^n} f(X_1^n) + \epsilon].$$

Using Markov inequality, we have for all positive λ

$$\delta \leq e^{-\lambda(\mathbf{E}_{X_1^n} f(X_1^n) + \epsilon)} \mathbf{E}_{X_1^n} e^{\lambda f(X_1^n)} \leq \exp\left[-\lambda\epsilon + \frac{\lambda^2}{8} \sum_{k=1}^n c_k^2\right].$$

Since $\lambda > 0$ is arbitrary, we conclude that

$$\ln \delta \leq \inf_{\lambda \geq 0} \left[\frac{\lambda^2}{8} \sum_{k=1}^n c_k^2 - \lambda\epsilon \right] = -\frac{2\epsilon^2}{\sum_{k=1}^n c_k^2}.$$

This implies the theorem. ■

McDiarmid's inequality is referred to as *concentration inequality* because it states that the sample dependent quantity $f(X_1, \dots, X_n)$ does not deviate significantly from its expectation $\mathbf{E}f(X_1, \dots, X_n)$.

Note that if we take

$$f(x_1, \dots, x_n) = \frac{1}{n} \sum_{i=1}^n x_i,$$

and assume that $x_i \in [0, 1]$, then we can take $c_i = 1/n$, which implies the additive Chernoff bound in Theorem 2.5. Therefore McDiarmid's inequality is a generalization of Chernoff bound.

We can apply Theorem 6.2 to empirical processes, and obtain the following result.

Corollary 6.2 *Assume that $g_w(z) \in [0, M]$. Then with probability at least $1 - \delta$: for all $w \in \Omega$,*

$$g_w(D) \leq g_w(S_n) + \epsilon_n(G, D) + M \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Proof Consider $S_n = \{Z_1, \dots, Z_n\}$ and $S'_n = \{Z_1, \dots, Z_{i-1}, Z'_i, Z_{i+1}, \dots, Z_n\}$. Let $f(S_n) = \epsilon(G, S_n)$. For simplicity, we assume that the sup in $\epsilon(G, S_n)$ can be achieved at $\hat{w}$ as

$$\hat{w} = \arg \max_{w \in \Omega} [g_w(D) - g_w(S_n)].$$

Then

$$\begin{aligned} f(S_n) - f(S'_n) &= [g_{\hat{w}}(D) - g_{\hat{w}}(S_n)] - \sup_{w \in \Omega} [g_w(D) - g_w(S'_n)] \\ &\leq [g_{\hat{w}}(D) - g_{\hat{w}}(S_n)] - [g_{\hat{w}}(D) - g_{\hat{w}}(S'_n)] \\ &\leq \frac{1}{n} [g_{\hat{w}}(Z'_i) - g_{\hat{w}}(Z_i)] \leq M/n. \end{aligned}$$

Similarly, $f(S'_n) - f(S_n) \leq M/n$. Therefore we may take $c_i = M/n$ in Theorem 6.2, which implies the desired result. ■

A direct combination of Corollary 6.1 and McDiarmid's inequality implies the following result. It shows that as long as we can obtain a good estimate of Rademacher complexity, we can obtain a concentration result with $O(1/\sqrt{n})$ convergence.

Corollary 6.3 Consider $g_w(Z) \in [0, M]$ with $Z = (X, Y)$, where $Z \sim D$. Let S_n be n iid samples from D . With probability at least $1 - \delta$, for all $w \in \Omega$:

$$g_w(D) \leq g_w(S_n) + 2R_n(G, D) + M\sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Moreover, for the approximate ERM method (3.3), we have with probability at least $1 - \delta$:

$$g_{\hat{w}}(D) \leq \inf_{w \in \Omega} g_w(D) + \epsilon' + 2R_n(G, D) + 2M\sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Proof The first inequality is a direct consequence of McDiarmid's inequality, with the estimate $\epsilon_n(G, D) \leq 2R_n(G, D)$ from Corollary 6.1.

The second inequality is because given any $w \in \Omega$, with probability $1 - \delta/2$, we have

$$g_w(S_n) \leq g_w(D) + M\sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Taking the union bound with the first inequality at probability $1 - \delta/2$, we obtain with probability $1 - \delta$:

$$\begin{aligned} g_{\hat{w}}(D) &\leq g_{\hat{w}}(S_n) + 2R_n(G, D) + M\sqrt{\frac{\ln(2/\delta)}{2n}} \\ &\leq g_w(S_n) + \epsilon' + 2R_n(G, D) + M\sqrt{\frac{\ln(2/\delta)}{2n}} \\ &\leq g_w(D) + \epsilon' + 2R_n(G, D) + 2M\sqrt{\frac{\ln(2/\delta)}{2n}}. \end{aligned}$$

This proves the desired bound. ■

The Rademacher complexity analysis (together with McDiarmid's inequality) is convenient to apply. Therefore we will focus on this analysis in later chapters.

However, one draw back of the Rademacher complexity analysis is that it only leads to convergence rates of no better than $O(1/\sqrt{n})$. In order to prove faster convergence rate, we will have to rely on more sophisticated analysis, referred to as *local Rademacher complexity analysis*, which we will briefly discuss in Section 6.5.

6.3 Estimating Rademacher Complexity

In this section provides a few useful results to estimate Rademacher complexity. For a finite function class, we have the following simple estimate.

Theorem 6.3 *If G is a finite function class with $|G| = N$, then*

$$R(G, S_n) \leq \sup_{g \in G} \|g\|_{L_2(S_n)} \cdot \sqrt{\frac{2 \ln N}{n}}.$$

Proof Let $B = \sup_{g \in G} \|g\|_{L_2(S_n)}$. Then we have for all $\lambda > 0$:

$$\begin{aligned} R(G, S_n) &= \mathbf{E}_\sigma \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i g(Z_i) \stackrel{(a)}{\leq} \mathbf{E}_\sigma \frac{1}{\lambda n} \ln \sum_{g \in G} \exp \left[\lambda \sum_{i=1}^n \sigma_i g(Z_i) \right] \\ &\stackrel{(b)}{\leq} \frac{1}{\lambda n} \ln \mathbf{E}_\sigma \sum_{g \in G} \exp \left[\lambda \sum_{i=1}^n \sigma_i g(Z_i) \right] = \frac{1}{\lambda n} \ln \sum_{g \in G} \prod_{i=1}^n \mathbf{E}_{\sigma_i} \exp [\lambda \sigma_i g(Z_i)] \\ &\stackrel{(c)}{\leq} \frac{1}{\lambda n} \ln \sum_{g \in G} \prod_{i=1}^n \exp [\lambda^2 g(Z_i)^2 / 2] \leq \frac{1}{\lambda n} \ln N \exp [\lambda^2 n B^2 / 2]. \end{aligned}$$

In (a), we use soft-max to bound the max. In (b), we used the Jensen's inequality and the concavity of logarithm. In (c), we used the moment generating function for bounded random variables (Chernoff bound). Now we can obtain the desired bound by optimizing over $\lambda > 0$. ■

Example 6.1 *Consider $g_w(Z) \in [0, 1]$ and $|G| = N$. With probability $1 - \delta$. We have the following uniform convergence results for all w . If we use the union of Chernoff bound (covering number) method, then*

$$g_w(D) \leq g_w(S_n) + \sqrt{\frac{\ln(N/\delta)}{2n}},$$

which implies that

$$g_w(D) \leq g_w(S_n) + \sqrt{\frac{\ln(N)}{2n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

If we use the Rademacher bound, then we can obtain

$$g_w(D) \leq g_w(S_n) + 4\sqrt{\frac{\ln(N)}{2n}} + \sqrt{\frac{\ln(1/\delta)}{2n}},$$

which is worse in constant. However, we may also obtain a more refined bound as follows

$$g_w(D) \leq g_w(S_n) + 4\mathbf{E}_{S_n} \sup_{g \in G} \|g\|_{L_2(S_n)} \sqrt{\frac{\ln(N)}{2n}} + \sqrt{\frac{\ln(1/\delta)}{2n}},$$

which is not directly available with the standard Chernoff bound.

The following result shows that Rademacher complexity can be estimated from the empirical L_2 covering number. Its proof employs an important technique called *chaining*, due to Dudley [33]. This is a better estimate than the single scale covering number bounds used in Chapter 3 and Chapter 4, which led to suboptimal complexity estimates:

$$\inf_{\epsilon > 0} \left[\epsilon + \sqrt{\ln N(\epsilon/2, G, L_\infty(S_n))/n} \right].$$

The improvement is obtained by considering multiple approximation scales instead of a single scale as in Chapter 3 and Chapter 4.

Theorem 6.4 *We have*

$$R(G, S_n) \leq \inf_{\epsilon_0 \geq 0} \left[4\epsilon_0 + 12 \int_{\epsilon_0}^{\infty} \sqrt{\frac{\ln N(\epsilon'/2, G, L_2(S_n))}{n}} d\epsilon' \right].$$

Proof Let $B = \sup_{g \in G} \|g\|_{L_2(S_n)}$, and let $\epsilon_\ell = 2^{-\ell}B$ for $\ell = 0, 1, \dots$. Let G_ℓ be an ϵ_ℓ -cover of G with metric $L_2(S_n)$. We may let $G_0 = \{0\}$ at scale $\epsilon_0 = B$.

For each $g \in G$, we consider $g_\ell(g) \in G_\ell$ so that $\|g - g_\ell(g)\|_{L_2(S_n)} \leq \epsilon_\ell$. The key idea in chaining is to rewrite $g \in G$ using the following multi-scale decomposition:

$$g = (g - g_L(g)) + \sum_{\ell=1}^L (g_\ell(g) - g_{\ell-1}(g)).$$

We also have

$$\|g_\ell(g) - g_{\ell-1}(g)\|_{L_2(S_n)} \leq \|g_\ell(g) - g\|_{L_2(S_n)} + \|g_{\ell-1}(g) - g\|_{L_2(S_n)} \leq 3\epsilon_\ell. \quad (6.1)$$

It implies that

$$\begin{aligned} R(G, S_n) &= \mathbf{E}_\sigma \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i \left[(g - g_L(g)) + \sum_{\ell=1}^L (g_\ell(g) - g_{\ell-1}(g)) \right] \\ &\leq \mathbf{E}_\sigma \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i (g - g_L(g)) + \sum_{\ell=1}^L \mathbf{E}_\sigma \sup_{g \in G} \frac{1}{n} \sum_{i=1}^n \sigma_i (g_\ell(g) - g_{\ell-1}(g)) \\ &\stackrel{(a)}{\leq} \epsilon_L + \sum_{\ell=1}^L \sup_{g \in G} \|g_\ell(g) - g_{\ell-1}(g)\|_{L_2(S_n)} \sqrt{\frac{2 \ln[N_\ell N_{\ell-1}]}{n}} \\ &\stackrel{(b)}{\leq} \epsilon_L + 3 \sum_{\ell=1}^L \epsilon_\ell \sqrt{\frac{2 \ln[N_\ell N_{\ell-1}]}{n}} \\ &\leq \epsilon_L + 12 \sum_{\ell=1}^L (\epsilon_\ell - \epsilon_{\ell+1}) \sqrt{\frac{\ln[N_\ell]}{n}} \\ &\leq \epsilon_L + 12 \int_{\epsilon_L/2}^{\infty} \sqrt{\ln N(\epsilon/2, G, L_2(S_n))/n} d\epsilon. \end{aligned}$$

The derivation of (a) used Theorem 6.3, and the derivation of (b) used (6.1). Now for each ϵ , we can choose ϵ_L so that $\epsilon \in [\epsilon_L/4, \epsilon_L/2]$. This leads to the desired result. ■

Example 6.2 *It is known (e.g. [89]) that if a binary-valued function class G has VC-dimension d , then*

$$\ln N_2(\epsilon, G, n) \leq \ln 2 + d \ln(4e/\epsilon^2).$$

Since $\int \sqrt{\ln N_2(\epsilon, G, n)} d\epsilon < \infty$, we know that there exists $C > 0$ such that

$$R(G, S_n) \leq \frac{C\sqrt{d}}{\sqrt{n}}.$$

This implies the following uniform convergence result: with probability at least $1 - \delta$, for all $w \in \Omega$,

$$g_w(D) \leq g_w(S_n) + \frac{2C\sqrt{d}}{\sqrt{n}} + \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

This bound removes a $\ln n$ factor from the additive uniform convergence bound in Theorem 4.8.

Example 6.3 *If $\ln N_2(\epsilon, G, n) \leq 1/\epsilon^q$ for $q \in (0, 2)$, then*

$$\int_0^\infty \sqrt{\ln N_2(\epsilon, G, n)} d\epsilon < \infty.$$

Therefore there exists $C > 0$ such that

$$R(G, S_n) \leq \frac{C}{\sqrt{n}}.$$

In comparison, if we use the discretized estimate of Theorem 4.5, then the convergence rate is

$$\inf_{\epsilon > 0} \left[\epsilon + \sqrt{\ln N(\epsilon/2, G, L_\infty(S_{2n}))/n} \right] = O((1/\sqrt{n})^{1/(q+1)}),$$

which has a sub-optimal rate of convergence.

One convenient fact about Rademacher average is the following result. Let $\{\phi_i\}$ be a set of functions, each characterized by a Lipschitz constant γ_i , namely $|\phi_i(\theta) - \phi_i(\theta')| \leq \gamma_i |\theta - \theta'|$. Then the result implies a bound on the Rademacher complexity of the function composition $\phi \circ f$.

Theorem 6.5 *Let $\{\phi_i\}_{i=1}^n$ be functions with Lipschitz constants γ_i , then*

$$\mathbf{E}_\sigma \left[\sup_{f \in F} \sum_{i=1}^n \sigma_i \phi_i(f(Z_i)) \right] \leq \mathbf{E}_\sigma \left[\sup_{f \in F} \sum_{i=1}^n \sigma_i \gamma_i f(Z_i) \right].$$

This result is an immediate consequence of the following comparison Lemma [54] by setting $c(w, Z) = 0$, $g_i(w) = \phi_i(f(Z_i))$, and $h_i(w) = \gamma_i f(Z_i)$. The proof presented here is from [61].

Lemma 6.1 *Let $\{g_i(w)\}$ and $\{h_i(w)\}$ be sets of functions defined for all w in some domain Ω . If for all i , w , w' , $|g_i(w) - g_i(w')| \leq |h_i(w) - h_i(w')|$, then for any function $c(z, w)$,*

$$\mathbf{E}_\sigma \mathbf{E}_Z \sup_{w \in \Omega} \left[c(Z, w) + \sum_{i=1}^n \sigma_i g_i(w) \right] \leq \mathbf{E}_\sigma \mathbf{E}_Z \sup_{w \in \Omega} \left[c(Z, w) + \sum_{i=1}^n \sigma_i h_i(w) \right].$$

Proof We prove this result by induction. The result holds for $n = 0$. Assume that the result hold for $n = k$, then when $n = k + 1$, we have:

$$\begin{aligned} & \mathbf{E}_{\sigma_1, \dots, \sigma_{k+1}} \mathbf{E}_Z \sup_w \left[c(Z, w) + \sum_{i=1}^{k+1} \sigma_i g_i(w) \right] \\ &= \mathbf{E}_{\sigma_1, \dots, \sigma_k} \mathbf{E}_Z \sup_{w_1, w_2} \left[\frac{c(Z, w_1) + c(Z, w_2)}{2} + \sum_{i=1}^k \sigma_i \frac{g_i(w_1) + g_i(w_2)}{2} \right. \\ & \quad \left. + \frac{g_{k+1}(w_1) - g_{k+1}(w_2)}{2} \right] \\ &= \mathbf{E}_{\sigma_1, \dots, \sigma_k} \mathbf{E}_Z \sup_{w_1, w_2} \left[\frac{c(Z, w_1) + c(Z, w_2)}{2} + \sum_{i=1}^k \sigma_i \frac{g_i(w_1) + g_i(w_2)}{2} \right. \\ & \quad \left. + \frac{|g_{k+1}(w_1) - g_{k+1}(w_2)|}{2} \right] \\ &\leq \mathbf{E}_{\sigma_1, \dots, \sigma_k} \mathbf{E}_Z \sup_{w_1, w_2} \left[\frac{c(Z, w_1) + c(Z, w_2)}{2} + \sum_{i=1}^k \sigma_i \frac{g_i(w_1) + g_i(w_2)}{2} \right. \\ & \quad \left. + \frac{|h_{k+1}(w_1) - h_{k+1}(w_2)|}{2} \right] \\ &= \mathbf{E}_{\sigma_1, \dots, \sigma_k} \mathbf{E}_{\sigma_{k+1}} \mathbf{E}_Z \sup_w \left[c(Z, w) + \sigma_{k+1} h_{k+1}(w) + \sum_{i=1}^k \sigma_i g_i(w) \right] \\ &\leq \mathbf{E}_{\sigma_1, \dots, \sigma_k} \mathbf{E}_{\sigma_{k+1}} \mathbf{E}_Z \sup_w \left[c(Z, w) + \sigma_{k+1} h_{k+1}(w) + \sum_{i=1}^k \sigma_i h_i(w) \right]. \end{aligned}$$

The last inequality follows from the induction hypothesis. ■

The following result is a direct consequence of Theorem 6.5 and Corollary 6.3.

Theorem 6.6 *Consider real-valued function class $F = \{f(w, \cdot) : w \in \Omega\}$, and $G = \{g_w(z) = L(f(w, x), y) : w \in \Omega, z = (x, y)\}$. Assume that $L(f, y) \in [0, M]$ is Lipschitz:*

$$|L(f, y) - L(f', y)| \leq \gamma |f - f'|.$$

Let S_n be n iid samples from D . With probability at least $1 - \delta$, for all $w \in \Omega$:

$$\mathbf{E}_D L(f(w, X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(w, X_i), Y_i) + 2\gamma R_n(F, D) + M \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Moreover, for the approximate ERM method (3.3), we have with probability at least $1 - \delta$:

$$\mathbf{E}_D L(f(\hat{w}, X), Y) \leq \inf_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y) + \epsilon' + 2\gamma R_n(F, D) + M \sqrt{\frac{2 \ln(2/\delta)}{n}}.$$

6.4 Data Dependent Bounds

An important property of the empirical Rademacher complexity is that it can be estimated from the training data S_n . The following result follows directly from McDiarmid's inequality.

Corollary 6.4 *If $g_w(Z) \in [0, M]$, then with probability at least $1 - \delta$, we have*

$$R_n(G, D) \leq R(G, S_n) + M \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Similarly, with probability at least $1 - \delta$, we have

$$R_n(G, D) \geq R(G, S_n) - M \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Proof Let $S_n = \{Z_1, \dots, Z_n\}$ and $S'_n = \{Z'_1, \dots, Z'_n\}$, where $Z_i = Z'_i$ except at some k . Let $f(S_n) = R(G, S_n)$, and for any σ , we assume the sup of the following can be achieved at

$$\hat{w}(\sigma, S_n) = \arg \max_{w \in \Omega} \sum_{i=1}^n \sigma_i g_w(Z_i).$$

Then we have

$$\begin{aligned}
R(G, S_n) - R(G, S'_n) &= \mathbf{E}_\sigma \left[\frac{1}{n} \sum_{i=1}^n \sigma_i g_{\hat{w}(\sigma, S_n)}(Z_i) - \frac{1}{n} \sum_{i=1}^n \sigma_i g_{\hat{w}(\sigma, S'_n)}(Z'_i) \right] \\
&\leq \mathbf{E}_\sigma \left[\frac{1}{n} \sum_{i=1}^n \sigma_i g_{\hat{w}(\sigma, S_n)}(Z_i) - \frac{1}{n} \sum_{i=1}^n \sigma_i g_{\hat{w}(\sigma, S_n)}(Z'_i) \right] \\
&\leq \frac{1}{n} \sup_{w, Z, Z'} |g_w(Z) - g_w(Z')|.
\end{aligned}$$

This implies that $|R(G, S_n) - R(G, S'_n)| \leq M/n$. The desired bound is a direct consequence of McDiarmid's inequality. $\blacksquare$

This result, combined with Corollary 6.3, leads to the following sample-dependent uniform convergence result.

Theorem 6.7 *Consider $G = \{g_w(z) = L(f(w, x), y) : w \in \Omega, z = (x, y)\}$. Assume that $L(f, y) \in [0, M]$. Let S_n be n iid samples from D . With probability at least $1 - \delta$, for all $w \in \Omega$:*

$$g_w(D) \leq g_w(S_n) + 2R(G, S_n) + 3M \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Note that in the sample-dependent bound of Theorem 6.7, the right hand side can be estimated using training data only. Although computationally the estimation of $R(G, S_n)$ can be quite challenging, it is statistically possible to do.

Sample dependent bounds can be used to design model selection algorithms on training data only, without splitting the training data into training versus validation partitions. The idea is similar to the model selection on validation data presented in Chapter 4, with data dependent uniform convergence bound replacing Chernoff bound.

Consider a countable sequence of models $G_j = \{g_w(Z) : w \in \Omega_j\}$, characterized by $\{\Omega_j : j = 1, 2, \dots\}$, and let $\{q_j \geq 0\}$ be a sequence of numbers such that $\sum_j q_j \leq 1$. Consider the following model selection algorithm, which finds a model $\hat{j}$ and $\hat{w} \in \Omega_j$:

$$Q(G_{\hat{j}}, S_n) \leq \inf_j Q(G_j, S_n) + \tilde{\epsilon}, \quad (6.2)$$

where

$$\begin{aligned}
Q(G_j, S_n) &= g_{\hat{w}(j)}(S_n) + 2R(G_j, S_n) + 3M \sqrt{\frac{\ln(2/q_j)}{2n}}, \\
g_{\hat{w}(j)}(S_n) &\leq \inf_{w \in \Omega_j} g_w(S_n) + \epsilon'.
\end{aligned}$$

If the direct computation of $R(G, S_n)$ is difficult, we may also replace it by an estimated upper bound that depends on S_n . For such a model selection method, we have the following result.

Theorem 6.8 *Assume $L(\cdot, \cdot) \in [0, M]$. Let S_n be n iid samples from D . Then with probability at least $1 - \delta$, we have for all j and $w \in \Omega_j$:*

$$g_w(D) \leq g_w(S_n) + 2R(G_j, S_n) + 3M\sqrt{\frac{\ln(2/q_j)}{2n}} + 3M\sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Consider the model selection algorithm in (6.2), we have the following oracle inequality for model selection. With probability of at least $1 - \delta$:

$$\begin{aligned} g_{\hat{w}(j)}(D) &\leq \inf_{j, w \in \Omega_j} \left[g_w(D) + 2R(G_j, D) + 3M\sqrt{\frac{\ln(2/q_j)}{2n}} \right] \\ &\quad + 6M\sqrt{\frac{\ln(3/\delta)}{2n}} + \epsilon' + \tilde{\epsilon}. \end{aligned}$$

,

Proof Given j , the claim holds uniformly for all $w \in \Omega_j$ with probability at least $1 - q_j\delta$. By taking the union bound over j , and note that $\sum_j q_j\delta \leq \delta$, we obtain the desired uniform convergence inequality with probability at least $1 - \delta$.

To derive the oracle inequality, we notice the following three inequalities. With probability at least $1 - \delta/3$,

$$g_{\hat{w}(\hat{j})}(D) \leq g_{\hat{w}(\hat{j})}(S_n) + 2R(G_{\hat{j}}, S_n) + 3M\sqrt{\frac{\ln(2/q_{\hat{j}})}{2n}} + 3M\sqrt{\frac{\ln(3/\delta)}{2n}}.$$

Given fixed w and j , with probability at least $1 - \delta/3$,

$$g_w(S_n) \leq g_w(D) + M\sqrt{\frac{\ln(3/\delta)}{2n}}.$$

With probability at least $1 - \delta/3$:

$$2R(G_j, S_n) \leq 2R(G_j, D) + 2M\sqrt{\frac{\ln(3/\delta)}{2n}}.$$

Taking the union bound, we know that with probability at least $1 - \delta$, all of the three inequalities hold. Combine these inequalities with (6.2), we obtain the desired oracle inequality. $\blacksquare$

Compared to Corollary 4.3, Theorem 6.8 achieves similar results. This sample depend uniform convergence bound approach avoids partitioning

data into training and validation parts, and can be used to design integrated algorithms that can tune both the model parameter w and the hyperparameter simultaneously on the same training data. However, the resulting model selection algorithm (6.2) requires knowing the specific sample dependent generalization bound for each model, which may not be practical in some real applications.

6.5 Local Rademacher Complexity Analysis

The technique to prove McDiarmid's inequality is called the *Martingale method*, and it can derive concentration inequalities with convergence rates of $O(1/\sqrt{n})$. One can prove faster convergence rates with the Martingale method by deriving Bernstein style concentration inequalities. However, more refined forms of Bernstein style concentration inequalities are needed to analyze empirical processes, and those refined forms are some times referred to as Talagrand's concentration inequality [80, 81]. To illustrate its consequences, we state the following result from [14] which employs the version of Talagrand's inequality from [22].

Theorem 6.9 *Assume there exist numbers $r > 0$ and $M > 0$ such that $\forall w \in \Omega: \mathbf{Var}(g_w(Z)) \leq r$, and $\sup_Z g_w(Z) \leq \inf_Z g_w(Z) + M$. Then with probability at least $1 - \delta$, for all $w \in \Omega$ and $\gamma > 0$:*

$$g_w(D) \leq g_w(S_n) + (2 + \gamma)R_n(G, D) + \sqrt{\frac{2r \ln(1/\delta)}{n}} + \frac{(6 + \gamma)M \ln(1/\delta)}{3\gamma n}.$$

While the Chernoff style result of Corollary 6.2 only implies a convergence rate no better than $O(1/\sqrt{n})$, the Bennett style result of Theorem 6.9 can lead to faster convergence rate.

However, in order to apply this theorem to derive generalization bounds, we also need to employ local Rademacher complexity analysis [14], which is more complex. The following result of [14] is an analogy of Corollary 6.3.

Theorem 6.10 ([14]) *Let $\mathcal{F} = \{f(w, x) : w \in \Omega\}$ be a convex class of real-valued functions with range in $[-1, 1]$ and let $L(f, y)$ be a loss function that is γ -Lipschitz in f :*

$$|L(f_1, y) - L(f_2, y)| \leq \gamma |f_1 - f_2|.$$

Consider a distribution D , and let

$$f_*(x) = f(w_*, x), \quad w_* = \arg \min_{w \in \Omega} \mathbf{E}_D L(f(w, X), Y).$$

Assume there exists $B \geq 1$ such that the following the variance condition for all $f \in \mathcal{F}$.

$$\mathbf{E}_{X \sim D}(f(X) - f_*(X))^2 \leq B \mathbf{E}_{Z \sim D}[L(f(X), Y) - L(f_*(X), Y)].$$

Let

$$\hat{f}(x) = f(\hat{w}, x), \quad \hat{w} = \arg \min_{w \in \Omega} \sum_{i=1}^n L(f(w, X_i), Y_i).$$

Define

$$\mathcal{F}(r) = \{f \in \mathcal{F} : \gamma^2 \mathbf{E}_X(f(X) - f_*(X))^2 \leq c_3 r\},$$

and

$$\hat{\psi}_n(r) = c_1 R(\mathcal{F}(r), S_n) + \frac{c_2 \ln(4/\delta)}{n},$$

where $c_1 = 2\gamma \max(B, 10\gamma)$, $c_2 = 11\gamma^2 + c_1$, and $c_3 = 2824 + 4B(11\gamma + 27B)/c_2$. Then with probability at least $1 - \delta$:

$$\mathbf{E}_D[L(\hat{f}(X), Y) - L(f_*(X), Y)] \leq 705 \frac{\hat{r}}{B} + \frac{(11\gamma + 27B) \ln(4/\eta)}{n},$$

where $\hat{r}$ satisfies $\hat{r} \geq \hat{\psi}_n(\hat{r})$.

Example 6.4 If we have

$$\hat{\psi}_n \leq \frac{\tilde{c}_1}{\sqrt{n}} r^p + \tilde{c}_2$$

for some $0 < p < 0.5$, then we obtain

$$\hat{r} = O(n^{0.5/(1-p)}),$$

which leads to a rate of convergence faster than $O(1/\sqrt{n})$.

We can analyze the ERM method for least squares regression using local Rademacher complexity. Consider a function class $\mathcal{F}$ and the ERM method for least squares regression:

$$\hat{f} = \arg \min_{f \in \mathcal{F}} \frac{1}{n} \sum_{i=1}^n (f(X_i) - Y_i)^2,$$

where (X_i, Y_i) are iid samples from D .

Assume that $|f(X) - Y| \in [0, 1]$ for all X and Y . Assume also that the empirical covering number of $\mathcal{F}$ satisfies:

$$\ln N_2(\epsilon, \mathcal{F}, n) \leq \frac{c}{\epsilon^q} \quad (6.3)$$

for some constant $c > 0$ and q . We have two situations with either $q \in (0, 2)$ or $q \geq 2$.

For $q \in (0, 2)$, the conditions of Theorem 6.10 hold with $L = 2$ and $B = 1$. We can use chaining to bound the empirical local Rademacher complexity, and obtain

$$\hat{\psi}_n(r) \leq \frac{\tilde{c}_1}{\sqrt{n}} r^{0.5-q/4} + \frac{c_2 x}{n}.$$

This is a sub-root function. We have This implies the following bound:

$$\hat{r} \leq \frac{\tilde{c}_1^{2/(1+q/2)}}{n^{2/(2+q)}} + \frac{c_2 x}{n},$$

and the following oracle inequality. We have with probability at least $1 - \eta$:

$$\mathbf{E}_D L(\hat{f}(X), Y) \leq \mathbf{E}_D L(f_*(X), Y) + \frac{\tilde{c}_1^{2/(1+q/2)}}{n^{2/(2+q)}} + \frac{(49 + c_2) \ln(4/\eta)}{n}.$$

In the case of $q > 2$, the entropy integral gives

$$\hat{\psi}_n(r) \leq \frac{\tilde{c}_1}{n^{1/q}} + \frac{c_2 x}{n}.$$

We thus obtain a rate of convergence of

$$\hat{r} = O(n^{-1/q})$$

using local Rademacher complexity. It can be shown that this is the same rate as that we can obtain from the standard non-localized Rademacher complexity.

6.6 Exercises

1. Consider $G = \{g_w(z) : w \in \Omega\}$ with $g_w(z) \in [0, 1]$. Given a dataset S_{n+m} , we consider random partitions of S_{n+m} into disjoint subsets $S_n \cup S_m$. Using notations in the lecture notes, we define

$$R(G, S_n, S'_m) = \sup_{w \in \Omega} [g_w(S_{n+m}) - g_w(S_n)].$$

If $|G| = N$ is finite, derive an upper bound of

$$\mathbf{E}_{S_n \cup S'_m} R(G, S_n, S'_m)$$

by using the proof technique of Theorem 6.3, where the expectation is with respect to the random partition.

2. For general G , with covering number $N(\epsilon, G, L_\infty(S_{n+m}))$, estimate

$$\mathbf{E}_{S_n \cup S'_m} R(G, S_n, S'_m)$$

defined in the previous problem by using the chaining technique of Theorem 6.4. Here the expectation is with respect to the random partition.

3. Let $F = \{f(w, x) : w \in \Omega, x \in \mathcal{X}\}$, and each function $f(w, x)$ takes binary values in $\{\pm 1\}$. Consider $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, where $X_i \in \mathcal{X}$ and $Y_i \in \{\pm 1\}$. Let $g_w(X, Y) = I(f(w, X) \neq Y)$. Let $R(F, S_n)$ be the Rademacher complexity of F on S_n , find $R(G, S_n)$.
4. Consider the least squares example after Example 6.4. If $q = 2$, derive an oracle inequality using local Rademacher complexity.
5. Assume the empirical covering number of a function class $\mathcal{F}$ satisfies (6.3) with $q \geq 2$. Estimate the Rademacher complexity $R(\mathcal{F}, S_n)$.

Chapter 7

Analysis of Kernel Methods

Kernel methods have appeared in statistics, and popularized in machine learning through support vector machines (SVMs) in 1990s [28]. In this chapter we review kernel methods, and investigate their theoretical analysis. A detailed treatment of kernel methods can be found in [73].

7.1 Introduction to Kernel Learning

In order to motivate kernel methods, we first consider classical linear models, with the following real-valued function class $\mathcal{X} \rightarrow \mathbb{R}$ indexed by w :

$$\mathcal{F} = \{f(w, x) : f(w, x) = w \cdot \psi(x)\}, \quad (7.1)$$

where $\psi(x)$ is a pre-defined (possibly infinite dimensional) feature vector for the input variable $x \in \mathcal{X}$, and $\cdot$ denotes the inner product in the feature vector space. In classical machine learning, the process of constructing the feature vector $\psi(x)$ is referred to as *feature engineering*, and is problem dependent, with features constructed by hand-crafted rules.

If $\psi(x)$ is infinite dimensional (or the dimension is very large), then it is not feasible to work with $\psi(x)$. The learning of the linear model $\mathcal{F}$ can be achieved via kernel methods. With a properly defined inner product $\cdot$ in this vector space, we can define a kernel function $k(x, x')$ (also called reproducing kernel) as the inner product of their feature vectors

$$k(x, x') = \psi(x) \cdot \psi(x'). \quad (7.2)$$

Definition 7.1 A symmetric function $k(x, x')$ is called a *positive-definite kernel* if for all $\alpha_1, \dots, \alpha_m \in \mathbb{R}$ and $x_1, \dots, x_m \in \mathcal{X}$, we have

$$\sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j k(x_i, x_j) \geq 0.$$

Definition 7.2 *Given a symmetric positive-definite kernel, we define a function space $\mathcal{H}_0$ of the form*

$$\mathcal{H}_0 = \left\{ f(x) : f(x) = \sum_{i=1}^m \alpha_i k(x_i, x) \right\},$$

with inner product defined as

$$\|f(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^m \sum_{j=1}^m \alpha_i \alpha_j k(x_i, x_j).$$

The completion of $\mathcal{H}_0$ with respect to this inner product, defined as $\mathcal{H}$, is called the reproducing kernel Hilbert space (RKHS) of kernel k .

We have the following result, which shows that one can represent RKHS either in kernel formulation or feature formulation. The result stated is slightly non-rigorous, as it assumes that each $f(x) \in \mathcal{H}$ can be represented to a single w . This may not always be the case, and in such case we simply choose the weight w with the smallest norm.

Theorem 7.1 *A symmetric kernel function $k(x, x')$ is positive-definite if and only if there exists $\psi(x)$ so that it can be written in the form of (7.2). Consequently, functions $f(x)$ in its RKHS $\mathcal{H}$ can be written in the form of (7.1), with $\|f(x)\|_{\mathcal{H}}^2 = w \cdot w$.*

Using the notation of RKHS, the function class of the linear model (7.1) can be regarded as the RKHS $\mathcal{H}$ of the reproducing kernel (7.2). If we have the representation of w in (7.1) as

$$w = \sum_{i=1}^{\infty} \alpha_i \psi(x_i),$$

then

$$w \cdot \psi(x) = \sum_{i=1}^{\infty} \alpha_i \psi(x_i) \cdot \psi(x) = \sum_{i=1}^{\infty} \alpha_i k(x_i, x).$$

Therefore we may represent the corresponding function $f(\cdot) \in \mathcal{H}$ either in its feature representation as in (7.1), or its kernel representation

$$f(x) = \sum_{i=1}^{\infty} \alpha_i k(x_i, x). \quad (7.3)$$

We will later show that the solution of kernel representation in the ERM formulation can be represented in finite sum over the training data $\{X_i : i = 1, \dots, n\}$.

The norm constrained ERM problem in RKHS is given by

$$\hat{f}(\cdot) = \arg \min_{f(\cdot) \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) \quad \text{subject to } \|f(\cdot)\|_{\mathcal{H}} \leq A, \quad (7.4)$$

and the corresponding soft-regularized formulation with appropriate $\lambda > 0$ is

$$\hat{f}(\cdot) = \arg \min_{f(\cdot) \in \mathcal{H}} \left[\frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \frac{\lambda}{2} \|f(\cdot)\|_{\mathcal{H}}^2 \right]. \quad (7.5)$$

A detailed representation of function $f(x) \in \mathcal{H}$ can equivalently be parametrized either by the kernel formulation (7.3) or by the feature formulation (7.1). In feature representation, we can write (7.5) as

$$\hat{w} = \arg \min_{w \in \mathcal{H}} \left[\frac{1}{n} \sum_{i=1}^n L(w \cdot \psi(X_i), Y_i) + \frac{\lambda}{2} \|w\|^2 \right]. \quad (7.6)$$

If $\psi(x)$ is infinite dimensional, then it is infeasible to compute in the feature space. In such case, we can represent the solution of (7.5) in the kernel representation, and it can be shown that the solution has a finite representation using training data.

Theorem 7.2 *For real valued functions $f(x)$, the solution of (7.5) have the following kernel representation:*

$$\hat{f}(x) = \sum_{i=1}^n \hat{\alpha}_i k(X_i, x),$$

Therefore the solution of (7.5) is equivalent to the solution of the following finite dimensional kernel optimization problem:

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^n} \left[\frac{1}{n} \sum_{i=1}^n L \left(\sum_{j=1}^n k(X_i, X_j) \alpha_j, Y_i \right) + \frac{\lambda}{2} \alpha^\top K_{n \times n} \alpha \right], \quad (7.7)$$

with

$$K_{n \times n} = \begin{bmatrix} k(X_1, X_1) & \cdots & k(X_1, X_n) \\ \vdots & \ddots & \vdots \\ k(X_n, X_1) & \cdots & k(X_n, X_n) \end{bmatrix}.$$

Proof We consider the feature representation of (7.6). The solution satisfies the following first order optimality condition:

$$\frac{1}{n} \sum_{i=1}^n L'_1(\hat{w} \cdot \psi(X_i), Y_i) \psi(X_i) + \lambda \hat{w} = 0.$$

Here $L'_1(p, y)$ is the derivative of $L(p, y)$ with respect to p . We thus obtain the following representation as its solution:

$$\hat{w} = \sum_{i=1}^n \hat{\alpha}_i \psi(X_i), \quad \hat{\alpha}_i = -\frac{1}{\lambda n} L'_1(\hat{w} \cdot \psi(X_i), Y_i) \quad (i = 1, \dots, n).$$

where

$$\hat{\alpha}_i = -\frac{1}{\lambda n} L'_1(\hat{w} \cdot \psi(X_i), Y_i) \quad (i = 1, \dots, n).$$

Using this notation, we obtain the following kernel representation of the solution.

$$\hat{f}(x) = \hat{w} \cdot \psi(x) = \sum_{i=1}^n \hat{\alpha}_i \psi(X_i) \cdot \psi(x) = \sum_{i=1}^n \hat{\alpha}_i k(X_i, x).$$

This proves the theorem. ■

If $x \in \mathbb{R}^d$, then a standard choice of kernel is the RBF (radial basis function) kernel:

$$k(x, x') = \exp \left[\frac{-\|x - x'\|_2^2}{2\sigma^2} \right],$$

which is often used in practice.

Example 7.1 In kernel ridge regression, we consider least squares problem, which in the feature space, can be written as:

$$\hat{w} = \arg \min_w \left[\frac{1}{n} \sum_{i=1}^n (w \cdot \psi(X_i) - Y_i)^2 + \frac{\lambda}{2} w \cdot w \right].$$

The primal kernel formulation is:

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^n} \left[\frac{1}{n} \sum_{i=1}^n \left(\sum_{j=1}^n k(X_i, X_j) \alpha_j - Y_i \right)^2 + \frac{\lambda}{2} \alpha^\top K_{n \times n} \alpha \right].$$

There is also a dual formulation:

$$\hat{\alpha} = \arg \max_{\alpha \in \mathbb{R}^n} \left[-\frac{\lambda}{2} \alpha^\top K_{n \times n} \alpha + \lambda \alpha^\top \mathbf{Y} - \frac{\lambda^2}{4} \alpha^\top \alpha \right],$$

where $\mathbf{Y}$ is the n dimensional vector with Y_i as its component.

Example 7.2 In support vector machines for binary classification, we assume that the label $Y_i \in \{\pm 1\}$, and consider the following method in feature space:

$$\hat{w} = \arg \min_w \left[\frac{1}{n} \sum_{i=1}^n \max(0, 1 - w \cdot \psi(X_i) Y_i) + \frac{\lambda}{2} w \cdot w \right].$$

The primal kernel formulation is:

$$\hat{w} = \arg \min_{\alpha} \left[\frac{1}{n} \sum_{i=1}^n \max \left(0, 1 - \sum_{j=1}^n k(X_i, X_j) \alpha_j Y_i \right) + \frac{\lambda}{2} \alpha^\top K_{n \times n} \alpha \right].$$

The dual kernel formulation is:

$$\hat{\alpha} = \arg \max_{\alpha \in \mathbb{R}^n} \left[-\frac{\lambda}{2} \alpha^\top K_{n \times n} \alpha + \lambda \alpha^\top \mathbf{Y} \right], \quad \text{subject to } \alpha_i Y_i \in [0, 1/(\lambda n)].$$

A drawback of kernel methods is that the computation requires the full kernel matrix $K_{n \times n}$, and thus it is at least quadratic in n . Moreover, the inference time is linear in n . Both are rather expensive if n is large.

7.2 Universal Approximation

An important question for learning methods is whether the underlying function class (model family) can approximate all measurable functions. A function class that can represent all functions is called an *universal approximator*. In order to prove a universal approximation result, we only need to show that the class can approximate any continuous or differentiable functions under some criteria. For kernel methods, the universal approximation result was established in [64]. We present an approximation bound for Lipschitz functions, with a special form of kernels on $\mathbb{R}^d$ which include the RBF kernel as a special case. Since Lipschitz functions are dense in the set measurable functions. This result implies universal approximation.

Theorem 7.3 Consider kernel of the form

$$k(x, x') = h(\|x - x'\|/\sigma),$$

where $\|\cdot\|$ is a norm on $\mathbb{R}^d$, and assume that $h(\cdot) \leq 1$, and

$$c_0 = \int h(\|x\|) dx < \infty, \quad c_1 = \int \|x\| h(\|x\|) dx < \infty.$$

Assume that f is Lipschitz with respect to the norm: $\exists \gamma > 0$ such that $|f(x) - f(x')| \leq \gamma \|x - x'\|$ for all $x, x' \in \mathbb{R}^d$. Define

$$\|f\|_1 = \int |f(x)| dx, \quad \|f\|_\mu = \int |f(x)| d\mu(x),$$

where μ is any given probability measure. For all $\epsilon > 0$, there exists $\sigma > 0$, and a finite kernel approximation $f_m(x)$ of $f(x)$ that satisfies

$$\int (f(x) - f_m(x))^2 d\mu(x) \leq \epsilon^2, \quad f_m(x) = \sum_{i=1}^m \alpha_i k(x, x_i),$$

when

$$m \geq \frac{(3/c_0) \cdot (2\gamma c_1/c_0)^d \|f\|_1 \|f\|_\mu}{\epsilon^{d+2}}.$$

Proof We want to use $f_m(x)$ to approximate f . To do so, we first approximate f by continuous approximation below:

$$\psi_\sigma(x) = (c_0 \sigma^d)^{-1} \int f(z) h(\|x - z\|/\sigma) dz.$$

We also have the expression

$$\psi_\sigma(x) = c_0^{-1} \int f(x - \sigma z) h(\|z\|) dz$$

and

$$f(x) = c_0^{-1} \int f(x) h(\|z\|) dz.$$

Since f is Lipschitz, we have

$$|f(x) - \psi_\sigma(x)| \leq \int |f(x) - f(x - \sigma z)| h(\|z\|) c_0^{-1} dz \leq \sigma \gamma c_0^{-1} c_1.$$

Let $q(z) = \frac{|f(z)|}{\|f\|_1}$. We draw $z_1, \dots, z_m \sim q(z)$ independently, and set

$$\alpha_i = \text{sgn}(f(z_i)) \|f\|_1 (m c_0 \sigma^d)^{-1} \quad (i = 1, \dots, m)$$

in the approximator $f_m(x)$. This implies that

$$\mathbf{E}_{z_1, \dots, z_m} f_m(x) = \int_z \frac{\text{sgn}(f(z)) \|f\|_1}{c_0 \sigma^d} h\left(\frac{\|x - z\|}{\sigma}\right) \frac{|f(z)|}{\|f\|_1} dz = \psi_\sigma(x). \quad (7.8)$$

It follows that

$$\begin{aligned} & \mathbf{E}_{z_1, \dots, z_m} (\psi_\sigma(x) - f_m(x))^2 \\ &= \frac{1}{m} \left[\int_z \left[\text{sgn}(f(z)) \|f\|_1 (c_0 \sigma^d)^{-1} h(\|x - z\|/\sigma) \right]^2 \frac{|f(z)|}{\|f\|_1} dz - \psi_\sigma(x)^2 \right] \\ &\leq \frac{\|f\|_1}{m c_0^2 \sigma^d} \int \frac{|f(z)|}{\sigma^d} h(\|x - z\|/\sigma)^2 dz \\ &= \frac{\|f\|_1}{m c_0^2 \sigma^d} \int |f(x - \sigma z)| h(\|z\|)^2 dz \\ &\leq \frac{\|f\|_1}{m c_0^2 \sigma^d} \int (|f(x)| + \gamma \sigma \|z\|) h(\|z\|) dz \leq \frac{\|f\|_1}{m c_0^2 \sigma^d} (c_0 |f(x)| + \gamma \sigma c_1). \end{aligned}$$

From (7.8), we obtain

$$\mathbf{E}_{z_1, \dots, z_m} (f_m(x) - f(x))^2 = \mathbf{E}_{z_1, \dots, z_m} (f_m(x) - \psi_\sigma(x))^2 + (\psi_\sigma(x) - f(x))^2.$$

Taking expectation over μ , we obtain

$$\mathbf{E}_{z_1, \dots, z_m} \int (f_m(x) - f(x))^2 d\mu(x) \leq \frac{\|f\|_1}{mc_0^2 \sigma^d} (c_0 \|f\|_\mu + \gamma c_1 \sigma) + (\sigma \gamma c_0^{-1} c_1)^2.$$

Note that the theorem holds trivially if $\epsilon \geq \|f\|_\mu$. Otherwise, we may set $\sigma = 0.5\epsilon c_0/(\gamma c_1)$, and as long as m satisfies

$$\frac{\|f\|_1}{mc_0^2 (0.5\epsilon c_0/(\gamma c_1))^d} (c_0 \|f\|_\mu + \gamma c_1 (0.5\epsilon c_0)/(\gamma c_1)) \leq \epsilon^2/2,$$

we can obtain the desired approximation bound of the theorem. The condition of m in the theorem, and the assumption $\epsilon \leq \|f\|_\mu$ implies the above inequality, and this finishes the proof. $\blacksquare$

7.3 Generalization Analysis

We will use the Rademacher complexity analysis, and in particular, we want to bound the Rademacher complexity of (7.4).

Theorem 7.4 *Let $\mathcal{F} = \{f(x) : \|f\|_{\mathcal{H}} \leq A\}$, then*

$$R(\mathcal{F}, S_n) \leq A \sqrt{\frac{1}{n^2} \sum_{i=1}^n k(X_i, X_i)}.$$

Proof We consider the feature representation of $\mathcal{F}$ in (7.1), so that $f(x)$ can be represented as

$$\mathcal{F} = \{f(x) = w \cdot \psi(x) : \|w\| \leq A\},$$

where $\|w\| = (w \cdot w)^{1/2}$. We have

$$\begin{aligned} R(\mathcal{F}, S_n) &= \mathbf{E}_\sigma \sup_{\|w\| \leq A} \frac{1}{n} \sum_{i=1}^n \sigma_i w \cdot \psi(X_i) \\ &= \mathbf{E}_\sigma \sup_{\|w\| \leq A} w \cdot \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(X_i) \\ &\leq \mathbf{E}_\sigma A \cdot \left\| \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(X_i) \right\| \leq A \sqrt{\mathbf{E}_\sigma \left\| \frac{1}{n} \sum_{i=1}^n \sigma_i \psi(X_i) \right\|^2} \\ &= A \sqrt{\frac{1}{n^2} \sum_{i=1}^n \|\psi(X_i)\|^2} = A \sqrt{\frac{1}{n^2} \sum_{i=1}^n k(X_i, X_i)}. \end{aligned}$$

■

Corollary 7.1 *Let $G = \{L(f(x), y) : \|f\|_{\mathcal{H}} \leq A\}$, where $L(p, y)$ is γ Lipschitz in p , then*

$$R(G, S_n) \leq A\gamma \sqrt{\frac{1}{n^2} \sum_{i=1}^n k(X_i, X_i)}$$

and

$$R_n(G, D) \leq A\gamma \sqrt{\frac{\mathbf{E}_D k(X, X)}{n}}.$$

Proof The first inequality follows from Theorem 7.4 and Theorem 6.5. The second inequality follows from the following derivation:

$$\begin{aligned} R_n(G, D) &= \mathbf{E}_{S_n} R(G, S_n) \leq A\gamma \mathbf{E}_{S_n} \sqrt{\frac{1}{n^2} \sum_{i=1}^n k(X_i, X_i)} \\ &\leq A\gamma \sqrt{\frac{1}{n^2} \mathbf{E}_{S_n} \sum_{i=1}^n k(X_i, X_i)} \\ &= A\gamma \sqrt{\frac{1}{n} \mathbf{E}_D k(X, X)}. \end{aligned}$$

■

Now using Theorem 6.6, we obtain the following result.

Corollary 7.2 *Assume that $L(p, y) \in [0, M]$ is γ Lipschitz with respect to p . Then with probability at least $1 - \delta$: for all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} \leq A$:*

$$\mathbf{E}_D L(f(X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + 2\gamma A \sqrt{\frac{1}{n} \mathbf{E}_D k(X, X)} + M \sqrt{\frac{\ln(1/\delta)}{2n}}.$$

Moreover, for (7.4), if we solve it approximately up to sub-optimality of ϵ' , then we have with probability at least $1 - \delta$:

$$\begin{aligned} \mathbf{E}_D L(\hat{f}(X), Y) &\leq \inf_{\|f\|_{\mathcal{H}} \leq A} \mathbf{E}_D L(f(X), Y) + \epsilon' + 2\gamma A \sqrt{\frac{1}{n} \mathbf{E}_D k(X, X)} \\ &\quad + M \sqrt{\frac{2 \ln(2/\delta)}{n}}. \end{aligned}$$

Note that as $A \rightarrow \infty$, we have

$$\inf_{\|f\|_{\mathcal{H}} \leq A} \mathbf{E}_D L(f(X), Y) \rightarrow \inf_{\|f\|_{\mathcal{H}} < \infty} \mathbf{E}_D L(f(X), Y).$$

The universal approximation result of Theorem 7.3 implies that

$$\lim_{A \rightarrow \infty} \inf_{\|f\|_{\mathcal{H}} \leq A} \mathbf{E}_D L(f(X), Y) \rightarrow \inf_{\text{measurable } f} \mathbf{E}_D L(f(X), Y).$$

Combine this with the generalization result of kernel method in Corollary 7.2, we know as $n \rightarrow \infty$, and we let $A \rightarrow \infty$ so that with probability 1:

$$\mathbf{E}_D L(\hat{f}(X), Y) \rightarrow \inf_{\text{measurable } f} \mathbf{E}_D L(f(X), Y).$$

Such a result is referred to as *consistency*.

Example 7.3 For binary classification problem with $y \in \{\pm 1\}$, we may consider a classifier induced by a real valued $f(x)$ such that we predict $y = 1$ if $f(x) > 0$ and $y = -1$ otherwise. If $f(x)$ is taken from an RKHS, then we have the following margin bound. With probability $1 - \delta$: for all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} \leq A$:

$$\begin{aligned} \mathbf{E}_D I(f(X)Y \leq 0) &\leq \frac{1}{n} \sum_{i=1}^n I(f(X_i)Y_i \leq \gamma) + \frac{2A}{\gamma} \sqrt{\frac{1}{n} \mathbf{E}_D k(X, X)} \\ &\quad + \sqrt{\frac{\ln(1/\delta)}{2n}}. \end{aligned}$$

It says that if we can find a classifier with a small margin error, then we can achieve a good test classification error. Unlike analysis using VC-dimension in Chapter 4, the bound does not depend on the dimensionality of the feature vector $\psi(x)$, but the classifiers RKHS norm A , under margin condition.

The bound can be obtained as a direct consequence of Corollary 7.2, using a loss function $L(p, y) = \min(1, \max(0, 1 - py/\gamma))$, which is γ^{-1} Lipschitz. In this case, $I(f(x)y \leq 0) \leq L(f(x), y) \leq I(f(x)y \leq \gamma)$.

Note that the Rademacher complexity analysis only leads to a convergence rate of $O(1/\sqrt{n})$. However, similar to the VC-analysis, it is possible to obtain a margin bound of $O(\ln n/n)$ when the margin error is zero. This requires a more careful analysis with empirical L_∞ covering numbers available from [94].

Example 7.4 For SVM loss, $\gamma = 1$. With hard regularization, we can take $M = (1 + AB)$. From Corollary 7.2, we obtain with probability at least $1 - \delta$:

$$\mathbf{E}_D L(\hat{f}(X), Y) \leq \inf_{\|f\|_{\mathcal{H}} \leq A} \mathbf{E}_D L(f(X), Y) + \epsilon' + \frac{2\gamma AB}{\sqrt{n}} + (1 + AB) \sqrt{\frac{2 \ln(2/\delta)}{n}}.$$

Given an arbitrary competitor $f \in \mathcal{H}$, the optimal value of A would be $A = \|f\|_{\mathcal{H}}$. However, this requires us to know the value of $\|f\|_{\mathcal{H}}$, which is not feasible. In practice, it is more convenient to use the soft-regularized version, where the choice of λ is much less sensitive, and can be set independently of $\|f\|_{\mathcal{H}}$.

We can further obtain a bound for soft-regularized SVM in (7.5) as follows. Since for ERM in the feature representation formulation, with $w = 0$, we have empirical risk of 1. Therefore we always have at the ERM solution:

$$\frac{\lambda}{2} \hat{w} \cdot \hat{w} \leq 1.$$

It follows that if we set $A = \sqrt{2/\lambda}$, then $\|\hat{f}\|_{\mathcal{H}}^2 = \hat{w} \cdot \hat{w} \leq A^2$. We can treat the solution of (7.5) as an approximation solution of (7.4) with $\epsilon' = 0.5\lambda[\|f\|_{\mathcal{H}}^2 - \|\hat{f}\|_{\mathcal{H}}^2]$ for any $f \in \mathcal{H}$. We thus obtain the following oracle inequality. With probability at least $1 - \delta$:

$$\begin{aligned} \mathbf{E}_D L(\hat{f}(X), Y) + \frac{\lambda}{2} \|\hat{f}\|_{\mathcal{H}}^2 &\leq \inf_{f \in \mathcal{H}} \left[\mathbf{E}_D L(f(X), Y) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 \right] \\ &\quad + \frac{2\sqrt{2}B}{\sqrt{\lambda n}} + (\sqrt{2\lambda} + 2B) \sqrt{\frac{\ln(2/\delta)}{\lambda n}}. \end{aligned}$$

If we take $\lambda = O(1/\sqrt{n})$, then we get a convergence no better than $O(1/n^{1/4})$. Note that this bound can be improved.

Corollary 7.3 Assume that $L(p, y) \geq 0$ is γ Lipschitz, and let $M_0 = \sup_Y L(0, Y)$, $B^2 = \sup_x k(x, x)$. Consider $A_0 > 0$, then with probability at least $1 - \delta$: for all $f \in \mathcal{H}$:

$$\begin{aligned} \mathbf{E}_D L(f(X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \frac{4\gamma}{\sqrt{n}} (A_0 + \|f\|_{\mathcal{H}}) B \\ &\quad + \gamma (M_0 + 2(A_0 + \|f\|_{\mathcal{H}}) B) \sqrt{\frac{\ln((2 + \log_2(1 + \|f\|_{\mathcal{H}}/A_0))^2/\delta)}{2n}}. \end{aligned}$$

Proof Let $A_j = 2^j A_0$, with $q_j = 1/(j+1)^2$ for $j = 1, 2, \dots$. Consider $\Omega_j = \{f \in \mathcal{H} : \|f\|_{\mathcal{H}} \leq A_j\}$. We know that $L(p, y) \leq L(0, y) + \gamma|p| \leq M_0 + \gamma A_j B$. Then with probability at least $1 - q_j \delta$: for all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} \leq A_j$:

$$\mathbf{E}_D L(f(X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \frac{2\gamma A_j B}{\sqrt{n}} + (M_0 + \gamma A_j B) \sqrt{\frac{\ln(1/(q_j \delta))}{2n}}.$$

Taking the union bound, we have with probability $1 - \delta$: for all $f \in \mathcal{H}$ and all $j \geq 1$ such that $\|f\|_{\mathcal{H}} \leq A_j$:

$$\mathbf{E}_D L(f(X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \frac{2\gamma A_j B}{\sqrt{n}} + (M_0 + \gamma A_j B) \sqrt{\frac{\ln(1/(q_j \delta))}{2n}}.$$

We let $j(f) \geq 1$ be the smallest index such that $\|f\|_{\mathcal{H}} \leq A_j$. Then $j(f) \leq \log_2(2 + 2\|f\|_{\mathcal{H}}/A_0)$ and $A_{j(f)} \leq 2(A_0 + \|f\|_{\mathcal{H}})$. This implies the theorem. $\blacksquare$

Corollary 7.4 *Assume that $L(p, y) \geq 0$ is γ Lipschitz, and let $M_0 = \sup_Y L(0, Y)$, $B^2 = \sup_x k(x, x)$. Consider $A_0 > 0$, then with probability at least $1 - \delta$: for (7.5),*

$$\mathbf{E}_D L(\hat{f}(X), Y) \leq \inf_{f \in \mathcal{H}} \left[\mathbf{E}_D L(f(X), Y) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2 + \gamma \|f\|_{\mathcal{H}} B \sqrt{\frac{\ln(2/\delta)}{2n}} \right] + \epsilon_n(\delta),$$

where

$$\begin{aligned} \epsilon_n(\delta) = \sup_{\hat{f} \in \mathcal{H}} & \left[-\frac{\lambda}{2} \|\hat{f}\|_{\mathcal{H}}^2 + \frac{4\gamma}{\sqrt{n}} (A_0 + \|\hat{f}\|_{\mathcal{H}}) B \right. \\ & \left. + 2(M_0 + \gamma(A_0 + \|\hat{f}\|_{\mathcal{H}})B) \sqrt{\frac{\ln(2(2 + \log_2(1 + \|\hat{f}\|_{\mathcal{H}}/A_0))^2/\delta)}{2n}} \right]. \end{aligned}$$

Proof We obtain from Corollary 7.3 that with probability $1 - \delta/2$:

$$\begin{aligned} \mathbf{E}_D L(\hat{f}(X), Y) & \leq \frac{1}{n} \sum_{i=1}^n L(\hat{f}(X_i), Y_i) + \frac{\lambda}{2} \|\hat{f}\|_{\mathcal{H}}^2 + \epsilon_n(\delta) \\ & \quad - M_0 \sqrt{\frac{\ln(2(2 + \log_2(1 + \|\hat{f}\|_{\mathcal{H}}/A_0))^2/\delta)}{2n}}. \end{aligned}$$

Moreover, for any $f \in \mathcal{H}$, we have with probability $1 - \delta/2$:

$$\frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) \leq \mathbf{E}_D L(f(X_i), Y_i) + (M_0 + \gamma \|f\|_{\mathcal{H}} B) \sqrt{\frac{\ln(2/\delta)}{2n}}$$

Taking union bound, both inequality holds with probability $1 - \delta$. We also have

$$\frac{1}{n} \sum_{i=1}^n L(\hat{f}(X_i), Y_i) + \frac{\lambda}{2} \|\hat{f}\|_{\mathcal{H}}^2 \leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \frac{\lambda}{2} \|f\|_{\mathcal{H}}^2.$$

By combining the three inequalities, we obtain the desired bound. $\blacksquare$

Example 7.5 Consider the soft-regularized SVM in (7.5). We can let $\gamma = 1$, $A_0 = 1/B$ and $M_0 = 1$. Therefore

$$\epsilon_n(\delta) = O\left(\sqrt{\frac{\ln((\lambda + B^2)/(\delta\lambda))}{n}}\right) + O\left(\frac{B^2 \ln((\lambda + B^2)/(\delta\lambda))}{\lambda n}\right)$$

It follows that if we take $\lambda = O(1/\sqrt{n})$, then we get a convergence rate of $O(\sqrt{\ln \ln n/n})$.

7.4 Vector Valued Functions

We now consider vector valued functions using kernels. In this case, we have $f(x) : \mathcal{X} \rightarrow \mathbb{R}^q$ for some $q > 1$. Let $f(x) = [f_1(x), \dots, f_q(x)]$. If we consider the feature space representation, then there are two possibilities. One is to treat these functions as sharing the same feature vector, but with different w :

$$f_\ell(x) = w_\ell \cdot \psi(x) \quad (\ell = 1, \dots, q).$$

Another view is to consider the same w , but with different features for different dimension:

$$f_\ell(x) = w \cdot \psi(x, \ell). \quad (7.9)$$

The second view is more general because we may write the first view as follows. We concatenate the feature representation by letting $w = [w_1, \dots, w_q]$ and $\psi(x, \ell) = [0, \dots, 0, \psi(x), 0, \dots, 0]$ with only ℓ -th concatenating component to be $\psi(x)$, and the other components as zeros. It follows in this representation that $w \cdot \psi(x, \ell) = w_\ell \cdot \psi(x)$. In the following discussions, we focus on (7.9).

Similar to (7.6), we have the following formulation in feature representation:

$$\hat{w} = \arg \min_{w \in \mathcal{H}} \left[\frac{1}{n} \sum_{i=1}^n L([w \cdot \psi(X_i, \ell)]_\ell, Y_i) + \frac{\lambda}{2} \|w\|^2 \right]. \quad (7.10)$$

Its solution has a kernel representation as follows.

Theorem 7.5 For vector valued function $f(x)$, let $\hat{f}(x) = [\hat{w} \cdot \psi(x, \ell)]_{\ell=1, \dots, q}$, with $\hat{w}$ being the solution of (7.10). Then

$$\hat{f}(x) = \sum_{i=1}^n \mathbf{k}(X_i, x) \hat{\alpha}_i,$$

with

$$\|\hat{w}\|^2 = \sum_{i=1}^n \sum_{j=1}^n \hat{\alpha}_i^\top \mathbf{k}(X_i, X_j) \hat{\alpha}_j,$$

where each $\hat{\alpha}_i \in \mathbb{R}^q$, and

$$\mathbf{k}(x) = \begin{bmatrix} k_{1,1}(x, X_1) & \cdots & k_{1,1}(x, X_n) \\ \vdots & & \vdots \\ k_{q,1}(x, X_1) & \cdots & k_{q,q}(x, X_n) \end{bmatrix},$$

and

$$k_{i,j}(x, x') = \psi(x, i) \cdot \psi(x', j) \quad (i, j = 1, \dots, q).$$

Therefore the solution of (7.10) is equivalent to

$$\hat{\alpha} = \arg \min_{\alpha \in \mathbb{R}^q \times n} \left[\frac{1}{n} \sum_{i=1}^n L \left(\sum_{j=1}^n \mathbf{k}(X_i, X_j) \alpha_j, Y_i \right) + \frac{\lambda}{2} \sum_{i=1}^n \sum_{j=1}^n \alpha_i^\top \mathbf{k}(X_i, X_j) \alpha_j \right]$$

Proof The solution satisfies the following first order optimality condition:

$$\frac{1}{n} \sum_{i=1}^n [\psi(X_i, 1), \dots, \psi(X_i, q)] \nabla L_1(\hat{f}(X_i), Y_i) + \lambda \hat{w} = 0,$$

where $\nabla L_1(p, y)$ is the gradient of $L(p, y)$ with respect to p as a column vector. We thus obtain the following representation as its solution:

$$\hat{w} = \sum_{i=1}^n [\psi(X_i, 1), \dots, \psi(X_i, q)] \hat{\alpha}_i,$$

where

$$\hat{\alpha}_i = -\frac{1}{\lambda n} \nabla L_1([\hat{w} \cdot \psi(X_i, \ell)]_\ell, Y_i) \quad (i = 1, \dots, n).$$

Using this representation, we have

$$\begin{aligned} \hat{f}(x) &= \begin{bmatrix} \hat{w} \cdot \psi(x, 1) \\ \vdots \\ \hat{w} \cdot \psi(x, q) \end{bmatrix} = \sum_{i=1}^n \begin{bmatrix} [\psi(X_i, 1), \dots, \psi(X_i, q)] \hat{\alpha}_i \cdot \psi(x, 1) \\ \vdots \\ [\psi(X_i, 1), \dots, \psi(X_i, q)] \hat{\alpha}_i \cdot \psi(x, q) \end{bmatrix} \\ &= \sum_{i=1}^n \mathbf{k}(X_i, x) \hat{\alpha}_i, \end{aligned}$$

and

$$\begin{aligned} \hat{w} \cdot \hat{w} &= \sum_{i=1}^n [\psi(X_i, 1), \psi(X_i, q)] \hat{\alpha}_i \cdot \sum_{j=1}^n [\psi(X_j, 1), \psi(X_j, q)] \hat{\alpha}_j \\ &= \sum_{i=1}^n \sum_{j=1}^n \hat{\alpha}_i^\top \mathbf{k}(X_i, X_j) \hat{\alpha}_j. \end{aligned}$$

This proves the result. ■

In order to obtain generalization analysis using Rademacher complexity for vector valued functions, we can employ following generalization of Theorem 6.5. The proof is similar to that of Lemma 6.1.

Theorem 7.6 *Consider $L(p, y)$ that is γ_1 -Lipschitz in p with respect to L_1 -norm:*

$$|L(p, y) - L(p', y)| \leq \gamma_1 \|p - p'\|_1.$$

Consider

$$G \subset \{L([f_1(x), \dots, f_q(x)], y) : f_j(x) \in \mathcal{F}_j \ (j = 1, \dots, q)\},$$

then

$$R(G, S_n) \leq \gamma \sum_{j=1}^q R(\mathcal{F}_j, S_n).$$

For vector kernel method, we may consider the hard-constrained version using feature representation as:

$$\hat{w} = \arg \min_{w \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n L([w \cdot \psi(X_i, \ell)]_\ell, Y_i) \quad \text{subject to } \|w\| \leq A. \quad (7.11)$$

We have the following estimate of Rademacher complexity

Lemma 7.1 *Let*

$$G = \{L([w \cdot \psi(x, \ell)]_\ell, y) : \|w\| \leq A\},$$

then with γ_1 defined in Theorem 7.6,

$$R(G, S_n) \leq \frac{\gamma_1 A}{n} \sum_{j=1}^q \sqrt{\sum_{i=1}^n k_{j,j}(X_i, X_i)}$$

and

$$R_n(G, D) \leq \gamma_1 A \sum_{j=1}^q \sqrt{\frac{1}{n} \mathbf{E}_D k_{j,j}(X, X)}.$$

Proof We can set and let $\mathcal{F}_j = \{w \cdot \psi(x, j) : \|w\| \leq A\}$. From Theorem 7.4, we obtain

$$R(\mathcal{F}_j, S_n) \leq \frac{A}{n} \sqrt{\sum_{i=1}^n k_{j,j}(X_i, X_i)},$$

and

$$R_n(\mathcal{F}_j, D) \leq A \sqrt{\frac{1}{n} \mathbf{E}_D k_{j,j}(X, X)}.$$

We can now apply Theorem 7.6 to obtain the desired bound. $\blacksquare$

If $k_{j,j}(x, x) \leq B^2$ for all j and x , then Corollary 7.1 implies a bound

$$R(G, S_n) \leq \frac{\gamma_1 q AB}{\sqrt{n}}.$$

One drawback of this bound is that the Rademacher complexity becomes linear in q , which is usually suboptimal when q is large. For such problems, one needs a more sophisticated estimation of Rademacher complexity.

Example 7.6 Consider the structured SVM loss function [87] for q -class classification problem, with $y \in \{1, \dots, q\}$, and for $f \in \mathbb{R}^q$:

$$L(f, y) = \max_{\ell} [\gamma(y, \ell) - (f_y - f_{\ell})],$$

where $\gamma(y, y) = 0$ and $\gamma(y, \ell) \geq 0$. This loss tries to separate the true class y from alternative $\ell \neq y$ with margin $\gamma(y, \ell)$. It is Lipschitz with respect to $\|f\|_1$ with $\gamma_1 = 1$. Therefore for problems with $k_{\ell, \ell}(x, x) \leq B^2$ for all x and ℓ , we have from Corollary 7.1 that

$$R(G, S_n) \leq \frac{qAB}{\sqrt{n}}.$$

However, a better bound can be obtained which depends on q only logarithmically. We leave it as an exercises.

7.5 Refined Analysis: Ridge Regression

Although Rademacher complexity analysis leads to useful generalization results, they are generally not tight, and only lead to a convergence rate of no better than $O(1/\sqrt{n})$. To obtain better rates, we may employ local Rademacher complexity analysis. For ridge regression, we compare the bound of a simple Rademacher complexity analysis to that of a refined analysis of [44] using a more advanced and specialized proof technique.

For simplicity, we consider a simplified case, where x is random, and y is generated as

$$y = w_* \cdot \psi(x) + \epsilon,$$

where ϵ is a zero-mean bounded noise $|\epsilon| \leq \sigma$. Moreover, we assume that $k(x, x) = \psi(x) \cdot \psi(x) \leq B^2$.

Under this assumption, and consider the following ridge regression estimator:

$$\hat{w} = \arg \min_w \left[\frac{1}{n} \sum_{i=1}^n (w \cdot \psi(X_i) - Y_i)^2 + \frac{\lambda}{2} w \cdot w \right].$$

The corresponding infinity data estimator is

$$w_\lambda = \arg \min_w \left[\mathbf{E}_D(w \cdot \psi(X) - Y)^2 + \frac{\lambda}{2} w \cdot w \right].$$

In order to obtain an oracle inequality, we can apply Corollary 7.4. With $\lambda = \Omega(1/\sqrt{n})$, we can obtain a convergence of $O(1/(\lambda n))$. Choosing λ of order $1/\sqrt{n}$, we obtain a rate of $O(1/\sqrt{n})$.

In comparison, the result of [44], in a simplified form, can be written roughly as:

$$\begin{aligned} \mathbf{E}_D((\hat{w} - w_*) \cdot \psi(X))^2 &\leq \left(1 + O\left(\frac{B^2}{(\lambda + \min_j \lambda_j)n} \right) \right) \mathbf{E}_D((w_\lambda - w_*) \cdot \psi(X))^2 \\ &\quad + O\left(\frac{\sigma^2 d_\lambda}{n} \right), \end{aligned}$$

when $B^2 = \tilde{O}((\lambda + \min_j \lambda_j)n)$, where $\tilde{O}(\cdot)$ hides a logarithmic factor,

$$d_\lambda = \sum_j \frac{\lambda_j^2}{(\lambda_j + \lambda)^2},$$

and λ_j is the j -th eigenvalue of $\mathbf{E}_D \psi(X) \psi(X)^\top$.

The quantity d_λ can be considered as effective dimension. For finite dimensional problem with dimensionality of $\psi(x)$ being d , then $d_\lambda \leq d$ for all λ . Therefore if $\min_j \lambda_j > 0$, we may let $\lambda \rightarrow 0$, and get a simplified bound

$$\mathbf{E}_D((\hat{w} - w_*) \cdot \psi(X))^2 = O\left(\frac{\sigma^2 d}{n} \right).$$

This achieves $O(1/n)$ rate.

In the more general case, the bound has faster convergence when d_λ is small, and it happens when the value of λ_j decays fast towards zero. In such case, we can choose λ at a order smaller than $1/\sqrt{n}$ and get a convergence rate faster than $O(1/\sqrt{n})$. For example, when $\lambda_j \leq B^2/j^2$, then we may choose $\lambda = O(B^2/n^{2/3})$ and obtain a convergence of $O(n^{-2/3})$.

7.6 Exercises

1. We consider linear models with L_2 regularization:

$$\mathcal{F}_A = \{f(w, x) = w \cdot \psi(x) : \|w\|_2^2 \leq A\}.$$

Assume that $k(x, x) = \psi(x) \cdot \psi(x) \leq B^2$ for all x . It is known from [94] that for any n data points S_n , the L_∞ covering number is bounded by

$$\log_2 N(\mathcal{F}_A, \epsilon, L_\infty(S_n)) \leq \frac{36A^2B^2}{\epsilon^2} \log_2[2\lceil(4AB/\epsilon) + 2\rceil n + 1]. \quad (7.12)$$

We would like to derive a multiplicative form of margin bound similar to the additive form margin bound of Example 7.2.

- Consider a $\gamma/2$ L_∞ cover $F_{\gamma/2}$ of $\mathcal{F}_A$ with sample S_{n+m} ($m = n$). Derive a uniform convergence bound (validation error versus training error) for $\{I(f(X)Y \leq \gamma/2) : f \in F_{\gamma/2}\}$ over random permutations of $S_{n+m} = S_n \cup S'_m$ using the multiplicative form of Chernoff bound.
 - Use this estimate, derive a multiplicative form of margin bound that bounds the classification error $\mathbf{E}_{(X,Y) \sim D} I(f(X)Y \leq 0)$ in terms of the margin training error $\frac{1}{n} \sum_{(X,Y) \in S_n} I(f(X)Y \leq \gamma)$.
2. Consider the structured SVM problem in Example 7.6.
 - Use the empirical L_∞ covering number bound of (7.12) to derive an L_∞ -covering number bound for the structured SVM loss $L(f, y)$ defined in the example, with a logarithmic dependence on q .
 - Use chaining to obtain an estimate of the Rademacher complexity $R(G, S_n)$, and compare the results to that of Example 7.6.
 3. Assume we want to solve the soft-regularized SVM of Example 7.2, with λ chosen based on the training data according to an appropriate learning bounds. Assume we obtain the SVM solutions $\hat{w}_1, \dots, \hat{w}_N$ at different choices $\lambda_1, \dots, \lambda_N$. Assume that $k(X, X) \leq B^2$.
 - Which bound do you use to select the best λ_j ? Corollary 7.2, Corollary 7.3, or Corollary 7.4? And explain the procedure.
 - Obtain an oracle inequality for your procedure.

Chapter 8

Additive Models and Sparsity Analysis

In this chapter, we focus on *additive model* of the following form:

$$f([w, \theta], x) = \sum_{j=1}^m w_j \psi(\theta_j, x), \quad (8.1)$$

where for simplicity, we consider real valued functions $\psi(\theta, \cdot) : \mathcal{X} \rightarrow \mathbb{R}$ first. In additive model, each $\psi(\theta, \cdot)$ may be regarded as a prediction function, which is parametrized by $\theta \in \Theta$. The goal of additive model is to find a combination of models $\psi(\theta, \cdot)$ so that the combined model $f(w, x)$ is more accurate.

8.1 Sparse Model Combination

In order to motivate sparse regularization, we will first consider the case that Θ is finite. Assume that Θ has m elements $\{\theta_1, \dots, \theta_m\}$, then (8.1) can be regarded as a linear model with respect to the model parameter w , and we can simply denote it as:

$$f(w, x) = \sum_{j=1}^m w_j \psi_j(x) = w^\top \psi(x),$$

with features $\psi_j(x) = \psi(\theta_j, x)$, and $\psi(x) = [\psi_1(x), \dots, \psi_m(x)]$. We further assume that each feature $\psi_j(x)$ is a prediction function so that

$$\psi_j(x) \in [0, M].$$

If we do not consider regularization, then the model complexity is determined by the model dimensionality m . For example, the VC dimension of

the linear model $I(f(w, x) \geq 0)$ is m . If we consider regularization, such as kernel method or L_2 regularization, then consider the function class

$$\Omega' = \{f(w, x) : \|w\|_2 \leq A\}$$

By the Rademacher complexity analysis, we have

$$R_n(\Omega', D) \leq A\sqrt{\mathbf{E}_D k(x, x)/n},$$

with

$$k(x, x) = \sum_{j=1}^m \psi_j(x)^2.$$

This means we have a bound of

$$\mathbf{E}_D k(x, x) \leq \sum_{j=1}^m \mathbf{E}_D \psi_j(x)^2 \leq mM^2,$$

which means that the kernel is proportional to the feature dimensionality m .

This gives a bound on Rademacher complexity of

$$R_n(\Omega', D) \leq \sqrt{\frac{m}{n}} AM,$$

which depends linearly on $\sqrt{m}$.

When the features are orthonormal, then it can be shown using the theory of Gaussian processes that when n is sufficiently large,

$$R_n(\Omega', D) \geq c\sqrt{\frac{m}{n}} A$$

for some constant $c > 0$. It follows that the factor $\sqrt{m}$ cannot be removed in general with L_2 regularization, or kernel methods. If m is large (or infinity), this can become a problem.

We may compare the model combination problem to the kernel formulation, using RBF kernel as an example. Note that $k(x, x) \leq 1$, which implies that $\sum_j \mathbf{E}_D \psi_j(x)^2 \leq 1$. This means that the value of $\mathbf{E}_D \psi_j(x)^2$ should be very small for most feature j . In comparison, this is not a reasonable assumption for model combination, as we expect that $\mathbf{E}_D \psi_j(x)$ is not small for most models $\psi_j(x)$.

In order to fix the dimensionality dependency problem of model combination, one needs to introduce regularization conditions other than L_2 regularization. One common assumption to achieve this is to impose an additional sparsity constraint. We may define the sparsity pattern, or support of a weight vector $w \in \mathbb{R}^m$ as:

$$\text{supp}(w) = \{j : w_j \neq 0\},$$

and the L_0 norm of w as

$$\|w\|_0 = |\text{supp}(w)|.$$

For each sparsity pattern $S \subset \{1, \dots, m\}$, we may define the $|S|$ -dimensional sparse function class

$$G_S = \left\{ L(w^\top \psi(x), y) : \text{supp}(w) \subset S \right\}.$$

We can now consider the following sparse learning method:

$$\hat{w} = \arg \min_w \left[\frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) + r(S) \right], \quad \text{supp}(w) \subset S \quad (8.2)$$

with a properly defined regularizer $r(S)$. We note that the solution of (8.2) may be computationally very challenging due to the nonconvex sparsity constraint. However, if we assume that we can find an approximate solution, then we can obtain a generalization error bound that only depends logarithmically on m , and linear only with respect to the sparsity. The following theorem is a direct consequence of Theorem 6.8.

Theorem 8.1 *Assume $L(\cdot, \cdot) \in [0, M]$. Let S_n be n iid samples from D . Then with probability at least $1 - \delta$, the following bound holds for all $w \in \mathbb{R}^m$ and S such that $\text{supp}(w) \subset S$:*

$$\begin{aligned} \mathbf{E}_D L(f(w, X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(f(w, X_i), Y_i) + 2R(G_S, S_n) \\ &\quad + 3M \sqrt{\frac{|S| \ln(em/|S|) + \ln 2(|S| + 1)^2}{2n}} + 3M \sqrt{\frac{\ln(1/\delta)}{2n}}. \end{aligned}$$

Consider the sparse learning algorithm in (8.2), with

$$r(S) \geq 2R(G_S, S_n) + 3M \sqrt{\frac{|S| \ln(em/|S|) + \ln 2(|S| + 1)^2}{2n}}$$

for all S_n . Then we have the following oracle inequality. With probability of at least $1 - \delta$:

$$\mathbf{E}_D L(f(w, X), Y) \leq \inf_{w \in \mathbb{R}^m} [\mathbf{E}_D L(f(w, X), Y) + r(S)] + 4M \sqrt{\frac{\ln(2/\delta)}{2n}},$$

where $\text{supp}(w) \subset S$.

Proof We note that

$$C_m^s \leq \frac{m^s}{s!} \leq \frac{m^s}{s^s} \cdot \frac{s^s}{s!} \leq \frac{m^s}{s^s} \cdot e^s = (me/s)^s.$$

We can now consider each G_S as a model, with

$$q_S = \frac{(|S| + 1)^{-2}}{(me/|S|)^{|S|}} \leq \frac{(|S| + 1)^{-2}}{C_m^{|S|}}.$$

Therefore we have

$$\sum_{S: |S| \geq 1} q_S \leq \sum_{s \geq 1} \frac{1}{(s+1)^2} \sum_{S: |S|=s} \frac{1}{C_m^{|S|}} = \sum_{s \geq 1} \frac{1}{(s+1)^2} < 1.$$

For each index S , we may consider $\hat{w}$ as the ERM solution under the constraint $\text{supp}(w) \subset S$. We can thus apply Theorem 6.8 with models indexed by $S \subset \{1, \dots, m\}$ to obtain the first desired bound (uniform convergence).

The second desired bound (oracle inequality) deviates from Theorem 6.8 in that we do not apply Rademacher concentration result, but rely on the uniform bound in the assumption of the theorem. We simply note that for all w and $\text{supp}(w) \subset S$, with probability $1 - \delta/2$:

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) + r(S) &\leq \mathbf{E}_D L(w^\top \psi(X), Y) + r(S) \\ &\quad + M \sqrt{\frac{\ln(2/\delta)}{2}}. \end{aligned}$$

Moreover, the uniform convergence bound holds with probability $1 - \delta/2$:

$$\mathbf{E}_D L(f(\hat{w}, X), Y) \leq \frac{1}{n} \sum_{i=1}^n L(f(\hat{w}, X_i), Y_i) + r(\text{supp}(\hat{w})) + 3M \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Taking the union bound, and combine with (8.2), we obtain the bound. ■

Example 8.1 Consider the linear binary classification problem, with the loss function

$$L(f(w, x), y) = I(w^\top \psi(x)y \leq 0).$$

In this problem, we have

$$R(G_S, S_n) \leq c \sqrt{\frac{|S|}{n}}$$

for some constant c . It follows that we may simply take

$$r(S) = c' \sqrt{\frac{|S| \ln m}{n}}$$

for a sufficiently large constant c' , which leads to the following sparsity constrained optimization:

$$\hat{w} = \arg \min_{w \in \mathbb{R}^m} \left[\frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) + \lambda \sqrt{\|w\|_0} \right]$$

with $\lambda = c' \sqrt{\frac{\ln m}{n}}$. We obtain the following oracle inequality. With probability at least $1 - \delta$:

$$\mathbf{E}_D L(f(\hat{w}, X), Y) \leq \inf_{w \in \mathbb{R}^m} \left[\mathbf{E}_D L(f(w, X), Y) + \lambda \sqrt{\|w\|_0} \right] + 4M \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

The bound is linear in $\sqrt{\|w\|_0}$, and logarithmic in m .

Note that we used Rademacher complexity to derive such a bound, which has a $O(\sqrt{1/n})$ convergence rate. For some more specialized problems such as least squares regression, where the variance condition holds, we can obtain a better rate of $O(1/n)$. The resulting bound will have a convergence rate of $O(\|w\|_0 \ln m/n)$, similar to the Bayesian information criterion (BIC) for parametric models in classical statistics [74].

8.2 L_1 Regularization

Sparse regularization can reduce the learning complexity. However, the optimization problem of (8.2) is usually hard. To alleviate this problem, practitioners often solve convex relaxation formulations, with L_0 regularization $\|w\|_0$ replaced by L_1 regularization $\|w\|_1$. The resulting method is also referred to as *Lasso* [85].

In the following, we will consider L_1 regularization for the model combination problem, and its relationship to sparse learning. We will consider the general situation that Θ is infinite. For notation simplicity, we can define the function class

$$\Psi = \{\psi(\theta, x) : \theta \in \Theta\},$$

and its convex hull can be derived as (see Definition 5.2

$$\text{CONV}(\Psi) = \left\{ \sum_{j=1}^m w_j \psi(\theta_j, x) : m > 0, \|w\|_1 = 1, w_j \geq 0, \theta_j \in \Theta \right\}.$$

The non-negative L_1 regularized additive models are:

$$\mathcal{F}_{A, L_1}^+(\Psi) = \{af(x) : a \in [0, A], f(x) \in \text{CONV}(\Psi)\}.$$

We may also consider L_1 regularized additive models as:

$$\mathcal{F}_{A, L_1}(\Psi) = \left\{ \sum_{j=1}^m w_j \psi(\theta_j, x) : \|w\|_1 \leq A, \theta_j \in \Theta, m > 0 \right\}.$$

Note that if $-f \in \Psi$ for all $f \in \Psi$, then $\mathcal{F}_{A,L_1}(\Psi) = \mathcal{F}_{A,L_1}^+(\Psi)$. The following result on Rademacher complexity for L_1 regularization is straight forward.

Theorem 8.2 *We have*

$$R(\text{CONV}(\Psi), S_n) = R(\Psi, S_n).$$

If $0 \in \Psi$, then the following equality holds:

$$R(\mathcal{F}_{A,L_1}^+(\Psi), S_n) = A \cdot R(\Psi, S_n).$$

If $\Psi = -\Psi$, then the following equality holds:

$$R(\mathcal{F}_{A,L_1}(\Psi), S_n) = A \cdot R(\Psi, S_n).$$

Proof We will prove the second inequality. Since $A \cdot \Psi \subset \mathcal{F}_{A,L_1}^+(\Psi, S_n)$, we have

$$A \cdot R(\Psi, S_n) \leq R(\mathcal{F}_{A,L_1}^+(\Psi), S_n).$$

Moreover, consider any

$$\sum_{j=1}^m w_j \psi(\theta_j, x) : \|w\|_1 \leq A, w_j \geq 0, \theta_j \in \Theta$$

and $\sigma_i \in \{\pm 1\}$, we know that

$$\begin{aligned} \sum_{i=1}^n \sigma_i \sum_{j=1}^m w_j \psi(\theta_j, X_i) &\leq \|w\|_1 \sup_j \sum_{i=1}^n \sigma_i \psi(\theta_j, X_i) \\ &\leq \|w\|_1 \sup_{\psi \in \Psi} \sum_{i=1}^n \sigma_i \psi(X_i) \\ &\leq A \sup_{\psi \in \Psi} \sum_{i=1}^n \sigma_i \psi(X_i), \end{aligned}$$

where the last inequality used the fact that $\|w\|_1 \leq A$ and $\sup_{\psi \in \Psi} \sum_{i=1}^n \sigma_i \psi(X_i) \geq 0$. This implies that

$$R(\mathcal{F}_{A,L_1}^+(\Psi), S_n) \leq A \cdot R(\Psi, S_n),$$

and we obtain the second desired bound. The proof of the first bound and the third bound are similar. ■

Example 8.2 Assume that $|\Psi| = N$, then $|\Psi \cup -\Psi| \leq 2N$. From Theorem 6.3, we have

$$R(\mathcal{F}_{A,L_1}(\Psi), S_n) \leq A \sup_{\psi \in \Psi} \|\psi\|_{L_2(S_n)} \cdot \sqrt{\frac{2 \ln(2N)}{n}}.$$

If $|\psi(x)| \leq B$ for all $\psi \in \Psi$, then

$$R(\mathcal{F}_{A,L_1}(\Psi), S_n) \leq AB \sqrt{\frac{2 \ln(2N)}{n}}.$$

Example 8.3 Assume that Ψ is binary function class with VC dimension d , then we know that there exists a constant $c > 0$ such that

$$R(\Psi, S_n) \leq c \sqrt{d/n}.$$

It follows that

$$R(\mathcal{F}_{A,L_1}(\Psi), S_n) \leq 2Ac \sqrt{d/n}.$$

Example 8.4 Assume that Ψ is L_2 -regularized ReLU function class:

$$\Psi = \{\psi(\theta, x) = \max(0, \theta \cdot x) \quad \|\theta\|_2 \leq \alpha, \|x\|_2 \leq \beta\}.$$

Then

$$R(\mathcal{F}_{A,L_1}(\Psi), S_n) \leq A\alpha\beta/\sqrt{n}.$$

We now consider the following hard-constrained L_1 regularized learning:

$$\hat{w} = \arg \min_w \frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) \quad \|w\|_1 \leq A. \quad (8.3)$$

Similarly, we may consider the soft-regularized version as:

$$\hat{w} = \arg \min_w \frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) + \lambda \|w\|_1. \quad (8.4)$$

Now using Theorem 6.6, we obtain the following result.

Corollary 8.1 Assume that $L(p, y) \in [0, M]$ is γ Lipschitz with respect to p . For fixed $A > 0$, with probability at least $1 - \delta$: for all $f(x) = w^\top \psi(x)$ such that $\|w\|_1 \leq A$:

$$\begin{aligned} \mathbf{E}_D L(w^\top \psi(X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(w^\top \psi(X_i), Y_i) \\ &\quad + 2\gamma A R_n(\Psi_\pm, D) + M \sqrt{\frac{\ln(1/\delta)}{2n}}, \end{aligned}$$

where $\Psi_{\pm} = \{\psi(x) : \psi(x) \in \Psi \text{ or } -\psi(x) \in \Psi\}$. Moreover, for (8.3), if we solve it approximately up to sub-optimality of ϵ' , then we have with probability at least $1 - \delta$:

$$\begin{aligned} \mathbf{E}_D L(\hat{w}^\top \psi(X), Y) &\leq \inf_{\|w\|_1 \leq A} \mathbf{E}_D L(w^\top \psi(X), Y) + 2\gamma A R_n(\Psi_{\pm}, D) + \epsilon' \\ &\quad + M \sqrt{\frac{2 \ln(2/\delta)}{n}}. \end{aligned}$$

Example 8.5 If Ψ contains m functions $\{\psi_1(x), \dots, \psi_m(x)\}$, each $|\psi_j(x)| \leq B$, then

$$R_n(\Psi_{\pm}, D) \leq B \sqrt{2 \ln(2m)/n}.$$

Therefore the bound of Corollary 8.1 implies the oracle inequality

$$\begin{aligned} \mathbf{E}_D L(\hat{w}^\top \psi(X), Y) &\leq \inf_{\|w\|_1 \leq A} \mathbf{E}_D L(w^\top \psi(X), Y) + 2\gamma A B \sqrt{2 \ln(2m)/n} \\ &\quad + M \sqrt{\frac{2 \ln(2/\delta)}{n}}. \end{aligned}$$

This has a logarithmic dependency on m , similar to that of the sparsity constraint in Theorem 8.1.

Similar to the analysis of kernel method, we may analyze the soft L_1 regularization method by considering the sample dependent bound below. The proof is similar to that of Corollary 7.3.

Corollary 8.2 Assume that $L(p, y) \geq 0$ is γ Lipschitz, and let $M_0 = \sup_Y L(0, Y)$, and $B = \sup_{x, \psi \in \Psi} |\psi(x)|$. Consider $A_0 > 0$, then with probability at least $1 - \delta$: for all $f(x) = w^\top \Psi(x)$:

$$\begin{aligned} \mathbf{E}_D L(f(X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + 4\gamma(A_0 + \|w\|_1) R_n(\Psi_{\pm}, D) \\ &\quad + \gamma(M_0 + 2(A_0 + \|w\|_1 B)) \sqrt{\frac{\ln((2 + \log_2(1 + \|w\|_1/A_0))^2/\delta)}{2n}}. \end{aligned}$$

Proof Let $A_j = 2^j A_0$, with $q_j = 1/(j+1)^2$ for $j = 1, 2, \dots$. Consider $\Omega_j = \{w^\top \Psi(x) : \|w\|_1 \leq A_j\}$. We know that $L(p, y) \leq L(0, y) + \gamma|p| \leq M_0 + \gamma A_j B$. Then with probability at least $1 - q_j \delta$: for all $f \in \mathcal{H}$ with $\|f\|_{\mathcal{H}} \leq A_j$:

$$\begin{aligned} \mathbf{E}_D L(f(X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \gamma A_j R_n(\Psi_{\pm}, D) \\ &\quad + (M_0 + \gamma A_j B) \sqrt{\frac{\ln(1/(q_j \delta))}{2n}}. \end{aligned}$$

Taking the union bound, we have with probability $1 - \delta$: for all $f(x) = w^\top \psi(x)$ and all $j \geq 1$ such that $\|w\|_1 \leq A_j$:

$$\begin{aligned} \mathbf{E}_D L(f(X), Y) &\leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + 2\gamma A_j R_n(\Psi_\pm, D) \\ &\quad + (M_0 + \gamma A_j B) \sqrt{\frac{\ln(1/(q_j \delta))}{2n}}. \end{aligned}$$

We let $j(f) \geq 1$ be the smallest index such that $\|w\|_1 \leq A_j$. Then $j(f) \leq \log_2(2 + 2\|f\|_{\mathcal{H}}/A_0)$ and $A_{j(f)} \leq 2(A_0 + \|w\|_1)$. This implies the theorem. ■

Using the sample dependent bound, we can then analyze (8.4).

Corollary 8.3 *Assume that $L(p, y) \geq 0$ is γ Lipschitz, and let $M_0 = \sup_Y L(0, Y)$, and $B = \sup_{x, \psi \in \Psi} |\psi(x)|$. Consider $A_0 > 0$, and*

$$\lambda \geq 4\gamma R_n(\Psi_\pm, D) + 2\gamma B \sqrt{\frac{\ln(2(2 + \log_2(1 + M_0/(\lambda A_0)))^2/\delta)}{2n}}.$$

Then with probability at least $1 - \delta$, for (8.4):

$$\begin{aligned} \mathbf{E}_D L(\hat{w}^\top \psi(X), Y) &\leq \inf_w \left[\mathbf{E}_D L(w^\top \psi(X), Y) + \left(\lambda + \gamma B \sqrt{\frac{\ln(2/\delta)}{2n}} \right) \|w\|_1 \right] \\ &\quad + \epsilon_n(\delta), \end{aligned}$$

where

$$\epsilon_n(\delta) = 4\gamma A_0 R_n(\Psi_\pm, D) + 2(M_0 + \gamma A_0) \sqrt{\frac{\ln(2(2 + \log_2(1 + M_0/(\lambda A_0)))^2/\delta)}{2n}}.$$

Proof From

$$\lambda \|\hat{w}\|_1 \leq M_0,$$

we know that $\|\hat{w}\|_1 \leq M_0/\lambda$. We obtain from Corollary 8.2 that with prob-

ability $1 - \delta/2$:

$$\begin{aligned}
& \mathbf{E}_D L(\hat{w}^\top \psi(X), Y) \\
& \leq \frac{1}{n} \sum_{i=1}^n L(\hat{w}^\top \psi(X_i), Y_i) + 4\gamma(A_0 + \|w\|_1) R_n(\Psi_\pm, D) \\
& \quad + \gamma(M_0 + 2(A_0 + \|w\|_1)B) \sqrt{\frac{\ln((2 + \log_2(1 + M_0/(\lambda A_0)))^2/\delta)}{2n}} \\
& = \frac{1}{n} \sum_{i=1}^n L(\hat{w}^\top \psi(X_i), Y_i) + \lambda \|\hat{w}\|_1 \\
& \quad + \sup_{\hat{w}} [-\lambda \|\hat{w}\|_1 + 4\gamma(A_0 + \|\hat{w}\|_1) R_n(\Psi_\pm, D) \\
& \quad + (M_0 + 2\gamma(A_0 + \|\hat{w}\|_1)B) \sqrt{\frac{\ln(2(2 + \log_2(1 + M_0/(\lambda A_0)))^2/\delta)}{2n}}] \\
& \leq \frac{1}{n} \sum_{i=1}^n L(\hat{w}^\top \psi(X_i), Y_i) + \lambda \|\hat{w}\|_1 + \epsilon_n(\delta) - M_0 \sqrt{\frac{\ln(2/\delta)}{2n}}.
\end{aligned}$$

Moreover, for any $f(x) = w^\top \psi(x)$, we have with probability $1 - \delta/2$:

$$\frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) \leq \mathbf{E}_D L(f(X_i), Y_i) + (M_0 + \gamma\|w\|_1 B) \sqrt{\frac{\ln(2/\delta)}{2n}}.$$

Taking union bound, both inequality holds with probability $1 - \delta$. We also have

$$\frac{1}{n} \sum_{i=1}^n L(\hat{w}^\top \psi(X_i), Y_i) + \lambda \|\hat{w}\|_1 \leq \frac{1}{n} \sum_{i=1}^n L(f(X_i), Y_i) + \lambda \|w\|_1.$$

By combining the three inequalities, we obtain the desired bound. $\blacksquare$

While we often use the regularizer $\lambda\|w\|_1$ for L_1 regularization, it is also possible to use $\lambda\|w\|_1^2$.

8.3 Entropy Regularization and Gibbs Algorithm

In the literature, one often encounters entropy regularization, which is closely related to L_1 regularization. We consider a more specialized case that the weights $w \geq 0$, and $\sum_j w_j = 1$. In such case, w can be considered as a distribution over models. Therefore we can use a different notation, and denote it by a general distribution $q(\theta)$ that may not necessarily be discrete. Our goal is to find a distribution q on Θ , such that we draw models $\psi(\theta, x) \in \Psi$ in the test time:

$$f(q, x) = \int \psi(\theta, x) q(\theta) d\theta = \mathbf{E}_{\theta \sim q(\cdot)} \psi(\theta, x). \quad (8.5)$$

In this case, $q(\theta)$ is often referred to as *posterior distribution*, using the Bayesian statistics analogy. Consider a prior $q_0(\theta)$ on Θ , we may consider the entropy regularization:

$$\text{KL}(q||q_0) = \int q(\theta) \ln \frac{q(\theta)}{q_0(\theta)} d\theta.$$

We have the following Rademacher complexity estimate.

Theorem 8.3 ([61]) *Let $\mathcal{F}_A = \{f(q, x) : \text{KL}(q||q_0) \leq A^2\}$ be entropy regularized functions of (8.5). Then*

$$R(\mathcal{F}_A, S_n) \leq \sqrt{\frac{2}{n}} A \sup_{\theta} \sqrt{\frac{1}{n} \sum_{i=1}^n \psi(\theta, X_i)^2}.$$

This bound holds for general function classes. In the case of finite family with $\Theta = \{\theta_1, \dots, \theta_m\}$, and $q_0(\theta) = 1/m$ for all θ , we have

$$\text{KL}(q||q_0) \leq \ln m$$

for all q . Therefore entropy regularization implies a bound for L_1 regularization with nonnegative constraint $\sum_{j=1}^m w_j = 1$ and $w_j \geq 0$. Since this is exactly the convex hull of $\Psi = \{\psi(\theta, x)\}$, Theorem 8.3 implies that

$$R(\text{CONV}(\Psi), S_n) \leq \sqrt{\frac{2 \ln m}{n}} \sup_{\theta} \|\psi(\theta, \cdot)\|_{L_2(S_n)},$$

which is identical to the Rademacher complexity of convex hull of finite function class obtained earlier.

A particularly interesting application of entropy regularization is the *PAC-Bayes* analysis [57], which randomly draws a model from the posterior q to make prediction, instead of forming the averaged model for prediction as in (8.5). The following uniform convergence result is a direct consequence of Theorem 6.6 and Theorem 8.3.

Theorem 8.4 *Assume that $L(p, y) \in [0, M]$. Let q_0 be a distribution on Θ . Consider fixed $A > 0$. Then with probability at least $1 - \delta$: for all distribution q on Θ with $\text{KL}(q||q_0) \leq A^2$,*

$$\begin{aligned} \mathbf{E}_{\theta \sim q} \mathbf{E}_{(X, Y) \sim D} L(f(\theta, X), Y) &\leq \mathbf{E}_{\theta \sim q} \frac{1}{n} \sum_{i=1}^n L(f(\theta, X_i), Y_i) \\ &\quad + AM \sqrt{\frac{8}{n}} + M \sqrt{\frac{\ln(1/\delta)}{2n}}. \end{aligned}$$

The uniform convergence result implies that if we randomly draw a sample from q , then the average loss of its test error is not much smaller than the average loss of its training error, as long as $\text{KL}(q||q_0)$ is small. This suggests the following hard constrained method for entropy regularization:

$$\hat{q} = \arg \min_q \mathbf{E}_{\theta \sim q} \frac{1}{n} \sum_{i=1}^n L(f(\theta, X_i), Y_i) \quad \text{KL}(q||q_0) \leq A^2. \quad (8.6)$$

The corresponding soft-regularized version is

$$\hat{q} = \arg \min_q \mathbf{E}_{\theta \sim q} \frac{1}{n} \sum_{i=1}^n L(f(\theta, X_i), Y_i) + \lambda \text{KL}(q||q_0), \quad (8.7)$$

which has the following closed form solution, referred to as the *Gibbs distribution*:

$$\hat{q}(\theta) \propto q_0(\theta) \exp \left(-\frac{1}{\lambda n} \sum_{i=1}^n L(f(\theta, X_i), Y_i) \right). \quad (8.8)$$

Theorem 6.6 and Theorem 8.3 imply the following oracle inequality for the Gibbs algorithm in (8.6).

Theorem 8.5 *Assume that $L(p, y) \in [0, M]$. Let q_0 be a distribution on Θ . Let $\hat{q}$ be the solution of (8.6). Then with probability at least $1 - \delta$:*

$$\begin{aligned} \mathbf{E}_{\theta \sim \hat{q}} \mathbf{E}_{(X,Y) \sim D} L(f(\theta, X), Y) &\leq \inf_{q: \text{KL}(q||q_0) \leq A} \mathbf{E}_{\theta \sim q} \mathbf{E}_{(X,Y) \sim D} L(f(\theta, X), Y) \\ &\quad + AM \sqrt{\frac{8}{n}} + M \sqrt{2 \frac{\ln(2/\delta)}{n}}. \end{aligned}$$

Similar to Corollary 7.4 and Corollary 8.3, we can derive oracle inequalities for the soft-regularized Gibbs algorithm in (8.8) with a fixed λ of order $O(1/\sqrt{n})$.

8.4 Boosting and Greedy Algorithm

We introduce the following definition of smoothness.

Definition 8.1 *A loss function $L(p, y)$ is γ -smooth in p if for all p and p' :*

$$L(p', y) \leq L(p, y) + (p' - p)L'_1(p, y) + \frac{\gamma}{2}(p' - p)^2,$$

where $L'_1(p, y)$ is the derivative of $L(p, y)$ with respect to p .

We show that the smooth-loss with L_1 regularized function can be sparsely approximated. Such a sparsification result was first shown for the least squares loss, and referred to as Maurey's lemma in [65].

Theorem 8.6 Assume $L(p, y)$ is a γ -smooth loss in p . Consider $f(x) = w^\top \Psi(x)$ such that $\|w\|_1$ is finite. Then there exists a sparse vector u so that $\|u\|_0 \leq N$, and

$$\begin{aligned} \mathbf{E}_D L(u^\top \Psi(x), y) &\leq \mathbf{E}_D L(w^\top \Psi(x), y) \\ &\quad + \frac{\gamma[\|w\|_1^2 \sup_{\theta \in \Theta} \mathbf{E}_D \psi(\theta, X)^2 - \mathbf{E}_D f(X)^2]}{2N}. \end{aligned}$$

Proof We may assume that for some $m > 0$:

$$f(x) = \sum_{j=1}^m w_j \psi(\theta_j, x),$$

and let $\psi_j(x) = \psi(\theta_j, x)$. Consider N iid random variable j_k ($k = 1, \dots, N$) that takes values in $\{1, \dots, m\}$, so that $\Pr(j_k = j) = |w_j|/\|w\|_1$. Let

$$f_J(x) = \frac{1}{N} \sum_{k=1}^N u_{j_k} \psi_{j_k}(x), \quad u_j = \text{sign}(w_j) \|w\|_1 \quad (j = 1, \dots, N)$$

where $J = \{j_1, \dots, j_N\}$. It is easy to check that for all x :

$$\mathbf{E}_J f_J(x) = f(x).$$

Note also from the smoothness assumption, we have

$$\begin{aligned} L(f_J(x), y) &\leq L(f(x), y) + (f_J(x) - f(x)) L'_1((f(x), y)) \\ &\quad + \frac{\gamma}{2} |f_J(x) - f(x)|^2. \end{aligned}$$

Taking expectation, we have

$$\begin{aligned} \mathbf{E}_J L(f_J(x), y) &\leq L(f(x), y) + \frac{\gamma}{2} \mathbf{E}_J |f_J(x) - f(x)|^2 \\ &= L(f(x), y) + \frac{\gamma}{2N} [\mathbf{Var}_{j_1} [u_{j_1} \psi_{j_1}(x)]] \\ &= L(f(x), y) + \frac{\gamma}{2N} \mathbf{E}_{j_1} [\|w\|_1^2 \psi_{j_1}(x)^2 - f(x)^2]. \end{aligned}$$

Therefore by taking expectation with respect to D , we obtain

$$\begin{aligned} \mathbf{E}_D L(f_J(X), Y) &\leq \mathbf{E}_D L(f(X), Y) + \frac{\gamma}{2N} \mathbf{E}_{j_1} \mathbf{E}_D [\|w\|_1^2 f_{j_1}(X)^2 - f(X)^2] \\ &\leq \mathbf{E}_D L(f(X), Y) + \frac{\gamma[\|w\|_1^2 \sup_j \mathbf{E}_D f_j(X)^2 - \mathbf{E}_D f(X)^2]}{2N}. \end{aligned}$$

Since each $f_J(X)$ can be expressed as a sparse combination of N functions, this implies the existence of u . ■

The above result shows that an L_1 regularized model can be approximated by a sparse model. Since sparse model has low complexity for model combination problems, the analysis explains why L_1 regularization has low complexity for model combination problems.

One method to obtain the sparse approximation is to use greedy algorithm such as [35, 95, 77], which is closely related to boosting [37, 38]. We present one version of greedy boosting algorithm from [77].

Algorithm 8.1: Fully Corrective Gradient Boosting

Input: $S_n, \Psi, L(\cdot, \cdot)$
Output: $f^{(T)}(x)$

- 1 Let $f^{(0)}(x) = 0$
- 2 **for** $t = 1, 2, \dots, T$ **do**
- 3 Let $g_i = L'_1(f^{(t-1)}(X_i), Y_i)$ ($i = 1, \dots, n$) be functional gradients
- 4 Solve for $[\alpha_t, \theta_t] = \arg \min_{\alpha \in \mathbb{R}, \theta \in \Theta} Q_t(\alpha, \theta)$,
- 5 where $Q_t(\alpha, \theta) = \sum_{i=1}^n [\alpha \psi(\theta, X_i) - g_i]^2$
- 6 Let $w = \arg \min_w \sum_{i=1}^n L(\sum_{s=1}^t w_j \psi(\theta_j, X_i), Y_i)$
- 7 Let $f^{(t)}(x) = \sum_{s=1}^t w_j \psi(\theta_j, X_i)$

Return: $f^{(T)}(x)$

Theorem 8.7 ([77]) Assume that $L(p, y)$ is convex and γ smooth in p . Then at time T , we have

$$\sum_{i=1}^n L(f^{(T)}(X_i), Y_i) \leq \inf_w \left[\sum_{i=1}^n L(w^\top \Psi(X_i), Y_i) + \frac{2\gamma \|w\|_1^2 B^2}{T} \right],$$

where

$$B^2 = \sup_{\theta} \sum_{i=1}^n \psi(\theta, X_i)^2.$$

One can use the relationship of greedy algorithm and L_1 regularization to analyze the generalization behavior of boosting. By combining a convergence analysis of greedy algorithm in terms of L_1 norm, with generalization analysis using L_1 regularization, one can show the consistency of boosting procedures [101]. It is also possible to investigate the generalization analysis of boosting using margin with respect to L_1 norm, as in [72]. One may use Rademacher complexity to derive such a data-dependent margin bound, as a direct consequence of Corollary 8.2.

Corollary 8.4 Assume that $Y \in \{\pm 1\}$, and $\psi(\theta, x) \in [-1, 1]$. Given $\gamma > 0$.

Then with probability $1 - \delta$: for all $f(x) = \sum_j w_j \psi(\theta_j, x)$, we have

$$\begin{aligned} \mathbf{E}_D I(f(X)Y \leq 0) &\leq \frac{1}{n} \sum_{i=1}^n L(f(X_i) \leq \gamma) + 4(1 + \gamma^{-1} \|w\|_1) R_n(\Psi_{\pm}, D) \\ &\quad + (\gamma^{-1} + 2 + 2\gamma^{-1} \|w\|_1) \sqrt{\frac{\ln((2 + \log_2(1 + \|w\|_1/\gamma))^2/\delta)}{2n}}. \end{aligned}$$

8.5 Sparse Recovery Analysis

We have shown that it is possible to approximate the L_1 regularized solution by sparse solutions using greedy boosting. On the other hand, people frequently use the convex L_1 regularization as an approximation to the nonconvex L_0 regularization. Under appropriate conditions, it can be shown that such a convex relaxation leads to exact or approximation solutions to the nonconvex problem.

We will briefly introduce such conditions. For convenience, we will consider the least squares problem, and use notations that are different from the previous sections, and more consistent with the literature. We write the Lasso optimization problem as follows

$$\hat{\beta} = \arg \min_{\beta} Q(\beta); \quad Q(\beta) = 0.5 \|X\beta - y\|_2^2 + \lambda \|\beta\|_1,$$

where X is an $n \times p$ matrix and y is n dimensional vector. The solution $\hat{\beta}$ is a p dimensional vector, and $\lambda > 0$ is a regularization parameter.

We allow $p > n$, and in the statistics literature, this situation is referred to as *high dimensional*, and for this situation, we are interested in the case that $\hat{\beta}$ is sparse.

We consider the case that y is generated according to the linear model

$$y = X\bar{\beta} + \epsilon,$$

where ϵ is a zero-mean noise vector. Here the target $\bar{\beta}$ is sparse, in that $\text{supp}(\bar{\beta}) \ll n$.

In sparse recovery, theoretical questions we ask for Lasso is whether the solution $\hat{\beta}$ of Lasso with appropriate λ can lead to good estimation of $\bar{\beta}$ in the following sense:

- (Support Recovery) Whether Lasso finds the correct features $\text{supp}(\hat{\beta}) = \text{supp}(\bar{\beta})$? Moreover, we are interested in whether the signs are consistent $\text{sign}(\hat{\beta}_j) = \text{sign}(\bar{\beta}_j)$ when $j \in \text{supp}(\bar{\beta})$ in addition to the requirement of correct feature selection.
- (Parameter Recovery) How good is the parameter estimation, or how small is $\|\hat{\beta} - \bar{\beta}\|$ in some norm?

To answer these questions, we need to analyze Lasso using the KKT condition at the optimal solution, which is more than the empirical process analysis studied earlier.

$$X^\top(X\hat{\beta} - y) + \lambda \nabla \|\hat{\beta}\|_1 = 0.$$

Here $\nabla \|\cdot\|_1$ denotes a subgradient of the 1-norm:

$$[\nabla \|\hat{\beta}\|_1]_j = \begin{cases} -1 & \hat{\beta}_j < 0 \\ 1 & \hat{\beta}_j > 0 \\ \in [-1, 1] & \hat{\beta}_j = 0 \end{cases}.$$

This KKT condition can be summarized as below.

Proposition 8.1 *Let $\hat{F} = \text{supp}(\hat{\beta})$, then $\hat{\beta}$ is a Lasso solution if and only if*

$$X_{\hat{F}}^\top(X_{\hat{F}}\hat{\beta}_{\hat{F}} - y) + \lambda \text{sign}(\hat{\beta})_{\hat{F}} = 0$$

and

$$X_j^\top(X_{\hat{F}}\hat{\beta}_{\hat{F}} - y) \in [-\lambda, \lambda] \quad (j \notin \hat{F}),$$

where X_j is the j -th column of X .

Note that the first condition in the proposition implies that $\hat{\beta}$ is the solution with support restricted to $\hat{F}$:

$$\hat{\beta} = \arg \min_{\beta} Q(\beta) \quad \text{subject to } \text{supp}(\beta) \subset \hat{F}.$$

In general, Lasso solution is not unique. For example, when $j \neq j' \in \hat{F}$ such that $X_j = X_{j'}$, then $\hat{\beta}_j$ and $\hat{\beta}_{j'}$ have the same sign, and any convex combination of $(\hat{\beta}_j + \hat{\beta}_{j'}, 0)$ and $(0, \hat{\beta}_j + \hat{\beta}_{j'})$ for the two variables keep the objective unchanged, and thus are valid solutions. However, under appropriate assumptions, we can obtain the uniqueness of solution, as follows.

Proposition 8.2 *Consider $\hat{F} = \text{supp}(\hat{\beta})$, and if $X_{\hat{F}}^\top X_{\hat{F}}$ is positive definite, and*

$$X_{\hat{F}}^\top(X_{\hat{F}}\hat{\beta}_{\hat{F}} - y) + \lambda \text{sign}(\hat{\beta})_{\hat{F}} = 0$$

and

$$X_j^\top(X_{\hat{F}}\hat{\beta}_{\hat{F}} - y) \in (-\lambda, \lambda) \quad (j \notin \hat{F}),$$

then $\hat{\beta}$ is the unique solution of Lasso.

Proof We only need to show that for any $\Delta\beta \neq 0$, $\hat{\beta} + \Delta\beta$ is not a solution. Let

$$\lambda'_j = |X_j^\top(X_{\hat{F}}\hat{\beta}_{\hat{F}} - y)| < \lambda,$$

and let

$$Q'(\beta) = 0.5\|X\beta - y\|_2^2 + \lambda \sum_{j \in \hat{F}} |\beta_j| + \sum_{j \notin \hat{F}} \lambda'_j |\beta_j|,$$

then since $\nabla Q'(\hat{\beta}) = 0$, we know that $\hat{\beta}$ achieves the minimum of $Q'(\beta)$, and thus $Q'(\hat{\beta} + \Delta\beta) \geq Q'(\hat{\beta}) = Q(\hat{\beta})$. Now, if $\Delta\beta_j \neq 0$ for some $j \notin \hat{F}$, then $Q(\hat{\beta} + \Delta\beta) > Q'(\hat{\beta} + \Delta\beta) \geq Q(\hat{\beta})$, and thus $\hat{\beta} + \Delta\beta$ is not a solution of the original Lasso problem. If $\Delta\beta_j = 0$ for all $j \notin \hat{F}$, then $\text{supp}(\hat{\beta} + \Delta\beta) \subset \hat{F}$. However, The positive-definiteness of $X_{\hat{F}}^\top X_{\hat{F}}$ means that $Q(\beta)$ is strictly convex when the support is restricted to $\hat{F}$, and hence the solution is unique, implying $\Delta\beta = 0$. ■

One important property of Lasso is that under suitable conditions, the method can find the set of nonzero elements $\text{supp}(\hat{\beta})$ that equals $\text{supp}(\bar{\beta})$. This property is referred to as *feature selection consistency* for the support recovery problem. The following result gives an answer to the problem with the requirement of sign consistency.

Proposition 8.3 *Let $\bar{F} = \text{supp}(\bar{\beta})$. Let*

$$\tilde{\beta}_{\bar{F}} = \bar{\beta}_{\bar{F}} + (X_{\bar{F}}^\top X_{\bar{F}})^{-1}(-\lambda \text{sign}(\bar{\beta})_{\bar{F}} + X_{\bar{F}}^\top \epsilon)$$

and $\text{supp}(\tilde{\beta}) \subset \bar{F}$. Assume that $(X_{\bar{F}}^\top X_{\bar{F}})$ is positive definite)

- $\text{sign}(\tilde{\beta}_{\bar{F}}) = \text{sign}(\hat{\beta}_{\bar{F}})$
- $|X_j^\top [-\lambda X_{\bar{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}} + (X_{\hat{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1} X_{\bar{F}}^\top - I)\epsilon]| < \lambda$ for all $j \notin \bar{F}$.

Then $\tilde{\beta}$ is the unique solution of Lasso that is sign consistent.

Proof We only need to verify that $\tilde{\beta}_{\bar{F}}$ satisfies the KKT condition inside $\bar{F}$ as:

$$X_{\hat{F}}^\top (X_{\hat{F}} \tilde{\beta}_{\hat{F}} - y) + \lambda \text{sign}(\tilde{\beta})_{\hat{F}} = 0,$$

and that

$$X_j^\top (X_{\hat{F}} \tilde{\beta}_{\hat{F}} - y) = X_j^\top [-\lambda X_{\bar{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}} + (X_{\hat{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1} X_{\bar{F}}^\top - I)\epsilon].$$

The result follows directly. ■

Theorem 8.8 *Let $\bar{F} = \text{supp}(\bar{\beta})$. Assume that*

$$\mu = \sup_{j \notin \bar{F}} |X_j^\top X_{\bar{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}}| < 1,$$

and assume that we choose a sufficiently large λ so that

$$\lambda > (1 - \mu)^{-1} \sup_{j \notin \bar{F}} |X_j^\top (X_{\bar{F}}^\top X_{\bar{F}})^{-1} X_{\bar{F}}^\top - I) \epsilon|,$$

and if

$$\min_{j \in \bar{F}} |\bar{\beta}_j| > \|(X_{\bar{F}}^\top X_{\bar{F}})^{-1}\|_{\infty \rightarrow \infty} (\lambda + \|X_{\bar{F}}^\top \epsilon\|_\infty),$$

then the solution of Lasso is unique and sign consistent. Here $\|M\|_{\infty \rightarrow \infty} = \sup_u [\|Mu\|_\infty / \|u\|_\infty]$ is the maximum row sum of M .

Proof The proof is just a verification of the previous proposition. Since

$$\|(X_{\bar{F}}^\top X_{\bar{F}})^{-1} (-\lambda \text{sign}(\bar{\beta})_{\bar{F}} + X_{\bar{F}}^\top \epsilon)\|_\infty \leq \|(X_{\bar{F}}^\top X_{\bar{F}})^{-1}\|_{\infty \rightarrow \infty} (\lambda + \|X_{\bar{F}}^\top \epsilon\|_\infty),$$

we know that $\text{sign}(\tilde{\beta}_j) = \text{sign}(\bar{\beta}_j)$ for all $j \in \bar{F}$; this verifies the first condition. The second condition can be similarly verified. ■

If $\mu < 1$, then in order for the theorem to apply, we only need to choose a small λ when ϵ is small for the first condition to hold. When λ is small enough, then the second condition will also hold. Therefore the result means that given any $\bar{\beta}$, if $\mu < 1$, then as long as the noise ϵ is sufficiently small, then we can obtain sign consistency using Lasso. The requirement of $\mu < 1$ is called *irrepresentable condition* [102], which is sufficient for the feature selection consistency of Lasso. The slightly weaker condition $\mu \leq 1$ is necessary for the feature selection consistency of Lasso with sufficiently small ϵ .

Proposition 8.4 Let $\bar{F} = \text{supp}(\bar{\beta})$. Assume that

$$\mu = \sup_{j \notin \bar{F}} |X_j^\top (X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}}| > 1,$$

then for sufficiently small ϵ ,

$$|X_j^\top [-\lambda X_{\bar{F}} (X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}} + X_{\bar{F}} (X_{\bar{F}}^\top X_{\bar{F}})^{-1} X_{\bar{F}}^\top \epsilon]| > \lambda$$

for some j , and thus Lasso is not sign consistent when ϵ is small enough.

If the noise ϵ is non-stochastic, Theorem 8.8 requires it to be small. For stochastic zero-mean noise, the condition can be satisfied when n is large, as can be seen in the Theorem 8.8 below.

Theorem 8.9 Let $\bar{F} = \text{supp}(\bar{\beta})$. Assume that the columns are normalized so that $\sup_j \|X_j\|_2 \leq B$, and assume that ϵ are n independent zero-mean sub-Gaussian noise: $\mathbf{E} e^{\lambda \epsilon_i} \leq \lambda^2 \sigma^2 / 2$. Assume that

$$\mu = \sup_{j \notin \bar{F}} |X_j^\top X_{\bar{F}} (X_{\bar{F}}^\top X_{\bar{F}})^{-1} \text{sign}(\bar{\beta})_{\bar{F}}| < 1,$$

then given $\delta \in (0, 1)$, if we choose a sufficiently large λ so that

$$\lambda_o > (1 - \mu)^{-1}; \quad \lambda = \lambda_o \sigma B \sqrt{2 \ln(2p/\delta)/n},$$

and if

$$\min_{j \in \bar{F}} |\bar{\beta}_j| > \|(X_{\bar{F}}^\top X_{\bar{F}})^{-1}\|_{\infty \rightarrow \infty} (\lambda_o + 1) \sigma B \sqrt{2 \ln(2p/\delta)/n}$$

then the solution of Lasso is unique and sign consistent with probability at least $1 - \delta$.

Proof Note that sub-Gaussian probability inequality implies that for each fixed X_j , we have: if $j \in \bar{F}$, then

$$\Pr[|X_j^\top \epsilon| \geq \epsilon_0] \leq 2e^{-n\epsilon_0^2/2B^2\sigma^2},$$

and if $j \notin \bar{F}$,

$$\Pr[|X_j^\top (P_{\bar{F}} - I)\epsilon| \geq \epsilon_0] \leq 2e^{-n\epsilon_0^2/2B^2\sigma^2},$$

with $P_{\bar{F}} = X_{\bar{F}}(X_{\bar{F}}^\top X_{\bar{F}})^{-1}X_{\bar{F}}^\top$ as projection matrix to subspace spanned by X_j ($j \in \bar{F}$). Therefore taking union bound for $j = 1, \dots, p$. We know that With probability $1 - \delta$, the following bound holds:

$$\|X_j^\top \epsilon\|_\infty \leq \sigma B \sqrt{2 \ln(2p/\delta)/n} \quad (j \in \bar{F})$$

and

$$\|X_j^\top (P_{\bar{F}} - I)\epsilon\|_\infty \leq \sigma B \sqrt{2 \ln(2p/\delta)/n} \quad (j \notin \bar{F}).$$

We can then apply the previous theorem. ■

It is also possible to study the support recovery question without sign consistency, which leads to a slightly weaker condition. For parameter recovery, we often employ a condition that is weaker than the irrepresentable condition, called *restricted isometry condition* (RIP) [23]. This condition has often been employed to derive parameter estimation bounds for Lasso. There are variations such as *restricted eigenvalue condition* [18] that are more or less similar to RIP. RIP requires the smallest eigenvalue of the matrix $X_F^\top X_F$ to be bounded away from zero for all subsets F such that $|F| \leq \text{const} * \|\bar{\beta}\|_0$. Under such a condition, the main theoretical consequences are as follows.

- For random matrices, RIP is satisfied with $n = O(\|\bar{\beta}\|_0 \ln p)$ examples.
- Under RIP, Lasso may not select features consistently (KKT condition may be violated).
- Under RIP, one can show that $\|\hat{\beta} - \bar{\beta}\|_2^2 = O(\lambda^2 \|\beta\|_0)$ with probability $1 - \delta$, if we choose $\lambda = \Omega(\sigma \sqrt{\ln(p/\delta)/n})$.

Note that similar sparse recovery results can also be obtained using greedy algorithms, both for support recovery and for parameter recovery. For example, see [86, 99, 100].

8.6 Exercises

1. Consider the uniform convergence bound for sparse learning in Theorem 8.1. Let $\{1, \dots, m\} = J_1 \cup \dots \cup J_g$ be a non-overlapping partition of $\{1, \dots, m\}$ into g groups of size m/g (we assume m/g is an integer). Let H contain the subsets of $\{1, \dots, m\}$ that can be expressed as the union of one or more groups $\{J_\ell\}$. That is, if $J \in H$, then for all $\ell = 1, \dots, g$: either $J_\ell \subset J$ or $J_\ell \cap J = \emptyset$.

In group sparsity, we consider only the sparse models indexed by $S \in H$.

- Derive a uniform convergence bound for group sparsity, where we consider a uniform bound that holds for all w and S with $\text{supp}(w) \subset S$.
 - Compare this bound to the sparse learning bound of Theorem 8.1, and derive an estimation method in (8.2) with $r(S)$ obtained using this bound. Explain when does group sparsity perform better.
2. Consider Theorem 8.4. Use the same techniques in the proofs of Corollary 8.2 and Corollary 8.3 for the following.
 - Derive a data dependent bound that holds uniformly for all $A > 0$.
 - Use the above bound to derive an oracle inequality for the soft regularization version of the Gibbs algorithm 8.7.

Chapter 9

Analysis of Neural Networks

This chapter presents the theoretical analysis for neural networks. We will discuss approximation bounds and generalization bounds, and compare kernel analysis to that of L_1 based analysis.

9.1 Introduction to Neural Networks

We start with two-layer neural networks with real-valued output, and d -dimensional input vector $x \in \mathbb{R}^d$. It has the following form, which is the same form as that of the additive model in (8.1)

$$f_m(w, x) = \sum_{j=1}^m u_j h(\theta_j^\top x + b_j), \quad (9.1)$$

where $x \in \mathbb{R}^d$, $\theta_j \in \mathbb{R}^d$, $b_j \in \mathbb{R}$, $u_j \in \mathbb{R}$, and $w = \{[u_j, \theta_j, b_j] : j = 1, \dots, m\}$. The function $h(\cdot)$ is referred to as activation function, and popular choices rectified linear unit (ReLU) $h(z) = \max(0, z)$ and sigmoid $h(z) = e^z / (1 + e^z)$.

In practice, the model parameters w are often trained using SGD. Since it has the same form of additive-model, generalization analysis discussed in Chapter 8 can be applied.

More generally, we may consider a K -layer deep neural networks as follows, with $m^{(0)} = d$ and $m^{(K)} = 1$:

$$\begin{aligned} x_j^{(0)} &= x_j \quad (j = 1, \dots, m^{(0)}), \\ x_j^{(k)} &= h \left(\sum_{j'=1}^{m^{(k-1)}} \theta_{j,j'}^{(k)} x_{j'}^{(k-1)} + b_j \right) \quad (j = 1, \dots, m^{(k)}), \quad k = 1, 2, \dots, K \\ f(x) &= x_1^{(K)}. \end{aligned}$$

9.2 Approximation Results

For two-layer neural networks, it is known that (9.1) is universal if $h(\cdot)$ is not a polynomial [55].

Theorem 9.1 ([55]) *If h is a non-polynomial continuous function, then the function class in (9.1) is dense in $C^0(K)$ for all compact subsets K of $\mathbb{R}^d$, where $C^0(K)$ denotes the set of continuous functions on K .*

A more refined result was obtained in [13], which considered functions with a certain smoothness property in Fourier representation.

Definition 9.1 *Consider a real valued function $f(x) : \mathbb{R}^d \rightarrow \mathbb{R}$. Assume that $f \in L_1(\mathbb{R}^d)$ has the following Fourier representation:*

$$f(x) = \int_{\mathbb{R}^d} e^{i\omega^\top x} \tilde{f}(\omega) d\omega,$$

where $\tilde{f}(\omega)$ is the Fourier transform of $f(x)$ that may be a complex function. Define

$$C(f) = \int_{\mathbb{R}^d} \|\omega\|_2 |\tilde{f}(\omega)| d\omega.$$

Theorem 9.2 ([13]) *If $h(z)$ is a bounded measurable function on the real line for which $\lim_{z \rightarrow -\infty} h(z) = 0$ and $\lim_{z \rightarrow \infty} h(z) = 1$. For every function f with $C(f) < \infty$, there exists a neural network (9.1) such that*

$$\int (f(x) - f(0) - f_m(w, x))^2 d\mu(x) \leq \frac{(2rC(f))^2}{m},$$

where μ is an arbitrary probability measure on $B_r = \{x \in \mathbb{R}^d : \|x\|_2 \leq r\}$.

Note that $f(0) = (f(0)/h(b))h(0^\top x + b)$ as long as $h(b) \neq 0$, and thus it can be absorbed into (9.1). It was shown in [13] that $f(x) - f(0)$ belongs to the convex closure of functions $\{uh(\theta^\top x + b) : |u| \leq 2rC(f)\}$. This result, together with the probabilistic argument of Theorem 8.6, implies Theorem 9.2.

Theorem 9.2 can also be applied to the ReLu activation function $h(z) = \max(0, z)$ via an intermediate function $h'(z) = h(z+1) - h(z)$, which satisfies the conditions of Theorem 9.2.

The universal approximation theory of [13] shows that a function can be efficiently represented by a two-layer neural network if $C(f)$ is small. However, for certain target functions with large $C(f)$, two layer neural networks may require exponentially many nodes to represent. In some of such cases, using deep representation can reduce the number of nodes needed to represent such functions. This phenomenon has been extensively investigated in

the literature. A representative result along this line of research is given in [83], which we state below. Note that the original results in [83] (Theorem 1.1 and Theorem 3.12) are more general. But we state a simplified result to avoid introducing the concept of semi-algebra. It demonstrates the benefit of using deep neural networks with multiple layers, which has been done in practice.

Theorem 9.3 ([83]) *Consider any integer $k \geq 1$ and any dimension $d \geq 1$. There exists $f(x) : \mathbb{R}^d \rightarrow \mathbb{R}$ computed by a neural network with standard ReLU gates in $2k^3 + 9$ layers, $3k^3 + 12$ total nodes, and $4 + d$ distinct parameters so that*

$$\inf_{g \in C} \int_{[0,1]^d} |f(x) - g(x)| dx \geq \frac{1}{64},$$

where C contains functions computed by ReLU networks of $\leq (k+1)$ layers with $\leq 2^k$ nodes.

In the following, we briefly explain the intuition of [83], which indicates what kind of functions are difficult to approximate with shallow neural networks. Consider the case of $d = 1$, a specific construction of a hard function $f(x)$ for shallow neural networks is via the function composition of the triangle function $f_0(x) = \max(0, \min(2x, 2(1-x))) = 2 \max(0, x) - 4 \max(0, x - 0.5)$ on $[0, 1]$. We may define $f_k(x) = f_0(f_{k-1}(x))$ with $k \geq 1$. Since $f_0(x)$ can be represented by a two-layer neural network with no more than two nodes per layer, $f_k(x)$ can be represented by a $2k$ -layer neural network with no more than two nodes per layer. However, it can be shown that $f_k(x)$ cannot be represented by a shallow neural network.

It can be seen that $f_k(x)$ contains 2^k points that reach values of 1, and $2^k + 1$ points that reach values of 0. It follows that the number of solution segments of $f_k(x) = 0.5$, referred to as its crossing number, is 2^{k+1} .

It is not difficult to show that $f_k(x)$ cannot be approximated well by a function $f'(x)$ with crossing number $\leq 2^{k-1}$. Therefore in order to show that $f_k(x)$ cannot be approximated well by shallow neural networks, we only need to show that the function of a shallow neural network cannot have many crossings unless it contains exponentially many nodes. This property was shown in [83]. Consider a neural network with ℓ -layers, m ReLU nodes per layer, then the resulting function has a crossing number of no larger than $2(2m/\ell)^\ell$. If the neural network can approximate $f_k(x)$ well, then $(2m) \geq 2^{(k^2-2)/\ell}$. Therefore the node number m is at least 2^{k-2} if $\ell \leq k$.

At a high level, we note that function composition allows a deep neural network to reuse learned patterns (functions with high crossing numbers) to form more complex patterns. This corresponds to the intuition that neural networks learn basic features, and then combination of basic features as high level features. In general, deep combinations of high level features cannot be easily learned using shallow networks.

9.3 Random Feature Method

If in two-layer neural networks, we do not train the parameters θ , but randomly draw these parameters from a distribution, then the resulting method becomes the random feature method, also referred to as *random kitchen sinks* in the literature [67, 66].

In the random feature approach, we may consider the continuous representation of (9.1) when $m \rightarrow \infty$ as a linear model, where the model parameter is denoted by $w = u$:

$$f(w, x) = w \cdot \psi(x) = \int w(\theta)h(\theta^\top x)p_0(\theta)d\theta = \mathbf{E}_{\theta \sim p_0} w(\theta)h(\theta^\top x), \quad (9.2)$$

where $p_0(\theta)$ is a probability density of $\theta \in \mathbb{R}^d$ with nonzero density everywhere. Usually we choose $p_0(\theta)$ as a Gaussian distribution.

Note that for notation simplicity, unlike in (9.1), we do not include a bias term b in (9.2). This does not affect generality because we may always change $x \in \mathbb{R}^d$ into $\tilde{x} = [x, 1] \in \mathbb{R}^{d+1}$, and change $\theta \in \mathbb{R}^d$ and $b \in \mathbb{R}$ into $\tilde{\theta} = [\theta, b] \in \mathbb{R}^{d+1}$, so that we have $\theta^\top x + b = \tilde{\theta}^\top \tilde{x}$. It follows from Theorem 9.2 that (9.2) is universal.

Here we may treat $w(\theta)$ as an infinite dimensional vector indexed by θ , and $\psi(x)$ as an infinite dimensional vector $h(\theta^\top x)$ indexed by θ .

The kernel of this function class is

$$k(x, x') = \mathbf{E}_{\theta \sim p_0} h(\theta^\top x)h(\theta^\top x').$$

We obtain the following result from Theorem 7.4.

Corollary 9.1 *Let*

$$\mathcal{F}_A^2 = \{\mathbf{E}_{\theta \sim p_0} w(\theta)h(\theta^\top x) : \mathbf{E}_{\theta \sim p_0} w(\theta)^2 \leq A^2\},$$

then

$$R_n(\mathcal{F}_A^2, D) \leq A \sqrt{\frac{\mathbf{E}_{x \sim D} \mathbf{E}_{\theta \sim p_0} h(\theta^\top x)^2}{n}}.$$

In practical implementations of the random features, we draw m samples $\theta_j \sim p_0$ for $j = 1, \dots, m$ to approximate the expectation over p_0 . This leads to a learning model with a finite number of random features:

$$f_m(u, x) = \sum_{j=1}^m u_j h(\theta_j^\top x) = u^\top \psi_m(x), \quad (9.3)$$

where $u_j = w(\theta_j)/m$ and $\psi_m(x) = [h(\theta_1^\top x), \dots, h(\theta_m^\top x)]$. As $m \rightarrow \infty$, we have $f_m(x) \rightarrow f(w, x)$. Note that due to the difference in scaling, we explicitly write the following result.

Corollary 9.2 *Let*

$$\mathcal{F}_{A,m}^2 = \left\{ \sum_{j=1}^m u_j h(\theta_j^\top x) : \|u\|_2^2 \leq A^2/m \right\},$$

then

$$R_n(\mathcal{F}_{A,m}^2, D) \leq A \sqrt{\frac{\mathbf{E}_{x \sim D} m^{-1} \sum_{j=1}^m h(\theta_j^\top x)^2}{n}}.$$

This means for the random feature approach, $\|u\|_2 = O(1/\sqrt{m})$ needs to be rather small in order to have a good generalization performance.

We may compare the result of kernel method to that of L_1 regularization, which requires an estimate of the Rademacher complexity of $h(\theta^\top x)$.

Proposition 9.1 *Let $\mathcal{F} = \{h(\theta^\top x) : \theta \in \mathbb{R}^d\}$ and $\mathcal{F}_\pm = \mathcal{F} \cup -\mathcal{F}$. There exists a universal constant $C > 0$ such that for all monotone function $h(\cdot)$ with $h(\cdot) \in [-M, M]$:*

$$R(\mathcal{F}_\pm, S_n) \leq CM \sqrt{\frac{d+1}{n}}$$

for all S_n .

Proof Since $h(\cdot)$ is monotone, we know that $\mathcal{F}$ is a VC-subgraph class with VC dimension $d+1$. Therefore from Theorem 5.3, we obtain the following covering number estimate

$$\ln N(\epsilon, \mathcal{F}, L_2(S_n)) \leq [\ln 2 + (d+1) \ln(4e(2M/\epsilon)^2)].$$

We can thus obtain an estimate of Rademacher complexity with chaining:

$$R(\mathcal{F}, S_n) \leq 12 \int_0^{2M} \sqrt{\frac{\ln N(\epsilon/2, \mathcal{F}, L_2(S_n))}{n}} d\epsilon.$$

This leads to the bound. ■

We have the following L_1 regularization result for the continuous random feature method from Theorem 8.2.

Corollary 9.3 *Let*

$$\mathcal{F}_A^1 = \{\mathbf{E}_{\theta \sim p_0} w(\theta) h(\theta^\top x) : \mathbf{E}_{\theta \sim p_0} |w(\theta)| \leq A\},$$

and Assume that $h(\cdot) \in [-M, M]$ and $h(\cdot)$ is monotone. Then

$$R(\mathcal{F}_A^1, S_n) \leq C \cdot AM \sqrt{\frac{d+1}{n}},$$

where C is the absolute constant in Proposition 9.1.

We note that the L_1 and L_2 regularized continuous random feature methods satisfy $\mathcal{F}_A^2 \subset \mathcal{F}_A^1$, and $\mathcal{F}_A^1$ can be significantly larger. In fact, given a fixed A , there exists $f \in \mathcal{F}_A^1$ so that $\mathbf{E}_{\theta \sim p_0} |w(\theta)| \leq A$ but $\mathbf{E}_{\theta \sim p_0} w(\theta)^2$ is arbitrarily large. This can be seen even more clearly with the finite random feature method (9.3), for which we have the following result.

Corollary 9.4 *Let*

$$\mathcal{F}_{A,m}^1 = \left\{ \sum_{j=1}^m u_j h(\theta_j^\top x) : \|u\|_1 \leq A \right\}.$$

Assume that $h(\cdot) \in [-M, M]$ and $h(\cdot)$ is monotone. Then

$$R(\mathcal{F}_{A,m}^1, S_n) \leq C \cdot AM \sqrt{\frac{d+1}{n}},$$

where C is the absolute constant in Proposition 9.1.

For the random kernel method with L_2 regularization, we require $\|u\|_2 \leq A/\sqrt{m}$, which implies that $\|u\|_1 \leq A$. However, the reverse is not true. For example, if $u = Ae_j$ (vector of zeros except the j -th component being 1), then $\|u\|_2 = A/\sqrt{m}$, which is significantly larger than $A/\sqrt{m}$. Note that for the finite m case, we may also replace the linear dependency on d by $\ln m$ using the standard L_1 Rademacher complexity bound.

In general, when m is large, for sparse target that can be represented by $f_m(u, x)$ with a sparse u of small L_1 norm $\|u\|_1$, the random kernel method with L_2 regularization can be inferior at least from the generalization analysis point of view.

We may also compare the result of random feature kernel to entropy regularization in the continuous limit as follows.

Corollary 9.5 *Assume $h(\cdot) \in [-M, M]$, and let*

$$\mathcal{F}_{A,B}^{\text{entro}} = \left\{ \mathbf{E}_{\theta \sim p_0} w(\theta) h(\theta^\top x) : \mathbf{E}_{\theta \sim p_0} |w(\theta)| \leq A, \quad \text{entro}(w) \leq B^2 \right\},$$

where

$$\text{entro}(w) = \mathbf{E}_{\theta \sim p_0} \frac{|w(\theta)|}{A} \ln \frac{|w(\theta)|}{A} + \ln 3.$$

Then

$$R(\mathcal{F}_{A,B}^{\text{entro}}, S_n) \leq \sqrt{\frac{2}{n}} ABM.$$

Proof We consider the following prior on the function family

$$\{Ah(\theta^\top x), -Ah(\theta^\top x), 0\},$$

where $\tilde{p}(f) = p(\theta)/3$ on $f = Ah(\theta^\top x)$, $\tilde{p}(f) = p(\theta)/3$ on $f = -Ah(\theta^\top x)$, and $\tilde{p}(f) = 1/3$ on $f = 0$.

Similarly, for each $f \in \mathcal{F}_{A,B}^{\text{entro}}$, we let $\tilde{q}(f) = w(\theta)/A$ on $f = Ah(\theta^\top x)$ for $w(\theta) > 0$, and $\tilde{q}(f) = -w(\theta)/A$ on $f = -Ah(\theta^\top x)$ for $w(\theta) \leq 0$, and $\tilde{q}(f) = 1 - \mathbf{E}_{\theta \sim p_0} |w(\theta)|/A$ on $f = 0$. It can be checked that

$$\begin{aligned} & \text{KL}(\tilde{q}||\tilde{p}) \\ &= \mathbf{E}_{\theta \sim p_0} \frac{|w|}{A} \ln \frac{3|w|}{A} + (1 - \mathbf{E}_{\theta \sim p_0} |w(\theta)|/A) \ln[3(1 - \mathbf{E}_{\theta \sim p_0} |w(\theta)|/A)] \\ &\leq \text{entro}(w) \leq B^2. \end{aligned}$$

Moreover,

$$\mathbf{E}_{\theta \sim p_0} w(\theta) h(\theta^\top x) = \mathbf{E}_{f \sim \tilde{q}} f(x).$$

The result follows from Theorem 8.3. ■

For the finite formulation (9.3), we obtain

Corollary 9.6 *Assume $h(\cdot) \in [-M, M]$, and let*

$$\mathcal{F}_{A,B,m}^{\text{entro}} = \left\{ \sum_{j=1}^m u_j h(\theta_j^\top x) : \|u\|_1 \leq A, \quad \text{entro}(u) \leq B^2 \right\},$$

where

$$\text{entro}(u) = \sum_{j=1}^m \frac{|u_j|}{A} \ln \frac{|u_j m|}{A} + \ln 3.$$

Then

$$R(\mathcal{F}_{A,B,m}^{\text{entro}}, S_n) \leq \sqrt{\frac{2}{n}} ABM.$$

We note that $\text{entro}(u) \leq \ln(3m)$, and thus we may take $B = \sqrt{\ln(3m)}$ in the worse case. This leads to

$$R(\mathcal{F}_{A,B,m}^{\text{entro}}, S_n) \leq \sqrt{\frac{2 \ln(3m)}{n}} AM.$$

The bound is logarithmic in m , consistent with L_1 regularization with finite function class. This bound is superior to the kernel regularization of the random feature method, which has a much worse dependency in m .

In particular, for a sparse target, which can be well approximated by L_1 or entropy regularization, both methods work well. However, the kernel view may perform poorly because the corresponding L_2 regularization of the target can grow linearly in m . This is due to the fact that L_1 and entropy regularization have the ability to select the most meaningful features from the random features, while L_2 regularization does not have this ability.

9.4 Neural Tangent Kernel

In the random feature approach, the bottom layer model parameter θ is fixed, and only top layer model parameter u is trained. However, in practice, both model parameter θ and parameter u are trained. It is possible to generalize the kernel view to handle this case, which leads to the concept of neural tangent kernel (NTK) [45].

Similar to (9.2), we first consider the continuous formulation of NTK, by introducing a scaling factor $\alpha > 0$:

$$f_{\text{nn}}(x) = \alpha \mathbf{E}_{\tilde{w} \sim p_0} u(\tilde{w}) h(\theta(\tilde{w})^\top x), \quad (9.4)$$

where the random distribution p_0 is placed on some random variable $\tilde{w}$. Usually we take p_0 as a normal distribution, and take $[u_0(\tilde{w}), \theta_0(\tilde{w})] \in \mathbb{R}^{d+1}$, which may be regarded as the initial distribution of $[u, \theta]$. Usually we simply let $\tilde{w} = [u_0, \theta_0]$, so that it is directly drawn from the random initialization.

We may now consider an expansion of $[u, \theta]$ around $[u_0, \theta_0]$ as

$$u(\tilde{w}) = u_0(\tilde{w}) + \Delta u(\tilde{w}), \quad \theta(\tilde{w}) = \theta_0(\tilde{w}) + \Delta \theta(\tilde{w}),$$

so that $\Delta u(\tilde{w})$ and $\Delta \theta(\tilde{w})$ are both small. For simplicity, we assume that $h(\cdot)$ is continuously differentiable with derivative $h'(\cdot)$. We can then expand (9.4) by the following linear approximation, which we may call the NTK approximation:

$$\begin{aligned} f_{\text{NTK}}([\Delta u, \Delta \theta], x) &= \alpha \mathbf{E}_{\tilde{w} \sim p_0} u_0(\tilde{w}) h(\theta_0(\tilde{w})^\top x) \\ &\quad + \alpha \mathbf{E}_{\tilde{w} \sim p_0} \Delta u(\tilde{w}) h(\theta_0(\tilde{w})^\top x) \\ &\quad + \alpha \mathbf{E}_{\tilde{w} \sim p_0} u_0(\tilde{w}) h'(\theta_0(\tilde{w})) (\Delta \theta(\tilde{w})^\top x). \end{aligned} \quad (9.5)$$

The scaling α is explicitly introduced so that as $\alpha \rightarrow \infty$, $\Delta u = O(1/\alpha)$, and $\Delta \theta = O(1/\alpha)$. In such regime, called *NTK regime*, the two-layer neural network (9.4) can be approximated by NTK $f_{\text{NTK}}(x)$ as $\alpha \rightarrow \infty$:

$$f_{\text{NTK}}([\Delta u, \Delta \theta], x) \rightarrow f_{\text{nn}}(x) \quad \text{as } \alpha \rightarrow \infty.$$

It can be shown that under appropriate conditions, when $m \rightarrow \infty$, the entire neural network training process stays in the NTK regime, and thus the NTK approximation is valid for such case.

Observe now that in the NTK formulation (9.5), $f_{\text{NTK}}(x)$ is linear with respect to $[\Delta u, \Delta \theta]$, and thus we can apply the linear model analysis to NTK. We may consider the following L_2 regularization

$$Q_{\text{NTK}}(\Delta u, \Delta \theta) := \mathbf{E}_{\tilde{w} \sim p_0} [\Delta u(\tilde{w})^2 + \nu \|\Delta \theta(\tilde{w})\|_2^2] \leq A^2/\alpha^2,$$

with kernel

$$\begin{aligned} k_{\text{NTK}}(x, x') &:= \mathbf{E}_{\tilde{w} \sim p_0} h(\theta_0(\tilde{w})^\top x) h(\theta_0(\tilde{w})^\top x') \\ &\quad + \nu^{-1} \mathbf{E}_{\tilde{w} \sim p_0} [u_0(\tilde{w})^2 h'(\theta_0(\tilde{w}))^2] x^\top x'. \end{aligned}$$

We obtain the following result from Theorem 7.4.

Corollary 9.7 *Let*

$$\mathcal{F}_A^{\text{NTK}} = \{f_{\text{NTK}}([\Delta u, \Delta \theta], x) : Q_{\text{NTK}}(\Delta u, \Delta \theta) \leq A^2/\alpha^2\},$$

then

$$R_n(\mathcal{F}_A^{\text{NTK}}, D) \leq A \sqrt{\frac{\mathbf{E}_{x \sim D} k_{\text{NTK}}(x, x)}{n}}.$$

We may also consider the following finite approximation of (9.5) using samples $\tilde{W}_n = \{\tilde{w}_1, \dots, \tilde{w}_m\}$, independently drawn from $p_0(\tilde{w})$:

$$\begin{aligned} f_{\text{NTK}}([\Delta u, \Delta \theta], x) &\approx \frac{\alpha}{m} \sum_{j=1}^m u_0(\tilde{w}_j) h(\theta_0(\tilde{w}_j)^\top x) \\ &\quad + \frac{\alpha}{m} \sum_{j=1}^m \Delta u(\tilde{w}_j) h(\theta_0(\tilde{w}_j)^\top x) \\ &\quad + \frac{\alpha}{m} \sum_{j=1}^m u_0(\tilde{w}_j) h'(\theta_0(\tilde{w}_j)) (\Delta \theta(\tilde{w}_j)^\top x). \end{aligned}$$

This can be simplified as follows:

$$f_{\text{NTK},m}([\Delta u, \Delta \theta], x) = f_0(x) + \sum_{j=1}^m \left[\Delta u_j h(\theta_{0,j}^\top x) + \beta_{0,j} (\Delta \theta_j^\top x) \right]. \quad (9.6)$$

In the discrete case, the scaling factor α should be chosen such as $\alpha = O(\sqrt{m})$. Assume that $\mathbf{E}_{\tilde{W}_n} f_0(x) = 0$, then this is to ensure that for randomly chosen $\tilde{W}_n$, we have

$$\mathbf{E}_{\tilde{W}_n} f_0(x)^2 = \frac{\alpha^2}{m} \mathbf{E}_{\tilde{w}_1} u_0(\tilde{w}_1)^2 h(\theta_0(\tilde{w}_1)^\top x)^2 = O(1).$$

We note that in this case, $\alpha \rightarrow \infty$ as $m \rightarrow \infty$, and thus the assumption of NTK holds in the continuous limit. The corresponding regularizer for the finite NTK formulation (9.6) is:

$$Q_{\text{NTK},m}(\Delta u, \Delta \theta) := \sum_{j=1}^m [\Delta u_j^2 + \nu \|\Delta \theta_j\|_2^2] \leq A^2/m,$$

with kernel

$$k_{\text{NTK},m}(x, x') := \frac{1}{m} \sum_{j=1}^m h(\theta_{0,j}^\top x) h(\theta_{0,j}^\top x') + \frac{1}{\nu m} \sum_{j=1}^m \beta_{0,j}^2 x^\top x'.$$

We have the following Rademacher complexity estimate.

Corollary 9.8 *Let*

$$\mathcal{F}_{A,m}^{\text{NTK}} = \{f_{\text{NTK},m}([\Delta u, \Delta \theta], x) : Q_{\text{NTK},m}(\Delta u, \Delta \theta) \leq A^2/m\},$$

then

$$R_n(\mathcal{F}_{A,m}^{\text{NTK}}, D) \leq A \sqrt{\frac{\mathbf{E}_{x \sim D} k_{\text{NTK},m}(x, x)}{n}}.$$

Similar to the random feature case, it is possible to derive L_1 regularization based results for the NTK formulation. It has advantage over the NTK formulation for sparse target functions because such functions can be directly approximated using L_1 regularization, but not the NTK regularization. This is due to the fact that L_1 regularization has the ability to select the most meaningful features, while the NTK method employs random features without the ability to select good features.

We will not repeat the L_1 regularization analysis here, because we may directly work with L_1 regularization for the original continuous formulation (9.4), instead of the NTK approximation (9.5).

9.5 Mean-Field Formulation

We consider the generalization analysis of (9.4). Unlike the NTK approximation, which requires $\alpha \rightarrow \infty$, we consider the case of $\alpha = 1$ for simplicity. Since $[u(\tilde{w}), \theta(\tilde{w})]$ can be regarded as a random variable with $\tilde{w} \sim p_0(\cdot)$, we may consider $[u, \theta] \sim q([u, \theta])$, which leads to the following reformulation of (9.4):

$$f_{\text{mf}}(q, x) = \mathbf{E}_{[u, \theta] \sim q} u h(\theta^\top x), \quad (9.7)$$

where the parameter of the function is the distribution q on $[u, \theta] \in \mathbb{R}^{d+1}$. This continuous formulation is some time referred to as the *mean field* formulation [60].

Because the parameter q of the mean-field model is a distribution, the two-layer NN in this continuous formulation can be regarded as a convex hull model combination method over the individual models $u h(\theta^\top x)$. We can thus apply the L_1 -regularization result and the entropy-regularization result from Chapter 8 to compute the corresponding Rademacher complexity.

The following result is a straight-forward consequence of Proposition 9.1 and Theorem 8.2.

Corollary 9.9 *Let*

$$\mathcal{F}_A^{\text{mf}} = \{\mathbf{E}_{[u, \theta] \sim q} u h(\theta^\top x) : \mathbf{E}_{u \sim q} |u| \leq A\}.$$

Assume that $h(\cdot) \in [-M, M]$ and $h(\cdot)$ is monotone. Then

$$R(\mathcal{F}_A^{\text{mf}}, S_n) \leq C \cdot A M \sqrt{\frac{d+1}{n}},$$

where C is the absolute constant in Proposition 9.1.

It can be shown that in the continuous limit, neural network training converges to the optimal solution of the mean field formulation (9.7), under suitable conditions [60, 25]. Therefore it makes sense to study properties of the global solution of (9.7), as the limiting solution of the ERM method for two-layer neural networks.

One interesting observation is that for the mean-field formulation (9.7), convex regularization in u is equivalent to L_1 regularization.

Proposition 9.2 *Consider the following optimization problem:*

$$\min_q \left[\mathbf{E}_{(X,Y) \sim S_n} L(\mathbf{E}_{[u,\theta] \sim q} uh(\theta^\top X), Y) + \mathbf{E}_{u \sim q} r(|u|) \right],$$

where $r(\cdot)$ is a strictly convex function of u . Let q be the optimal solution, there exists $A > 0$ such that $|u| = A$ a.e. $u \sim q(\cdot)$, and q is a solution with respect to

$$\min_q \mathbf{E}_{(X,Y) \sim S_n} L(\mathbf{E}_{[u,\theta] \sim q} uh(\theta^\top X), Y) \quad \mathbf{E}_{u \sim q} |u| \leq A.$$

Proof For notation simplicity, we consider discrete q (the continuous case is analogous except for more complex notation). Consider discrete values of $[u_1, \theta_1]$ with probability $q_1 > 0$ and $[u_2, \theta_2]$ with probability $q_2 > 0$. Without loss of generality, we may assume that $|u_1| \geq |u_2| \geq 0$.

We prove the first statement by contradiction. If $|u_1| > |u_2|$, then we can find $\delta > 0$, and let $q'_1 = q_1 + \delta$, $q'_2 = q_2 - \delta$, so that $|u'_1| = |u'_2|$ with $u'_1 = u_1 q_1 / q'_1$ and $u'_2 = u_2 q_2 / q'_2$. Let q' be the distribution on $[u, \theta]$ which equals q on all other points, except for a change of probability mass q_j on $[u_j, \theta_j]$ to a probability mass q'_j on $[u'_j, \theta'_j]$ ($j = 1, 2$). The construction implies that $\mathbf{E}_{[u,\theta] \sim q'} uh(\theta^\top x) = \mathbf{E}_{[u,\theta] \sim q} uh(\theta^\top x)$ and $\mathbf{E}_{u \sim q'} r(|u|) < \mathbf{E}_{u \sim q} r(|u|)$. This is a contradiction to the optimality of p . Therefore we must have $|u_1| = |u_2|$. This implies that $|u| = \text{const}$ when $u \sim q(\cdot)$.

We prove the second statement by contradiction. If q is not a solution of the L_1 regularization formulation, then we can find q' with smaller objective and $\mathbf{E}_{u \sim q'} |u| \leq A$. From the previous analysis, we can also find such a q' such that $|u| = A'$ for some $A' \leq A$ with smaller objective. However, this is not possible because q' would lead to a smaller regularized loss than q with respect to the $r(\cdot)$ regularization. ■

This result can be interpreted as follows. Even if we use L_2 regularization on u in a two-layer neural network, which seemingly is related to the kernel method, it actually solves an L_1 regularization problem for the mean-field formulation, when we allow both u and θ to be optimized simultaneously. The optimization of θ modifies the L_2 regularization on u into the L_1 regularization with respect to the function class $\{h(\theta^\top x)\}$.

The discrete formulation of (9.7) is given by (9.1), with $b_j = 0$. Here $\{[u_j, \theta_j]\}$ may be regarded as m independent samples from q . For such a formulation, both $[u_j, \theta_j]$ are trained simultaneously. The following result holds true.

Corollary 9.10 *Let*

$$\mathcal{F}_{A,m}^{\text{mf}} = \left\{ \sum_{j=1}^m u_j h(\theta_j^\top x) : \|u\|_1 \leq A \right\}.$$

Assume that $h(\cdot) \in [-M, M]$ and $h(\cdot)$ is monotone. Then

$$R(\mathcal{F}_{A,m}^{\text{mf}}, S_n) \leq C \cdot AM \sqrt{\frac{d+1}{n}},$$

where C is the absolute constant in Proposition 9.1.

Note that unlike Corollary 9.4, where $\{\theta_j\}$ are random, Corollary 9.10 allows $\{\theta_j\}$ to be trained together with $\{u_j\}$.

Next, we consider the entropy regularization condition, which was employed in [59]. The following result is a direct consequence of Theorem 8.3.

Corollary 9.11 *Assume that we choose parameters so that $uh(\theta^\top x) \in [-M, M]$, and let*

$$\mathcal{F}_A^{\text{mf,entro}} = \{\mathbf{E}_{[u,\theta] \sim q} uh(\theta^\top x) : \text{KL}(q||p_0) \leq A^2\}.$$

Then

$$R(\mathcal{F}_A^{\text{mf,entro}}, S_n) \leq \sqrt{\frac{2}{n}} AM.$$

The entropy regularization bound in Corollary 9.11 is potentially better than the L_1 regularization bound of Corollary 9.9 when the KL-divergence $\text{KL}(q||p_0)$ is small.

9.6 Analysis of Deep Neural Networks

It is possible to extend the NTK analysis to deep neural networks using an extension of the linear approximation 9.5 for two-layer neural networks without too much difficulty. We will not consider it here for simplicity. It has similar flaws as the kernel approaches for random features and for two-layer neural networks, in that the linearization requires model parameters to be restricted in a small region around the initialization.

An additional disadvantage of the NTK approximation is that it can be regarded as a shallow network approximation of a deep network using

a linear model. According to [83], highly oscillating functions with large crossing numbers cannot be approximated efficiently with NTK, although they can be well approximated using deep neural networks. This is related to the fact that NTK cannot learn features efficiently, thus they cannot learn feature combinations effectively. However, the efficient representation of such feature combinations is the key advantage of deep neural networks. Therefore in this section we will investigate the deep neural network method directly without considering the NTK approximation.

Here we consider a formulation of deep neural networks which may be considered as an extension of the mean field formulation of two-layer neural networks, and then investigate its generalization. The deep function class is defined recursively as follows:

$$\mathcal{F}^{(1)} = \{\theta^\top x : \theta \in \Theta\},$$

and for $k = 2, \dots, K$, using the notation of Section 8.2, we define

$$\begin{aligned} \mathcal{F}^{(k)} &= \mathcal{F}_{A_k, L_1}(h \circ \mathcal{F}^{(k-1)}) \\ &= \left\{ \sum_{j=1}^m w_j h(f_j(x)) : \|w\|_1 \leq A_k, f_j \in \mathcal{F}^{(k-1)}, m > 0 \right\}. \end{aligned} \quad (9.8)$$

When $A_1, \dots, A_k$ are sufficiently large, then it is clear that any function $f(x)$ that can be represented by a deep K layer neural networks belongs to $\mathcal{F}^{(k)}$. The representation allows arbitrary large m , and thus can handle continuous deep neural networks. Therefore it can be regarded as an extension of the two-layer mean field formulation.

For simplicity, we consider the case that $h \circ \mathcal{F}^{(1)}$ is bounded, h is monotone and Lipschitz. The following result, first presented in [50], is a direct consequence of Theorem 8.2.

Theorem 9.4 *Consider K -layer neural networks defined by (9.8). Assume that $h(\theta^\top x) \in [-M, M]$ for all $\theta \in \Theta$ and $x \in \mathcal{X}$. Assume also that h is 1-Lipschitz and monotone. Then there exists a constant C , such that for all distribution D on $\mathcal{X}$, we have*

$$R_n(\mathcal{F}^{(k)}, D) \leq C \cdot AM \sqrt{\frac{d+1}{n}},$$

where $A = 2^{k-2} \prod_{\ell=2}^k A_\ell$, and C is the absolute constant in Proposition 9.1.

Proof We prove the statement by induction. The case $k = 2$ follows from Corollary 9.4. Assume the statement holds at layer $k - 1$, then at layer k . We have

$$\begin{aligned} R_n(\mathcal{F}^{(k)}, D) &\leq A_k [R_n(h \circ \mathcal{F}^{(k-1)}, D) + R_n(-h \circ \mathcal{F}^{(k-1)}, D)] \\ &\leq 2A_k R_n(\mathcal{F}^{(k-1)}, D). \end{aligned}$$

Using induction, we obtain the desired bound. ■

The result implies that if we use L_1 regularization for every layer of a deep neural network, then the Rademacher complexity can be easily bounded using the multiplications of the layer-wise L_1 regularization parameters. For the ReLu function, the L_1 regularization can be moved to the last layer. It was also worth noticing that the learning complexity for the function composition example considered in [83] is still high if we measure it by L_1 regularization. This is not surprising because functions with exponentially many crossing numbers are complex. However, it can be compactly represented by the deep neural network with a moderately small network, and thus gain efficiency in computation.

One benefit of the generalization bound in Theorem 9.4 is that the generalization performance does not depend on the number of neurons. Therefore the more neurons we use, the better. This is consistent with the practical observations. If we allow the number of neurons go to infinity, then we obtain a continuous formulation of deep neural networks with the same learning bound measured by Rademacher complexity.

9.7 Exercises

1. Consider ReLu neural networks for approximating one-dimensional functions on $[0, 1]$. Prove that any piecewise linear function can be represented by a two-layer ReLu network.
2. For a k -layer ReLu neural network, with m nodes per layer, show that its output is a piecewise linear function. Estimate the maximum possible number of linear segments, hence the crossing number it may contain. How large should m be to achieve the same crossing number as that of the $f_{k^2}(x)$ function defined below Theorem 9.3?
3. Let $w_* = e_1 \in \mathbb{R}^d$. Consider $\psi(x) = [\psi_1(x), \dots, \psi_d(x)] \in \mathbb{R}^d$ such that for observations $\{(X_1, Y_1), \dots, (X_n, Y_n)\}$, $n^{-1/2}[\psi_j(X_1), \dots, \psi_j(X_n)]$ are d orthonormal vectors in $\mathbb{R}^n$ ($j = 1, \dots, d$), and $Y_i = w_*^\top \psi(X_i) + \epsilon_i$, where $\epsilon_i \sim N(0, \sigma^2)$ ($i = 1, \dots, n$) are zero-mean noises. Consider ridge regression:

$$\hat{w}(\lambda) = \arg \min_w \left[\frac{1}{n} \sum_{i=1}^n (w^\top \psi(X_i) - Y_i)^2 + \lambda \|w\|_2^2 \right].$$

Find the optimal λ so that

$$\mathbf{E}_\epsilon \|\hat{w}(\lambda) - w_*\|_2^2$$

is minimized. Compute the minimum value.

4. In the previous example. Consider Lasso regression:

$$\hat{w}(\lambda) = \frac{1}{n} \sum_{i=1}^n (w^\top \psi(X_i) - Y_i)^2 + \lambda \|w\|_1.$$

For $\lambda > 0$, find an upper of

$$\mathbf{E}_\epsilon \|\hat{w}(\lambda) - w_*\|_2^2.$$

Find λ to approximately minimize the upper bound with logarithmic dependency on d , and compare to that of the ridge regression.

Chapter 10

Statistical Lower and Upper Bounds

In this chapter, we consider lower bounds for empirical processes and statistical estimation problems. We know upper bounds for empirical processes and empirical risk minimization can be obtained from the covering number analysis. We show that under suitable conditions, lower bounds can also be obtained using covering numbers.

10.1 Lower Bounds for Empirical Processes

In previous chapters, such as Chapter 6, we showed that uniform convergence of empirical processes in terms of Rademacher complexity can be obtained using covering numbers and the chaining technique.

This section considers the lower bounds. For simplicity, we will only consider empirical processes associated with a function family $F = \{f_w(z) : w \in \Omega\}$, defined on the empirical measure $S_n = \{Z_1, \dots, Z_n\}$.

We shall first introduce the notation of Gaussian complexity, which is useful for obtaining lower bounds.

Definition 10.1 *The empirical Gaussian complexity of F is defined as*

$$G(F, S_n) = \mathbf{E}_g \sup_{f \in F} \frac{1}{n} \sum_{i=1}^n g_i f(Z_i),$$

where $[g_1, \dots, g_n]$ are independent normal random variables: $g_i \sim N(0, 1)$ for $i = 1, \dots, n$.

The following result shows that Gaussian complexity and Rademacher complexity are equivalent up to a logarithmic factor in n [15].

Proposition 10.1 *There exists an absolute constant $C > 0$ such that if $F = -F$, then*

$$C^{-1}R(F, S_n) \leq G(F, S_n) \leq C \ln n R(F, S_n).$$

Both Rademacher complexity and Gaussian complexity can be used to obtain expected uniform convergence for empirical processes. Upper bounds for both Rademacher complexity and Gaussian can be obtained from covering numbers and Dudley's entropy integral in Theorem 6.4. However, the Gaussian complexity also has a lower bound using covering numbers, called Sudakov Minoration. Its proof relies on Slepian's lemma, which is a comparison lemma for Gaussian complexity, similar to the comparison lemma for Rademacher complexity in Lemma 6.1. The proof of this result can be found in [79, 46].

Lemma 10.1 (Slepian's Lemma) *Let $[X_1, \dots, X_n]$ and $[Y_1, \dots, Y_n]$ denote two zero-mean multivariate normal random vectors. Assume that*

$$\forall i \neq j, \quad \mathbf{E}(X_i - X_j)^2 \geq \mathbf{E}(Y_i - Y_j)^2.$$

Then

$$\mathbf{E} \max_i X_i \geq \mathbf{E} \max_i Y_i.$$

For lower bounds, we consider a special function class cover called packing subset, which is defined below.

Definition 10.2 *Consider a function class F . An ϵ -packing subset of F in d -metric is a finite subset $F_M = \{f_1, \dots, f_M\}$ of F of the largest cardinality M , so that for all $i \neq j$:*

$$d(f_i, f_j) \geq \epsilon.$$

The number M , denoted as $M(\epsilon, F, d)$, is called the ϵ packing number of F .

It is clear from the definition that an ϵ packing subset of F is also an ϵ cover of F with the same metric.

Proposition 10.2 *We have the following bounds:*

$$N(\epsilon, F, d) \leq M(\epsilon, F, d), \quad M(\epsilon, F, d) \leq N(\epsilon/2, F, d).$$

Using the notation of packing subset, we can obtain the following result, referred to as *Sudakov minoration*, as a direct consequence of Slepian's lemma.

Theorem 10.1 (Sudakov Minoration) *For any $\epsilon > 0$:*

$$\sqrt{n}G(F, S_n) \geq 0.45\epsilon \sqrt{\ln M(\epsilon, F, L_2(S_n))} - 0.9\epsilon.$$

Proof Let $F_M = \{f_1, \dots, f_M\} \subset F$ be an ϵ packing subset of F under the $L_2(S_n)$ metric. Consider independent standard Gaussian random variables $[g_1, \dots, g_n]$.

$$\mathbf{E} \sup_{f \in F} \frac{1}{n} \sum_{i=1}^n g_i f(Z_i) \geq \mathbf{E} \sup_j \frac{1}{n} \sum_{i=1}^n g_i f_j(Z_i).$$

Let $g'_1, \dots, g'_M$ be independent zero-mean normal random variables with variance $\epsilon^2/(2n)$ each. We then have for each $j \neq k$:

$$\begin{aligned} & \mathbf{E} \left(\frac{1}{n} \sum_{i=1}^n g_i f_j(Z_i) - \frac{1}{n} \sum_{i=1}^n g_i f_k(Z_i) \right)^2 \\ &= \frac{1}{n^2} \sum_{i=1}^n (f_j(Z_i) - f_k(Z_i))^2 \geq \frac{1}{n} \epsilon^2 = \mathbf{E} (g'_j - g'_k)^2. \end{aligned}$$

Using Slepian's lemma, we have

$$G(\epsilon, F, L_2(S_n)) \geq \mathbf{E} \sup_j g'_j.$$

We know from Section 2.1 that for all $z \geq 0$,

$$\begin{aligned} \Pr(g'_j \leq \epsilon z / \sqrt{2n}) &\leq 1 - 0.5e^{-(z+1)^2/2} \\ \Pr(g'_j \leq -\epsilon z / \sqrt{2n}) &\leq 0.5e^{-z^2/2}, \end{aligned}$$

and thus

$$\begin{aligned} \Pr(\sup_j g'_j \leq \epsilon z / \sqrt{2n}) &\leq (1 - 0.5e^{-(z+1)^2/2})^M \\ \Pr(\sup_j g'_j \leq -\epsilon z / \sqrt{2n}) &\leq 0.5e^{-Mz^2/2}. \end{aligned}$$

It follows that for $M \geq 2$:

$$\begin{aligned} & \mathbf{E} \sup_j \frac{\sqrt{2n} g'_j}{\epsilon} \\ &= - \int_{-\infty}^0 \Pr \left(\sup_j g'_j \leq \frac{\epsilon z}{\sqrt{2n}} \right) dz + \int_0^{\infty} \Pr \left(\sup_j g'_j \geq \frac{\epsilon z}{\sqrt{2n}} \right) dz \\ &\geq \int_0^{\infty} (1 - (1 - 0.5e^{-(z+1)^2/2})^M) dz - \int_{-\infty}^0 0.5e^{-Mz^2/2} dz \\ &\geq \int_0^{\sqrt{2 \ln M} - 1} (1 - 0.5(1 - 0.5/M)^M) dz - \int_{-\infty}^0 0.5e^{-Mz^2/2} dz \\ &\geq \int_0^{\sqrt{2 \ln M} - 1} (1 - 0.5/\sqrt{e}) dz - 0.25M^{-1/2} \int_{-\infty}^{\infty} e^{-z^2/2} dz \\ &\geq 0.65(\sqrt{2 \ln M} - 2). \end{aligned}$$

This implies the desired bound. ■

A more precise characterization of Gaussian complexity, due to [82], is to consider a generalization of covering numbers, referred to as *majorization measures*, with a generalization of the chaining technique referred to as *generic chaining*. We consider any measure μ on F , and let

$$\mu(f, \epsilon, L_2(S_n)) = \mu(\{f' \in F : \|f' - f\|_{L_2(S_n)} \leq \epsilon\}).$$

Then $-\ln \mu(f, \epsilon)$ may be regarded as a generalization of the entropy number of F localized around f . If we define

$$\gamma_2(F, S_n) = \inf_{\mu} \sup_{f \in F} \int_0^\infty \sqrt{\frac{-\ln \mu(f, \epsilon, L_2(S_n))}{n}} d\epsilon,$$

then the following result shows the equivalence of Gaussian complexity and the corresponding entropy-integral in terms of majorization measure [82].

Theorem 10.2 (The majorizing measure theorem) *There exists an absolute constant $C > 0$ so that*

$$C^{-1} \gamma_2(F, S_n) \leq G(F, S_n) \leq C \gamma_2(F, S_n).$$

The result shows that the entropy integral bound for empirical processes is generally tight, although one may need to replace covering numbers by majorization measures in certain situations. In other cases, the difference between covering number and majorization measures is not significant.

10.2 Minimax Analysis for Statistical Estimation

We consider the statistical estimation problem, where we want to find a certain function $\hat{f}$ from training examples $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, which are iid samples from an underlying distribution D via a learning algorithm $\mathcal{A}$:

$$\hat{f} = \mathcal{A}(S_n).$$

Assume that f_* is the true function which we want to estimate. The quality of the estimator $\hat{f}$ can be measured by a non-negative function

$$Q(\hat{f}, f_*).$$

Example 10.1 *For least squares problem, $f(x)$ is a real valued regression function. Let $f_*(x) = \mathbf{E}[Y|X = x]$. We may define*

$$Q(f, f_*) = \mathbf{E}_{X \sim D}(f(X) - f_*(X))^2.$$

Example 10.2 For conditional density estimation with K classes $y \in \{1, \dots, K\}$, we may consider vector valued density functions

$$f(x) = [p(y = 1|x), \dots, p(y = K|x)],$$

and assume that $f_*(x) = p_*(\cdot|x)$ is the true conditional density.

For density estimation, the estimation quality can be measured by the KL-divergence:

$$Q(f, f_*) = \mathbf{E}_{X \sim D} \mathbf{E}_{Y \sim p_*(Y|X)} \ln \frac{p_*(Y|X)}{p(Y|X)}.$$

If we are interested in classification accuracy, then we may use the excess classification error over the Bayes classification error as quality measure. Here

$$f_*(x) = \arg \max_{\ell} p_*(Y = \ell | X = x)$$

is the optimal Bayes classifier. Let $f(x) \in \{1, \dots, K\}$ be any classifier, then we can define

$$Q(f, f_*) = \mathbf{E}_{X \sim D} [\Pr(Y = f_*(X) | X) - \Pr(Y = f(X) | X)].$$

Given a predefined function class F , we can associate it with a family of distributions $\mathcal{D}$ over $Z = (X, Y)$. For each $D \in \mathcal{D}$, we can assume that the corresponding optimal function $f_*(D) \in F$. In statistical analysis, we often use the worst case expected risk of a learning algorithm $\mathcal{A}$ to measure the ability of the algorithm to learn functions in F from a family of distributions $\mathcal{D}$.

Definition 10.3 Consider distribution family $\mathcal{D}$, and a function class F such a distribution dependent quantity $f_*(D) \in F$ for all $D \in \mathcal{D}$. Then the worst case expected risk of a learning algorithm (i.e., a statistical estimator) $\mathcal{A}$ with respect to $\mathcal{D}$ is given by

$$r_n(\mathcal{A}, \mathcal{D}) = \sup_{D \in \mathcal{D}} \mathbf{E}_{S_n \sim D} Q(\mathcal{A}(S_n), f_*(D)).$$

Moreover, the minimax risk for learning $f_*(\cdot)$ on $\mathcal{D}$ is defined as:

$$r_n(\mathcal{D}) = \inf_{\mathcal{A}} r_n(\mathcal{A}, \mathcal{D}).$$

The standard statistical framework for optimality is to find an algorithm with worse case risk $r_n(\mathcal{A}, \mathcal{D})$ as small as possible. This type of analysis is referred to as *minimax analysis*. The minimax risk $r_n(\mathcal{D})$ depends only on the sample size n and distribution family $\mathcal{D}$. In nonparametric statistics, it is difficult to find the precise minimax risk. Therefore we often consider rate optimal algorithms $\mathcal{A}$, which achieves the optimal minimax risk $r_n(\mathcal{D})$ up to a constant factor for all n .

In minimax analysis, we are interested in estimating the minimax risk (up to a constant factor). To establish an upper bound of $r_n(\mathcal{D})$, we consider specific learning algorithms and analyze their convergence rates. For example, if we consider the ERM method $\mathcal{A}_{erm}$ for the least squares regression problem, then we may obtain an upper bound of

$$r_n(\mathcal{D}) \leq r_n(\mathcal{A}_{erm}, \mathcal{D}) = O(n^{-r}),$$

for some $r > 0$, based on the analysis of Section 6.5.

On the other hand, if we can show a lower bound $R_n(\mathcal{D}) \geq cn^{-r}$ for some constant c that may depend on $\mathcal{D}$ but independent of n , then we know that the ERM method achieves the optimal minimax lower bound. That is, no statistical estimator can perform much better than ERM up to a constant factor for this distribution family. If the lower bound is $O(n^{-r'})$ for $r' > r$, then ERM may not be optimal. As we will see, both situation can happen depending on the entropy of the function class F .

10.3 General Nonparametric Lower Bound

In parametric statistics, with suitable regularity conditions, the classical asymptotic lower bound for unbiased parameter estimation of a probability model is given by the Cramer-Rao inequality [29], and the matching upper bound can be achieved by the maximum likelihood method. This leads to the asymptotic optimality of maximum likelihood method (with appropriate regularity conditions).

In non-parametric statistics, one needs to consider other techniques from information theory to establish lower bounds for statistical convergence rate, usually up to a constant. Common techniques for lower bounds include the Fano's inequality [34] and Assouad's Lemma [5]. For example, see [19, 92, 93] for applications in nonparametric statistical estimation.

In this section, we will consider a closely related but more direct information theoretical technique, often referred *data-processing inequality*. The analysis follows that of [98]. In general, information theoretical lower bound applies to general density estimation problems. It can be used to derive other lower bounds such as least squares regression, simply by treating such problems as density estimation problems. For least squares regression, we may consider it as density estimation with Gaussian noise.

Consider two random variables A and B on the same domain. Let $p_A(z)$ and $p_B(z)$ be the densities of A and B with respect to some underlying measure $d\mu(z)$. Recall that the KL-divergence (Kullback-Leibler divergence)

between A and B is defined as follows:

$$\begin{aligned} \text{KL}(A||B) &= \text{KL}(p_A||p_B) = \mathbf{E}_{z \sim p_A} \ln \frac{p_A(z)}{p_B(z)} \\ &= \int p_A(z) \ln \frac{p_A(z)}{p_B(z)} d\mu(z). \end{aligned}$$

We always have $\text{KL}(A||B) \geq 0$. We first state the following lemma, which says that any data process procedure (including supervised learning procedure) never increases KL divergence.

Lemma 10.2 (Data Processing Inequality for KL-divergence) *Consider random variables A and B , and a (possibly random) processing function h , then the inequality $\text{KL}(A||B) \geq \text{KL}(h(A)||h(B))$ holds.*

Proof Given random variables A_1, A_2 and B_1, B_2 , It is easy to check that

$$\begin{aligned} &\text{KL}((A_1, A_2)|| (B_1, B_2)) \\ &= \mathbf{E}_{z_1 \sim p_{A_1}} \mathbf{E}_{z_2 \sim p_{A_2|A_1}(z_2|z_1)} \ln \frac{p_{A_1}(z_1)p_{A_2}(z_2|A_1 = z_1)}{p_{B_1}(z_1)p_{B_2}(z_2|B_1 = z_1)} \\ &= \text{KL}(A_1||B_1) + \mathbf{E}_{z_1 \sim p_{A_1}} \text{KL}(p_{A_2}(\cdot|A_1 = z_1)||p_{B_2}(\cdot|B_1 = z_1)) \\ &\geq \text{KL}(A_1||B_1). \end{aligned}$$

Now let $A_1 = A$, $A_2 = h(A_1)$, $B_1 = B$, and $B_2 = h(B_1)$, then $P_{A_2}(\cdot|A_1 = z_1) = P_{B_2}(\cdot|B_1 = z_1)$ because the conditional probability only depends on the processing function h . Therefore

$$\text{KL}(p_{A_2}(\cdot|A_1 = z_1)||p_{B_2}(\cdot|B_1 = z_1)) = 0,$$

which implies that

$$\text{KL}(A||B) = \text{KL}((A, h(A))|| (B, h(B))).$$

Now, we change the way to evaluate the right hand side of the above equation by setting $A_2 = A$, $A_1 = h(A)$, $B_2 = B$, and $B_1 = h(B)$, which implies that

$$\text{KL}((h(A), A)|| (h(B), B)) \geq \text{KL}(h(A)||h(B)).$$

This proves the desired result. ■

The data processing lemma can be used to derive the following information theoretic lower bound for density estimation.

Theorem 10.3 *Consider a set of densities $G = \{p_w(z) : w \in \Omega\}$, which contains a subset $p_1(z), \dots, p_N(z) \in G$ where*

$$\ln N \geq \ln 4 + 2n \sup_{j,k} \text{KL}(p_j||p_k).$$

Assume that each $p_j(z)$ is associated with a function $f_j(\cdot)$. Let $Q(f, f')$ be an arbitrary non-negative function, and define

$$\epsilon = \inf_{j \neq k} \inf_f \{\max(Q(f, f_j), Q(f, f_k))\}.$$

For $j = 1, \dots, N$ and $S_n = \{Z_1, \dots, Z_n\}$ with iid data $Z_i \sim p_j(\cdot)$, we have

$$\frac{1}{N} \sum_{j=1}^N \mathbf{E}_{S_n \sim \prod_{i=1}^n p_j(Z_i)} Q(\mathcal{A}(S_n), f_j) \geq 0.5\epsilon$$

for any learning algorithm $\mathcal{A}$.

Proof Consider uniform distribution on $j \in \{1, \dots, N\}$, and then draw Z_i according to p_j . Then the joint distribution of (j, S_n) is given by

$$p(j, S_n) = \frac{1}{N} \prod_{i=1}^n p_j(Z_i).$$

Denote by $I(j, S_n)$ the mutual information between j and S_n :

$$I(j, S_n) = \text{KL}((j, S_n) || (j, \tilde{S}_n)),$$

which is the KL-divergence between the joint distributions of (j, Z_1^n) and $(j, \tilde{S}_n)$, where $\tilde{S}_n = \{\tilde{Z}_1, \dots, \tilde{Z}_n\}$ has the same marginal distribution as S_n , but is independent of j :

$$p(\tilde{S}_n) = \frac{1}{N} \sum_{j=1}^N \prod_{i=1}^n p_j(\tilde{Z}_i).$$

Now, consider an arbitrary possibly random estimator $\hat{f} = \mathcal{A}(S_n)$. Let $\hat{f}' = \mathcal{A}(\tilde{S}_n)$. By the data processing inequality for KL-divergence, with input (j, S_n) and binary output $h(j, S_n) = I(Q(\hat{f}, f_j) < \epsilon)$, we obtain

$$\begin{aligned} & \text{KL}(I(Q(\hat{f}, f_j) < \epsilon) || I(Q(\hat{f}', f_j) < \epsilon)) \\ & \leq \text{KL}((j, S_n) || (j, \tilde{S}_n)) = I(j, S_n) \\ & = \frac{1}{N} \sum_{j=1}^N \mathbf{E}_{S_n \sim \prod_{i=1}^n p_j(Z_i)} \ln \frac{\prod_{i=1}^n p_j(Z_i)}{N^{-1} \sum_{k=1}^N \prod_{i=1}^n p_k(Z_i)} \\ & \leq \frac{1}{N^2} \sum_{j=1}^N \sum_{k=1}^N \mathbf{E}_{S_n \sim \prod_{i=1}^n p_j(Z_i)} \ln \frac{\prod_{i=1}^n p_j(Z_i)}{\prod_{i=1}^n p_k(Z_i)} \\ & = \frac{n}{N^2} \sum_{j=1}^N \sum_{k=1}^N \text{KL}(p_j || p_k) \leq 0.5 \ln(N/4). \end{aligned}$$

The first inequality is the data processing inequality. The second inequality is a consequence of Jensen's inequality and the concavity of logarithm.

Now let $q = \Pr(Q(\hat{f}, f_j) < \epsilon)$ and $q' = \Pr(Q(\hat{f}', f_j) < \epsilon)$, then the above inequality can be rewritten as:

$$\text{KL}(q||q') = q \ln \frac{q}{q'} + (1 - q) \ln \frac{1 - q}{1 - q'} \leq 0.5 \ln(N/4).$$

Since $\hat{f}'$ is independent of j , and $\epsilon \leq \inf_{j \neq k} \inf_f \{\max(Q(f, f_j), Q(f, f_k))\}$, for each $\hat{f}'$, there can only be no more than one $j \in \{1, \dots, N\}$ such that $Q(\hat{f}', f_j) < \epsilon$. Therefore we have

$$q' \leq N^{-1} \leq 1/4.$$

We now show that in this case, $q \leq 1/2$. Since $\text{KL}(q||q')$ as a function of q is increasing in $[q', 1]$, we only need to show that $\text{KL}(0.5||q') \geq 0.5 \ln(N/4)$. This easily follows from the inequality

$$\text{KL}(0.5||q) \geq 0.5 \ln \frac{0.5}{q'} + 0.5 \ln \frac{0.5}{1} \geq 0.5 \ln(N/4).$$

Now, we have shown that $q \leq 0.5$, which implies that $\Pr(Q(\hat{f}, f_j) \geq \epsilon) \geq 0.5$. Therefore desired inequality is a consequence of

$$\frac{1}{N} \sum_{j=1}^N \mathbf{E}_{S_n \sim \prod_i p_j(Z_i)} Q(\hat{f}, f_j) \geq \frac{1}{N} \sum_{j=1}^N \epsilon \Pr(Q(\hat{f}, f_j) \geq \epsilon),$$

which follows from the Markov inequality. ■

In the condition of the theorem, covering numbers are sufficiently large compared to the KL-divergence between densities. If the densities are nearly indistinguishable (that is, the KL-divergences among two densities are small), and n is not sufficiently large compared to the covering numbers, we will make mistakes with constant probability in identifying which density is the true underlying distribution. This implies that we make nearly as much mistake as a random guessing of f_j .

If we let $\mathcal{D} = \{p_1, \dots, p_N\}$, then Theorem 10.3 implies a lower bound for the minimax risk:

$$r_n(\mathcal{D}) \geq \inf_{\mathcal{A}} \frac{1}{N} \sum_{j=1}^N \mathbf{E}_{S_n \sim \prod_{i=1}^n p_j(Z_i)} Q(\mathcal{A}(S_n), f_j) \geq 0.5\epsilon.$$

10.4 Lower Bound for Least Squares Regression

For the least squares regression problem in Example 10.1, we consider a function class F that contains the optimal prediction rule $f_*(X) = \mathbf{E}[Y|X]$,

and

$$Q(f, f') = \mathbf{E}_X(f(X) - f'(X))^2.$$

The following result is a direct consequence of Theorem 10.3.

Theorem 10.4 *Consider the regression model, where $X \sim D_X$ with known D_X , and*

$$Y = f_*(X) + \epsilon,$$

where ϵ is zero-mean noise that may depend on $f_(\cdot) \in F$. Assume there exists $\sigma > 0$ so that*

$$Q(f, f') \geq 2\sigma^2 \text{KL}(p_f || p_{f'}),$$

where p_f is the distribution of (X, Y) when $f_ = f$. If F contains N functions $f_1, \dots, f_N$ such that*

$$\ln N \geq \ln 4 + n\sigma^{-2} \sup_{j \neq k} Q(f_j, f_k),$$

then

$$r_n(\mathcal{D}) \geq 0.125 \inf_{j \neq k} Q(f_j, f_k).$$

Proof Note that for each $f \in F$, we associate a $p_f \in \mathcal{D} = \{p_f : f \in F\}$. We also let $\epsilon = 0.25 \min_{j,k} Q(f_j, f_k)$, and it can be checked that $\max(Q(f, f_j), Q(f, f_k)) \geq \epsilon$ for all $j \neq k$. The theorem is now a direct consequence of Theorem 10.3. ■

The following result implies that the condition of Theorem 10.4 holds for Gaussian noise.

Proposition 10.3 *Consider $p_f(X, Y)$ so that $X \sim D_X$ is identical for all $f \in F$, and $Y \sim N(f(X), \sigma^2)$ for some constant $\sigma > 0$. Then*

$$Q(f, f') = 2\sigma^2 \text{KL}(p_f || p_{f'}).$$

Proof We consider a distribution p over $Z = (X, Y)$ associated with each $f \in F$ as

$$p_f(Z) = \Pr(X) \Pr_{N(0,1)}((Y - f(X))/\sigma),$$

where $X \sim D_X$ and $\Pr_{N(0,1)}$ is the density of standard normal distribution. Using this notation, we have

$$\text{KL}(p_{\hat{f}} || p_f) = \frac{\mathbf{E}_X(\hat{f}(X) - f(X))^2}{2\sigma^2} = 0.5Q(\hat{f}, f)/\sigma^2.$$

This proves the desired result. ■

Similar results can also be obtained for other noise models such as Bernoulli noise, where $Y \in \{0, 1\}$.

Proposition 10.4 Assume that $Y \in \{0, 1\}$ such that

$$\mathbf{E}[Y|X] = f(X),$$

for some $f \in F$, and let $p_f(Z)$ be the distribution of $Z = (X, Y)$. If there exists $c \in (0, 0.5)$, and for all $f \in F$ and X : $f(X) \in [c, 1 - c]$, then there exists $\sigma > 0$ such that for all $f, f' \in F$:

$$Q(f, f') \geq 2\sigma^2 \text{KL}(p_f || p_{f'}).$$

Consider a distribution D_X over X , with metric $d = L_2(D_X)$:

$$d(f, f') = (\mathbf{E}_{X \sim D_X} (f(X) - f'(X))^2)^{1/2}.$$

The following result shows that metric entropy leads to a lower bound on the minimax risk.

Corollary 10.1 If for some $C > 0$ and $\epsilon > 0$:

$$C^{-1}\epsilon^{-q} \leq \ln M(\epsilon, F, L_2(D_X)) \leq C\epsilon^{-q}.$$

For noise model that satisfies the condition of Theorem 10.4, we have

$$r_n(\mathcal{D}) \geq C'n^{-2/(2+q)}$$

for some $C' > 0$.

Proof We consider an ϵ packing subset F' of F with size of at least $C^{-1}\epsilon^{-q}$. Since for some $C_0 > 0$,

$$\ln N(0.5C_0\epsilon, F, L_2(D_X)) \leq 0.5C^{-1}\epsilon^{-q},$$

it implies that there exists a ball of size $0.5C_0\epsilon$, which contains at least $0.5C^{-1}\epsilon^{-q}$ members of F' .

This means we can find $N \geq 0.5C^{-1}\epsilon^{-q}$ functions $\{f_1, \dots, f_N\}$ such that

$$\sup_{j \neq k} Q(f_j, f_k) \leq C_0^2 \epsilon^2,$$

and

$$\inf_{j \neq k} Q(f_j, f_k) \geq \epsilon^2.$$

Therefore if

$$\ln N \geq 0.5C^{-1}\epsilon^{-q} \geq \ln 4 + n\sigma^{-2}C_0^2\epsilon^2, \quad (10.1)$$

then

$$r_n(F) \geq 0.125\epsilon^2.$$

The condition (10.1) is satisfied with $n = 8C'\epsilon^{-(q+2)}$ for some constant C' , which implies the desired bound. $\blacksquare$

If F is bounded, and has a uniform covering number of $O(\epsilon^{-q})$ for $q < 2$, then we know from the local Rademacher complexity analysis of Section 6.5 that for the empirical risk minimization method

$$\hat{f}_{erm} = \arg \min_{f \in F} \sum_{i=1}^n (f(X_i) - Y_i)^2,$$

the following risk bound holds:

$$\mathbf{E}_{S_n} Q(\hat{f}_{erm}, f_*) = O(n^{-2/(q+2)}).$$

This matches the lower bound of Corollary 10.1.

For $q > 2$, the generalization bound for empirical risk minimization method is

$$\mathbf{E}_{S_n} Q(\hat{f}_{erm}, f_*) = O(n^{-1/q}),$$

and since $1/q < 2/(q+2)$, the rate is inferior to the minimax rate. In general, this rate cannot be improved without additional assumptions [20]. Therefore when the nonparametric family has a large entropy, ERM can be suboptimal.

It is possible to achieve the optimal rate of $O(n^{-2/(q+2)})$. One of the optimal method is least squares on sieves, and another is the exponential model aggregation method, studied in Section 12.3.

In the following, we consider the sieve method [41, 40, 21], which in our setting, is simply least squares regression on discrete ϵ -net.

Given a function class F , instead of running least squares on F with

$$\hat{f} = \arg \min_{f \in F} \sum_{i=1}^n (f(X_i) - Y_i)^2,$$

the sieve method considers a subset $F_n \subset F$, and then perform least squares regression restricted to this subset:

$$\hat{f}_{F_n} = \arg \min_{f \in F_n} \sum_{i=1}^n (f(X_i) - Y_i)^2.$$

The following result shows that with appropriate F_n , we may achieve the minimax risk of Corollary 10.1.

Proposition 10.5 *Assume that the distribution of X is D_X . Let F_n be an ϵ packing subset of F with the $L_2(D_X)$ metric. Assume $|f(X) - Y|$ is bounded for all $f \in F$. Then there exists $C' > 0$ so that given any $f_* \in F$ such that $\mathbf{E}[Y|X] = f_*(X)$:*

$$\mathbf{E}_{S_n} Q(\hat{f}_{F_n}, f_*) \leq C' \left[\epsilon^2 + \frac{1}{n} \ln M(\epsilon, F, L_2(D_X)) \right].$$

Proof Note that $g_f(Z) = (f(X) - Y)^2 - (f_*(X) - Y)^2$ is a bounded real-valued function, which satisfies the variance condition $\mathbf{E}_Z g(Z)^2 \leq c \mathbf{E} g(Z)$ for some constant $c > 0$. Moreover, we have

$$\mathbf{E}_Z g_f(Z) = Q(f, f_*).$$

Let $G_n = \{g_f(Z) : f \in F_n\}$. There exist $c'_1 > 0$ and $c'_2 > 0$ so that with appropriate $\gamma > 0$:

$$\begin{aligned} & \Pr \left[\sup_{g \in G_n} \left(-\frac{1}{n} \sum_{i=1}^n g(Z_i) + (1 - \gamma) \mathbf{E}_Z g(Z) \right) \geq \epsilon \right] \\ & \leq \sum_{g \in G_n} \Pr \left[-\frac{1}{n} \sum_{i=1}^n g(Z_i) + (1 - \gamma) \mathbf{E}_Z g(Z) \geq \epsilon \right] \\ & \leq \sum_{g \in G_n} \exp \left[\frac{-n(\gamma \mathbf{E}_Z g(Z) + \epsilon)^2}{\mathbf{E}_Z g(Z)^2 + c'_1 \epsilon} \right] \\ & \leq |G_n| \sup_{g \in G_n} \exp \left[\frac{-n(\gamma \mathbf{E}_Z g(Z) + \epsilon)^2}{c \mathbf{E}_Z g(Z) + c'_1 \epsilon} \right] \\ & \leq |G_n| \exp[-nc'_2 \epsilon]. \end{aligned}$$

In the above derivations, the first inequality is union bound, the second inequality is Bernstein's inequality, and the last two inequalities follow from simple algebra.

Note that $|G_n| \leq M(\epsilon, F, L_2(D_X))$. This implies that

$$\mathbf{E}_{S_n} \sup_{g \in G_n} \left(-\frac{1}{n} \sum_{i=1}^n g(Z_i) + (1 - \gamma) \mathbf{E}_Z g(Z) \right) \leq c'_3 \epsilon_n^2$$

for some $c'_3 > 0$, where

$$\epsilon_n^2 = \left[\epsilon^2 + \frac{1}{n} \ln M(\epsilon, F, L_2(D_X)) \right].$$

Similarly, for any $f \in F_n$, we have

$$\mathbf{E}_{S_n} \frac{1}{n} \sum_{i=1}^n g_f(Z_i) \leq \mathbf{E}_{S_n} (1 + \gamma) \mathbf{E}_Z g_f(Z) + c'_4 \epsilon_n^2$$

for some $c'_4 > 0$. Therefore if we let $\hat{g} = g_{\hat{f}_{F_n}}$, then for any $f \in F_n$:

$$\begin{aligned}
 \mathbf{E}_{S_n}(1 - \gamma)Q(\hat{f}_{F_n}, f_*) &= \mathbf{E}_{S_n}(1 - \gamma)\mathbf{E}\hat{g}(Z) \\
 &\leq \mathbf{E}_{S_n} \frac{1}{n} \sum_{i=1}^n \hat{g}(Z_i) + c'_3 \epsilon_n^2 \\
 &\leq \mathbf{E}_{S_n} \frac{1}{n} \sum_{i=1}^n g_f(Z_i) + c'_3 \epsilon_n^2 \\
 &\leq \mathbf{E}_{S_n}(1 + \gamma)\mathbf{E}_Z g_f(Z) + (c'_3 + c'_4)\epsilon_n^2 \\
 &= (1 + \gamma)Q(f, f_*) + (c'_3 + c'_4)\epsilon_n^2.
 \end{aligned}$$

We can take $f \in F_n$ be the closest function to f_* . Then this implies the desired bound. $\blacksquare$

Under the condition of Corollary 10.1, we note that

$$\inf_{\epsilon} \left[\epsilon^2 + \frac{1}{n} \ln M(\epsilon, F, L_2(D_X)) \right] = O(n^{-2/(2+q)}).$$

Therefore the upper bound of Proposition 10.5 matches the lower bound of Corollary 10.1.

The method of sieves can be regarded as a regularization method for ERM. The optimality of using a sieve versus ERM on the original function class implies that ERM can easily overfit if not used properly. It is suitable only for certain function classes but not others.

The following example gives a simple but intuitive illustration of such overfitting phenomenon.

Example 10.3 *Consider the following function class*

$$F = \{f_0(x)\} \cup \{f_1(x) + \Delta f_1(x) : |\Delta f_1(x)| \leq 1/\sqrt{n}\},$$

where $f_1(x) = f_0(x) + n^{-1/4}$.

If we consider ERM with sieve $F_n = \{f_k(x) : k = 0, 1\}$, then we have a convergence rate no worse than $O(1/n)$.

However, we have a overfitting problem with ERM on F . To see this, we may consider the model

$$Y = f_0(X) + \epsilon, \quad \epsilon \sim N(0, 1),$$

and training data $(X_1, Y_1), \dots, (X_n, Y_n)$. Let $\delta_i = \Delta f_1(X_i)$ and $\epsilon_i = Y_i -$

$f_0(X_i)$. Then we have

$$\begin{aligned}
 & \sum_{i=1}^n \min_{|\delta_i| \leq 1/\sqrt{n}} (f_1(X_i) + \delta_i - Y_i)^2 - \sum_{i=1}^n (f_0(X_i) - Y_i)^2 \\
 &= \sum_{i=1}^n \min_{|\delta_i| \leq 1/\sqrt{n}} (n^{-1/4} + \delta_i - \epsilon_i)^2 - \sum_{i=1}^n \epsilon_i^2 \\
 &\leq \sqrt{n}(1 + n^{-1/4})^2 - 2n^{-1/4} \sum_{i=1}^n \epsilon_i - 2n^{-1/2} \sum_{i=1}^n |\epsilon_i|.
 \end{aligned}$$

When n is large, this quantity is negative with large probability. It implies that ERM gives an estimator $\hat{f}(x) = f_1(x) + \Delta f_1(x)$ with $|\Delta f(x)| \leq 1/\sqrt{n}$. This means that

$$\mathbf{E}_{S_n} \mathbf{E}_X (\hat{f}(X) - f(X))^2 \geq c/\sqrt{n}$$

for some $c > 0$. Note that this is a suboptimal rate.

In addition to the sieve method, one may also consider model aggregation which is studied in Section 12.3. The generalization bound there implies that using model aggregation on a sieve $F_n = \{f_1, \dots, f_N\}$, with $N = M(\epsilon, F, L_2(D_X))$, we have

$$\mathbf{E}_{S_n} Q(\hat{f}, f_*) = O\left(\inf_j Q(f_j, f_*)^2 + n^{-1} \ln M(\epsilon, F, L_2(D_X))\right).$$

This achieves the same rate as that of Proposition 10.5. It follows that the exponential aggregation algorithm is also an optimal algorithm for least squares regression. Moreover, exponential model aggregation is superior to empirical risk minimization for misspecified models [53, 6]. We do not consider misspecified models in the minimax rate analysis.

10.5 Exercises

1. Prove Proposition 10.1.
2. Consider the function class

$$F = \{w^\top x : \|w\|_2 \leq 1\}.$$

Consider empirical distribution $S_n = \{X_1, \dots, X_n\}$ so that $\|X_i\|_2 \leq 1$ for all i .

Derive an upper bound for the Gaussian complexity $G(F, S_n)$, and obtain an upper bound for $N(\epsilon, F, L_2(S_n))$ using Sudakov minoration.

3. Consider a function class defined on $[0, 1]$, defined as

$$F = \{f_0(x)\} \cup \{f_0(x) + 0.1n^{-r/3} + n^{-r/2}\Delta f_u(x) : u \in \{\pm 1\}^n\},$$

where $f_0(x) = 0$ and $\Delta f_u(x) = u_j : x \in [(j-1)/n, j/n]$, where $r \in (0, 1]$.

Let

$$Y = f_*(x) + \epsilon, \quad \epsilon \sim N(0, 1),$$

for $f_* \in F$, and X drawn uniformly from $[0, 1]$.

Consider least squares problem

$$Q(f, f') = \mathbf{E}_X(f(X) - f'(X))^2,$$

and consider n samples $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$.

- Find the minimax risk (up to a constant) for estimating f_* using these n samples.
- Show that there is a small enough constant c so that with probability at least 0.5, there are at least $0.1n$ buckets $[(j-1)/n, j/n]$ so that one and only one X_i belongs to $[(j-1)/n, j/n]$.
- Find a risk lower bound on the risk for the ERM method on F for sufficiently large n .

Chapter 11

Basic Concepts of Online Learning

In online learning, we consider a learning model that is different from that of the supervised learning, in that we make predictions sequentially and obtain feedbacks after predictions are made. In this section, we introduce this learning model as well as some first order algorithms.

11.1 Online Learning Model

We will use the notations that are consistent with earlier chapters on supervised learning.

In online learning, we consider observing $Z_t = (X_t, Y_t)$ one by one in a time sequence from $t = 1, 2, \dots$, and tries to make predictions for the next item in the sequence, and then obtain the feedback of the prediction.

The online learning learning model can be considered as a repeated game. For $t = 1, 2, \dots$,

- An adversary picks (X_t, Y_t) , and reveals X_t only.
- An online learning algorithm $\mathcal{A}$ predicts $\hat{f}_{t-1}(X_t)$.
- The value of Y_t is revealed and a loss $L(\hat{f}_{t-1}(X_t), Y_t)$ is computed.

The goal of online learning is to minimize the aggregated loss

$$\sum_{t=1}^T L(\hat{f}_{t-1}(X_t), Y_t).$$

In general, an online algorithm $\mathcal{A}$ picks a prediction model $f(x)$ from $\mathcal{F} = \{f(w, x) : w \in \Omega\}$, and by learning a model parameter $w_{t-1} \in \Omega$ at time t based on previously observed data $S_{t-1} = \{Z_1, \dots, Z_{t-1}\}$. That is, $w_{t-1} = \mathcal{A}(S_{t-1})$. The algorithm then makes the prediction $f(w_{t-1}, X_t)$. In

online learning, we are interested in the following inequality, referred to as *regret bound*, where the aggregated loss of an online algorithm is compared to the optimal aggregated loss:

$$\sum_{t=1}^T L(f(w_{t-1}, X_t), Y_t) - \inf_{w \in \Omega} \sum_{t=1}^T L(f(w, X_t), Y_t) \leq \epsilon_T. \quad (11.1)$$

In the online learning literature, the standard supervised learning setting is also referred to as *batch learning*. Regret bound analysis can be considered as a generalization of the oracle inequality in batch learning. We will show later that a regret bound directly implies an oracle inequality using a simple technique called online to batch conversion.

11.2 Perceptron Algorithms

In this section, we will analyze an online learning algorithm called Perceptron Algorithm.

Consider the binary classification problem with $Y \in \{\pm 1\}$, and linear functions

$$f(w, X) = w^\top X,$$

with prediction rule:

$$\begin{cases} 1 & f(w, X) \geq 0 \\ -1 & \text{otherwise} \end{cases}.$$

The loss function is binary classification error: $I(f(w, X)Y \leq 0)$. The Perceptron algorithm [68, 69, 63] is an algorithm which works with data coming in sequentially as in Algorithm 11.1. It is mistake-driven, and only update the weight when the prediction makes a mistake.

Algorithm 11.1: Perceptron Algorithm

Input: Sequence $(X_1, Y_1), \dots, (X_T, Y_T)$

Output: w_s

```

1 Let  $w_0 = 0$ 
2 for  $t = 1, 2, \dots, T$  do
3   Observe  $X_t$  and predict label  $\text{sign}(w_{t-1}^\top X_t)$ 
4   Observe  $Y_t$  and compute mistake  $I(w_{t-1}^\top X_t Y_t \leq 0)$ 
5   if  $w_{t-1}^\top X_t Y_t > 0$  then
6     // No mistake
7     Let  $w_t = w_{t-1}$ 
8   else
9     // A mistake is observed
10    Let  $w_t = w_{t-1} + X_t Y_t$ 
11 Randomly pick  $s$  from  $0$  to  $T - 1$ 
Return:  $w_s$ 
```

Note that the randomly choice of the returned predictor in the Perceptron algorithm is for the purpose of online to batch conversion, which we will discuss later. In the following, we first state the mistake bound of Perceptron as follows.

Theorem 11.1 *Consider the Perceptron Algorithm in Algorithm 11.1. Consider $\gamma > 0$ and weight vector w_* such that for all t*

$$w_*^\top X_t Y_t \geq \gamma.$$

Then we have the following mistake bound:

$$\sum_{t=1}^T I(w_{t-1}^\top X_t Y_t \leq 0) \leq \frac{\|w_*\|_2^2 \sup\{\|X_t\|_2^2\}}{\gamma^2}.$$

Proof Let $M = \sup\{\|X_t\|_2\}$, and let $\eta = \gamma/M^2$. Assume that we have a mistake at time step t , then we have

$$(\eta w_{t-1} - w_*)^\top X_t Y_t \leq 0 - w_*^\top X_t Y_t \leq -\gamma.$$

Therefore by taking

$$\begin{aligned} \|\eta w_t - w_*\|_2^2 &= \|\eta w_{t-1} + \eta X_t Y_t - w_*\|_2^2 \\ &= \|\eta w_{t-1} - w_*\|_2^2 + 2\eta(\eta w_{t-1} - w_*)^\top X_t Y_t + \eta^2 \|X_t\|_2^2 \\ &\leq \|\eta w_{t-1} - w_*\|_2^2 - 2\eta\gamma + \eta^2 M^2 \\ &\leq \|\eta w_{t-1} - w_*\|_2^2 - \frac{\gamma^2}{M^2}. \end{aligned}$$

Note also that $\|\eta w_t - w_*\|_2^2 = \|\eta w_{t-1} - w_*\|_2^2$ if there is no mistake at time step t . Therefore by summing over $t = 1$ to $t = t$, we obtain

$$0 \leq \|\eta w_t - w_*\|_2^2 \leq \|\eta w_0 - w_*\|_2^2 - \frac{\gamma^2}{M^2} k,$$

where k is the number of mistakes. This implies the bound. ■

The mistake bound can be regarded as an online version of the margin bound discussed in chapter 7, where empirical process techniques were used to obtain such margin bounds for ERM estimators. The technique required for analyzing the ERM method is much more complicated.

Using essentially the same proof, we can also obtain a mistake bound for multi-class classification problems [27]. Note that for ERM, the analysis of multi-class problems is more complex, as discussed in Section 7.4.

For multi-class prediction with q classes $y \in \{1, \dots, q\}$, we may use the notations of section 7.4, and consider a vector prediction function $f(x) \in \mathbb{R}^q$, with linear prediction model for class ℓ in (7.9) defined as:

$$f_\ell(x) = w^\top \psi(x, \ell).$$

The predicted class for each x is

$$\hat{y}(w, x) \in \arg \max_{\ell} w^\top \psi(x, \ell),$$

and the error (or mistake) for an instance x with true label y is

$$I(\hat{y}(w, x) \neq y).$$

Algorithm 11.2: Multi-Class Perceptron Algorithm

Input: Sequence $(X_1, Y_1), \dots, (X_T, Y_T)$

Output: w_s

```

1 Let  $w_0 = 0$ 
2 for  $t = 1, 2, \dots, T$  do
3   Observe  $X_t$  and predict label  $\hat{Y}_t \in \arg \max_{\ell} \{w_{t-1}^\top \psi(X_t, \ell)\}$ 
4   Observe  $Y_t$  and compute mistake  $I(\hat{Y}_t \neq Y_t)$ 
5   if  $\hat{Y}_t = Y_t$  then
6     // No mistake
7     Let  $w_t = w_{t-1}$ 
8   else
9     // A mistake is observed
10    Let  $w_t = w_{t-1} + [\psi(X_t, Y_t) - \psi(X_t, \hat{Y}_t)]$ 
11 Randomly pick  $s$  from  $0$  to  $T - 1$ 
    Return:  $w_s$ 

```

Theorem 11.2 Consider Algorithm 11.2. We have the following mistake bound:

$$\sum_{t=1}^T I(\hat{Y}_t \neq Y_t) \leq \inf_{\gamma > 0, w} \left[\sum_{t=1}^T 2 \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y_t} w^\top [\psi(X_t, Y_t) - \psi(X_t, \ell)] \right) + \frac{\|w\|_2^2 \sup\{\|\psi(X_t, Y_t) - \psi(X_t, Y_\ell)\|_2^2\}}{\gamma^2} \right].$$

Proof The proof is basically the same as that of the binary case. Given any $\gamma > 0$ and w . We let $\psi_t = \psi(X_t, Y_t) - \psi(X_t, \hat{Y}_t)$, $M = \sup\{\|\psi_t\|_2\}$, and $\eta = \gamma/M^2$. Assume that we have a mistake at time step t , then we have $\hat{Y}_t \neq Y_t$, and $w_{t-1}^\top \psi_t \leq 0$. It implies that

$$(\eta w_{t-1} - w_*)^\top \psi_t \leq 0 - w_*^\top \psi_t \leq \max(0, \gamma - w_*^\top \psi_t) - \gamma.$$

Therefore by taking

$$\begin{aligned}
\|\eta w_t - w_*\|_2^2 &= \|\eta w_{t-1} + \eta \psi_t - w_*\|_2^2 \\
&= \|\eta w_{t-1} - w_*\|_2^2 + 2\eta(\eta w_{t-1} - w_*)^\top \psi_t + \eta^2 \|\psi_t\|_2^2 \\
&\leq \|\eta w_{t-1} - w_*\|_2^2 + 2\eta \max(0, \gamma - w_*^\top \psi_t) - 2\eta\gamma + \eta^2 M^2 \\
&\leq \|\eta w_{t-1} - w_*\|_2^2 + 2\eta \max(0, \gamma - w_*^\top \psi_t) - \frac{\gamma^2}{M^2}.
\end{aligned}$$

Note also that $\|\eta w_t - w_*\|_2^2 = \|\eta w_{t-1} - w_*\|_2^2$ if there is no mistake at time step t . Therefore by summing over $t = 1$ to $t = t$, we obtain

$$0 \leq \|\eta w_T - w_*\|_2^2 \leq 2\eta \sum_{t=1}^T \max(0, \gamma - w_*^\top \psi_t) + \|\eta w_0 - w_*\|_2^2 - \frac{\gamma^2}{M^2} k,$$

where k is the number of mistakes. This implies the bound. $\blacksquare$

Although the above derivation requires that there is a margin $\gamma > 0$ so that $w_*^\top \psi(X, Y) \geq w_*^\top \psi(X, \ell) + \gamma$ for all $\ell \neq Y$. It is not difficult to see that we can also replace this margin γ by the average margin encountered in the Perceptron algorithm defined as $w_*^\top [\psi(X_t, Y_t) - \psi(X_t, \hat{Y}_t)]$ for $\hat{Y}_t \neq Y_t$.

11.3 Online to Batch Conversion

In online learning, each time, we train a model using historic data, and then test on the next datum. This naturally corresponds to the generalization performance.

If we assume that in online learning, the observed data are random, with $Z_t = (X_t, Y_t) \sim D$, then by taking expectation, we obtain the following expected generation bound from the Perceptron mistake bound of Theorem 11.2, with the prediction rule

$$\hat{y}(w, x) \in \arg \max_{\ell} w \cdot \psi(x, \ell)$$

chosen consistently as in Algorithm 11.2.

Corollary 11.1 *If Z_t are iid: $Z_t \sim D$. Then for Algorithm 11.2, we have*

$$\begin{aligned}
&\mathbf{E}_{S_{T-1}} \sum_{t=1}^T \mathbf{E}_{Z \in D} I(\hat{y}(w_{t-1}, X) \neq Y) \\
&\leq \inf_{\gamma > 0, w} \left[2T \mathbf{E}_Z \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y_t} w^\top [\psi(X, Y) - \psi(X, \ell)] \right) \right. \\
&\quad \left. + \frac{\|w\|_2^2 \sup \{ \|\psi(X, Y) - \psi(X, Y')\|_2^2 \}}{\gamma^2} \right],
\end{aligned}$$

This implies that for s chosen in Algorithm 11.2, we have

$$\begin{aligned} & \mathbf{E}_{S_{t-1}} \mathbf{E}_s \mathbf{E}_{Z \in D} I(\hat{y}(w_s, X) \neq Y) \\ & \leq \inf_{\gamma > 0, w} \left[2 \mathbf{E}_Z \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y_t} w^\top [\psi(X, Y) - \psi(X, \ell)] \right) \right. \\ & \quad \left. + \frac{\|w\|_2^2 \sup\{\|\psi(X, Y) - \psi(X, Y')\|_2^2\}}{\gamma^2 T} \right]. \end{aligned}$$

Proof We simply note that

$$\mathbf{E}_{S_{T-1}} \sum_{t=1}^T \mathbf{E}_{Z \in D} I(\hat{y}(w_{t-1}, X) \neq Y) = \mathbf{E}_{S_T} \sum_{t=1}^T I(\hat{y}(w_{t-1}, X_t) \neq Y_t),$$

and

$$\begin{aligned} & \sum_{t=1}^T 2 \mathbf{E}_Z \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y_t} w^\top [\psi(X, Y) - \psi(X, \ell)] \right) \\ & = \mathbf{E}_{S_T} \sum_{t=1}^T 2 \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y_t} w^\top [\psi(X_t, Y_t) - \psi(X_t, \ell)] \right). \end{aligned}$$

Now the result is a straight-forward application of Theorem 11.2. ■

More generally, consider an online algorithm $\mathcal{A}$ such that it returns a weight vector $w_t = \mathcal{A}(S_t)$ with $S_t = \{Z_1, \dots, Z_t\}$. Assume that we have a regret bound of the general form:

$$\sum_{t=1}^T g_{w_{t-1}}(Z_t) \leq \epsilon(S_T). \quad (11.2)$$

For example, for the Perceptron algorithm, we may let

$$g_w(Z) = I(\hat{y}(w, X) \neq Y) - 2 \max \left(0, 1 - \gamma^{-1} \min_{\ell \neq Y} w^\top [\psi(X, Y) - \psi(X, \ell)] \right).$$

By taking expectations, we obtain an expected generalization bound of

$$\mathbf{E}_{S_T} \sum_{t=1}^T \mathbf{E}_Z g_{w_{t-1}}(Z) \leq \mathbf{E}_{S_T} \epsilon(S_T).$$

This implies that if we select s uniformly from 0 to $T-1$, then

$$\mathbf{E}_{S_T} \mathbf{E}_s \mathbf{E}_Z g_{w_s}(Z) \leq \mathbf{E}_{S_T} \epsilon(S_T) / T.$$

It is also possible to obtain a high probability result of the following form: with probability at least $1 - \delta$ over S_T ,

$$\sum_{t=1}^T \mathbf{E}_Z g_{w_{t-1}}(Z) \leq \sum_{t=1}^T g_{w_{t-1}}(Z_t) + \epsilon(\delta). \quad (11.3)$$

Combine this bound with (11.2), we obtain the following probability bound for the randomized estimator s , uniformly chosen from 0 to $T - 1$. With probability at least $1 - \delta$ over S_T :

$$\mathbf{E}_s \mathbf{E}_Z g_{w_s}(Z) \leq \frac{\epsilon(\delta) + \epsilon(S_T)}{T}.$$

It is known [97] that by using the technique of logarithmic moment generating functions, we can obtain bounds of (11.3) similar to those of Chapter 2. We use the following notation

$$\xi_t(S_t) = g_{w_{t-1}}(Z_t).$$

Then we have the following lemma.

Lemma 11.1 *Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$. We have*

$$\mathbf{E}_{S_n} \exp \left(\sum_{i=1}^n \xi_i - \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{\xi_i} \right) = 1.$$

Proof We prove the lemma by induction on n . When $n = 1$, the equality is easy to verify. Assume that the claim holds for all $n \leq k$. Now for $n = k + 1$, we have

$$\begin{aligned} & \mathbf{E}_{Z_1, \dots, Z_n} \exp \left(\sum_{i=1}^n \xi_i - \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{\xi_i} \right) \\ &= \mathbf{E}_{Z_1, \dots, Z_{n-1}} \left[\exp \left(\sum_{i=1}^{n-1} \xi_i - \sum_{i=1}^{n-1} \ln \mathbf{E}_{Z_i} e^{\xi_i} \right) \mathbf{E}_{Z_n} \exp(\xi_n - \ln \mathbf{E}_{Z_n} e^{\xi_n}) \right] \\ &= \mathbf{E}_{Z_1, \dots, Z_{n-1}} \exp \left(\sum_{i=1}^{n-1} \xi_i - \sum_{i=1}^{n-1} \ln \mathbf{E}_{Z_i} e^{\xi_i} \right) = 1. \end{aligned}$$

Note that the last equation follows from the induction hypothesis. ■

The following result is a direct consequence of Lemma 11.1, which generalizes Theorem 2.2 for sum of independent variables to the so-called *martingales*, which correspond to the sum of random variables that can depend on previous observations with zero conditional-mean, such as $\xi_i(S_i) - \mathbf{E}_{Z_i} \xi_i(S_i)$.

Theorem 11.3 Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$. We have for any $\delta \in (0, 1)$ and $\lambda > 0$:

$$\Pr \left[-\sum_{i=1}^n \xi_i \geq \frac{\ln(1/\delta)}{\lambda} + \frac{1}{\lambda} \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{-\lambda \xi_i} \right] \leq \delta.$$

Then for any $n \forall z$, and event A :

$$\ln \Pr \left[\sum_{i=1}^n \xi_i \leq z \text{ and } S_n \in A \right] \leq \inf_{\lambda > 0} \sup_{S_n \in A} \left[\lambda z + \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{-\lambda \xi_i} \right].$$

Proof Let $\xi(\lambda) = -\sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{-\lambda \xi_i} - \lambda \sum_{i=1}^n \xi_i$, then we have from Lemma 11.1: $\mathbf{E} e^{\xi(\lambda)} = 1$. Now $\forall t(S_n)$, we have

$$\Pr(\xi(\lambda) \geq -t(S_n) \text{ \& } S_n \in A) \inf_{S_n \in A} e^{-t(S_n)} \leq \mathbf{E} e^{\xi(\lambda)} = 1.$$

Therefore we obtain

$$\ln \Pr \left[-\lambda \sum_{i=1}^n \xi_i \geq -t(S_n) + \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{-\lambda \xi_i} \text{ \& } S_n \in A \right] \leq \sup_{S_n \in A} t(S_n).$$

Let $t(S_n) = \ln \delta$, we obtain the first inequality. Let

$$t(S_n) = \lambda z + \sum_{i=1}^n \ln \mathbf{E}_{Z_i} e^{-\lambda \xi_i},$$

we obtain the second inequality. ■

Using the logarithmic moment generating functions of Chapter 2, we obtain the following results.

Theorem 11.4 (Martingale sub-Gaussian Inequality) Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$. Assume each ξ_i is sub-Gaussian with respect to Z_i :

$$\ln \mathbf{E}_{Z_i} e^{\lambda \xi_i} \leq \lambda \mathbf{E}_{Z_i} \xi_i + \frac{\lambda^2 \sigma_i^2}{2}$$

for some σ_i that may depend on S_{i-1} . Then with probability at least $1 - \delta$,

$$\frac{1}{n} \sum_{i=1}^n \mathbf{E}_{Z_i} \xi_i \leq \frac{1}{n} \sum_{i=1}^n \xi_i + \sigma \sqrt{\frac{2 \ln(1/\delta)}{n}} \text{ or } \frac{1}{n} \sum_{i=1}^n \sigma_i^2 > \sigma^2,$$

where $\sigma > 0$ is a constant.

Proof We can let $A = \{S_n : \sum_{i=1}^n \sigma_i^2 \leq n\sigma^2\}$, and apply Theorem 11.3 with sub-Gaussian estimate of the rate function $I(\cdot)$ from Chapter 2. ■

A direct consequence of the sub-Gaussian inequality is the additive Chernoff bound, which can be stated below. It is often referred to as the Azuma's Inequality [11].

Theorem 11.5 (Azuma' Inequality) *Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$. If for each i : $\sup \xi_i - \inf \xi_i \leq M_i$ for some constant M_i , then with probability at least $1 - \delta$,*

$$\frac{1}{n} \sum_{i=1}^n \mathbf{E}_{Z_i} \xi_i \leq \frac{1}{n} \sum_{i=1}^n \xi_i + \sqrt{\frac{\sum_{i=1}^n M_i^2 \ln(1/\delta)}{2n^2}}.$$

For the analogy of multiplicative Chernoff bound, we can obtain a weaker version than Corollary 2.1.

Theorem 11.6 *Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$ such that $\xi_i \in [0, 1]$ for all i . We have with probability at least $1 - \delta$:*

$$\frac{1}{n} \sum_{i=1}^n \mathbf{E}_{Z_i} \xi_i \leq \frac{\lambda}{1 - e^{-\lambda}} \frac{1}{n} \sum_{i=1}^n \xi_i + \frac{\ln(1/\delta)}{(1 - e^{-\lambda}) n}.$$

Proof We obtain from Lemma 2.1 and Theorem 11.3 that with probability at least $1 - \delta$

$$\begin{aligned} -\sum_{i=1}^n \xi_i &\leq \frac{\ln(1/\delta)}{\lambda} + \frac{1}{\lambda} \sum_{i=1}^n \ln(1 + (e^{-\lambda} - 1) \mathbf{E}_{Z_i} \xi_i) \\ &\leq \frac{\ln(1/\delta)}{\lambda} + \frac{1}{\lambda} \sum_{i=1}^n (e^{-\lambda} - 1) \mathbf{E}_{Z_i} \xi_i. \end{aligned}$$

This implies the bound. ■

We can also obtain the following Bennett/Bernstein Inequality. This inequality is often referred to as the Freedman's Inequality [36].

Theorem 11.7 (Freedman's Inequality) *Consider a sequence of random functionals $\xi_1(S_1), \dots, \xi_n(S_n)$. Assume that $\xi_i \geq \mathbf{E}_{Z_i} \xi_i - b$ for some constant $b > 0$. Then with probability at least $1 - \delta$:*

$$\frac{1}{n} \sum_{i=1}^n \mathbf{E}_{Z_i} \xi_i \leq \frac{1}{n} \sum_{i=1}^n \xi_i + \sigma \sqrt{\frac{2 \ln(1/\delta)}{n}} + \frac{2b \ln(1/\delta)}{3n} \text{ or } \sum_{i=1}^n \mathbf{Var}_{Z_i}(\xi_i) > n\sigma^2,$$

where σ is a constant.

Example 11.1 Consider Perceptron in Algorithm 11.1, with mistake bound in Theorem 11.1. We have the following generalization error bound by using Theorem 11.6 with $\xi_i = I(w_{i-1}^\top X_i Y_i \leq 0)$. Consider any $\lambda > 0$. With probability at least $1 - \delta$:

$$\mathbf{E}_s \mathbf{E}_D I(w_s^\top XY \leq 0) \leq \frac{\lambda}{1 - e^{-\lambda}} \frac{\|w_*\|_2^2 \sup\{\|X\|_2^2\}}{\gamma^2} + \frac{\ln(1/\delta)}{(1 - e^{-\lambda}) n}.$$

Since now the right hand side does not depend on the data, we can optimize it over λ .

11.4 Online Convex Optimization

One can extend the analysis of the Perceptron algorithms to general convex loss functions, leading to the so-called online convex optimization, which was introduced into machine learning in [103]. The stochastic optimization setting was investigated by [96]. The proof techniques used there are similar to the those of non-stochastic gradient descent methods in the optimization literature [62].

Definition 11.1 A function $\ell(w) : \Omega \rightarrow \mathbb{R}$ is convex if for all $\lambda \in (0, 1)$ and $w_1, w_2 \in \Omega$:

$$\ell(\lambda w_1 + (1 - \lambda)w_2) \leq \lambda \ell(w_1) + (1 - \lambda)\ell(w_2).$$

One important property of convex function is that

$$\ell(w') \geq \ell(w) + \nabla \ell(w)^\top (w' - w),$$

which can be used to obtain regret bound.

A general first-order algorithm for online convex optimization can be found in Algorithm 11.3, and its analysis is given in Theorem 11.8.

Algorithm 11.3: Online Convex Optimization

Input: Sequence of loss functions $\ell_1, \dots, \ell_T$ defined on Ω

Output: $\hat{w}$

1 Let $w_0 = 0$

2 **for** $t = 1, 2, \dots, T$ **do**

3 Observe loss $\ell_t(w_{t-1})$

4 Let $\tilde{w}_t = w_{t-1} - \eta_t \nabla \ell_t(w_{t-1})$

5 Let $w_t = \arg \min_{w \in \Omega} \|w - \tilde{w}_t\|_2^2$

6 Let $\hat{w} = T^{-1} \sum_{t=1}^T w_{t-1}$ or $\hat{w} = w_s$ for a random s from 0 to $T - 1$

Return: $\hat{w}$

Theorem 11.8 Consider convex loss functions $\ell_t(w) : \Omega \rightarrow \mathbb{R}$, which are G -Lipschitz (that is, $\|\nabla \ell_t(w)\|_2 \leq G$). If we let $\eta_t = \eta > 0$ be a constant in Algorithm 11.3. Then for all $w \in \Omega$, we have

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \sum_{t=1}^T \ell_t(w) + \frac{\|w_0 - w\|_2^2}{2\eta} + \frac{\eta T}{2} G^2.$$

Proof We have the following inequality:

$$\begin{aligned} \|\tilde{w}_t - w\|_2^2 &= \|w_{t-1} - w - \eta \nabla \ell_t(w_{t-1})\|_2^2 \\ &= \|w_{t-1} - w\|_2^2 - 2\eta \nabla \ell_t(w_{t-1})^\top (w_{t-1} - w) + \eta^2 \|\nabla \ell_t(w_{t-1})\|_2^2 \\ &\leq \|w_{t-1} - w\|_2^2 - 2\eta \nabla \ell_t(w_{t-1})^\top (w_{t-1} - w) + G^2 \eta^2 \\ &\leq \|w_{t-1} - w\|_2^2 - 2\eta [\ell_t(w_{t-1}) - \ell_t(w)] + G^2 \eta^2, \end{aligned}$$

where the first inequality used the Lipschitz condition, and the second inequality used the convexity condition.

Since $w_t \in \Omega$ is the projection of $\tilde{w}_t$ onto Ω and $w \in \Omega$. We also have

$$\|w_t - w\|_2^2 \leq \|\tilde{w}_t - w\|_2^2.$$

Therefore, we have

$$\|w_t - w\|_2^2 \leq \|w_{t-1} - w\|_2^2 - 2\eta [\ell_t(w_{t-1}) - \ell_t(w)] + G^2 \eta^2.$$

Now we may sum over $t = 1$ to $t = T$, and obtain

$$\|w_T - w\|_2^2 \leq \|w_0 - w\|_2^2 - 2\eta \sum_{t=1}^T [\ell_t(w_{t-1}) - \ell_t(w)] + TG^2 \eta^2.$$

Rearrange the terms, we obtain the desired bound. ■

Compared to the Perceptron mistake bound, the analysis technique is quite similar. More generally, if we can find ℓ_t , $\tilde{\ell}_t$, and g_t so that $g_t^\top (w - w_{t-1}) \leq \tilde{\ell}_t(w) - \ell_t(w_{t-1})$, then we may obtain a regret bound using update rule $w_t = w_{t-1} - \eta_t g_t$. In online convex optimization, we take $\tilde{\ell}_t(w) = \ell_t(w)$ and $g_t = \nabla \ell_t(w_{t-1})$. In Theorem 11.1, we take $\ell_t(w) = \gamma I(w^\top X_t Y_t \leq 0)$, $\tilde{\ell}_t(w) = \max(0, \gamma - w^\top X_t Y_t)$, and $g_t = -I(w_{t-1}^\top X_t Y_t \leq 0) X_t Y_t$.

We may also obtain an oracle inequality for Algorithm 11.3 as follows.

Theorem 11.9 Consider loss function $g_w(Z) \in [0, M]$ with $Z \sim D$, and $w \in \Omega$, where Ω is a convex set. Assume that $g_w(Z)$ is convex and G -Lipschitz with respect to w . Let $Z_1, \dots, Z_T \sim D$ be independent samples,

and consider $\hat{w}$ obtained from Algorithm 11.3, with $\ell_i(w) = g_w(Z_i)$ and $\eta_t = \eta > 0$. Then with probability at least $1 - \delta$,

$$\mathbf{E}_D g_{\hat{w}}(Z) \leq \inf_{w \in \Omega} \left[\mathbf{E}_D g_w(Z) + \frac{1}{2\eta T} \|w - w_0\|_2^2 \right] + \frac{\eta}{2} G + M \sqrt{\frac{2 \ln(2/\delta)}{T}}.$$

Proof Note that the convexity and Jensen's inequality implies that

$$\mathbf{E}_Z g_{\hat{w}}(Z) \leq \frac{1}{T} \sum_{t=1}^T \mathbf{E}_{Z_t} g_{w_{t-1}}(Z_t).$$

Moreover, using the Azuma's inequality, we have with probability $1 - \delta/2$,

$$\frac{1}{T} \sum_{t=1}^T \mathbf{E}_{Z_t} g_{w_{t-1}}(Z_t) \leq \frac{1}{T} \sum_{t=1}^T g_{w_{t-1}}(Z_t) + M \sqrt{\frac{\ln(2/\delta)}{2T}}.$$

Using Theorem 11.8, we obtain

$$\frac{1}{T} \sum_{t=1}^T g_{w_{t-1}}(Z_t) \leq \frac{1}{T} \sum_{t=1}^T g_w(Z_t) + \frac{\|w_0 - w\|_2^2}{2\eta T} + \frac{\eta}{2} G^2.$$

Using Chernoff bound, we have with probability $1 - \delta/2$:

$$\frac{1}{T} \sum_{t=1}^T g_w(Z_t) \leq \mathbf{E}_D g_w(Z) + M \sqrt{\frac{\ln(2/\delta)}{2T}}.$$

By taking the union bound, and combine the above four inequalities, we obtain the following. With probability at least $1 - \delta$:

$$\begin{aligned} \mathbf{E}_D g_{\hat{w}}(Z) &\leq \frac{1}{T} \sum_{t=1}^T \mathbf{E}_{Z_t} g_{w_{t-1}}(Z_t) \\ &\leq \frac{1}{T} \sum_{t=1}^T g_{w_{t-1}}(Z_t) + M \sqrt{\frac{\ln(2/\delta)}{2T}} \\ &\leq \frac{1}{T} \sum_{t=1}^T g_w(Z_t) + \frac{\|w_0 - w\|_2^2}{2\eta T} + \frac{\eta}{2} G^2 + M \sqrt{\frac{\ln(2/\delta)}{2T}} \\ &\leq \mathbf{E}_D g_w(Z) + \frac{\|w_0 - w\|_2^2}{2\eta T} + \frac{\eta}{2} G^2 + M \sqrt{\frac{2 \ln(2/\delta)}{T}}. \end{aligned}$$

This proves the desired result. ■

If we take $\eta = O(1/\sqrt{T})$, then we obtain a convergence result of $O(1/\sqrt{T})$ in Theorem 11.9. This result can be compared to that of Corollary 7.4, which has a similar convergence rate if we set $\lambda = 1/(\eta T)$.

In Corollary 7.4, the loss function does not have to be convex, and the theorem applies to ERM, which implies the following bound for the ERM estimator:

$$\begin{aligned} \mathbf{E}_D g_{\hat{w}}(Z) &\leq \mathbf{E}_D g_w(Z) + \frac{\lambda}{2} \|w\|_2^2 \\ &\quad + O\left(\sqrt{\frac{\ln((\lambda + B^2)/(\delta\lambda))}{n}}\right) + O\left(\frac{B^2 \ln((\lambda + B^2)/(\delta\lambda))}{\lambda n}\right). \end{aligned}$$

This result is similar to that of Theorem 11.9 with $\eta = 1/(\lambda T)$.

However, the result in Theorem 11.9 does not apply to ERM, but rather the specific online learning procedure for convex functions described in Algorithm 11.3. We note that the definition of Lipschitz are different, with $G \leq \gamma B$, where γ is the Lipschitz of the loss function, and $B = \sup\{\|X\|_2\}$.

In addition, one may regard the regret bound in the online learning analysis as an analogy of the Rademacher complexity bound for ERM estimator. Azuma's inequality is the counterpart of the McDiarmid's Inequality to obtain the concentration results.

For vector valued functions, the Rademacher complexity analysis is trickier unless one works with covering numbers. This is due to the fact that the corresponding comparison lemma based on loss Lipschitz assumption may depend on the dimensionality of the vector function, as discussed in Section 7.4. However, the Lipschitz parameter of $g_w(\cdot)$ in Theorem 11.9 is with respect to the model parameter, this issue does not exist. In particular, we may directly analyze the structured-SVM loss of Example 7.6 without much difficulty.

Example 11.2 Consider the structured-SVM loss of Example 7.6, where $g_w(z) = \max_{\ell} [\gamma(y, \ell) - w \cdot (\psi(x, y) - \psi(x, \ell))]$. If $\|\psi(x, y) - \psi(x, \ell)\|_2 \leq B$, then we can take $G = B$ in Theorem 11.9. Using Theorem 11.8, we obtain

$$\frac{1}{T} \sum_{t=1}^T g_{w_{t-1}}(Z_t) \leq \frac{1}{T} \sum_{t=1}^T g_w(Z_t) + \frac{\|w_0 - w\|_2^2}{2\eta T} + \frac{\eta}{2} B^2.$$

By taking expectation, and use Jensen's inequality with $\lambda = 1/(\eta T)$, we obtain

$$\mathbf{E}_{S_T} \mathbf{E}_D g_{\hat{w}}(Z) \leq \mathbf{E}_D g_w(Z) + \frac{\lambda}{2} \|w_0 - w\|_2^2 + \frac{1}{2\lambda T} B^2.$$

For L_2 regularization (or kernel methods), one can obtain a better bound using strong convexity.

Definition 11.2 A convex function $\ell(w)$ is λ strongly convex for some $\lambda > 0$ if

$$\ell(w') \geq \ell(w) + \nabla \ell(w)^\top (w' - w) + \frac{\lambda}{2} \|w' - w\|_2^2.$$

Observe that for regularized loss, we take

$$\ell_t(w) = g_w(Z_t) + \frac{\lambda}{2} \|w - w_0\|_2^2, \quad (11.4)$$

and $g_w(z)$ is convex in w , then $\ell_t(w)$ is λ strongly convex.

The following result holds for strongly convex loss functions, and the learning rate has been proposed to solve SVMs [76].

Theorem 11.10 Consider convex loss functions $\ell_t(w) : \Omega \rightarrow \mathbb{R}$, which are G -Lipschitz (that is, $\|\nabla \ell_t(w)\|_2 \leq G$) and λ strongly convex. If we let $\eta_t = 1/(\lambda t) > 0$ in Algorithm 11.3, then for all $w \in \Omega$, we have

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \sum_{t=1}^T \ell_t(w) + \frac{1 + \ln T}{2\lambda} G^2.$$

Proof Similar to the proof of Theorem 11.8, we have

$$\begin{aligned} \|w_t - w\|_2^2 &\leq \|\tilde{w}_t - w\|_2^2 \\ &= \|w_{t-1} - w - \eta_t \nabla \ell_t(w_{t-1})\|_2^2 \\ &= \|w_{t-1} - w\|_2^2 - 2\eta_t \nabla \ell_t(w_{t-1})^\top (w_{t-1} - w) + \eta_t^2 \|\nabla \ell_t(w_{t-1})\|_2^2 \\ &\leq \|w_{t-1} - w\|_2^2 + 2\eta_t [\ell_t(w) - \ell_t(w_{t-1})] - \eta_t \lambda \|w_{t-1} - w\|_2^2 + G^2 \eta_t^2 \\ &= (1 - \lambda \eta_t) \|w_{t-1} - w\|_2^2 + 2\eta_t [\ell_t(w) - \ell_t(w_{t-1})] + G^2 \eta_t^2, \end{aligned}$$

where strong-convexity is used to derive the second inequality. Note that $1 - \eta_t \lambda = \eta_t / \eta_{t-1}$ and we take $1/\eta_0 = 0$. This implies that

$$\eta_t^{-1} \|w_t - w\|_2^2 \leq \eta_{t-1}^{-1} \|w_{t-1} - w\|_2^2 + 2[\ell_t(w) - \ell_t(w_{t-1})] + G^2 \eta_t.$$

By summing over $t = 1$ to $t = T$, we obtain

$$\eta_T^{-1} \|w_T - w\|_2^2 \leq \eta_0^{-1} \|w_0 - w\|_2^2 + 2 \sum_{t=1}^T [\ell_t(w) - \ell_t(w_{t-1})] + G^2 \sum_{t=1}^T \frac{1}{\lambda t}.$$

Using $\sum_{t=1}^T (1/t) \leq 1 + \ln T$, we obtain the desired bound. ■

It is possible to remove the $\ln T$ factor if we use weighted regret. This has been discussed in [78]. For example, we have the following result.

Theorem 11.11 Consider convex loss functions $\ell_t(w) : \Omega \rightarrow \mathbb{R}$, which are G -Lipschitz (that is, $\|\nabla \ell_t(w)\|_2 \leq G$) and λ strongly convex. If we let $\eta_t = 2/(\lambda(t+1)) > 0$ in Algorithm 11.3, then for all $w \in \Omega$, we have

$$\sum_{t=1}^T \frac{2(t+1)}{T(T+3)} \ell_t(w_{t-1}) \leq \sum_{t=1}^T \frac{2(t+1)}{T(T+3)} \ell_t(w) + \frac{2G^2}{\lambda(T+3)}.$$

Proof As in the proof of Theorem 11.10, we have

$$\|w_t - w\|_2^2 \leq (1 - \eta_t \lambda) \|w_{t-1} - w\|_2^2 + 2\eta_t [\ell_t(w) - \ell_t(w_{t-1})] + G^2 \eta_t^2.$$

This implies that $\eta_t^{-2}(1 - \eta_t \lambda) \leq \eta_{t-1}^{-2}$, where we set $\eta_0^{-2} = 0$:

$$\eta_t^{-2} \|w_t - w\|_2^2 \leq \eta_{t-1}^{-2} \|w_{t-1} - w\|_2^2 + 2\eta_t^{-1} [\ell_t(w) - \ell_t(w_{t-1})] + G^2.$$

By summing over $t = 1$ to $t = T$, we obtain

$$\eta_T^{-2} \|w_T - w\|_2^2 \leq \eta_0^{-2} \|w_0 - w\|_2^2 + 2 \sum_{t=1}^T \eta_t^{-1} [\ell_t(w) - \ell_t(w_{t-1})] + G^2 T.$$

This leads to the bound. ■

The following expected oracle inequality is a straight-forward result of Theorem 11.10 and Theorem 11.11.

Corollary 11.2 Consider the regularized loss function (11.4) with $w_0 = 0$, where $g_w(z)$ is convex in w , and G Lipschitz in w . Moreover assume that $d(\Omega) = \sup\{\|w\|_2 : w \in \Omega\}$. If $Z_1, \dots, Z_T \sim D$ are independent samples, then we can obtain the following expected oracle inequality for Algorithm 11.3 if we take learning rate in Theorem 11.10:

$$\mathbf{E}_{S_T} \mathbf{E}_{Dg_{\hat{w}}}(Z) + \frac{\lambda}{2} \|\hat{w}\|_2^2 \leq \inf_{w \in \Omega} \left[\mathbf{E}_{Dg_w}(Z) + \frac{\lambda}{2} \|w\|_2^2 \right] + \frac{\ln(eT)}{2\lambda T} [G + \lambda d(\Omega)]^2.$$

We can also obtain the following expected oracle inequality for Algorithm 11.3 if we take learning rate in Theorem 11.11 with

$$\hat{w}' = \sum_{t=1}^T \frac{2(t+1)}{T^2 + 3T} w_{t-1},$$

then

$$\mathbf{E}_{S_T} \mathbf{E}_{Dg_{\hat{w}'}}(Z) + \frac{\lambda}{2} \|\hat{w}'\|_2^2 \leq \inf_{w \in \Omega} \left[\mathbf{E}_{Dg_w}(Z) + \frac{\lambda}{2} \|w\|_2^2 \right] + \frac{2[G + \lambda d(\Omega)]^2}{\lambda(T+3)}.$$

Proof Note that $\ell_t(w)$ is λ -strongly convex in w . Moreover, it is $G + \lambda d(\Omega)$ Lipschitz. We simply take expectation and apply Jensen's inequality to obtain the desired bounds. ■

In order to obtain concentration bounds, or obtain an oracle inequality without the assumption of $d(\Omega)$, then one can apply the technique of sample dependent bounds similar to the analysis of kernel methods as in Corollary 7.3 and Corollary 7.4.

11.5 Exercises

1. In Freedman's inequality, we want to remove the condition

$$\text{or } \sum_{i=1}^n \mathbf{Var}_{Z_i}(\xi_i) > n\sigma^2$$

by deriving a data-dependent probability bound that depends on the empirical conditional standard deviation (instead of a predefined σ) as

$$\hat{\sigma} = \sqrt{\frac{1}{n} \sum_{i=1}^n \mathbf{Var}_{Z_i}(\xi_i)}.$$

- Show how to derive such an inequality.
- Apply the above result to $\xi_i \in [0, b]$ that satisfies the variance condition of Definition 3.4:

$$\mathbf{Var}_{Z_i}(\xi_i) \leq \beta \mathbf{E}_{Z_i} \xi_i.$$

Obtain a probability inequality to upper bound $\sum_{i=1}^n \mathbf{E}_{Z_i} \xi_i$ in terms of $\sum_{i=1}^n \xi_i$, with probability at least $1 - \delta$.

2. Consider Corollary 11.2. Show that even if $d(\Omega) = \infty$, we can still derive an oracle inequality for $\hat{w}$ of the form

$$\mathbf{E}_{S_T} \mathbf{E}_D g_{\hat{w}}(Z) \leq \inf_{w \in \Omega} \left[\mathbf{E}_D g_w(Z) + \frac{\lambda}{2} \|w\|_2^2 \right] + \epsilon_T. \quad (11.5)$$

Derive such a bound.

3. For the previous problem, obtain a large probability statement of an oracle inequality which is of the form (11.5).

Hint: First derive a sample dependent version of the sub-Gaussian inequality with boundedness condition $|g_{\hat{w}_{t-1}}(z) - g_0(z)| \leq G \|\hat{w}_{t-1}\|_2$, where the concentration term depends on the sample dependent quantity

$$O \left(G \sqrt{\sum_{t=1}^T \|w_{t-1}\|_2^2} \right).$$

Then apply the technique used in Corollary 7.4 to obtain the desired oracle inequality.

Chapter 12

Online Aggregation and Second Order Algorithms

In Chapter 11, we introduced the basic definitions of online learning, and analyzed a few first order algorithms. In this chapter, we consider a number of online learning algorithms that are more advanced, and inherently exploit second order information.

12.1 Bayesian Posterior Averaging

Consider conditional probability density estimation with log-loss (negative-log-likelihood), where the loss function is

$$g_w(Z) = -\ln p(Y|w, X).$$

Although we consider conditional probability models for online learning, in the regret bound analysis, we do not have to assume that the true conditional model of $P(Y|X)$ is given by $p(Y|w, X)$ for some w .

Example 12.1 For discrete $y \in \{1, \dots, K\}$, we have

$$P(y|w, x) = \frac{\exp(f_y(w, x))}{\sum_{k=1}^K \exp(f_k(w, x))},$$

and

$$g_w(Z) = -f_y(w, x) + \ln \sum_{k=1}^K \exp(f_k(w, x)).$$

Example 12.2 For least squares regression with noise variance σ^2 , we may have

$$P(y|w, x) = \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{(y - f(w, x))^2}{2\sigma^2}\right),$$

and

$$g_w(Z) = \frac{(y - f(w, x))^2}{2\sigma^2} + \ln(\sqrt{2\pi}\sigma).$$

We may also consider the noise as part of the model parameter, and let

$$P(y|[w, \sigma], x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y - f(w, x))^2}{2\sigma^2}\right)$$

$$g_{[w, \sigma]}(Z) = \frac{(y - f(w, x))^2}{2\sigma^2} + \ln(\sqrt{2\pi}\sigma).$$

In the Bayesian setting, the optimal statistical estimation of posterior density for log-loss is the Bayesian posterior average estimator. Here we consider a prior $p_0(w)$ on Ω . Given the training data $S_n = \{Z_1, \dots, Z_n\}$, the posterior distribution is

$$p(w|S_n) = \frac{\prod_{i=1}^n p(Y_i|w, X_i)p_0(w)}{\int_{\Omega} \prod_{i=1}^n p(Y_i|w', X_i)p_0(w') dw'}. \quad (12.1)$$

Unlike ERM, the Bayesian posterior averaging method does not use $p(y|\hat{w}, x)$ for a specific $\hat{w} \in \Omega$ that is estimated from the training data, but instead use the averaged probability estimate over the posterior as follows:

$$\hat{p}(y|x, S_n) = \int_{\Omega} p(y|w, x)p(w|S_n) dw. \quad (12.2)$$

The following proposition is useful. We leave it as an exercise.

Proposition 12.1 *For all function $\phi(w)$ define on Ω , we have*

$$-\ln \mathbf{E}_{w \sim p_0} \exp(-\phi(w)) = \inf_q \left[\mathbf{E}_{w \sim q} \phi(w) + \mathbf{E}_{w \sim q} \frac{q(w)}{p_0(w)} \right],$$

where $q(w)$ is a probability distribution over Ω . The optimal solution is achieved at

$$\hat{q}(w) = \frac{\exp(-\phi(w))p_0(w)}{\int \exp(-\phi(w))p_0(w) dw}.$$

In the online setting, we have the following result, which may be regarded as the general version of regret bound for the Bayesian posterior model averaging method, as in [93].

Theorem 12.1 *We have*

$$-\sum_{t=1}^T \ln \hat{p}(Y_t, |X_t, S_{t-1}) = -\ln \mathbf{E}_{w \sim p_0} \prod_{i=1}^T p(Y_i|w, X_i)$$

$$= \inf_q \left[-\mathbf{E}_{w \sim q} \sum_{t=1}^T \ln p(Y_t|w, X_t) + \mathbf{E}_{w \sim q} \ln \frac{q(w)}{p_0(w)} \right],$$

where $q(\cdot)$ is a probability distribution over Ω .

Proof We have

$$\begin{aligned}
\ln \hat{p}(Y_t, |X_t, S_{t-1}) &= \ln \int_{\Omega} p(Y_t|w, X_t) p(w|S_{t-1}) dw \\
&= \ln \frac{\int_{\Omega} \prod_{i=1}^t p(Y_i|w, X_i) p_0(w) dw}{\int_{\Omega} \prod_{i=1}^{t-1} p(Y_i|w', X_i) p_0(w') dw'} \\
&= \ln \mathbf{E}_{w \in p_0} \prod_{i=1}^t p(Y_i|w, X_i) - \ln \mathbf{E}_{w \in p_0} \prod_{i=1}^{t-1} p(Y_i|w, X_i).
\end{aligned}$$

By summing over $t = 1$ to $t = T$, we obtain

$$\begin{aligned}
\sum_{t=1}^T \ln \hat{p}(y_t, |x_t, S_{t-1}) &= \ln \mathbf{E}_{w \in p_0} \prod_{i=1}^T p(Y_i|w, X_i) \\
&= \sup_q \left[\mathbf{E}_{w \sim q} \ln \prod_{i=1}^T p(Y_i|w, X_i) - \mathbf{E}_{w \sim q} \ln \frac{q(w)}{p_0(w)} \right],
\end{aligned}$$

where the second equality is Proposition 12.1. This implies the desired result. $\blacksquare$

The following result is a direct consequence of Theorem 12.1.

Corollary 12.1 *If $\Omega = \{w_1, \dots\}$ is countable, then*

$$-\sum_{t=1}^T \ln \hat{p}(Y_t, |X_t, S_{t-1}) \leq \inf_{w \in \Omega} \left[-\sum_{t=1}^T \ln p(Y_t|w, X_i) - \ln p_0(w) \right].$$

Proof Given any $w' \in \Omega$, if we choose $q(w) = 1$ when $w = w'$, and $q(w) = 0$ when $w \neq w'$, then

$$-\mathbf{E}_{w \sim q} \sum_{t=1}^T \ln p(Y_t|w, X_i) + \mathbf{E}_{w \sim q} \ln \frac{q(w)}{p_0(w)} = -\sum_{t=1}^T \ln p(Y_t|w', X_i) - \ln p_0(w').$$

$\blacksquare$

If $|\Omega| = N$ is finite, and $p_0(w) = 1/N$ is the uniform distribution on Ω , then we have

$$-\sum_{t=1}^T \ln \hat{p}(Y_t, |X_t, S_{t-1}) \leq \inf_{w \in \Omega} \left[-\sum_{t=1}^T \ln p(Y_t|w, X_i) + \ln N \right].$$

This means that we have a constant regret which is independent of T .

If we use batch to online conversion, then by taking expectation, we have

$$-\frac{1}{T} \sum_{t=1}^T \mathbf{E}_Z \ln \hat{p}(Y|X, S_{t-1}) \leq \inf_{w \in \Omega} \left[-\mathbf{E}_Z \ln p(Y|w, X) + \frac{\ln N}{T} \right].$$

Therefore the statistical convergence rate of the Bayesian posterior averaging algorithm for density estimation problem is naturally $\ln N/T$. This rate is generally superior to that of ERM. In fact, under suitable conditions, the Bayesian posterior averaging algorithm is statistically optimal, and the method can be used to achieve minimax optimal convergence rate [93].

12.2 Ridge Regression Analysis

The general regret bound for Bayesian model averaging can be used to analyze the ridge regression method. Consider the following linear prediction problem with least squares loss:

$$f(w, x) = w^\top \psi(x),$$

with loss function

$$(y - f(w, x))^2.$$

Consider the following ridge regression estimator:

$$\hat{w}(S_n) = \arg \min_w \left[\sum_{i=1}^n (Y_i - w^\top \psi(X_i))^2 + \lambda \|w\|_2^2 \right]. \quad (12.3)$$

We have the following Bayesian interpretation of ridge regression.

Proposition 12.2 *Consider probability model*

$$p(y|w, x) = N(w^\top \psi(x), \sigma^2),$$

with prior

$$p_0(w) = N(0, \sigma'^2),$$

where $\sigma' = \sigma/\sqrt{\lambda}$. Then given $S_n = \{(X_1, Y_1), \dots, (X_n, Y_n)\}$, we have

$$p(w|S_n) = N(\hat{w}(S_n), \hat{\Sigma}(S_n)),$$

where $\hat{w}(S_n)$ is given by (12.3) and

$$\hat{\Sigma}(S_n)^{-1} = \sigma^{-2} \sum_{i=1}^n \psi(X_i) \psi(X_i)^\top + \lambda \sigma^{-2} I.$$

Given x , the posterior distribution of y is

$$\hat{p}(y|x, S_n) = N\left(\hat{w}(S_n)^\top \psi(x), \sigma^2 + \psi(x)^\top \hat{\Sigma}(S_n) \psi(x)\right).$$

Proof It is clear that

$$p(w|S_n) \propto \exp \left(- \sum_{i=1}^n \frac{(w^\top \psi(X_i) - Y_i)^2}{2\sigma^2} - \frac{\|w\|_2^2}{2\sigma'^2} \right).$$

Note that (12.3) implies that

$$\sum_{i=1}^n \frac{(w^\top \psi(X_i) - Y_i)^2}{2\sigma^2} + \frac{\|w\|_2^2}{2\sigma'^2} \quad (12.4)$$

$$= \sum_{i=1}^n \frac{(\hat{w}^\top \psi(X_i) - Y_i)^2}{2\sigma^2} + \frac{\|\hat{w}\|_2^2}{2\sigma'^2} + \frac{1}{2}(w - \hat{w})^\top \hat{\Sigma}(S_n)^{-1}(w - \hat{w}). \quad (12.5)$$

This implies the first desired result. Moreover, given x , and let $u = w^\top \psi(x)$, we have

$$u|x, S_n \sim N(\hat{w}^\top \psi(x), \psi(x)^\top \hat{\Sigma} \psi(x)).$$

Since $y = u + \epsilon$ with $\epsilon \sim N(0, \sigma^2)$, we know that $y \sim N(\hat{w}^\top \psi(x), \psi(x)^\top \hat{\Sigma} \psi(x) + \sigma^2)$. This implies the desired result. ■

Theorem 12.1 implies the following predictive loss bound for ridge regression.

Theorem 12.2 *Consider the ridge regression method of (12.3). We have the following bound:*

$$\begin{aligned} & \sum_{t=1}^T \left[\frac{(Y_t - \hat{w}(S_{t-1})^\top \psi(X_t))^2}{b_t} + \sigma^2 \ln b_t \right] \\ & \leq \inf_w \left[\sum_{t=1}^T (Y_t - w^\top \psi(X_t))^2 + \lambda \|w\|_2^2 \right] + \sigma^2 \ln \left| I + \lambda^{-1} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top \right|, \end{aligned}$$

where $b_t = 1 + \psi(X_t)^\top (\sum_{i=1}^{t-1} \psi(X_i) \psi(X_i)^\top + \lambda I)^{-1} \psi(X_t)$.

Proof Assume $w \in \mathbb{R}^d$. We note that from Gaussian integration that

$$\begin{aligned} & \mathbf{E}_{w \sim p_0} \prod_{i=1}^T p(Y_i | w, X_i) \\ &= \int \frac{1}{(2\pi)^{(T+d)/2} \sigma^T \sigma'^d} \exp \left(- \frac{1}{2\sigma^2} \sum_{i=1}^T (Y_i - w^\top \psi(X_i))^2 - \frac{\|w\|_2^2}{2\sigma'^2} \right) dw \\ &= \frac{|\hat{\Sigma}(S_T)|^{1/2}}{(2\pi)^{T/2} \sigma^T (\sigma')^d} \exp \left(- \frac{1}{2\sigma^2} \sum_{i=1}^T (Y_i - \hat{w}(S_T)^\top \psi(X_i))^2 - \frac{\|\hat{w}(S_T)\|_2^2}{2\sigma'^2} \right), \end{aligned}$$

where the last equation used Gaussian integration with decomposition (12.5). That is,

$$\begin{aligned} -\ln \mathbf{E}_{w \sim p_0} \prod_{i=1}^T p(Y_i | w, X_i) &= T \ln(\sqrt{2\pi}\sigma) + \frac{1}{2} \ln \left| (\sigma')^2 \hat{\Sigma}(S_T)^{-1} \right| \\ &\quad + \frac{1}{2\sigma^2} \sum_{i=1}^T (Y_i - \hat{w}(S_T)^\top \psi(X_i))^2 + \frac{\|\hat{w}(S_T)\|_2^2}{2\sigma'^2}. \end{aligned}$$

Moreover, from Proposition 12.2, we have

$$\begin{aligned} &\sum_{i=1}^T -\ln p(Y_i | \hat{w}(S_{i-1}), X_i) \\ &= \sum_{t=1}^T \left[\frac{(Y_t - \hat{w}(S_{t-1})^\top \psi(X_t))^2}{2\sigma^2 b_t} + \frac{1}{2} \ln(b_t) + \ln(\sqrt{2\pi}\sigma) \right]. \end{aligned}$$

The desired result now follows from Theorem 12.1. ■

Theorem 12.2 implies the following result. This result can also be found in [91].

Corollary 12.2 *Assume that $Y_t \in [0, M]$ for $t \geq 1$. Consider the ridge regression method of (12.3), and let*

$$\hat{Y}_t = \max(0, \min(M, \hat{w}(S_{t-1})^\top \psi(X_t))).$$

We have

$$\begin{aligned} \sum_{t=1}^T (Y_t - \hat{Y}_t)^2 &\leq \inf_w \left[\sum_{t=1}^T (Y_t - w^\top \psi(X_t))^2 + \lambda \|w\|_2^2 \right] \\ &\quad + M^2 \ln \left| I + \lambda^{-1} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top \right|. \end{aligned}$$

Proof We can apply Theorem 12.2 by taking $\sigma^2 = M^2$. By using the following inequality

$$0 \leq \frac{1 - b_t}{b_t} + \ln b_t,$$

we obtain

$$\begin{aligned}
\sum_{t=1}^T (Y_t - \hat{Y}_t)^2 &\leq \sum_{t=1}^T \left[(Y_t - \hat{Y}_t)^2 + M^2 \frac{1 - b_t}{b_t} + M^2 \ln b_t \right] \\
&\leq \sum_{t=1}^T \left[(Y_t - \hat{Y}_t)^2 + (Y_t - \hat{Y}_t)^2 \frac{1 - b_t}{b_t} + M^2 \ln b_t \right] \\
&\leq \sum_{t=1}^T \left[\frac{(Y_t - \hat{w}(S_{t-1})^\top \psi(X_t))^2}{b_t} + M^2 \ln b_t \right],
\end{aligned}$$

where b_t is defined in Theorem 12.2. The result is now a direct application of Theorem 12.2. $\blacksquare$

The following result can be used to give a more precise upper bound estimate for the log-determinant term of the regret bound. This leads to a regret bound of $O(d \ln T/T)$ in Theorem 12.2.

Proposition 12.3 *If $\psi(X) \in \mathbb{R}^d$, and $\|\psi(X)\|_2 \leq B$ for all X , then*

$$\ln \left| I + \lambda^{-1} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top \right| \leq d \ln(1 + TB^2/(d\lambda)).$$

Proof Let $A = I + \lambda^{-1} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top$, then

$$\text{trace}(A) \leq d + \lambda^{-1} \sum_{i=1}^T \text{trace}(\psi(X_i) \psi(X_i)^\top) \leq d + B^2 T / \lambda.$$

Using the geometric and algebraic mean inequality, we have

$$|A| \leq [\text{trace}(A)/d]^d \leq (1 + B^2 T/(d\lambda))^d.$$

This implies the desired bound. $\blacksquare$

If we consider the case of online to batch conversion, where $Z_t = (X_t, Y_t)$ are coming from an underlying distribution, then we can replace $Y_t \in [0, M]$ by $\mathbf{E}_D[Y|X] \in [0, M]$ and $\mathbf{Var}_D[Y|X] \leq \sigma^2$. This naturally leads to the following result.

Corollary 12.3 *Assume that (X_t, Y_t) are from a distribution D , so that $\mathbf{E}_D[Y|X] \in [0, M]$ and $\mathbf{Var}_D[Y|X] \leq \sigma^2$. Consider the ridge regression method of (12.3), and let*

$$\hat{f}(w, x) = \max(0, \min(M, w^\top \psi(X))),$$

and

$$\bar{f}(x|S_T) = \frac{1}{T} \sum_{t=1}^T \hat{f}(\hat{w}(S_{t-1}), x).$$

We have

$$\begin{aligned} \mathbf{E}_{S_T} \mathbf{E}_D(Y - \bar{f}(X|S_T))^2 &\leq \inf_w \left[\mathbf{E}_D(Y - w^\top \psi(X))^2 + \frac{\lambda}{T} \|w\|_2^2 \right] \\ &\quad + \frac{M^2 + \sigma^2}{T} \ln \left| I + \lambda^{-1} T \mathbf{E}_D \psi(X) \psi(X)^\top \right|. \end{aligned}$$

Proof We have the following derivation, which is similar to that of Corollary 12.2:

$$\begin{aligned} &\mathbf{E}_{S_T} \mathbf{E}_D(Y - \bar{f}(X|S_T))^2 \\ &\leq \frac{1}{T} \mathbf{E}_{S_T} \sum_{t=1}^T [\sigma^2 + (\mathbf{E}[Y_t|X_t] - \hat{f}(\hat{w}_{t-1}(S_{t-1}), X_t))^2] \\ &\leq \frac{1}{T} \mathbf{E}_{S_T} \sum_{t=1}^T \left[[\sigma^2 + (\mathbf{E}[Y_t|X_t] - \hat{f}(\hat{w}_{t-1}(S_{t-1}), X_t))^2] \right. \\ &\quad \left. + [\sigma^2 + (\mathbf{E}[Y_t|X_t] - \hat{f}(\hat{w}_{t-1}(S_{t-1}), X_t))^2] \frac{1 - b_t}{b_t} + [M^2 + \sigma^2] \ln b_t \right] \\ &\leq \frac{1}{T} \mathbf{E}_{S_T} \sum_{t=1}^T \left[\frac{(Y_t - \hat{f}(\hat{w}_{t-1}(S_{t-1}), X_t))^2}{b_t} + [M^2 + \sigma^2] \ln b_t \right]. \end{aligned}$$

Now using Theorem 12.2, we obtain

$$\begin{aligned} \mathbf{E}_{S_T} \mathbf{E}_D(Y - \bar{f}(X|S_T))^2 &\leq \inf_w \left[\mathbf{E}_{S_T} \sum_{t=1}^T (Y_t - w^\top \psi(X_t))^2 + \lambda \|w\|_2^2 \right] \\ &\quad + \sigma^2 \mathbf{E}_{S_T} \ln \left| I + \lambda^{-1} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top \right| \\ &\leq \inf_w \left[T \mathbf{E}_D(Y - w^\top \psi(X))^2 + \lambda \|w\|_2^2 \right] \\ &\quad + \sigma^2 \ln \left| I + \lambda^{-1} \mathbf{E}_{S_T} \sum_{i=1}^T \psi(X_i) \psi(X_i)^\top \right|, \end{aligned}$$

where the second inequality is due to the concavity of the log-determinant function (see Proposition 12.7). This implies the result. $\blacksquare$

The bound obtained in Corollary 12.3 is superior to the Rademacher complexity bound, and the best convergence rate can be achieved is $O(\ln T/T)$.

12.3 Exponential Model Aggregation

Consider a more general loss function

$$g_w(Z) = L(f(w, x), y),$$

where $L(f, y)$ is convex with respect to f . Consider a prior $p_0(w)$ on Ω , and the following form of posterior

$$p(w|S_n) \propto \exp \left[-\eta \sum_{i=1}^n g_w(Z_i) \right] p_0(w), \quad (12.6)$$

where $\eta > 0$ is a learning rate parameter. The exponential model aggregation algorithm computes

$$\hat{f}(x|S_n) = \int_{\Omega} f(w, x) p(w|S_n) dw, \quad (12.7)$$

where $p(w|S_n)$ is given by (12.6).

Algorithm 12.1: Online Exponential Model Aggregation

Input: $\eta > 0$, $\{f(w, x) : w \in \Omega\}$, prior $p_0(w)$

Output: $\hat{f}(\cdot|S_T)$

```

1 for  $t = 1, 2, \dots, T$  do
2   Observe  $X_t$ 
3   Let  $\hat{f}_t = \hat{f}(X_t|S_{t-1})$  according to (12.7)
4   Observe  $Y_t$ 
5   Compute  $L(\hat{f}_t, Y_t)$ 
```

Return: $\hat{f}(\cdot|S_T)$

In order to analyze Algorithm 12.1, we need to introduce the following definition.

Definition 12.1 A convex function $\phi(f)$ is α -exponential concave for some $\alpha > 0$ if

$$e^{-\alpha\phi(f)}$$

is concave in f .

The following property is convenient to show exponential concavity [48].

Proposition 12.4 A convex function $\phi(f)$ is α exponentially concave if

$$\alpha \nabla \phi(f) \nabla \phi(f)^\top \leq \nabla^2 \phi(f).$$

Proof We have

$$\nabla^2 e^{-\alpha\phi(f)} = e^{-\alpha\phi(f)} \left[-\alpha \nabla^2 \phi(f) + \alpha^2 \nabla \phi(f) \nabla \phi(f)^\top \right] \leq 0.$$

This implies the concavity of $\exp(-\alpha\phi(f))$. ■

Proposition 12.5 *If a convex function $\phi(f)$ is α -exponentially concave for some $\alpha > 0$, then it is also β exponentially concave for $\beta \in (0, \alpha]$.*

Proof Let $h(f) = \exp(-\alpha\phi(f))$. We note that $\exp(-\beta\phi(f)) = g(h(f))$, where $g(z) = z^{\beta/\alpha}$ is an increasing and concave function. Therefore for all f_1, f_2 and $\theta \in [0, 1]$, the exponential concavity implies that

$$h(\theta f_1 + (1 - \theta)f_2) \geq \theta h(f_1) + (1 - \theta)h(f_2).$$

The increasing of g implies that

$$g(h(\theta f_1 + (1 - \theta)f_2)) \geq g(\theta h(f_1) + (1 - \theta)h(f_2)).$$

The concavity of g implies that

$$g(\theta h(f_1) + (1 - \theta)h(f_2)) \geq \theta g(h(f_1)) + (1 - \theta)g(h(f_2)),$$

which implies the concavity of $g \circ h$. ■

We note that if $\phi(f)$ is both Lipschitz $\|\nabla\phi(f)\|_2 \leq G$, and λ -strongly convex, then it is λ/G^2 exponentially concave. Moreover, the following proposition says that the property of exponential concavity holds under a linear transformation of the model parameter. Therefore in general, the exponential concavity criterion is easier to satisfy than strong convexity.

Proposition 12.6 *Assume $\phi(f)$ is α exponentially concave in f . Let $\tilde{\phi}(w) = \phi(w^\top X)$, then $\tilde{\phi}$ is also α exponentially concave in w .*

Example 12.3 *Consider the loss function $L(f, y) = (f - y)^2$. If $|f - y| \leq M$, then $L(f, y)$ is α -exponentially concave with $\alpha \leq 1/(2M^2)$.*

Example 12.4 *Consider $L(f, y) = -\ln f_y$, then $L(f, y)$ is α exponentially concave for $\alpha \leq 1$.*

For exponentially concave loss functions, we have the following result for the exponential model aggregation algorithm.

Theorem 12.3 *Assume that $L(f, y)$ is η -exponentially concave. Then (12.7) satisfies the following regret bound:*

$$\sum_{t=1}^T L(\hat{f}(X_t|S_{t-1}), Y_t) \leq \inf_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T L(f(w, X_t), Y_t) + \eta^{-1} \mathbf{E}_{w \sim q} \ln \frac{q(w)}{p_0(w)} \right].$$

Proof Since $e^{-\eta L(f,y)}$ is concave in f , we obtain

$$\ln \int e^{-\eta L(f(w,x),y)} p(w|S_n) dw \leq \ln e^{-\eta L(\hat{f}(x|S_n),y)}.$$

This implies that

$$L(\hat{f}(X_t|S_{t-1}), Y_t) \leq \frac{-1}{\eta} \ln \frac{\int_{\Omega} \exp(-\eta \sum_{i=1}^t L(f(w, X_i), Y_i)) p_0(w) dw}{\int_{\Omega} \exp(-\eta \sum_{i=1}^{t-1} L(f(w, X_i), Y_i)) p_0(w) dw}.$$

By summing over $t = 1$ to $t = T$, we obtain

$$\sum_{t=1}^T L(\hat{f}(X_t|S_{t-1}), Y_t) \leq \frac{-1}{\eta} \ln \int_{\Omega} \exp \left(-\eta \sum_{i=1}^T L(f(w, X_i), Y_i) \right) p_0(w) dw.$$

Using Proposition 12.1, we obtain the desired result. $\blacksquare$

We note that Theorem 12.1 is a special case of Theorem 12.3, with $\eta = 1$, $L(f(w, x), y) = -\ln P(y|w, x)$ and $f(w, x) = P(y|w, x)$. In this case, $\exp(-L(f, y)) = f_y$ is concave in f .

The following result holds for nonlinear least squares regression, which generalizes that of the ridge least squares regression.

Example 12.5 Assume that $L(f, y) = (f - y)^2$, and $\sup |f(w, x) - y| \leq M$. Then for $\eta \leq 1/(2M^2)$, $L(f, y)$ is η exponentially concave. Therefore we have

$$\sum_{t=1}^T (\hat{f}(X_t|S_{t-1}) - Y_t)^2 \leq \inf_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T (f(w, X_t) - Y_t)^2 + \eta^{-1} \mathbf{E}_{w \sim q} \ln \frac{q(w)}{p_0(w)} \right].$$

In particular, if Ω is countable, then

$$\sum_{t=1}^T (\hat{f}(X_t|S_{t-1}) - Y_t)^2 \leq \inf_{w \in \Omega} \left[\sum_{t=1}^T (f(w, X_t) - Y_t)^2 + \eta^{-1} \ln \frac{1}{p_0(w)} \right].$$

Example 12.6 In Example 12.5, if we assume further that $f(w, x) = w^\top \psi(x)$ is linear, $\sigma^2 = 1/(2\eta)$ and $p_0(w) \sim N(0, \sigma^2/\lambda)$. Let $q = N(\hat{w}, \hat{\Sigma})$ be the posterior distribution of ridge regression. Then one can obtain a bound from Theorem 12.3 similar to that of Corollary 12.2. We leave it as an exercise.

12.4 Second Order Methods

We note that ridge regression may be regarded as a second order method. The analysis can be extended to more general second order methods for α exponentially concave loss functions.

In general, we consider online learning with loss functions $\ell_1(w), \dots, \ell_T(w)$, where we assume that each $\ell_t(w)$ is α -exponentially concave. Algorithm 12.2, referred to as *online Newton step* (ONS) [42], may be regarded as a generalization of the online ridge regression analysis.

Algorithm 12.2: Online Newton Step

Input: $\eta > 0$, w_0 , A_0 , and a sequence of loss functions $\ell_t(w)$

Output: w_T

```

1 for  $t = 1, 2, \dots, T$  do
2   Observe loss  $\ell_t(w_{t-1})$ 
3   Let  $g_t = \nabla \ell_t(w_{t-1})$ 
4   Let  $A_t = A_{t-1} + g_t g_t^\top$ 
5   Let  $\tilde{w}_t = w_{t-1} - \eta A_t^{-1} g_t$ 
6   Let  $w_t = \arg \min_{w \in \Omega} (w - \tilde{w}_t)^\top A_t (w - \tilde{w}_t)$ 

```

Return: w_T

The following result shows that ONS can achieve logarithmic regret under exponential concavity. The result is analogous to that of the first order gradient descent for strong convex loss functions.

Theorem 12.4 *Assume that for all t , the loss function $\ell_t : \Omega \rightarrow \mathbb{R}$ is α exponentially concave and Lipschitz: $\|\nabla \ell_t\|_2 \leq G$. Let $\eta^{-1} = \beta \leq 0.5 \min(\alpha, 1/(4GD))$. Then ONS has the following regret bound:*

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \inf_{w \in \Omega} \left[\sum_{t=1}^T \ell_t(w) + \frac{1}{2\eta} (w - w_0)^\top A_0 (w - w_0) \right] + \frac{\eta}{2} \ln \frac{|A_T|}{|A_0|}.$$

where $D = \sup\{\|w' - w\|_2 : w, w' \in \Omega\}$ is the diameter of Ω , and d is the dimensionality of w .

We will first prove the following lemma.

Lemma 12.1 *For a function $\ell_t : \Omega \rightarrow \mathbb{R}$ that satisfies the conditions of Theorem 12.4, we have*

$$\ell_t(w') \geq \ell_t(w) + \nabla \ell_t(w)^\top (w' - w) + \frac{\beta}{2} (w' - w)^\top \nabla \ell_t(w) \nabla \ell_t(w)^\top (w' - w)$$

for all $w, w' \in \Omega$.

Proof Since $\exp(-\alpha \ell_t(w))$ is concave and $2\beta \leq \alpha$, the function $h(x) = \exp(-2\beta \ell_t(w))$ is concave, which implies that

$$h(w') \leq h(w) + \nabla h(w)^\top (w' - w).$$

That is

$$\exp(-2\beta \ell_t(w')) \leq \exp(-2\beta \ell_t(w)) [1 - 2\beta \nabla \ell_t(w)^\top (w' - w)].$$

By taking logarithm on each side, we obtain

$$\begin{aligned} -2\beta\ell_t(w') &\leq -2\beta\ell_t + \ln[1 - 2\beta\nabla\ell_t(w)^\top(w' - w)] \\ &\leq -2\beta\ell_t - 2\beta\nabla\ell_t(w)^\top(w' - w) + \frac{1}{4}(2\beta\nabla\ell_t(w)^\top(w' - w))^2, \end{aligned}$$

where the second inequality is because $\ln(1 - z) \leq -z + z^2/4$ when $|z| \leq 1/4$, and $|2\beta\nabla\ell_t(w)^\top(w' - w)| \leq 2\beta GD \leq 1/4$. $\blacksquare$

Proposition 12.7 *Consider a symmetric positive definite matrix A . The log-determinant function*

$$h(A) = \ln |A|$$

is concave in A . Moreover, $\nabla h(A) = A^{-1}$, which implies that

$$\text{trace}(A^{-1}(A - B)) \leq \ln |A| - \ln |B|.$$

Proof [of Theorem 12.4] Consider $w \in \Omega$, then from Lemma 12.1, we obtain

$$-(w_{t-1} - w)^\top g_t \leq \ell_t(w) - \ell_t(w_{t-1}) - \frac{1}{2\eta}[(w - w_{t-1})^\top g_t]^2.$$

Using this inequality, we obtain:

$$\begin{aligned} &(\tilde{w}_t - w)^\top A_t(\tilde{w}_t - w) \\ &= (w_{t-1} - \eta A_t^{-1} g_t - w)^\top A_t(w_{t-1} - \eta A_t^{-1} g_t - w) \\ &= (w_{t-1} - w)^\top A_t(w_{t-1} - w) - 2\eta(w_{t-1} - w)^\top g_t + \eta^2 g_t^\top A_t^{-1} g_t \\ &\leq (w_{t-1} - w)^\top A_t(w_{t-1} - w) + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] - [(w - w_{t-1})^\top g_t]^2 \\ &\quad + \eta^2 g_t^\top A_t^{-1} g_t \\ &\stackrel{(a)}{=} (w_{t-1} - w)^\top A_{t-1}(w_{t-1} - w) + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] \\ &\quad + \eta^2 \text{trace}(A_t^{-1}(A_t - A_{t-1})) \\ &\stackrel{(b)}{\leq} (w_{t-1} - w)^\top A_{t-1}(w_{t-1} - w) + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] \\ &\quad + \eta^2 [\ln |A_t| - \ln |A_{t-1}|]. \end{aligned}$$

Note that in (a), we used $g_t g_t^\top = A_t - A_{t-1}$. In (b), we used Proposition 12.7.

Now use the fact that $(w_t - w)^\top A_t(w_t - w) \leq (\tilde{w}_t - w)^\top A_t(\tilde{w}_t - w)$, we obtain

$$\begin{aligned} (w_t - w)^\top A_t(w_t - w) &\leq (w_{t-1} - w)^\top A_{t-1}(w_{t-1} - w) \\ &\quad + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] + \eta^2 [\ln |A_t| - \ln |A_{t-1}|]. \end{aligned}$$

We can now sum over $t = 1$ to $t = T$:

$$\begin{aligned} (w_T - w)^\top A_T (w_T - w) &\leq (w_0 - w)^\top A_0 (w_0 - w) \\ &\quad + 2\eta \sum_{t=1}^T [\ell_t(w) - \ell_t(w_{t-1})] + \eta^2 [\ln |A_T| - \ln |A_0|]. \end{aligned}$$

This implies the desired bound. $\blacksquare$

Using Proposition 12.3, we obtain a regret bound that is logarithmic in T for ONS. If we let $A_0 = \lambda I$, then

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \inf_{w \in \Omega} \left[\sum_{t=1}^T \ell_t(w) + \frac{\lambda}{2\eta} \|w - w_0\|_2^2 \right] + \frac{\eta d}{2} \ln(1 + TG^2/(d\lambda)).$$

If $\ell_t(w) = L(f(w, X_t), Y_t)$ with $(X_t, Y_t) \sim D$, then this implies the following online-to-batch conversion bound for s randomly chosen from 0 to $T - 1$:

$$\begin{aligned} \mathbf{E}_{S_T} \mathbf{E}_s \mathbf{E}_D L(f(w_s, X), Y) &\leq \inf_{w \in \Omega} \left[\mathbf{E}_D L(f(w, X), Y) + \frac{\lambda}{2\eta T} \|w - w_0\|_2^2 \right] \\ &\quad + \frac{\eta d}{2T} \ln(1 + TG^2/(d\lambda)). \end{aligned}$$

A method related to ONS, referred to as *Adaptive subGradient Method* (AdaGrad), was studied in [31]. The diagonal matrix inversion version is described in Algorithm 12.3, which is similar to ONS, except that we use $\text{diag}(A_t^{p-1})$ in the matrix inversion, instead of using A_t^{-1} in ONS. The analysis of this method does not require exponential concavity. The corresponding full matrix version, uses $G_t = A_t^{1-p}$ instead of $G_t = \text{diag}(A_t^{1-p})$.

Algorithm 12.3: Adaptive SubGradient Method (AdaGrad)

Input: $\eta > 0$, w_0 , A_0 , $p \in [0.5, 1)$, and a sequence of loss functions $\ell_t(w)$

Output: w_T

```

1 for  $t = 1, 2, \dots, T$  do
2   Observe loss  $\ell_t(w_{t-1})$ 
3   Let  $g_t = \nabla \ell_t(w_{t-1})$ 
4   Let  $A_t = A_{t-1} + g_t g_t^\top$ 
5   Let  $G_t = \text{diag}(A_t^{1-p})$  (full matrix version:  $G_t = A_t^{1-p}$ )
6   Let  $\tilde{w}_t = w_{t-1} - \eta G_t^{-1} g_t$ 
7   Let  $w_t = \arg \min_{w \in \Omega} (w - \tilde{w}_t)^\top G_t (w - \tilde{w}_t)$ 
```

Return: w_T

Theorem 12.5 Assume that for all t , the loss function $\ell_t : \Omega \rightarrow \mathbb{R}$ is convex. Then AdaGrad method (diagonal version) has the following regret

bound:

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \inf_{w \in \Omega} \sum_{t=1}^T \ell_t(w) + \frac{\eta}{2p} \text{trace}(A_T^p) + \frac{D_q^2}{2\eta} \text{trace}(A_T^p)^{(1-p)/p},$$

where $q = 2p/(2p-1)$, $D_q = \sup\{\|w' - w\|_q : w, w' \in \Omega\}$ is the L_q -diameter of Ω .

Proof Consider $w \in \Omega$. The convexity of ℓ_t implies that

$$-2\eta(w_{t-1} - w)^\top g_t \leq 2\eta[\ell_t(w) - \ell_t(w_{t-1})].$$

Let $G_t = \text{diag}(A_t^{1/2})$. We obtain the following result:

$$\begin{aligned} & (\tilde{w}_t - w)^\top G_t (\tilde{w}_t - w) \\ &= (w_{t-1} - \eta G_t^{-1} g_t - w)^\top G_t (w_{t-1} - \eta G_t^{-1} g_t - w) \\ &= (w_{t-1} - w)^\top G_t (w_{t-1} - w) - 2\eta(w_{t-1} - w)^\top g_t + \eta^2 g_t^\top G_t^{-1} g_t \\ &\leq (w_{t-1} - w)^\top G_t (w_{t-1} - w) + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] + \eta^2 g_t^\top G_t^{-1} g_t \\ &= (w_{t-1} - w)^\top G_{t-1} (w_{t-1} - w) + (w_{t-1} - w)^\top (G_t - G_{t-1}) (w_{t-1} - w) \\ &\quad + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] + \eta^2 \text{trace}((G_t^{1/(1-p)})^{p-1} (G_t^{1/(1-p)} - G_{t-1}^{1/(1-p)})) \\ &\stackrel{(a)}{\leq} (w_{t-1} - w)^\top G_{t-1} (w_{t-1} - w) + (w_{t-1} - w)^\top (G_t - G_{t-1}) (w_{t-1} - w) \\ &\quad + 2\eta[\ell_t(w) - \ell_t(w_{t-1})] + \eta^2 p^{-1} [\text{trace}(G_t^{p/(1-p)}) - \text{trace}(G_{t-1}^{p/(1-p)})]. \end{aligned}$$

In (a), we used the fact that $h(B) = p^{-1} \text{trace}(B^p)$ is concave in B , and $\nabla(h)(B) = B^{p-1}$. Thus $\text{trace}(B^{p-1}(B-B')) \leq p^{-1} [\text{trace}(B^p) - \text{trace}((B')^p)]$, where we let $B = G_t^{1/(1-p)}$ and $B' = G_{t-1}^{1/(1-p)}$. Here $p' = p/(1-p)$.

Now we can use the fact that $(w_t - w)^\top G_t (w_t - w) \leq (\tilde{w}_t - w)^\top G_t (\tilde{w}_t - w)$, and then sum over $t = 1$ to $t = T$. This implies

$$\sum_{t=1}^T \ell_t(w_{t-1}) \leq \sum_{t=1}^T \ell_t(w) + \frac{R_T}{2\eta} + \frac{\eta}{p} [\text{trace}(G_T^{p'}) - \text{trace}(G_0^{p'})],$$

where

$$\begin{aligned} R_T &= (w_0 - w)^\top G_0 (w_0 - w) + \sum_{t=1}^T (w_{t-1} - w)^\top (G_t - G_{t-1}) (w_{t-1} - w) \\ &\leq D_q^2 \text{trace}(G_0^{p'})^{1/p'} + \sum_{t=1}^T D_q^2 \text{trace}((G_t - G_{t-1})^{p'})^{1/p'} \\ &\leq D_q^2 \text{trace}(G_T^{p'})^{1/p'}. \end{aligned}$$

In the derivation, we note that $\text{trace}(|G|^{p'})^{1/p'}$ is the vector p' -norm of the diagonal of G . The first inequality is Holder's inequality, and the second inequality is from the triangle inequality of p' -norm. This implies the bound. ■

Note that compared to ONS, AdaGrad does not require either strong convexity or exponential concavity. Therefore the rate is slower, because we have

$$\text{trace}(A_T^p) = O(T^p).$$

Therefore if we take optimal $\eta = O(D_q \text{trace}(A_T^p)^{-1+1/(2p)})$, then the regret bound is $O(\sqrt{T})$, similar to that of Theorem 11.8. However, the bound is in terms of L_q -diameter for $q > 2$, which can be significantly smaller than L_2 -diameter for gradient descent.

Note that the original version in [31] was only with $p = 0.5$, and a version of AdaGrad with full matrix $G_t = A^{1/2}$ was also considered. While that algorithm is closer to ONS because both methods rely on full matrix inversion, the resulting bound stated in [31] relied on the L_2 diameter of Ω (otherwise, the bound and the proof are identical to that of Theorem 12.5), which has no advantage if compared to Theorem 11.8. However, a small modification of the proof can lead to a better bound using the spectral norm of G_T .

12.5 Exercises

1. In Example 12.5, assume further that $f(w, x) = w^\top \psi(x)$ is linear, $\sigma^2 = 1/(2\eta)$ and $p_0(w) \sim N(0, \sigma^2/\lambda)$. Let $q = N(\hat{w}, \hat{\Sigma})$ be the posterior distribution of ridge regression. Obtain a regret bound from Theorem 12.3, and compare to that of Corollary 12.2.
2. Prove Proposition 12.1.
3. Prove Proposition 12.7.
4. In Algorithm 12.3. Let $G_t = \text{trace}(\text{diag}(A_t))^{1/2}I$. Derive a regret bound by mimicking the proof of Theorem 12.5.

Chapter 13

Multi-armed Bandits

In supervised learning and online learning, we assume that the outcome y of an observation is fully observed. For example, for online image classification, we first predict the label $\hat{y}$ of an image x , and then the true label y is revealed.

In comparison, for bandit problems, we consider the so-called *partial information* setting, where only the outcome of the action taken is observed. For the online classification example, where we assume an image can be multiply classified, and we only predict one label, which is correct if the image contains the label. Assume the feedback we receive only says whether the predicted label $\hat{y}$ is correct or not, but not what the true labels are, then it becomes a bandit problem. In this chapter, we will investigate some bandit algorithms that are commonly used.

13.1 Multi-armed Bandit Problem

Multi-armed bandit can be regarded as a repeated game, similar to online learning.

Definition 13.1 *In the multi-armed bandit (or MAB) problem, we consider K arms. The environment generates a sequence of reward vectors for time steps $t \geq 1$ as $r_t = [r_t(1), \dots, r_t(K)]$. Each $r_t(a)$ is associated with an arm $a \in \{1, \dots, K\}$. At each time step $t = 1, 2, \dots, T$,*

- *The player pulls one of the arms $a_t \in \{1, \dots, K\}$*
- *The environment returns the reward $r_t(a_t)$, but does not reveal information on any other arm $a \neq a_t$.*

At each time t , a (randomized) bandit algorithm takes the historic observations observed so far, and maps it into a distribution $\hat{p}_{t-1}$ over actions $a \in \{1, \dots, K\}$, we then draw a random action a_t (arm) from $\hat{p}_{t-1}$.

In the adversarial bandit problem, we are given an arbitrary reward sequence $\{[r_1(1), \dots, r_t(K)] : t \geq 1\}$ before hand. The adversary in this setting is called *oblivious*, who decides the rewards of all the rounds before the game starts (but may take advantage of the bandit algorithm's weakness). For this reward sequence, the expected cumulative reward of a randomized bandit algorithm is

$$\mathbf{E} \sum_{t=1}^T r_t(a_t),$$

where the randomization is over the internal randomization of the algorithm. The expected regret is defined as

$$R_T = \max_a \sum_{t=1}^T r_t(a) - \mathbf{E} \sum_{t=1}^T r_t(a_t). \quad (13.1)$$

If the algorithm is deterministic: i.e. pulling a single arm a_t only at any time step, then the regret is

$$R_T = \max_a \sum_{t=1}^T r_t(a) - \sum_{t=1}^T r_t(a_t).$$

In the adversarial bandit problem, our goal is to minimize the expected regret R_T (and the expectation is with respect to the internal randomization of the algorithm, not data).

Another form of bandit problem, called *stochastic bandit*, assumes that the reward $r_t(a)$ is drawn independently from a distribution D_a , with mean $\mu(a) = \mathbf{E} r_t(a)$. In this setting, the goal of bandit is to find a that maximizes the expected reward $\mu(a)$. We are interested in the expected regret defined as:

$$R_T = T \max_a \mu(a) - \mathbf{E} \sum_{t=1}^T \mu(a_t). \quad (13.2)$$

Again, this regret is defined for each realization of the stochastic rewards, and the expectation is with respect to the randomness of the algorithm. If algorithm is deterministic and outputs a_t at each time t , then

$$R_T = T \max_a \mu(a) - \sum_{t=1}^T \mu(a_t).$$

We can further include randomization over data into the regret of stochastic bandit, by considering expected regret:

$$\mathbf{E} R_T = T \max_a \mu(a) - \mathbf{E} \sum_{t=1}^T \mu(a_t),$$

where the regret is both with respect to the data, and with respect to the internal randomization of the learning algorithm.

13.2 Upper Confidence Bound for Stochastic MAB

An important algorithm for stochastic bandit is *Upper Confidence Bound* (UCB) [51, 1, 9], which is a deterministic algorithm presented in Algorithm 13.1. In this method, we define for each $a = 1, \dots, K$

$$\hat{n}_t(a) = \sum_{s=1}^t I(a_s = a), \quad \hat{\mu}_t(a) = \frac{1}{\hat{n}_t(a)} \sum_{s=1}^t r_s(a_s) I(s_s = a), \quad (13.3)$$

and a properly defined $c_t(a)$, so that the following upper confidence bound holds for an optimal arm $a_* \in \arg \max_a \mu(a)$ with high probability:

$$\mu(a_*) \leq \hat{\mu}_{t-1}(a_*) + \hat{c}_{t-1}(a_*).$$

Algorithm 13.1: UCB Algorithm

Input: K and $T \geq K$

```

1 for  $a = 1, \dots, K$  do
2   | Let  $\hat{n}_0(a) = 0$ 
3   | Let  $\hat{\mu}_0(a) = 0$ 
4 for  $t = 1, 2, \dots, T$  do
5   | if  $t \leq K$  then
6   |   | Let  $a_t = t$ 
7   | else
8   |   | Let  $a_t \in \arg \max_a [\hat{\mu}_{t-1}(a) + \hat{c}_{t-1}(a)]$  according to (13.3)
9   |   Pull arm  $a_t$  and observe reward  $r_t(a_t)$ 
```

We first establish a regret bound for UCB which holds in high probability. The corresponding analysis is more intuitive. A generic result can be obtained as follows.

Lemma 13.1 *Let $a_* \in \arg \max_a \mu(a)$. Let*

$$\delta_1 = \Pr [\exists t \geq K : \mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)].$$

Let

$$\delta_2 = \Pr [\exists t \geq K \ \& \ a \in \{1, \dots, K\} : \mu(a) \leq \hat{\mu}_t(a) - \hat{c}'_t(a)].$$

Then for Algorithm 13.1, we have with probability $1 - \delta_1 - \delta_2$:

$$R_T \leq \sum_{a=1}^K [\mu(a_*) - \mu(a)] + \sum_{t=K+1}^T [\hat{c}_{t-1}(a_t) + \hat{c}'_{t-1}(a_t)].$$

Proof We have with probability $1 - \delta_1 - \delta_2$, the following hold for all $t > K$:

$$\begin{aligned}\mu(a_*) &\leq \hat{\mu}_{t-1}(a_*) + \hat{c}_{t-1}(a_*), \\ \hat{\mu}_{t-1}(a_t) &\leq \mu(a_t) + \hat{c}'_{t-1}(a_t).\end{aligned}$$

It follows that for all $t > K$:

$$\begin{aligned}\mu(a_*) &\leq \hat{\mu}_{t-1}(a_*) + \hat{c}_{t-1}(a_*) \\ &\stackrel{(a)}{\leq} \hat{\mu}_{t-1}(a_t) + \hat{c}_{t-1}(a_t) \\ &\leq \mu(a_t) + \hat{c}'_{t-1}(a_t) + \hat{c}_{t-1}(a_t),\end{aligned}$$

where (a) is due to the UCB algorithm. For $t \leq K$, we have

$$\mu(a_*) = \mu(a_t) + [\mu(a_*) - \mu(t)].$$

We obtain the bound by summing over $t = 1$ to $t = T$. ■

The analysis of bandit problem in Lemma 13.1 is similar to the empirical process analysis of ERM. This technique can be used in other bandit problems such as linear bandits. We note that although the algorithm uses an upper confidence bound, it only requires the bound to hold for the optimal arm a_* . The regret bound relies on both upper confidence bound, and lower confidence bound. The lower confidence bound needs to hold for all arms a , which implies that it holds for a_t . However, the upper confidence bound does not have to satisfy for all a_t . Given upper and lower confidence bounds, the regret bound for MAB becomes an estimation of the summation of the confidence bounds.

Theorem 13.1 *Assume that the reward $r_t(a) \in [0, 1]$. Let $a_* \in \arg \max_a \mu(a)$. With a choice of*

$$\hat{c}_t(a) = \sqrt{\frac{\ln(2(\hat{n}_t(a) + 1)^2/\delta)}{2\hat{n}_t(a)}},$$

we have with probability at least $1 - \delta$:

$$R_T \leq K + \sqrt{\frac{\ln(2KT^2/\delta)(T - K)K}{2}}.$$

Proof In the proof, we let $\hat{n}_{K-1}(a) = 0$ for all a . Therefore $\hat{n}_K(a) = 1 \neq$

$\hat{n}_{K-1}(a)$. We know from Chernoff bound that

$$\begin{aligned}
& \Pr [\exists t \geq K : \mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)] \\
& \leq \Pr [\exists t \geq K \ \& \ \hat{n}_t(a_*) \neq \hat{n}_{t-1}(a_*) : \mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)] \\
& \leq \sum_{t \geq K : \hat{n}_t(a_*) \neq \hat{n}_{t-1}(a_*)} \Pr [\mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)] \\
& \leq \sum_{t \geq K : \hat{n}_t(a_*) \neq \hat{n}_{t-1}(a_*)} 0.5\delta / (\hat{n}_t(a_*) + 1)^2 \\
& \leq \sum_{n \geq 1} 0.5\delta / (n + 1)^2 \leq \delta/2.
\end{aligned}$$

Moreover, let

$$\hat{c}'_t(a) = \sqrt{\frac{\ln(2K(\hat{n}_t(a) + 1)^2/\delta)}{2\hat{n}_t(a)}}.$$

From the Chernoff bound,

$$\begin{aligned}
& \Pr [\exists t \geq K \ \& \ a \in \{1, \dots, K\} : \mu(a) \leq \hat{\mu}_t(a) - \hat{c}'_t(a)] \\
& \leq \sum_{a=1}^K \Pr [\exists t \geq K \ \& \ \hat{n}_t(a) \neq \hat{n}_{t-1}(a) : \mu(a) \geq \hat{\mu}_t(a) + \hat{c}_t(a)] \\
& \leq \sum_{a=1}^K \sum_{t \geq K : \hat{n}_t(a) \neq \hat{n}_{t-1}(a)} \Pr [\mu(a) \geq \hat{\mu}_t(a) + \hat{c}_t(a)] \\
& \leq \sum_{a=1}^K \sum_{t \geq K : \hat{n}_t(a) \neq \hat{n}_{t-1}(a)} \frac{\delta}{2K(\hat{n}_t(a) + 1)^2} \\
& \leq \sum_{a=1}^K \sum_{n \geq 1} \frac{\delta}{2K(n + 1)^2} \leq \delta/2.
\end{aligned}$$

The following bound follows from Lemma 13.1.

$$\begin{aligned}
& \sum_{t=1}^T [\mu(a_*) - \mu(a_t)] - \sum_{a=1}^K [\mu(a_*) - \mu(a)] \leq \sum_{t=K+1}^T [\hat{c}_{t-1}(a_t) + \hat{c}'_{t-1}(a_t)] \\
& \leq 2 \sum_{t=K+1}^T \sqrt{\frac{\ln(2K(\hat{n}_{t-1}(a_t) + 1)^2/\delta)}{2\hat{n}_{t-1}(a_t)}} \\
& \leq 2 \sum_{a=1}^K \sum_{t=1}^{\hat{n}_T(a)-1} \sqrt{\frac{\ln(2K(t+1)^2/\delta)}{2t}} \\
& \leq 2 \sum_{a=1}^K \int_0^{\hat{n}_T(a)-1} \sqrt{\frac{\ln(2K(t+1)^2/\delta)}{2t}} dt \\
& \leq \sum_{a=1}^K \sqrt{\frac{\ln(2K\hat{n}_T(a)^2/\delta)(\hat{n}_T(a) - 1)}{2}} \\
& \leq \sqrt{\frac{\ln(2KT^2/\delta)(T - K)K}{2}},
\end{aligned}$$

where in the derivation of the last inequality, we used

$$\sum_{a=1}^K \hat{n}_T(a) = T,$$

which implies that

$$\sum_{a=1}^K \sqrt{\hat{n}_T(a) - 1} \leq \sqrt{(T - K)K}.$$

This implies the desired bound. ■

Example 13.1 Assume in Theorem 13.1, $\mu(a_*) \approx 0$. In such case, we can use the following multiplicative Chernoff bound to obtain an upper confidence bound. with probability $1 - \delta$:

$$\mu(a) \leq \hat{\mu}_t(a) + \sqrt{\frac{2\mu(a) \ln(1/\delta)}{\hat{n}_t(a)}}.$$

This implies an upper confidence bound of:

$$\mu(a) \leq \left(\sqrt{\hat{\mu}} + \sqrt{\frac{2 \ln(1/\delta)}{\hat{n}(a)}} \right)^2.$$

In order to obtain a better bound than that of Theorem 13.1, one also needs to use the multiplicative Chernoff bound to estimate the lower confidence interval.

Example 13.2 Assume for each arm a , the reward $r_t(a)$ is a sub-Gaussian random variable, but different arms have different reward distributions:

$$\ln \mathbf{E} \exp(\lambda r_t(a)) \leq \lambda \mathbf{E} r_t(a) + \frac{\lambda^2}{2} M(a)^2,$$

where $M(a)$ is known. Then one can obtain a bound similar to Theorem 13.1, with arm dependent UCB estimate involving $M(a)$.

One can also obtain a refined version of Lemma 13.1 that replies on the gap between a_* and any other a .

Lemma 13.2 Let $a_* \in \arg \max_a \mu(a)$. Let

$$\delta_1 = \Pr [\exists t \geq K : \mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)],$$

$$\delta_2 = \Pr [\exists t \geq K, a \in \{1, \dots, K\} : \mu(a) \leq \hat{\mu}_t(a) - \hat{c}'_t(a)].$$

Define $\Delta(a) = \mu(a_*) - \mu(a)$, and

$$T(a) = \max\{\hat{n}_t(a) : \Delta(a) \leq \hat{c}_t(a) + \hat{c}'_t(a), \quad K \leq t \leq T-1\}.$$

Then for Algorithm 13.1, we have with probability $1 - \delta_1 - \delta_2$:

$$R_T \leq \inf_{\Delta_0 > 0} \left[\sum_{a=1}^K \max(T(a), 1) \Delta(a) I(\Delta(a) > \Delta_0) + T \Delta_0 \right] + \sup_a \Delta(a).$$

Proof The proof of Lemma 13.1 shows that with probability at least $1 - \delta$, for all $t \geq K + 1$:

$$\mu(a_*) \leq \mu(a_t) + \hat{c}'_{t-1}(a_t) + \hat{c}_{t-1}(a_t),$$

which implies that $\hat{n}_{t-1}(a_t) \leq T(a)$. It follows that for all a , if $\hat{n}_{T-1}(a) \geq 2$, then there exists $K + 1 \leq t \leq T$ such that $\hat{n}_{T-1}(a) = \hat{n}_{t-1}(a_t)$. Therefore $\hat{n}_{T-1}(a) \leq \max(1, T(a))$ for all a . Let $M = \sup_a \Delta(a)$, then $\mu(a_*) - \mu(a_T) \leq M$. We note that the regret bound satisfies:

$$\begin{aligned} \sum_{t=1}^T [\mu(a_*) - \mu(a_t)] &= \Delta(a_T) + \sum_{t=1}^{T-1} [\mu(a_*) - \mu(a_t)] \\ &= \Delta(a_T) + \sum_{a=1}^K \hat{n}_{T-1}(a) \Delta(a) \\ &\leq M + \sum_{a=1}^K \hat{n}_{T-1}(a) \Delta_0 I(\Delta(a) \leq \Delta_0) + \sum_{a=1}^K \hat{n}_{T-1}(a) \Delta(a) I(\Delta(a) > \Delta_0) \\ &\leq M + T \Delta_0 + \sum_{a=1}^K \hat{n}_{T-1}(a) \Delta(a) I(\Delta(a) > \Delta_0), \end{aligned}$$

where $\Delta_0 \geq 0$. This implies the bound. $\blacksquare$

We can obtain a gap dependent bound using Lemma 13.2, as in Theorem 13.2.

Theorem 13.2 *Assume that the reward $r_t(a) \in [0, 1]$. Let $a_* \in \arg \max_a \mu(a)$. With a choice of*

$$\hat{c}_t(a) = \sqrt{\frac{\ln(2(\hat{n}_t(a) + 1)^2/\delta)}{2\hat{n}_t(a)}},$$

we have

$$R_T \leq \inf_{\Delta_0 > 0} \left[T\Delta_0 + \sum_{a=1}^K \frac{2\ln(2KT^2/\delta)}{\Delta(a)} I(\Delta(a) > \Delta_0) \right] + \sup_a \Delta(a).$$

Proof From the proof of Theorem 13.1, we know that with the choice of

$$\tilde{c}'_t(a) = \sqrt{\frac{\ln(2K(\hat{n}_t(a) + 1)^2/\delta)}{2\hat{n}_t(a)}},$$

the inequality

$$\Delta(a) \leq \hat{c}_{t-1}(a) + \tilde{c}'_{t-1}(a)$$

implies that

$$\Delta(a) \leq \sqrt{\frac{2\ln(2KT^2/\delta)}{\hat{n}_{t-1}(a)}}.$$

This implies that

$$\hat{n}_{t-1}(a) \leq \frac{2\ln(2KT^2/\delta)}{\Delta(a)^2}.$$

Therefore

$$T(a) \leq \frac{2\ln(2KT^2/\delta)}{\Delta(a)^2}.$$

This implies the desired result from Lemma 13.2. $\blacksquare$

The gap dependent regret bound of Theorem 13.2 implies the gap independent regret bound of Theorem 13.1. If we take

$$\Delta_0 = \sqrt{\frac{K \ln(KT/\delta)}{T}},$$

then we obtain a regret of

$$R_T = O\left(\sqrt{KT \ln(KT/\delta)}\right)$$

from Theorem 13.2.

In the following, we show that using a similar analysis, it is possible to obtain a bound in expectation, as in [9]. In the proof of the expectation version, we do not have to consider events that hold uniformly for all t , but rather focus on each specific t .

Lemma 13.3 *Let $a_* \in \arg \max_a \mu(a)$. Let*

$$\begin{aligned}\delta_1(t) &= \Pr [\mu(a_*) \geq \hat{\mu}_t(a_*) + \hat{c}_t(a_*)], \\ \delta_2(t) &= \Pr [\exists a \in \{1, \dots, K\} : \mu(a) \leq \hat{\mu}_t(a) - \hat{c}'_t(a)].\end{aligned}$$

Define $\Delta(a) = \mu(a_*) - \mu(a)$, $M = \sup_a \Delta(a)$,

$$T(a) = \max\{\hat{n}_t(a) : \Delta(a) \leq \hat{c}_t(a) + \hat{c}'_t(a), K \leq t \leq T-1\},$$

and

$$\delta = \sum_{t \geq K} [\delta_1(t) + \delta_2(t)].$$

Then for Algorithm 13.1, we have

$$\mathbf{E} R_T \leq \inf_{\Delta_0 > 0} \left[\sum_{a=1}^K (1 + T(a)) \Delta(a) I(\Delta(a) > \Delta_0) + T \Delta_0 \right] + \delta M.$$

Proof We obtain for any fixed $t > K$ that with probability $1 - \delta_1(t-1)$:

$$\mu(a_*) \leq \hat{\mu}_{t-1}(a_t) + \hat{c}_{t-1}(a_t).$$

Moreover, with probability $1 - \delta_2(t-1)$:

$$\hat{\mu}_{t-1}(a_t) \leq \mu(a_t) + \hat{c}'_{t-1}(a_t).$$

It follows that with probability $1 - \delta_1(t-1) - \delta_2(t-1)$:

$$\Delta(a_t) = \mu(a_*) - \hat{\mu}_{t-1}(a_t) \leq \hat{c}_{t-1}(a_t) + \hat{c}'_{t-1}(a_t).$$

This implies that $\hat{n}_{t-1}(a_t) \leq T(a_t)$. It follows that for all $t > K$:

$$\Pr(\hat{n}_{t-1}(a_t) > T(a_t)) \leq \delta_1(t-1) + \delta_2(t-1).$$

Now let

$$\delta(a) = \sum_{t=K+1}^T \mathbf{E} I(a = a_t, \hat{n}_{t-1}(a) > T(a)),$$

then

$$\begin{aligned} \sum_{a=1}^K \delta(a) &= \sum_{a=1}^K \sum_{t=K+1}^T \Pr(a = a_t, \hat{n}_{t-1}(a) > T(a)) \\ &= \sum_{t=K+1}^T \Pr(\hat{n}_{t-1}(a_t) > T(a_t)) \leq \delta. \end{aligned}$$

Now we have for any fixed a :

$$\begin{aligned} \mathbf{E} \hat{n}_T(a) &= 1 + \mathbf{E} \sum_{t=K+1}^T I(a = a_t) \\ &= 1 + \sum_{t=K+1}^T \mathbf{E} I(a = a_t, \hat{n}_{t-1}(a) \leq T(a)) \\ &\quad + \sum_{t=K+1}^T \mathbf{E} I(a = a_t, \hat{n}_{t-1}(a) > T(a)) \\ &\leq 1 + T(a) + \delta(a). \end{aligned}$$

Since expected regret bound satisfies

$$\begin{aligned} \mathbf{E} \sum_{t=1}^T [\mu(a_*) - \mu(a_t)] &= \sum_{a=1}^K \Delta(a) \mathbf{E} \hat{n}_T(a) \\ &\leq \sum_{a=1}^K (1 + T(a) + \delta(a)) \Delta(a) I(\Delta(a) > \Delta_0) + \sum_{a=1}^K \Delta_0 \mathbf{E} \hat{n}_T(a) \\ &\leq \sum_{a=1}^K \Delta(a) (1 + T(a)) I(\Delta(a) > \Delta_0) + T \Delta_0 + \delta \cdot M. \end{aligned}$$

This implies the desired bound. ■

Theorem 13.3 Assume that the reward $r_t(a) \in [0, 1]$. Let $a_* \in \arg \max_a \mu(a)$. With a choice of

$$\hat{c}_t(a) = \sqrt{\frac{\alpha \ln t}{2\hat{n}_t(a)}}$$

for $\alpha > 1$. We have

$$\mathbf{E} R_T \leq \inf_{\Delta_0 > 0} \left[\sum_{a=1}^K \left(\Delta(a) + \frac{2\alpha \ln(T)}{\Delta(a)} \right) I(\Delta(a) > \Delta_0) + T \Delta_0 \right] + \frac{2\alpha K^{2-\alpha}}{\alpha - 1} M.$$

Proof Using the same analysis of Theorem 13.1, we know that for any fixed $t = K + 1, \dots, T$, with probability at least $1 - (t - 1)^{-\alpha}$:

$$\mu(a_*) \leq \hat{\mu}_{t-1}(a_*) + \hat{c}_{t-1}(a_*).$$

Moreover, with probability at least $1 - K(t - 1)^{-\alpha}$: for all a ,

$$\hat{\mu}_{t-1}(a) \leq \mu(a) + \hat{c}'_{t-1}(a) \quad c'_{t-1}(a) = \sqrt{\frac{\alpha \ln(t - 1)}{2\hat{n}_{t-1}(a)}}.$$

Therefore we can choose $\delta_1(t) = t^{-\alpha}$ and $\delta_2(t) = Kt^{-\alpha}$. We have

$$\begin{aligned} \delta &= \sum_{t=K}^{\infty} [\delta_1(t) + \delta_2(t)] \leq (K + 1) \sum_{t=K}^{\infty} t^{-\alpha} \\ &\leq (K + 1)K^{-\alpha} + (K + 1) \int_K^{\infty} t^{-\alpha} dt \leq \frac{2\alpha}{\alpha - 1} K^{2-\alpha}. \end{aligned}$$

Since

$$\Delta(a) \leq \hat{c}_{t-1}(a) + \hat{c}'_{t-1}(a)$$

implies that

$$\hat{n}_{t-1}(a) \leq \frac{2\alpha \ln(T)}{\Delta(a)^2},$$

It follows that

$$T(a) \leq \frac{2\alpha \ln(T)}{\Delta(a)^2}.$$

We can now apply Lemma 13.3 to obtain the desired bound. ■

The following result gave a lower bound on any strongly consistent algorithm for MAB.

Theorem 13.4 ([51]) *Assume $r_t(a) \in \{0, 1\}$. Consider an algorithm such that $\forall \beta \in (0, 1)$, $\lim_{T \rightarrow \infty} (R_T / T^\beta) = 0$, then*

$$\liminf_{T \rightarrow \infty} \frac{\mathbf{E} [\sum_{t=1}^T I(a_t = a)]}{\ln T} \geq \frac{1}{\text{KL}(\mu(a), \mu_*)}$$

for all a , where $\mu_* = \min_a \mu(a)$.

Since $\text{KL}(\mu(a), \mu_*)^{-1} = \Omega(\Delta(a)^2)$, it follows that the UCB bound in Theorem 13.3 has the worst case optimal dependency on the gap $\Delta(a)$ up to a constant. One may also obtain more refined dependency on $\Delta(a)$ using the KL-divergence version of the Hoeffding's inequality in Theorem 2.6. This leads to KL-UCB as in [39, 56], with gap-dependency matching the lower bound of Theorem 13.4.

For gap-independent expected regret bounds, a result similar to that of Theorem 13.1, can be obtained from Theorem 13.3, where we take $\Delta_0 = O(\sqrt{K \ln(T)/T})$ to obtain

$$R_T = O(\sqrt{KT \ln(T)}).$$

It is known that the lower bound is $\Omega(\sqrt{KT})$, as stated in Theorem 13.5. The regret bound for UCB in Theorem 13.3 only achieves this lower bound up to a $\log(T)$ factor. It is known that there exists an algorithm that can remove this $\log T$ factor, and achieve the lower bound [7].

Theorem 13.5 ([10]) *Assume $r_t(a) \in [0, 1]$ and $K \geq 2$. There exists a distribution over the assignment of rewards such that the expected weak regret of any algorithm (where the expectation is taken with respect to both the randomization over rewards and the algorithms internal randomization) is at least*

$$\frac{1}{20} \min[\sqrt{KT}, T].$$

13.3 Thompson Sampling for Stochastic MAB

Thompson sampling [84] is another popular algorithm for bandit problems with a Bayesian interpretation. The basic idea is to consider a prior distribution on the mean of the reward distribution of every arm, and at any time step, sample a mean from the posterior for each arm, then pick the arm with the highest sampled mean.

In practice, it will be convenient to use a model so that the posterior is simple. This can be achieved with conjugate priors. For example, we can assume that a Gaussian prior and reward likelihood:

$$\mu(a) \sim N(0, 1), \quad r_t(a) \sim N(\mu(a), 1).$$

Then the posterior for arm a after time step $t - 1$ is given by the normal distribution

$$N(\hat{\mu}_{t-1}(a), \hat{V}_{t-1}(a)),$$

where

$$\hat{\mu}_{t-1}(a) = \frac{\sum_{s=1}^{t-1} I(a_s = a) r_s(a_s)}{1 + \sum_{s=1}^{t-1} I(a_s = a)}, \quad \hat{V}_{t-1}(a) = 1 + \sum_{s=1}^{t-1} I(a_s = a). \quad (13.4)$$

This leads to Algorithm 13.2.

Algorithm 13.2: Thompson Sampling (Gaussian)

Input: K and T

```

1 for  $t = 1, 2, \dots, T$  do
2   for  $a = 1, \dots, K$  do
3      $\tilde{\mu}(a) \sim N(\hat{\mu}_{t-1}(a), V_{t-1}(a))$  according to (13.4)
4   Let  $a_t = \arg \max_a \tilde{\mu}(a)$ 
5   Pull arm  $a_t$  and observe reward  $r_t(a_t)$ 

```

Alternatively, for binary rewards $r_t(a) \in \{0, 1\}$, we can use Beta prior with Bernoulli likelihood:

$$\mu(a) \sim \text{Beta}(1, 1), \quad r_t(a) \sim \text{Bernoulli}(\mu(a)).$$

The posterior for arm a after time step $t-1$ is given by the Beta distribution

$$\text{Beta}(1 + \hat{n}_{t-1}^1(a), 1 + \hat{n}_{t-1}^0(a))$$

where for $y \in \{0, 1\}$:

$$\hat{n}_{t-1}^y(a) = \sum_{s=1}^{t-1} I(a_s = a) I(r_s(a_s) = y). \quad (13.5)$$

This leads to Algorithm 13.3.

Algorithm 13.3: Thompson Sampling (Beta)

Input: K and T

```

1 for  $t = 1, 2, \dots, T$  do
2   for  $a = 1, \dots, K$  do
3      $\tilde{\mu}(a) \sim \text{Beta}(1 + \hat{n}_{t-1}^1(a), 1 + \hat{n}_{t-1}^0(a))$  according to
       (13.5)
4   Let  $a_t = \arg \max_a \tilde{\mu}(a)$ 
5   Pull arm  $a_t$  and observe reward  $r_t(a_t) \in \{0, 1\}$ 

```

Regret bounds for Thompson sampling have been established in the literature [2, 3, 47] and it is known [3] that we have a bound of

$$R_T = O(\sqrt{TK \ln K})$$

for Gaussian Thompson Sampling, and a bound of

$$R_T = O(\sqrt{TK \ln T})$$

for Beta Thompson Sampling. Moreover, near optimal gap-dependent bound can also be obtained [3].

13.4 EXP3 for Adversarial MAB

The UCB algorithm can only be applied to stochastic bandit problems. For adversarial bandits, one can use an exponential weighting method motivated from aggregation algorithm for online learning. The algorithm, referred to as EXP3 [10], is given in Algorithm 13.4.

Algorithm 13.4: EXP3

```

Input:  $K, T, \gamma \in (0, 1]$ 
1 for  $a = 1, \dots, K$  do
2    $\lfloor$  Let  $w_0(a) = 1$ 
3 for  $t = 1, 2, \dots, T$  do
4   Let  $w_{t-1} = \sum_{a=1}^K w_{t-1}(a)$ 
5   for  $a = 1, \dots, K$  do
6      $\lfloor$  Let  $p_{t-1}(a) = (1 - \gamma)w_{t-1}(a)/w_{t-1} + \gamma/K$ 
7   Sample  $a_t$  according to  $p_{t-1}(\cdot)$ 
8   Pull arm  $a_t$  and observe reward  $r_t(a_t) \in [0, 1]$ 
9   for  $a = 1, \dots, K$  do
10    Let  $\hat{r}_t(a) = r_t(a_t)I(a = a_t)/p_{t-1}(a_t)$ 
11    Let  $w_t(a) = w_{t-1}(a) \exp(\gamma \hat{r}_t(a)/K)$ 

```

Conditioned on the observations made before time step t , the reward estimator $\hat{r}_t(a)$ is a random estimator with respect to the random selection of the next arm $a_t \sim p_{t-1}(\cdot)$. It converts a partial information problem with only the reward $r_t(a_t)$ at a_t into a full information problem with reward $\hat{r}_t(a)$ defined for all arms $a \in \{1, \dots, K\}$. The choice of $\hat{r}_t(a)$ is that $\hat{r}_t(a)$ is an unbiased estimator of the full information reward vector $[r_t(a) : a \in \{1, \dots, K\}]$:

$$r_t(a) = \mathbf{E}_{a_t \sim p_{t-1}(\cdot)} \hat{r}_t(a).$$

The following regret bounds hold for the EXP3 algorithm, with regret defined in (13.1).

Theorem 13.6 *Consider Algorithm 13.4. Let*

$$G_* = \max_a \sum_{i=1}^T r_i(a).$$

We have

$$R_T \leq (e - 1)\gamma G_* + \frac{K \ln K}{\gamma}.$$

Proof We have

$$\gamma \hat{r}_t(a)/K \leq 1.$$

Therefore

$$\begin{aligned} \sum_{a=1}^K p_{t-1}(a) \hat{r}_t(a) &= r_t(a_t) \\ \sum_{a=1}^K p_{t-1}(a) \hat{r}_t(a)^2 &\leq \sum_{a=1}^K \hat{r}_t(a). \end{aligned}$$

It follows that for all $t \geq 1$:

$$\begin{aligned} \frac{w_t}{w_{t-1}} &= \sum_{a=1}^K \frac{w_t(a)}{w_{t-1}} \\ &= \sum_{a=1}^K \frac{w_{t-1}(a)}{w_{t-1}} \exp(\gamma \hat{r}_t(a)/K) \\ &= \sum_{a=1}^K \frac{p_{t-1}(a) - \gamma/K}{1 - \gamma} \exp(\gamma \hat{r}_t(a)/K) \\ &\stackrel{(a)}{\leq} \sum_{a=1}^K \frac{p_{t-1}(a) - \gamma/K}{1 - \gamma} [1 + \gamma \hat{r}_t(a)/K + (e - 2)(\gamma \hat{r}_t(a)/K)^2] \\ &\leq 1 + \frac{1}{1 - \gamma} \left[\sum_{a=1}^K p_{t-1}(a) \gamma \hat{r}_t(a)/K + \sum_{a=1}^K p_{t-1}(a) (e - 2)(\gamma \hat{r}_t(a)/K)^2 \right] \\ &\leq 1 + \frac{\gamma/K}{1 - \gamma} r_t(a_t) + \frac{(e - 2)(\gamma/K)^2}{1 - \gamma} \sum_{a=1}^K \hat{r}_t(a). \end{aligned}$$

In (a), we used $\exp(z) \leq 1 + z + (e - 2)z^2$ when $z \leq 1$.

From $\ln(1 + z) \leq z$, we obtain

$$\ln \frac{w_t}{w_{t-1}} \leq \frac{\gamma/K}{1 - \gamma} r_t(a_t) + \frac{(e - 2)(\gamma/K)^2}{1 - \gamma} \sum_{a=1}^K \hat{r}_t(a).$$

Now, by summing over t , and let

$$G_T = \sum_{t=1}^T r_t(a_t),$$

we get

$$\ln \frac{w_T}{w_0} \leq \frac{\gamma/K}{1 - \gamma} G_T + \frac{(e - 2)(\gamma/K)^2}{1 - \gamma} \sum_{t=1}^T \sum_{a=1}^K \hat{r}_t(a).$$

Note that for all a :

$$\ln \frac{w_T}{w_0} \geq \ln \frac{w_T(a)}{w_0} = \frac{\gamma}{K} \sum_{t=1}^T \hat{r}_t(a) - \ln K.$$

It follows that

$$\frac{\gamma}{K} \sum_{t=1}^T \hat{r}_t(a) \leq \frac{\gamma/K}{1-\gamma} G_T + \frac{(e-2)(\gamma/K)^2}{1-\gamma} \sum_{t=1}^T \sum_{a=1}^K \hat{r}_t(a) + \ln K.$$

Observe that conditioned on the history,

$$\mathbf{E}_{a_t \sim p_{t-1}} \hat{r}_t(a) = r_t(a).$$

We obtain the following bound by taking expectation:

$$\begin{aligned} \mathbf{E} \frac{\gamma}{K} \sum_{t=1}^T r_t(a) &\leq \frac{\gamma/K}{1-\gamma} \mathbf{E} G_T + \frac{(e-2)(\gamma/K)^2}{1-\gamma} \mathbf{E} \sum_{t=1}^T \sum_{a=1}^K r_t(a) + \ln K \\ &\leq \frac{\gamma/K}{1-\gamma} \mathbf{E} G_T + \frac{(e-2)(\gamma/K)^2}{1-\gamma} K G_* + \ln K. \end{aligned}$$

Let a be the optimal arm, and rearrange, we obtain

$$G_* \leq \frac{1}{1-\gamma} \mathbf{E} G_T + \frac{(e-2)\gamma}{(1-\gamma)} G_* + \frac{K}{\gamma} \ln K.$$

This implies the desired result. ■

If we take

$$\gamma = \min(1, \sqrt{K \ln K / ((e-1)g)})$$

for some $g \geq G_*$ in Theorem 13.6, then we obtain a bound

$$R_T \leq 2\sqrt{e-1} \sqrt{gK \ln K}.$$

In particular, with $g = T$, we have

$$R_T \leq 2\sqrt{e-1} \sqrt{TK \ln K}.$$

13.5 Exercises

1. Finish Example 13.2 by deriving the corresponding UCB algorithm and its regret bound in the style of Theorem 13.1.

Chapter 14

Contextual Bandits

In the standard multi-armed bandit problem, one observes a fixed number of arms. To achieve optimal regret bounds, one estimates confidence intervals of the arms by counting. In the contextual bandit problem, one observes side information for each arm, which can be used as features for more accurate confidence interval estimation.

The first explicit definition of this problem in the stochastic bandit setting was given in [52] as follows.

Definition 14.1 (Contextual bandit problem) *In a stochastic contextual bandit problem, there is a distribution D over $(x, r(1), \dots, r(K)) \in \mathcal{X} \times \mathbb{R}^K$, where $x \in \mathcal{X}$ is side information, $a \in \{1, \dots, K\}$ is one of the K arms to be pulled, and $r(a) \in \mathbb{R}$ is the reward for arm a . The problem is a repeated game: at each time step t :*

- *A sample $(x_t, r_t(1), \dots, r_t(K))$ is drawn from D , and the side information x_t is announced*
- *The player chooses precisely one arm $a_t \in \{1, \dots, K\}$*
- *The reward $r_t(a_t)$ is revealed.*

The problem includes the multi-armed bandit problem as special case if we take $x = x_0$ to be a fixed context for all time steps. Similar to the case of multi-armed bandit problem, one may also assume a pre-determined contextual sequence of $\{(x_t, r_t(1), \dots, r_t(K)) : t \geq 1\}$ without assuming a distribution D . This leads to the adversarial contextual bandit problem (with oblivious adversary).

For contextual bandit problems, a contextual bandit algorithm determines an arm $a \in \{1, \dots, K\}$ to pull at each time step t , based on the previous observation sequence $\{(x_1, (a_1, r_1(a_1)), \dots, (x_{t-1}, a_{t-1}, r_{t-1}(a_{t-1})))\}$, and the current side information x_t .

14.1 EXP4 Algorithm

The EXP4 algorithm is a generalization of the EXP3 algorithm for the adversarial multi-armed bandit problem [10]. It can be regarded as a policy based method for adversarial contextual bandit.

EXP4 works in the setting of “experts”. A deterministic expert may be considered a function of $\mathcal{X} \rightarrow \{1, \dots, K\}$, or a classifier. An expert can also be a random classifier, which maps $x \in \mathcal{X}$ to a distribution $p(\cdot|x)$ on $\{1, \dots, K\}$, and then one draws an arm $a \sim p(a|x)$. In this setting, an expert can be regarded as a *policy* using the notation of reinforcement learning.

Assume that we have a family of random policies (or experts) indexed by w :

$$G = \{p(a|w, x) : w \in \Omega\}.$$

Each policy $p(\cdot|w, \cdot)$ is a conditional distribution on $\{1, \dots, K\}$.

Let $p_0(w)$ be a prior on Ω , then the EXP4 algorithm, presented in Algorithm 14.1, has a regret bound that is logarithmic in $|G|$ for finite G , if the regret is to compete with the best policy in G . EXP4 can also be regarded as a generalization of EXP3, which has K policies, each pulls one arm $a \in \{1, \dots, K\}$ constantly.

Example 14.1 *As an example, we consider logistic policy defined as*

$$p(a|w, x) = \frac{\exp(w^\top \psi(a, x))}{\sum_{\ell=1}^K \exp(w^\top \psi(\ell, x))},$$

with Gaussian prior:

$$w \sim N(0, \sigma^2).$$

Algorithm 14.1: EXP4

Input: $K, T, G, p_0(\cdot), \gamma \in (0, 1], \eta > 0, b \geq 0$

- 1 Let $u_0(w) = 1$
- 2 **for** $t = 1, 2, \dots, T$ **do**
- 3 Observe x_t
- 4 **for** $a = 1, \dots, K$ **do**
- 5 Let $p_t(a) = (1 - \gamma)\mathbf{E}_{w \sim p_{t-1}(w)} p(a|w, x_t) + \gamma/K$
- 6 Sample a_t according to $p_t(\cdot)$
- 7 Pull arm a_t and observe reward $r_t(a_t) \in [0, 1]$
- 8 Let $\hat{r}_t(w, x_t) = p(a_t|w, x_t)(r_t(a_t) - b)/p_t(a_t)$
- 9 Let $u_t(w) = u_{t-1}(w) \exp(\eta \hat{r}_t(w, x_t))$
- 10 Let $p_t(w) = u_t(w)p_0(w)/\mathbf{E}_{w \sim p_0(w)} u_t(w)$

Note that conditioned on the history, the estimator $\hat{r}_t(w, x_t)$ is a random estimator that depends on the partial reward $r_t(a_t)$ received for $a_t \sim p_t(a)$.

Moreover, it is an unbiased estimator of

$$r_t(w, x_t) - b = \sum_{a=1}^k p(a|w, x_t)(r_t(a) - b),$$

which relies on the full reward vector $[r_t(a)]$ at time step t over all arms a .

Both parameter γ and b control exploration. The original EXP4 set $b = 0$, and in such case we need to set $\gamma > 0$. The larger γ is, the more uniform $p_t(a)$ becomes, and thus we explore more.

Similarly, the larger b is, the more penalty we put on arms that have been observed, and this favors arms that are not observed in the future. In the full information case, where all $r_t(a)$ are observed at every time step, the parameter b doesn't affect the algorithm. However, in the partial information case, if we set $b = 0$, we need $\gamma > 0$ to perform exploration. On the other hand, if we choose $b = 1$, then we can set $\gamma = 0$.

Theorem 14.1 *For any $K, T \geq 0$, and any $\gamma \in (0, 1]$, $\eta > 0$ and $b \geq 0$. Consider any family of policies $G = \{p(a|w, x) : w \in \Omega\}$ with prior $p_0(w)$. Then the expected reward of Algorithm 14.1 satisfies:*

$$\begin{aligned} \mathbf{E} \sum_{t=1}^T r_t(a_t) &\geq (1 - \gamma) \max_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T r_t(w, x_t) - \frac{1}{\eta} \text{KL}(q||p_0), \right] \\ &\quad - c(\eta, b) \eta \sum_{t=1}^T \sum_{a=1}^K |r_t(a) - b|, \end{aligned}$$

where the expectation is with respect to the randomization of the algorithm,

$$c(\eta, b) = \psi(z_0) \max(b, 1 - b), \quad z_0 = \max(0, \eta(1 - b)K/\gamma),$$

and $\psi(z) = (e^z - 1 - z)/z^2$.

Proof The proof is similar to that of Theorem 13.6. By construction, we have

$$\mathbf{E}_{w \sim p_{t-1}(w)} p(a_t|w, x_t) \leq p_t(a_t)/(1 - \gamma).$$

Therefore

$$\begin{aligned} \mathbf{E}_{w \sim p_{t-1}(w)} \hat{r}_t(w, x_t) &= \mathbf{E}_{w \sim p_{t-1}(w)} p(a_t|w, x_t)[r_t(a_t) - b]/p_t(a_t) \\ &\leq \frac{1}{1 - \gamma} r_t(a_t) - q_t(a_t)b, \end{aligned} \tag{14.1}$$

where $q_t(a) = \mathbf{E}_{w \sim p_{t-1}(w)} p(a|w, x_t)/p_t(a)$, and

$$\begin{aligned} &\mathbf{E}_{w \sim p_{t-1}(w)} \hat{r}_t(w, x_t)^2 \\ &= \mathbf{E}_{w \sim p_{t-1}(w)} p(a_t|w, x_t)^2 ((r_t(a_t) - b)/p_t(a_t))^2 \\ &\leq \max(b, 1 - b) \mathbf{E}_{w \sim p_{t-1}(w)} p(a_t|w, x_t) (|r_t(a_t) - b|/p_t(a_t))^2 \\ &\leq \frac{\max(b, 1 - b)}{1 - \gamma} (|r_t(a_t) - b|/p_t(a_t)). \end{aligned} \tag{14.2}$$

If we let

$$W_t = \mathbf{E}_{w \sim p_0(w)} u_t(w),$$

then

$$\begin{aligned} \ln \frac{W_t}{W_{t-1}} &= \ln \mathbf{E}_{w \sim p_0(w)} \frac{u_t(w)}{W_{t-1}} \\ &= \ln \mathbf{E}_{w \sim p_{t-1}(w)} \exp(\eta \hat{r}_t(w, x_t)) \\ &\leq \ln \mathbf{E}_{w \sim p_{t-1}(w)} [1 + (\eta \hat{r}_t(w, x_t)) + \psi(z_0)(\eta \hat{r}_t(w, x_t))^2] \\ &\leq \mathbf{E}_{w \sim p_{t-1}(w)} (\eta \hat{r}_t(w, x_t)) + \psi(z_0) \mathbf{E}_{w \sim p_{t-1}(w)} (\eta \hat{r}_t(w, x_t))^2 \\ &\leq \frac{\eta}{1-\gamma} r_t(a_t) - \eta q_t(a_t) b + \frac{c(\eta, b) \eta^2}{(1-\gamma)} \frac{|r_t(a_t) - b|}{p_t(a_t)}, \end{aligned}$$

where the first inequality used $\exp(z) \leq 1 + z + \psi(z)z^2$ when $z = \eta \hat{r}_t(w, x_t) \leq z_0$; the second inequality used $\ln(1 + z) \leq z$; and the third inequality used (14.1) and (14.2).

Now we can sum over $t = 1$ to $t = T$, and obtain:

$$\ln \frac{W_T}{W_0} \leq \frac{\eta}{1-\gamma} \sum_{t=1}^T r_t(a_t) - \eta b \sum_{t=1}^T q_t(a_t) + \frac{c(\eta, b) \eta^2}{(1-\gamma)} \sum_{t=1}^T \frac{|r_t(a_t) - b|}{p_t(a_t)}.$$

Note that $W_0 = 1$. We can take expectation with respect to the randomization of the algorithm, and note that $\mathbf{E} q_t(a_t) = 1$ to obtain

$$\begin{aligned} &\mathbf{E} \ln \mathbf{E}_{w \sim p_0(w)} \exp \left(\eta \sum_{t=1}^T \hat{r}_t(w, x_t) \right) \\ &\leq \frac{\eta}{1-\gamma} \mathbf{E} \sum_{t=1}^T r_t(a_t) - \eta T b + \frac{c(\eta, b) \eta^2}{(1-\gamma)} \mathbf{E} \sum_{t=1}^T \sum_{a=1}^K |r_t(a) - b|. \end{aligned}$$

We can apply Proposition 12.1 to obtain

$$\begin{aligned} &\mathbf{E} \ln \mathbf{E}_{w \sim p_0(w)} \exp \left(\eta \sum_{t=1}^T \hat{r}_t(w, x_t) \right) \\ &= \mathbf{E} \max_q \left[\mathbf{E}_{w \sim q} \eta \sum_{t=1}^T \hat{r}_t(w, x_t) - \text{KL}(q \| p_0) \right] \\ &\geq \max_q \mathbf{E} \left[\mathbf{E}_{w \sim q} \eta \sum_{t=1}^T \hat{r}_t(w) - \text{KL}(q \| p_0) \right] \\ &= \max_q \mathbf{E} \left[\mathbf{E}_{w \sim q} \eta \sum_{t=1}^T [r_t(w, x_t) - b] - \text{KL}(q \| p_0) \right]. \end{aligned}$$

This implies the result. ■

The following result is a direct consequence of Theorem 14.1. It can be regarded as a direct generalization of Theorem 13.6.

Corollary 14.1 *Let $\eta = \gamma/K$ and $b = 0$. Assumes that the uniform random policy belongs to Ω and $|\Omega| = N < \infty$. Let $p_0(w)$ be the uniform prior over Ω , then*

$$G_* - \mathbf{E} \sum_{t=1}^T r_t(a_t) \leq (e-1)\gamma G_* + \frac{K \ln N}{\gamma},$$

where the expectation is with respect to the randomization of the algorithm, and

$$G_* = \arg \max_w \sum_{t=1}^T r_t(w, x_t).$$

Proof We have $\eta \hat{r}_t(w, x_t) \leq 1$, and thus $c(\eta, b) = e - 2$. Note that the uniform random policy belongs to Ω implies that

$$\frac{1}{K} \sum_{t=1}^T \sum_{a=1}^K r_t(a) \leq G_*.$$

We consider Theorem 14.1, with q defined as $q(w) = I(w = w_*)$, where w_* achieves the maximum of G_* . This implies

$$\mathbf{E} \sum_{t=1}^T r_t(a_t) \geq (1-\gamma) \left[G_* - \frac{K}{\gamma} \ln N \right] - (e-2)\gamma G_*.$$

This implies the bound. ■

Note that $\psi(z)$ is an increasing function of z . If we take $\gamma = 0$ and $b = 1$, then we have $\eta(1-b) \leq 0$, and we may take

$$c(\eta, b) = 0.5.$$

Theorem 14.1 implies that

$$\mathbf{E} \sum_{t=1}^T r_t(a_t) \geq \max_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T r_t(w, x_t) - \frac{1}{\eta} \text{KL}(q \| p_0) \right] - 0.5\eta KT.$$

In the finite policy case as in Corollary 14.1, with the choice of $\gamma = 0$ and $b = 1$, then

$$G_* - \mathbf{E} \sum_{t=1}^T r_t(a_t) \leq \frac{\ln N}{\eta} + 0.5\eta KT.$$

It was argued in [52] that EXP4 is not an efficient algorithm for contextual bandit because the computation of

$$\mathbf{E}_{w \sim p_{t-1}(w)} p(a|w, x_t)$$

is difficult to implement efficiently. An empirical risk minimization oracle based method called epoch-greedy was investigated in [52]. However, epoch greedy has suboptimal regret of $O(T^{2/3})$ instead of $O(\sqrt{T})$. The gap was closed in a subsequent work [32] using a more complicated algorithm that can also use an empirical risk minimization oracle.

We may compare EXP4 to its full information counterpart, often referred to as the *Hedge Algorithm* [37]. For comparison purposes, we use the same notation as bandit in Algorithm 14.2. In essence, because we can observe the information for all arms, it is possible to replace $\hat{r}_t(w, x_t)$ by $r_t(w, x_t)$, which removes the K -dependency in the resulting regret bound because we do not have to explore in order to obtain rewards for different arms.

Algorithm 14.2: Hedge

Input: $K, T, G, p_0(\cdot), \eta$

```

1 Let  $u_0(w) = 1$ 
2 for  $t = 1, 2, \dots, T$  do
3   Observe  $x_t$ 
4   for  $a = 1, \dots, K$  do
5     Let  $p_t(a) = \mathbf{E}_{w \sim p_{t-1}(w)} p(a|w, x_t)$ 
6   Sample  $a_t$  according to  $p_t(\cdot)$ 
7   Pull arm  $a_t$  and observe reward  $r_t(a) \in [0, 1]$  for all  $a \in \{1, \dots, K\}$ 
8   Let  $r_t(w, x_t) = \sum_{a=1}^K p(a|w, x_t) r_t(a)$ 
9   Let  $u_t(w) = u_{t-1}(w) \exp(\eta r_t(w, x_t))$ 
10  Let  $p_t(w) = u_t(w) p_0(w) / \mathbf{E}_{w \sim p_0(w)} u_t(w)$ 

```

We have the following regret bound for Algorithm 14.2, which is independent of K .

Theorem 14.2 *For any $K, T \geq 0$, and any $\eta > 0$. Consider any family of policies $G = \{p(a|w, x) : w \in \Omega\}$ with prior $p_0(w)$. Then the expected reward of Algorithm 14.2 satisfies:*

$$\mathbf{E} \sum_{t=1}^T r_t(a_t) \geq \frac{1}{1 + \psi(\eta)\eta} \max_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T r_t(w, x_t) - \frac{1}{\eta} \text{KL}(q||p_0) \right],$$

where $\psi(z) = (e^z - z - 1)/z^2$, and the expectation is with respect to the randomization of the algorithm.

Proof The proof is analogous to that of Theorem 14.1.

If we let

$$W_t = \mathbf{E}_{w \sim p_0(w)} u_t(w),$$

then

$$\begin{aligned}
\ln \frac{W_t}{W_{t-1}} &= \ln \mathbf{E}_{w \sim p_0(w)} \frac{u_t(w)}{W_{t-1}} \\
&= \ln \mathbf{E}_{w \sim p_{t-1}(w)} \exp(\eta r_t(w, x_t)) \\
&\leq \ln \mathbf{E}_{w \sim p_{t-1}(w)} [1 + (\eta r_t(w, x_t)) + \psi(\eta)(\eta r_t(w, x_t))^2] \\
&\leq \mathbf{E}_{w \sim p_{t-1}(w)} (\eta r_t(w, x_t)) + \psi(\eta) \mathbf{E}_{w \sim p_{t-1}(w)} (\eta r_t(w, x_t))^2 \\
&\leq \eta(1 + \psi(\eta)\eta) \mathbf{E}_{w \sim p_{t-1}(w)} r_t(w, x_t) \\
&= \eta(1 + \psi(\eta)\eta) \mathbf{E}_{a \sim p_t(a)} r_t(a)
\end{aligned}$$

where the first inequality used $\exp(z) \leq 1 + z + \psi(\eta)z^2$ when $z \leq \eta$; the second inequality used $\ln(1 + z) \leq z$; and the third inequality used (14.1) and (14.2).

Now by summing over $t = 1$ to $t = T$, and note that $W_0 = 1$, we obtain

$$\ln W_T \leq \eta(1 + \psi(\eta)\eta) \mathbf{E} \sum_{t=1}^T r_t(a_t).$$

Using the same estimate of $\ln W_T$ as that of Theorem 14.1, we obtain the desired bound. $\blacksquare$

Corollary 14.2 *Assumes that $|\Omega| = N < \infty$. Let $p_0(w)$ be the uniform prior over Ω , then*

$$G_* - \mathbf{E} \sum_{t=1}^T r_t(a_t) \leq \frac{\psi(\eta)\eta}{1 + (e - 2)\eta} G_* + \frac{\eta^{-1}}{1 + \psi(\eta)\eta} \ln N,$$

where the expectation is with respect to the randomization of the algorithm, and

$$G_* = \arg \max_w \sum_{t=1}^T r_t(w, x_t).$$

Proof In Theorem 14.2, we simply take q such that $q(w) = I(w = w_*)$, where w_* achieves the maximum of G_* . This leads to the desired bound. $\blacksquare$

Example 14.2 *We consider each “policy” $p(a|w, x)$ as a deterministic classifier that returns $a = f(w, x) \in \{1, \dots, K\}$. At each time t , let the true label of x_t be $y_t \in \{1, \dots, K\}$. The loss for a prediction $f(w, x)$ is the classification error $\ell_t(w) = I(f(w, x_t) \neq y_t)$.*

We can take the reward $r_t(a) = -I(a \neq y_t)$. In this case,

$$r_t(w, x_t) = -I(f(w, x_t) \neq y_t) = -\ell_t(w).$$

Then Algorithm 14.2 becomes

$$a_t = f(w, x_t) \quad w \sim p_{t-1}(w) \propto \exp \left(-\eta \sum_{s=1}^{t-1} \ell_s(w) \right) p_0(w).$$

The regret bound is

$$\sum_{t=1}^T \mathbf{E}_{w \sim p_{t-1}(\cdot)} \ell_t(w) \leq \frac{1}{1 + \psi(\eta)\eta} \min_q \left[\mathbf{E}_{w \sim q} \sum_{t=1}^T \ell_t(w) + \frac{1}{\eta} \text{KL}(q||p_0) \right].$$

The algorithm and the resulting bound can be regarded as an application of the online aggregation algorithm discussed in Chapter 12.

14.2 Linear UCB

EXP4 is a policy based algorithm for contextual bandits. We can also obtain a value function based algorithm, where we consider a model of the expected reward of each arm given the context (also referred to as the value function).

Definition 14.2 *Stochastic Linear Bandit (or stochastic contextual bandit with linear payoff) is a contextual bandit problem, where the reward at each time step t is given by*

$$r_t(a) = r_t(a, x_t) = w_*^\top \psi(a, x_t) + \epsilon_t(a, x_t),$$

where $\epsilon_t(a, x)$ is a zero-mean random variable. Here $w_* \in \mathbb{R}^d$, and the feature vector $\psi(a, x) \in \mathbb{R}^d$.

In the stochastic linear bandit model, the number of arms can be either infinite or finite, and UCB style algorithms for this model have been analyzed in [8, 30, 26]. One version of the resulting UCB algorithm is described in Algorithm 14.3, and was analyzed in [30].

Algorithm 14.3: Ridge Regression UCB Algorithm

Input: $\lambda, T, \{\beta_t\}$

- 1 Let $A_0 = \lambda I$
- 2 Let $w_0 = 0$
- 3 Let $b_0 = 0$
- 4 **for** $t = 1, 2, \dots, T$ **do**
- 5 Observe x_t
- 6 Let $a_t \in \arg \max_a \left[w_{t-1}^\top \psi(a, x_t) + \beta_{t-1} \sqrt{\psi(a, x_t)^\top A_{t-1}^{-1} \psi(a, x_t)} \right]$
- 7 Pull arm a_t and observe reward $r_t(a_t, x_t)$
- 8 Let $b_t = b_{t-1} + r_t(a_t, x_t) \psi(a_t, x_t)$
- 9 Let $A_t = A_{t-1} + \psi(a_t, x_t) \psi(a_t, x_t)^\top$
- 10 Let $w_t = A_t^{-1} b_t$

We note that w_t is the solution of the ridge regression problem

$$w_t = \arg \min_w \left[\sum_{s=1}^t (w^\top \psi(a_s, x_s) - r_s(a_s, x_s))^2 + \lambda \|w\|_2^2 \right].$$

Therefore the regret analysis of ridge regression can be applied. The key lemma to bound confidence interval is the following lemma, which employs the regret analysis of ridge regression in Corollary 12.2.

Lemma 14.1 *Assume that in the stochastic linear bandit model, $r_s(a_s, x_s) \in [0, M]$ for all s , then for any t and $\delta_t \in (0, 1)$, Consider Algorithm 14.3, we have with probability at least $1 - \delta_t$:*

$$(w_t - w_*)^\top A_t(w_t - w_*) \leq \lambda \|w_*\|_2^2 + \frac{M^2 \ln(1/\delta_t)}{2} + M^2 \ln |A_t/\lambda|.$$

Proof We start with Corollary 12.2, which implies that

$$\sum_{s=1}^t (\hat{r}_s - r_s(a_s, x_s))^2 \leq \sum_{s=1}^t (w_t^\top \psi(a_s, x_s) - r_s(a_s, x_s))^2 + \lambda \|w_t\|_2^2 + M^2 \ln |A_t/\lambda|,$$

where $\hat{r}_s = \max(0, \min(M, w_{s-1}^\top \psi(a_s, x_s)))$. Using $\bar{r}_s = w_*^\top \psi(a_s, x_s) \in [0, M]$ and $r_s(a_s, x_s) = \bar{r}_s + \epsilon_s(a_s, x_s)$, we obtain

$$\begin{aligned} & \sum_{s=1}^t (w_t^\top \psi(a_s, x_s) - r_s(a_s, x_s))^2 + \lambda \|w_t\|_2^2 \\ &= \sum_{s=1}^t (\bar{r}_s - r_s(a_s, x_s))^2 + \lambda \|w_*\|_2^2 - (w_* - w_t)^\top A_t(w_* - w_t), \end{aligned}$$

where we use the fact that w_t is the solution of ridge regression. It follows that

$$\begin{aligned} & \sum_{s=1}^t (\hat{r}_s - r_s(a_s, x_s))^2 \\ & \leq \sum_{s=1}^t (\bar{r}_s - r_s(a_s, x_s))^2 + \lambda \|w_*\|_2^2 - (w_* - w_t)^\top A_t(w_* - w_t) + M^2 \ln |A_t/\lambda|. \end{aligned}$$

This is equivalent to:

$$\begin{aligned} & \sum_{s=1}^t [(\hat{r}_s - \bar{r}_s)^2 - 2(\hat{r}_s - \bar{r}_s)\epsilon_s(a_s, x_s)] \\ & \leq \lambda \|w_*\|_2^2 - (w_* - w_t)^\top A_t(w_* - w_t) + M^2 \ln |A_t/\lambda|. \end{aligned} \quad (14.3)$$

We note that

$$\sum_{s=1}^t (\hat{r}_s - \bar{r}_s) \epsilon_s(a_s, x_s)$$

is a Martingale, and the conditional exponential moment generating function is sub-Gaussian

$$\ln \mathbf{E}_{\epsilon_s(a_s, x_s)} \exp(\gamma \alpha_s \epsilon_s(a_s, x_s)) \leq \frac{\gamma^2 (\hat{r}_s - \bar{r}_s)^2 M^2}{8}.$$

We can apply Theorem 11.3 and obtain for any $\gamma > 0$, with probability at least $1 - \delta_t$:

$$2 \sum_{s=1}^t (\hat{r}_s - \bar{r}_s) \epsilon_s(a_s, x_s) \leq \frac{2 \ln(1/\delta_t)}{\gamma} + \frac{\gamma M^2 \sum_{s=1}^t (\hat{r}_s - \bar{r}_s)^2}{4}.$$

Let $\gamma = 4/M^2$. Then combined with (14.3), we obtain the desired bound. ■

We can obtain a uniform confidence interval for the linear bandit problem using Lemma 14.1. The bound holds uniformly for all arms (possibly infinitely many) and both for lower and upper bounds.

Lemma 14.2 *Assume that in the stochastic linear bandit model, $r_t(a_t, x_t) \in [0, M]$ for all t . Consider Algorithm 14.3. Assume further that $\|w_*\|_2 \leq B$ for some constant B . Then with probability at least $1 - \delta$, for all $t \geq 0$ and $u \in \mathbb{R}^d$:*

$$|u^\top (w_t - w_*)| \leq \beta_t \sqrt{u^\top A_t^{-1} u},$$

where $\{\beta_t\}$ is any sequence so that

$$\beta_t^2 \geq (\lambda B^2 + 0.5 M^2 \ln((t+1)^2/\delta) + M^2 \ln |A_t/\lambda|).$$

Proof Lemma 14.1 implies that with probability $\delta/(t+1)^2$, we have

$$(w_t - w_*)^\top A_t (w_t - w_*) \leq \beta_t^2.$$

Therefore taking union bound, we know that with probability $1 - \delta$, for all $t \geq 1$: $(w_t - w_*)^\top A_t (w_t - w_*) \leq \beta_t^2$. Moreover, the bound also holds trivially at $t = 0$.

Since for all $u \in \mathbb{R}^d$, the Cauchy-Schwarz inequality implies that

$$|u^\top (w_t - w_*)|^2 \leq u^\top A_t^{-1} u \beta_t^2,$$

we obtain the desired result. ■

We also need the following lemma, which can be used to bound the sum of confidence intervals.

Lemma 14.3 *In Algorithm 14.3, Assume that $\sup_t \|\psi(a_t, x_t)\|_2^2 \leq \rho\lambda$ for some $\rho > 0$, then we have the following estimate*

$$\sum_{t=1}^T \psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t) \leq (1 + \rho) \ln |A_T/A_0|.$$

Proof We note that the assumption implies that

$$A_t = A_{t-1} + \psi(a_t, x_t)\psi(a_t, x_t)^\top \leq A_{t-1} + \rho A_0 \leq (1 + \rho)A_{t-1}.$$

Therefore

$$\begin{aligned} \psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t) &\leq (1 + \rho) \psi(a_t, x_t)^\top A_t^{-1} \psi(a_t, x_t) \\ &= (1 + \rho) \text{trace}(A_t^{-1} (A_t - A_{t-1})) \\ &\leq (1 + \rho) [\ln |A_t| - \ln |A_{t-1}|], \end{aligned}$$

where the last inequality is due to Proposition 12.7. We can now sum over $t = 1$ to $t = T$, and obtain the desired bound. $\blacksquare$

The confidence interval estimate in Lemma 14.2 can be used to obtain regret bound for the upper confidence bound method in Algorithm 14.3, and the proof is similar to that of Theorem 13.1.

Theorem 14.3 *Assume that in the stochastic linear bandit model, $r_t(a_t, x_t) \in [0, M]$ for all t . Let $\mu_t(a, x) = \mathbf{E}_{\epsilon_t(a, x)} r_t(a, x) = w_*^\top \psi(a, x)$. Let $a_*(x) \in \arg \max_a \mu_t(a, x)$ be the optimal arm for each context x .*

Consider Algorithm 14.3. Assume further that $\|w_\|_2 \leq B$ for some constant B and $\sup_t \|\psi(a_t, x_t)\|_2^2 \leq \rho\lambda$ for some $\rho > 0$. Then with probability at least $1 - \delta$:*

$$\mathbf{E} \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] \leq 2\sqrt{(1 + \rho)T} \max_{t \leq T-1} \beta_t \sqrt{\ln |A_T/\lambda|},$$

where the expectation is with respect to the random in $\epsilon_t(a_t, s_t)$, and $\{\beta_t\}$ is any sequence so that

$$\beta_t^2 \geq (\lambda B^2 + 0.5M^2 \ln((t+1)^2/\delta) + M^2 \ln |A_t/\lambda|).$$

Proof We have for $t \geq 1$:

$$\begin{aligned} &w_*^\top \psi(a_*(x_t), x_t) \\ &\leq w_{t-1}^\top \psi(a_*(x_t), x_t) + \beta_{t-1} \sqrt{\psi(a_*(x_t), x_t)^\top A_{t-1}^{-1} \psi(a_*(x_t), x_t)} \\ &\leq w_{t-1}^\top \psi(a_t, x_t) + \beta_{t-1} \sqrt{\psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t)} \\ &\leq w_*^\top \psi(a_t, x_t) + 2\beta_{t-1} \sqrt{\psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t)}, \end{aligned}$$

where the first and the third inequalities use Lemma 14.2 . The second inequality is due to the definition of a_t in Algorithm 14.3. By summing over $t = 1$ to $t = T$, we obtain

$$\begin{aligned}
& \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] \\
& \leq 2 \sum_{t=1}^T \beta_{t-1} \sqrt{\psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t)} \\
& \leq 2 [\max_{t \leq T-1} \beta_t] \sum_{t=1}^T \sqrt{\psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t)} \\
& \leq 2\sqrt{T} [\max_{t \leq T-1} \beta_t] \sqrt{\sum_{t=1}^T \psi(a_t, x_t)^\top A_{t-1}^{-1} \psi(a_t, x_t)}.
\end{aligned}$$

We can now apply Lemma 14.3 to obtain the desired bound. ■

In Theorem 14.3, if we simply take

$$\beta_t = \sqrt{\lambda B^2 + 0.5M^2 \ln((t+1)^2/\delta) + M^2 \ln |A_t/\lambda|},$$

then

$$\begin{aligned}
& \mathbf{E} \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] \\
& \leq \sqrt{4(1+\rho)T(\lambda B^2 + 0.5M^2 \ln(T^2/\delta) + M^2 \ln |A_T/\lambda|) \ln |A_T/\lambda|}.
\end{aligned}$$

If we employ the log-determinant estimator of Proposition 12.3, then we obtain

$$\begin{aligned}
& \mathbf{E} \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] \\
& \leq \sqrt{4(1+\rho)T(\lambda B^2 + 0.5M^2 \ln(T^2/\delta) + dM^2 \ln(1+T/d))d \ln(1+T/d)}.
\end{aligned}$$

Ignoring the logarithmic factor, we obtain a worse case regret bound of

$$\mathbf{E} \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] = O(d\sqrt{T}).$$

For stochastic linear bandit with possibly infinity many arms, this bound is optimal (up to logarithmic factor) [30]. However, for finite arm problems, this bound is suboptimal.

Example 14.3 *The stochastic multi-armed bandit with K arms and rewards in $[0, 1]$ can be considered as a stochastic linear bandit, where we take $w_* = [\mu(1), \dots, \mu(K)]$, and $\psi(a, x) = e_a$ for $a \in \{1, \dots, K\}$. Therefore we may choose $B = \sqrt{K}$, so that $\|w_*\|_2 \leq B$. We can also choose $\lambda = 1$ and $M = 1$. Theorem 14.3 implies a suboptimal bound of (ignoring log factors):*

$$\mathbf{E} \sum_{t=1}^T [\mu_t(a_*(x_t), x_t) - \mu_t(a_t, x_t)] = O(K\sqrt{T}).$$

The extra $\sqrt{K}$ factor is due to the fact that the analysis does not take advantage of the fact that only finite number of arms are available.

The extra dependency on d (and K) is due to the uniform confidence interval over all $u \in \mathbb{R}^d$ in Lemma 14.2. To obtain better dependence on K , one needs to obtain confidence interval which is for a fixed u , instead of uniform over all $u \in \mathbb{R}^d$. One difficulty for obtaining non-uniform result for in Algorithm 14.3 is that A_t depends on $\epsilon_s(a_s, t_s)$ for $s \leq t$, which breaks the Martingale argument.

A more complex scheme, presented in [8, 26], can be used to derive such a non-uniform convergence result, which leads to improve bound for stochastic linear bandits with finitely many arms. We will briefly explain the key idea of the method below. We note that the confidence interval $u^\top A_t^{-1} u$ depends only on A_t , but not $\epsilon_s(a_s, x_s)$ for $s \leq t$. This means that if we select arms with large confidence interval, then the selection is independent of the noise $\epsilon_s(a_s, x_s)$, and Martingale argument can be applied to obtain a sharper estimate of $|u^\top (w_t - w_*)|$ as $O(\sqrt{u^\top A_t^{-1} u})$, which is without d -dependency, and better than the uniform convergence result in Lemma 14.2 (which has $\sqrt{d}$ dependency in β_t). Taking union bound over K arms will introduce $\log K$ factor, but saves a $\sqrt{d}$ factor.

This observation allows us to construct an arm elimination algorithm in S stages with shrinking confidence intervals. Each stage s maintains a confidence interval of size $\Delta_s = 2^{-s}$ and eliminates arms that are at least $2\Delta_s$ suboptimal from a subset of arms $A_{s,t}$ at each time step t , which is passed from stage $s - 1$:

- Select arms with confidence intervals larger than Δ_s until no such arms can be found. This requires no more than $\tilde{O}(d/\Delta_s^2)$ trials until all arms have confidence intervals no larger than Δ_s . Since the selection does not involve upper confidence bound, this maintains the independence of $\epsilon_t(a_t, x_t)$.
- Eliminate arms in $A_{s,t}$ with upper confidence bound that is Δ_s from optimal. They can be eliminated because such arms cannot be optimal.
- The remaining arms in $A_{s,t}$ forms $A_{s+1,t}$, and passed to stage $s + 1$.

Note that the regret cost of each trial at stage s is $O(\Delta_s)$, and thus total regret cost at stage s is $O(d/\Delta_s)$. The total elimination algorithm has regret of

$$\sum_{s=1}^S \tilde{O}(d/\Delta_s) + O(T\Delta_S),$$

and with appropriate S , the regret is $\tilde{O}(\sqrt{dT})$ ignoring log factor. This multi-stage elimination process can be used to obtain a more refined results with $\ln K$ dependency on the number of arms.

If we apply the bound to the multi-armed bandit problem, we obtain $\tilde{O}(\sqrt{KT})$ which is comparable to bounds presented in Chapter 13.

14.3 Thompson Sampling

It is also relatively easy to apply Thompson Sampling to contextual bandit [4, 70]. For example, it can be applied to the stochastic linear bandit problem as follows. Note that ridge regression method used in Algorithm 14.3 has a Bayesian interpretation: the prior is

$$w \sim p_0(w) = N(0, (\sigma^2/\lambda)I),$$

and the reward likelihood is:

$$r_t(a, x) \sim N(w^\top \psi(a, x), \sigma^2).$$

At the beginning of any time step t , after we have observed data up to time $t - 1$, the posterior is

$$w \sim N(w_{t-1}, \Sigma_{t-1}),$$

where

$$\sigma^2 \Sigma_{t-1}^{-1} = \lambda I + \sum_{s < t} \psi(a_s, x_s) \psi(a_s, x_s)^\top.$$

The Thompson sampling method samples $\tilde{w}$ from the posterior and plays an arm that is optimal with respect to $\tilde{w}$. The algorithm is presented in Algorithm 14.4. The regret bound obtained in [4] is superlinear in d , and hence inferior to that of the ridge regression with UCB. However, practical behavior of Thompson sampling can be better than what's indicated by the theoretical result [24]. It is also relatively easy to apply Thompson sampling to non-Gaussian models such as logistic regression, with posterior estimated

using approximation methods.

Algorithm 14.4: Ridge Regression Thompson Sampling Algorithm

Input: λ, σ, T

- 1 Let $A_0 = \lambda I$
- 2 Let $w_0 = 0$
- 3 Let $b_0 = 0$
- 4 **for** $t = 1, 2, \dots, T$ **do**
- 5 Sample $\tilde{w}_{t-1} \sim N(w_{t-1}, \sigma^2 A_{t-1}^{-1})$
- 6 Observe x_t
- 7 Let $a_t \in \arg \max_a [\tilde{w}_{t-1}^\top \psi(a, x_t)]$
- 8 Pull arm a_t and observe reward $r_t(a_t, x_t)$
- 9 Let $b_t = b_{t-1} + r_t(a_t, x_t) \psi(a_t, x_t)$
- 10 Let $A_t = A_{t-1} + \psi(a_t, x_t) \psi(a_t, x_t)^\top$
- 11 Let $w_t = A_t^{-1} b_t$

14.4 Exercises

1. Consider $\psi(a, x) = [\psi_1(a, x), \dots, \psi_d(a, x)]$, so that

$$\psi_k(a, x) \leq 2^{-k/2}.$$

Theorem 14.3, let $\lambda = 1$. Use Courant minimax principle to upper bound eigenvalues of A_T , and show that

$$\ln |A_T/\lambda| = O((\ln T)^2),$$

which is independent of d . What is the resulting regret bound?

2. In Algorithm 14.3 for stochastic linear contextual bandits, we consider a full exploration scheme where we choose arm in line 6 according to confidence as

$$a_t \in \arg \max_a \psi(a, x_t)^\top A_{t-1}^{-1} \psi(a, x_t).$$

Similar to Lemma 14.2, we would like to derive an upper confidence bound after step T for any fixed $u \in \mathbb{R}^d$ rather than uniformly for all u . Given any fixed $u \in \mathbb{R}^d$, we want to show that with probability at least $1 - \delta$:

$$|u^\top (w_* - w_T)| \leq \tilde{\beta}_T \sqrt{u^\top A_T^{-1} u}$$

with $\tilde{\beta}_T$ independent of d .

- Show that for appropriately defined γ_t ,

$$u^\top (w_* - w_T) = \lambda u^\top A_T^{-1} w_* + \sum_{t=1}^T \gamma_t \epsilon_t(a_t, x_t),$$

where $\{\gamma_t\}$ and $\{\epsilon_t(a_t, x_t)\}$ are independent.

- Use Azuma's inequality to obtain a concentration bound so that

$$\Pr \left[|u^\top (w_* - w_T)| > \tilde{\beta}_T \sqrt{u^\top A_T^{-1} u} \right] \leq \delta,$$

with appropriately chosen $\tilde{\beta}_T$ independent of d . As in Lemma 14.2, we assume that $\|w_*\|_2 \leq B$ with known B , and for all t , $r_t(a_t, x_t) \in [0, M]$.

- Assume we have K arms. Define $\tilde{\beta}_T$ so that with probability at least $1 - \delta$, the following inequality holds uniformly for all $a \in \{1, \dots, K\}$

$$|\psi(a, x_{T+1})^\top (w_* - w_T)| \leq \tilde{\beta}_T \sqrt{\psi(a, x_{T+1})^\top A_T^{-1} \psi(a, x_{T+1})}.$$

Compare this bound to that of Lemma 14.2. Why this better $\tilde{\beta}_T$ cannot replace β_T for the original Algorithm 14.3?

Bibliography

- [1] Rajeev Agrawal. Sample mean based index policies with $o(\log n)$ regret for the multi-armed bandit problem. *Advances in Applied Probability*, pages 1054–1078, 1995.
- [2] Shipra Agrawal and Navin Goyal. Analysis of Thompson sampling for the multi-armed bandit problem. In *Conference on learning theory*, 2012.
- [3] Shipra Agrawal and Navin Goyal. Further optimal regret bounds for Thompson sampling. In *Artificial intelligence and statistics*, pages 99–107, 2013.
- [4] Shipra Agrawal and Navin Goyal. Thompson sampling for contextual bandits with linear payoffs. In *International Conference on Machine Learning*, pages 127–135, 2013.
- [5] Patrice Assouad. Deux remarques sur l’estimation. *Comptes rendus des séances de l’Académie des sciences. Série 1, Mathématique*, 296(23):1021–1024, 1983.
- [6] Jean-Yves Audibert. Progressive mixture rules are deviation suboptimal. *Advances in neural information processing systems*, 20:41–48, 2007.
- [7] Jean-Yves Audibert and Sébastien Bubeck. Minimax policies for adversarial and stochastic bandits. In *Conference on learning theory*, 2009.
- [8] Peter Auer. Using confidence bounds for exploitation-exploration trade-offs. *Journal of Machine Learning Research*, 3(Nov):397–422, 2002.
- [9] Peter Auer, Nicolo Cesa-Bianchi, and Paul Fischer. Finite-time analysis of the multiarmed bandit problem. *Machine learning*, 47(2-3):235–256, 2002.

- [10] Peter Auer, Nicolo Cesa-Bianchi, Yoav Freund, and Robert E Schapire. The nonstochastic multiarmed bandit problem. *SIAM journal on computing*, 32(1):48–77, 2002.
- [11] Kazuoki Azuma. Weighted sums of certain dependent random variables. *Tohoku Math. J. (2)*, 19(3):357–367, 1967.
- [12] Remi Bardenet and Odalric-Ambrym Maillard. Concentration inequalities for sampling without replacement. *Bernoulli*, 21(3):1361–1385, 08 2015.
- [13] Andrew R Barron. Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information theory*, 39(3):930–945, 1993.
- [14] Peter L Bartlett, Olivier Bousquet, Shahar Mendelson, et al. Local rademacher complexities. *The Annals of Statistics*, 33(4):1497–1537, 2005.
- [15] Peter L Bartlett and Shahar Mendelson. Rademacher and gaussian complexities: Risk bounds and structural results. *Journal of Machine Learning Research*, 3(Nov):463–482, 2002.
- [16] George Bennett. Probability inequalities for the sum of independent random variables. *Journal of the American Statistical Association*, 57(297):33–45, 1962.
- [17] Sergei Bernstein. On a modification of chebyshev’s inequality and of the error formula of laplace. *Ann. Sci. Inst. Sav. Ukraine, Sect. Math*, 1(4):38–49, 1924.
- [18] Peter Bickel, Yaacov Ritov, and Alexandre Tsybakov. Simultaneous analysis of Lasso and Dantzig selector. *Annals of Statistics*, 37(4):1705–1732, 2009.
- [19] Lucien Birgé. On estimating a density using hellinger distance and some other strange facts. *Probability theory and related fields*, 71(2):271–291, 1986.
- [20] Lucien Birgé and Pascal Massart. Rates of convergence for minimum contrast estimators. *Probability Theory and Related Fields*, 97(1-2):113–150, 1993.
- [21] Lucien Birgé, Pascal Massart, et al. Minimum contrast estimators on sieves: exponential bounds and rates of convergence. *Bernoulli*, 4(3):329–375, 1998.

- [22] Olivier Bousquet. A bennett concentration inequality and its application to suprema of empirical processes. *Comptes Rendus Mathématique*, 334(6):495–500, 2002.
- [23] Emmanuel J. Candes and Terence Tao. Decoding by linear programming. *IEEE Trans. on Information Theory*, 51:4203–4215, 2005.
- [24] Olivier Chapelle and Lihong Li. An empirical evaluation of Thompson sampling. In *Advances in neural information processing systems*, pages 2249–2257, 2011.
- [25] Lenaïc Chizat and Francis Bach. On the global convergence of gradient descent for over-parameterized models using optimal transport. In *Advances in neural information processing systems*, pages 3036–3046, 2018.
- [26] Wei Chu, Lihong Li, Lev Reyzin, and Robert Schapire. Contextual bandits with linear payoff functions. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, pages 208–214, 2011.
- [27] Michael Collins. Discriminative training methods for hidden markov models: Theory and experiments with perceptron algorithms. In *Proceedings of the 2002 conference on empirical methods in natural language processing (EMNLP 2002)*, pages 1–8, 2002.
- [28] Corinna Cortes and Vladimir Vapnik. Support-vector networks. *Machine learning*, 20(3):273–297, 1995.
- [29] Harald Cramér. *Mathematical methods of statistics*, volume 43. Princeton university press, 1999.
- [30] Varsha Dani, Thomas P Hayes, and Sham M Kakade. Stochastic linear optimization under bandit feedback, 2008.
- [31] John Duchi, Elad Hazan, and Yoram Singer. Adaptive subgradient methods for online learning and stochastic optimization. *Journal of machine learning research*, 12(7), 2011.
- [32] Miroslav Dudík, Daniel Hsu, Satyen Kale, Nikos Karampatziakis, John Langford, Lev Reyzin, and Tong Zhang. Efficient optimal learning for contextual bandits. 2011.
- [33] Richard M Dudley. A course on empirical processes. In *Ecole d’été de Probabilités de Saint-Flour XII-1982*, pages 1–142. Springer, 1984.
- [34] Robert M Fano. Transmission of information: A statistical theory of communications. *American Journal of Physics*, 29(11):793–794, 1961.

- [35] Marguerite Frank, Philip Wolfe, et al. An algorithm for quadratic programming. *Naval research logistics quarterly*, 3(1-2):95–110, 1956.
- [36] David A Freedman. On tail probabilities for martingales. *the Annals of Probability*, pages 100–118, 1975.
- [37] Yoav Freund and Robert E Schapire. A decision-theoretic generalization of on-line learning and an application to boosting. *Journal of computer and system sciences*, 55(1):119–139, 1997.
- [38] Jerome H. Friedman. Greedy function approximation: a gradient boosting machine. *Ann. Statist.*, 29(5):1189–1232, 2001.
- [39] Aurélien Garivier and Olivier Cappé. The kl-ucb algorithm for bounded stochastic bandits and beyond. In *Proceedings of the 24th annual conference on learning theory*, pages 359–376, 2011.
- [40] Stuart Geman and Chii-Ruey Hwang. Nonparametric maximum likelihood estimation by the method of sieves. *The Annals of Statistics*, pages 401–414, 1982.
- [41] Ulf Grenander. Abstract inference. Technical report, 1981.
- [42] Elad Hazan, Amit Agarwal, and Satyen Kale. Logarithmic regret algorithms for online convex optimization. *Machine Learning*, 69(2-3):169–192, 2007.
- [43] W. Hoeffding. Probability inequalities for sums of bounded random variables. *Journal of the American Statistical Association*, 58(301):13–30, March 1963.
- [44] Daniel Hsu, Sham M Kakade, and Tong Zhang. Random design analysis of ridge regression. In *Conference on learning theory*, 2012.
- [45] Arthur Jacot, Franck Gabriel, and Clément Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *Advances in neural information processing systems*, 2018.
- [46] Kumar Joag-Dev, Michael D Perlman, and Loren D Pitt. Association of normal random variables and slepian’s inequality. *The Annals of Probability*, pages 451–455, 1983.
- [47] Emilie Kaufmann, Nathaniel Korda, and Rémi Munos. Thompson sampling: An asymptotically optimal finite-time analysis. In *International conference on algorithmic learning theory*, pages 199–213. Springer, 2012.

- [48] Jyrki Kivinen and Manfred K Warmuth. Averaging expert predictions. In *European Conference on Computational Learning Theory*, pages 153–167. Springer, 1999.
- [49] Vladimir Koltchinskii. Rademacher penalties and structural risk minimization. *IEEE Transactions on Information Theory*, 47(5):1902–1914, 2001.
- [50] Vladimir Koltchinskii and Dmitry Panchenko. Empirical margin distributions and bounding the generalization error of combined classifiers. *The Annals of Statistics*, 30(1):1–50, 2002.
- [51] Tze Leung Lai and Herbert Robbins. Asymptotically efficient adaptive allocation rules. *Advances in applied mathematics*, 6(1):4–22, 1985.
- [52] John Langford and Tong Zhang. The Epoch-Greedy algorithm for multi-armed bandits with side information. In *NIPS’07*, 2007.
- [53] Guillaume Lecué. Suboptimality of penalized empirical risk minimization in classification. In *International Conference on Computational Learning Theory*, pages 142–156. Springer, 2007.
- [54] Michel Ledoux and Michel Talagrand. *Probability in Banach Spaces: isoperimetry and processes*. Springer Science & Business Media, 2013.
- [55] Moshe Leshno, Vladimir Ya Lin, Allan Pinkus, and Shimon Schocken. Multilayer feedforward networks with a nonpolynomial activation function can approximate any function. *Neural networks*, 6(6):861–867, 1993.
- [56] Odalric-Ambrym Maillard, Rémi Munos, and Gilles Stoltz. A finite-time analysis of multi-armed bandits problems with kullback-leibler divergences. In *Proceedings of the 24th annual Conference On Learning Theory*, pages 497–514, 2011.
- [57] David McAllester. PAC-Bayesian model averaging. In *COLT’99*, pages 164–170, 1999.
- [58] Colin McDiarmid. On the method of bounded differences. *Surveys in combinatorics*, 141(1):148–188, 1989.
- [59] Song Mei, Theodor Misiakiewicz, and Andrea Montanari. Mean-field theory of two-layers neural networks: dimension-free bounds and kernel limit. In *Annual Conference on Learning Theory*, 2019.
- [60] Song Mei, Andrea Montanari, and Phan-Minh Nguyen. A mean field view of the landscape of two-layer neural networks. *Proceedings of the National Academy of Sciences*, 115(33):E7665–E7671, 2018.

- [61] Ron Meir and Tong Zhang. Generalization error bounds for bayesian mixture algorithms. *Journal of Machine Learning Research*, 4(Oct):839–860, 2003.
- [62] Yurii Nesterov. *Introductory lectures on convex optimization: A basic course*, volume 87. Springer Science & Business Media, 2013.
- [63] Albert B Novikoff. On convergence proofs for perceptrons. Technical report, STANFORD RESEARCH INST MENLO PARK CA, 1963.
- [64] Jooyoung Park and Irwin W Sandberg. Universal approximation using radial-basis-function networks. *Neural computation*, 3(2):246–257, 1991.
- [65] G. Pisier. Remarques su un resultat non publié de b. maurey. In *Séminaire d'Analyse Fonctionnelle*, volume 1, École Polytechnique, Centre de Mathematiques, Palaiseau, 1980-1981.
- [66] Ali Rahimi and Benjamin Recht. Uniform approximation of functions with random bases. In *2008 46th Annual Allerton Conference on Communication, Control, and Computing*, pages 555–561. IEEE, 2008.
- [67] Ali Rahimi and Benjamin Recht. Weighted sums of random kitchen sinks: Replacing minimization with randomization in learning. In *Advances in neural information processing systems*, 2009.
- [68] Frank Rosenblatt. *The perceptron, a perceiving and recognizing automaton Project Para*. Cornell Aeronautical Laboratory, 1957.
- [69] Frank Rosenblatt. *Principles of Neurodynamics: Perceptrons and the Theory of Brain Mechanisms*. Spartan, New York, 1962.
- [70] Daniel Russo and Benjamin Van Roy. An information-theoretic analysis of thompson sampling. *The Journal of Machine Learning Research*, 17(1):2442–2471, 2016.
- [71] Norbert Sauer. On the density of families of sets. *Journal of Combinatorial Theory, Series A*, 13(1):145–147, 1972.
- [72] R.E. Schapire, Y. Freund, P.L. Bartlett, and W.S. Lee. Boosting the margin: a new explanation for the effectiveness of voting methods. *Ann. Statist.*, 26(5):1651–1686, 1998.
- [73] Bernhard Scholkopf, Alexander J Smola, and Francis Bach. *Learning with kernels: support vector machines, regularization, optimization, and beyond*. the MIT Press, 2018.

- [74] Gideon Schwarz et al. Estimating the dimension of a model. *The annals of statistics*, 6(2):461–464, 1978.
- [75] R. J. Serfling. Probability inequalities for the sum in sampling without replacement. *Ann. Statist.*, 2(1):39–48, 01 1974.
- [76] Shai Shalev-Shwartz, Yoram Singer, Nathan Srebro, and Andrew Cotter. Pegasos: Primal estimated sub-gradient solver for svm. *Mathematical programming*, 127(1):3–30, 2011.
- [77] Shai Shalev-Shwartz, Nathan Srebro, and Tong Zhang. Trading accuracy for sparsity in optimization problems with sparsity constraints. *Siam Journal on Optimization*, 20:2807–2832, 2010.
- [78] Ohad Shamir and Tong Zhang. Stochastic gradient descent for non-smooth optimization: Convergence results and optimal averaging schemes. In *International conference on machine learning*, pages 71–79, 2013.
- [79] David Slepian. The one-sided barrier problem for gaussian noise. *Bell System Technical Journal*, 41(2):463–501, 1962.
- [80] Michel Talagrand. Sharper bounds for gaussian and empirical processes. *The Annals of Probability*, pages 28–76, 1994.
- [81] Michel Talagrand. New concentration inequalities in product spaces. *Inventiones mathematicae*, 126(3):505–563, 1996.
- [82] Michel Talagrand et al. Majorizing measures: the generic chaining. *The Annals of Probability*, 24(3):1049–1103, 1996.
- [83] Matus Telgarsky. Benefits of depth in neural networks. In *Annual Conference on Learning Theory*, 2016.
- [84] William R Thompson. On the likelihood that one unknown probability exceeds another in view of the evidence of two samples. *Biometrika*, 25(3/4):285–294, 1933.
- [85] Robert Tibshirani. Regression shrinkage and selection via the lasso. *Journal of the Royal Statistical Society: Series B (Methodological)*, 58(1):267–288, 1996.
- [86] Joel A. Tropp. Greed is good: Algorithmic results for sparse approximation. *IEEE Trans. Info. Theory*, 50(10):2231–2242, 2004.
- [87] Ioannis Tsochantaridis, Thorsten Joachims, Thomas Hofmann, and Yasemin Altun. Large margin methods for structured and interdependent output variables. *Journal of machine learning research*, 6(Sep):1453–1484, 2005.

- [88] Leslie G Valiant. A theory of the learnable. *Communications of the ACM*, 27(11):1134–1142, 1984.
- [89] Vaart Aad W Van and Wellner John A. *Weak Convergence and Empirical Processes with Application to Statistics*. Springer Verlage, 1996.
- [90] Vladimir N. Vapnik and Aleksei Y. Chervonenkis. The uniform convergence of frequencies of the appearance of events to their probabilities. In *Doklady Akademii Nauk*, volume 181, pages 781–783. Russian Academy of Sciences, 1968.
- [91] Volodya Vovk. Competitive on-line statistics. *International Statistical Review*, 69(2):213–248, 2001.
- [92] Yuhong Yang. Minimax nonparametric classification. i. rates of convergence. *IEEE Transactions on Information Theory*, 45(7):2271–2284, 1999.
- [93] Yuhong Yang and Andrew Barron. Information-theoretic determination of minimax rates of convergence. *Annals of Statistics*, pages 1564–1599, 1999.
- [94] Tong Zhang. Covering number bounds of certain regularized linear function classes. *Journal of Machine Learning Research*, 2:527–550, 2002.
- [95] Tong Zhang. Sequential greedy approximation for certain convex optimization problems. *IEEE Transaction on Information Theory*, 49:682–691, 2003.
- [96] Tong Zhang. Solving large scale linear prediction problems using stochastic gradient descent algorithms. In *Proceedings of the twenty-first international conference on Machine learning*, page 116, 2004.
- [97] Tong Zhang. Data dependent concentration bounds for sequential prediction algorithms. In *COLT 05*, 2005.
- [98] Tong Zhang. Information theoretical upper and lower bounds for statistical estimation. *IEEE Trans. Info. Theory*, 52:1307–1321, 2006.
- [99] Tong Zhang. On the consistency of feature selection using greedy least squares regression. *Journal of Machine Learning Research*, 10:555–568, 2009.
- [100] Tong Zhang. Sparse recovery with orthogonal matching pursuit under RIP. *IEEE Transactions on Information Theory*, 57:6215 – 6221, 2011.
- [101] Tong Zhang and Bin Yu. Boosting with early stopping: Convergence and consistency. *The Annals of Statistics*, 33:1538–1579, 2005.

- [102] Peng Zhao and Bin Yu. On model selection consistency of Lasso. *Journal of Machine Learning Research*, 7:2541–2567, 2006.
- [103] Martin Zinkevich. Online convex programming and generalized infinitesimal gradient ascent. In *Proceedings of the 20th international conference on machine learning (icml-03)*, pages 928–936, 2003.